中国科学技术大学 USTC

智能决策博弈与数字创新经济安徽省重点实验室

Anhui Province Key Laboratory of Intelligent Decision Games and Digital Economic Advancements

IDEA 重点实验室内部论文阅读集

优化学习讨论班

2026 年 11 月

目录

1. 柔性车间调度：Learning Memory-Enhanced Improvement Heuristics for Flexible Job Shop Scheduling (NeurIPS, 2025) 3

2. 数据驱动决策：From Contextual Data to Newsvendor Decisions: On the Actual Performance of Data-Driven Algorithms (MS, 2026) 4

3. 库存学习：Learning in Lost-Sales Inventory Systems with Stochastic Lead Times and Random Supplies (MS) 5

4. 学习分支：Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly (OR, 2026) 6

5. 排序与选择：Data-Driven Ranking and Selection Under Input Uncertainty (OR, 2024) 7

6. 混流装配线：Mixed-model assembly lines with variable takt and open stations (POM, 2023) 8

7. 异质环境决策：Beyond IID: Data-Driven Decision Making in Heterogeneous Environments (MS, 2025) 9

1. 柔性车间调度：Learning Memory-Enhanced Improvement Heuristics for Flexible Job Shop Scheduling (NeurIPS, 2025)

来源：Advances in Neural Information Processing Systems. Vol. 38, 2025. https://papers.neurips.cc/paper_files/paper/2025/hash/77629f2df5ef97ad967157c69cd8af7b-Abstract-Conference.html

作者：Jiaqi Wang, Zhiguang Cao, Peng Zhao, Rui Cao, Yubin Xiao, Yuan Jiang, You Zhou

关键词：柔性作业车间；深度强化学习；异构图；记忆增强；改进启发式

推荐人：李淑宇

推荐理由：现有深度强化学习调度方法多从头构造解，较难逼近最优；迭代改进法则在柔性机器分配下遇到状态表征与搜索困难。本文提出MIStar框架，以异构析取图显式描述机器上的工序顺序，并用记忆增强异构图神经网络利用历史搜索轨迹进行特征提取。框架结合并行贪心搜索，减少达到优质解所需的迭代次数。在合成数据与公开基准上的实验中，该方法优于所比较的人工设计改进启发式和先进的强化学习构造方法。

arXiv:2603.02846v1 [cs.LG] 3 Mar 2026

Learning Memory-Enhanced Improvement Heuristics for Flexible Job Shop Scheduling

Jiaqi Wang[1,2], Zhiguang Cao[4], Peng Zhao[1,3]*, Rui Cao[1,3], Yubin Xiao[1,3], Yuan Jiang[5]*, You Zhou[1,2,3]*

[1]Key Laboratory of Symbolic Computation and Knowledge Engineering of Ministry of Education, Jilin University
[2]College of Software, Jilin University
[3]College of Computer Science and Technology, Jilin University
[4]Singapore Management University [5]Nanyang Technological University

{jqwang24, pengzhao23, ruicao24, xiaoyb21}@mails.jlu.edu.cn
zhiguangcao@outlook.com, yuan005@e.ntu.edu.sg, zyou@jlu.edu.cn

Abstract

The rise of smart manufacturing under Industry 4.0 introduces mass customization and dynamic production, demanding more advanced and flexible scheduling techniques. The flexible job-shop scheduling problem (FJSP) has attracted significant attention due to its complex constraints and strong alignment with real-world production scenarios. Current deep reinforcement learning (DRL)-based approaches to FJSP predominantly employ constructive methods. While effective, they often fall short of reaching (near-)optimal solutions. In contrast, improvement-based methods iteratively explore the neighborhood of initial solutions and are more effective in approaching optimality. However, the flexible machine allocation in FJSP poses significant challenges to the application of this framework, including accurate state representation, effective policy learning, and efficient search strategies. To address these challenges, this paper proposes a **M**emory-enhanced **I**mprovement **S**earch framework with he**t**erogeneous gr**a**ph **r**epresentation—*MIStar*. It employs a novel heterogeneous disjunctive graph that explicitly models the operation sequences on machines to accurately represent scheduling solutions. Moreover, a memory-enhanced heterogeneous graph neural network (MHGNN) is designed for feature extraction, leveraging historical trajectories to enhance the decision-making capability of the policy network. Finally, a parallel greedy search strategy is adopted to explore the solution space, enabling superior solutions with fewer iterations. Extensive experiments on synthetic data and public benchmarks demonstrate that *MIStar* significantly outperforms both traditional handcrafted improvement heuristics and state-of-the-art DRL-based constructive methods.

1 Introduction

As an emerging paradigm integrating advanced manufacturing and digital technologies, Industry 4.0 is transforming production toward dynamic, large-scale, and customized manufacturing [1, 2]. The job-shop scheduling problem (JSP) [3], a classic NP-hard combinatorial optimization problem (COP), has been widely studied and plays a critical role in manufacturing systems [4]. As an extension of JSP [5, 6], the flexible job-shop scheduling problem (FJSP) allows each operation to be assigned to one of several eligible machines [7], making it more suitable in handling the flexibility and diversity of

*Corresponding author.

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

task–resource relations in new manufacturing paradigms [8]. While this increased flexibility enables FJSP to better meet the demands of mass customization and automation in smart manufacturing (SM) [5, 9, 10], it also introduces greater challenges in developing effective solution strategies [11].

Recent advances in neural models have shown remarkable effectiveness in solving complex COPs [12, 13, 14, 15, 16, 17, 18]. As one promising paradigm among them, Deep Reinforcement Learning (DRL) formulates the scheduling task as a Markov decision process (MDP), learning a parameterized policy network that receives scheduling state as input and outputs feasible actions. DRL approaches to scheduling can be categorized into *construction* and *improvement* [19, 20]. *Construction* methods build schedules by assigning an operation to a machine at each step, progressively extending partial solutions to complete ones [21]. However, their performance heavily relies on state representations [22]. Many studies [23, 24, 25, 26] model scheduling states using disjunctive graphs, but the incompleteness of partial solutions often leads to the omission of important components [21] (e.g., the disjunctive arcs among undispatched operations [23]). Additionally, disjunctive graphs struggle to incorporate essential work-in-progress information required during construction [21] (e.g., current machine load and job status). These limitations lead to suboptimal schedules and reduce their adaptability in SM. In contrast, *improvement* methods start from a complete initial solution and iteratively refine it through small adjustments [27]. As the MDP state represents a fully scheduled solution, it avoids the information loss inherent in partial ones. Moreover, the structure of disjunctive graphs naturally encodes topological relationships among operations [28], enabling complete schedules to be effectively represented with all necessary information [21]. This better supports the high-performance demands of complex scheduling scenarios in SM.

Despite the promising results, most existing DRL-based improvement methods focus only on nonflexible problems, such as the JSP [21, 29]. The increased complexity of FJSP over JSP poses significant challenges for designing effective improvement approaches. First, the one-to-many relationships between operations and machines make the scheduling state more intricate [8], challenging its representation and demanding more expressive neural networks for encoding. Second, improvement methods explore solutions through local moves within neighborhoods. In FJSP, due to the complex decisions involving both operation sequencing and machine assignment [6], this process requires tailored local moves as actions in the MDP [27]. Moreover, the existence of flexibility in FJSP significantly enlarges the solution space, increasing the risk of local optima entrapment and necessitating more efficient search strategies to reduce the extensive number of iterations for convergence [27, 30].

To address the above challenges, we propose a DRL-based improvement heuristic framework for FJSP. First, to represent complex scheduling states, we design a novel heterogeneous disjunctive graph by adding machine nodes with directed hyper-edges to encode the complete solution, effectively capturing critical information about operation sequencing on machines. We also propose a heterogeneous graph neural network and enhance its exploratory capability via a memory module, which stores compact representations of visited solutions and retrieves relevant information to enrich the current state embedding. This module leverages historical schedules to improve decision-making and alleviate local optima issues. Second, we construct the action space based on the Nopt2 neighborhood structure, enabling simultaneous adjustment to operation sequences and machine assignments, and further reduce its dimensionality via constraint relationships to enhance search efficiency. Finally, we propose a parallel greedy exploration strategy that evaluates multiple candidate actions at each step, achieving comparable solution quality with fewer iterations. Extensive experiments on synthetic data and public benchmarks demonstrate the superior performance of our approach in solving FJSP.

The contributions of this paper are summarized as follows: 1) The first DRL-based improvement heuristic framework for FJSP—*MIStar*, capable of learning size-agnostic policies that outperform both traditional handcrafted improvement and state-of-the-art DRL construction methods. 2) An MDP formulation with a heterogeneous disjunctive graph for state representation of complete scheduling solutions in FJSP, and an action space that enables simultaneous adjustments to operation sequences and machine assignments. 3) A memory-enhanced heterogeneous graph neural network (MHGNN) that leverages historical solutions to improve decision-making and alleviate local optima issues. 4) A parallel greedy exploration strategy that improves efficiency of solution space exploration.

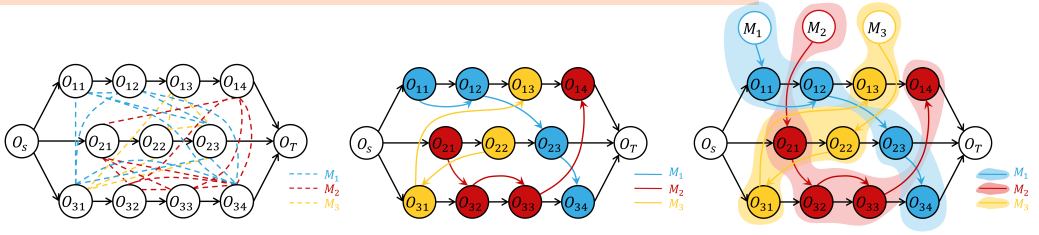
2 Related work

This section reviews the limitations of DRL-based construction methods and recent advances in improvement methods, highlighting the challenges in solving FJSP.

Construction methods construct complete solutions by assigning an operation to a machine at each step. Compared to the heuristic guidance offered by conventional dispatching rules, DRL employs DNNs to score actions, but their effectiveness heavily relies on state representations [22, 31, 32]. Early approaches [33, 34, 35, 36] extract multiple general state features as input, which overcompresses state information and neglects the structural nature of FJSP [37]. Recent works [8, 23, 24, 38, 39] integrate GNNs with DRL by modeling scheduling states as graphs, learning rich graph embeddings with structural information for decision-making. However, the incompleteness of partial solutions often results in the omission of important components [21] (e.g., the disjunctive arcs among undispatched operations [23]), and work-in-progress information required during construction is hard to explicitly represent in disjunctive graphs [21], leading to suboptimal performance.

Improvement methods iteratively refine solutions by performing local moves within the neighborhood. Conventional methods use handcrafted rules to greedily select the best solution at each step, but suffer from myopia and costly neighborhood evaluation [21]. Recent studies [21, 29, 40, 41] have addressed these limitations by leveraging deep policy networks in DRL to directly generate local moves. Such DRL-based improvement heuristics were initially applied to routing problems [42, 43, 44] and have since been adapted to scheduling domains. By representing complete solutions as MDP states through disjunctive graphs, these approaches circumvent the inaccuracies in state representation that arise from partial solutions [21]. Falkner et al. [40] train a GNN-based JSP solver using DRL to adapt local search strategies according to problem-specific features. However, their design of local operators relies heavily on expertise, limiting the generalizability. Closer to our work, Zhang et al. [21, 29] represent JSP solutions with disjunctive graphs and define the action space as a set of exchangeable operation pairs based on the N5 neighborhood structure [45], training a GNN-based policy network via DRL. While effective for JSP, extending this framework to FJSP—whose flexible machine assignments better suit mass customization and dynamic production in Industry 4.0 [9, 10]—faces critical challenges. First, conventional disjunctive graphs lack machine nodes, making them insufficient for complex states of FJSP. In addition, the N5-based action design does not account for machine assignment adjustments(See Appendix A for analysis). Furthermore, the increased flexibility of FJSP over JSP enlarges the solution space and aggravates local optima issues in local search [46].

An effective approach to this issue is leveraging memory mechanisms by incorporating past experience to promote exploration [47]. Garmendia et al. [48] maintain a tabu memory to filter redundant visits in routing problems, yet such methods fail to integrate historical data into the decision-making of the policy, leaving historical information underutilized. A closely related work is MARCO [47], which aggregates contextually relevant information from the memory module to enhance state embeddings and improve policy decisions. However, its application in improvement methods is limited to simple binary optimization problems (i.e., maximum cut and maximum independent set), and its approach to information aggregation is unsuitable for the complex constraints in FJSP. In contrast, we simplify scheduling representations, which are stored in the memory module and aggregated via a soft voting mechanism [49] to better accommodate the intricate constraints of FJSP.



(a) (b) (c)

Figure 1 **Graph representations of FJSP.** (a) FJSP instance in a disjunctive graph; (b) feasible solution in a disjunctive graph; (c) feasible solution in our graph.

3 Preliminaries

This section presents the formulation of FJSP, disjunctive graphs, and Nopt2 neighborhood structure.

Flexible job shop scheduling problem. FJSP is defined as follows: Given a set of jobs $\mathcal{J} = \{J_1, J_2, \cdots, J_n\}$ and a set of machines $\mathcal{M} = \{M_1, M_2, \cdots, M_m\}$, each job J_i consists of n_i operations denoted by $\mathcal{O}_i = \{O_{i_1}, O_{i_2}, \cdots, O_{in_i}\}$. The operations within a job must be processed in a specific order, known as precedence constraints. Each operation O_{ij} can be processed by any machine M_k for a processing time p_{ij}^k from its compatible set $\mathcal{M}_{ij} \subseteq \mathcal{M}$, and the processing is non-preemptive. Each machine can process only one operation at a time. The objective of FJSP is to assign each operation to a compatible machine and determine its processing sequence on the selected machine to minimize the makespan, defined as the maximum completion time $C_{\max} = \max_{i,j}\{C_{ij}\}$, where C_{ij} denotes the completion time of O_{ij}.

Disjunctive graph. An FJSP instance can be represented by a disjunctive graph $G = (\mathcal{O}, \mathcal{C}, \mathcal{D})$ [50]. The operation set $\mathcal{O} = \{O_{ij} \mid \forall i, j\} \cup \{O_S, O_T\}$ includes all operation nodes and two dummy ones representing the start and end states. The conjunctive arc set $\mathcal{C}$ consists of directed arcs that describe the precedence constraints between operations of the same job. The disjunctive arc set $\mathcal{D} = \cup_k \mathcal{D}_k$ comprises undirected edges, and each $\mathcal{D}_k$ connects operations processed on the same machine M_k. In FJSP, each operation can be connected to multiple disjunctive arcs. Therefore, solving an FJSP instance is equivalent to selecting a disjunctive arc for each node and fixing its direction to construct a DAG [28]. In the graph, the longest path from O_S to O_T is called the critical path (CP), whose length represents the makespan of the solution. See Figure 1a and 1b for examples.

The Nopt2 neighbourhood Structure. Given a solution s, the Nopt2 constructs the neighbourhood $Nopt2(s)$ as follows. First, the critical path CP(s) is identified. Then, an operation O_{ij}^k on the critical path is chosen and deleted from its current processing sequence of M_k. It is reinserted into the optimal insertion interval within the sequence of an alternative compatible machine $M_{k'}$ yielding a new solution s'. The optimal insertion interval is determined based on the precedence relations between the newly inserted operation and existing operations on $M_{k'}$, ensuring that all constraints are satisfied and the makespan of s' is potentially reduced. We adopt the Nopt2 neighborhood structure [51] (Figure 2) to define the local move during the search process of *MIStar*. By identifying optimal insertion interval, this structure significantly reduces the size of neighborhood. The neighborhood size $|Nopt2(s)|$ is influenced by both the instance scale and the flexibility of machine selection.

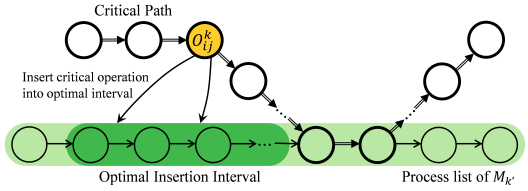


Figure 2 **Nopt2 neighborhood structure.**

4 Methodology

This section details our framework, illustrated in Figure 3. Given an FJSP instance, an initial solution is generated and transformed into a heterogeneous graph, from which a memory-enhanced GNN learns embeddings. The policy network samples several local moves from the neighborhood and evaluates in parallel. The best one is executed to update the current solution, yielding a new one. The process terminates after a predefined search horizon. We formulate the iterative solution optimization as a MDP. This section first formalizes the MDP, then presents a novel heterogeneous graph representation and the memory-enhanced GNN-based policy network. Finally, we describe the parallel greedy exploration strategy and the training algorithm.

4.1 MDP Formulation

We formulate the iterative optimization process of the FJSP as a discrete MDP. At each step t, the agent, the decision-maker based on a given policy, selects an action (local move) from the action space (neighborhood), and the state transitions accordingly to represent a new solution. The MDP is defined as follows.

State. The state s_t encodes a complete solution and is defined by a set of feature vectors, including $\mathbf{x}_{o_{ij}} \in \mathbb{R}^9$ for each operation O_{ij} and $\mathbf{y}_{M_k} \in \mathbb{R}^4$ for each machine M_k, detailed in Appendix B.
Action. The action space $\mathcal{A}_t$ comprises local moves in the Nopt2 neighborhood, changing with

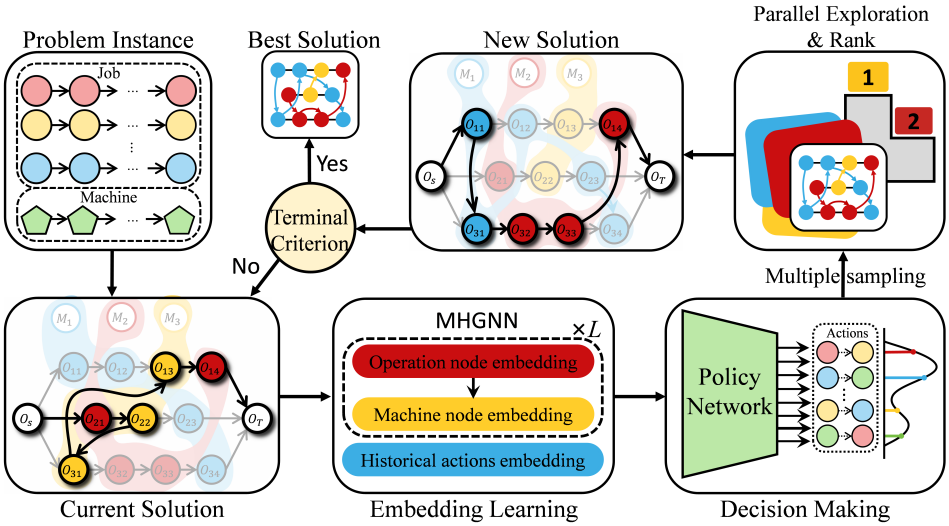


Figure 3 **The architecture of *MIStar*.**

the state s_t. Each action $a_t = [O_m, O_n, M_k]$ denotes inserting critical operation O_m before operation O_n in the processing sequence of machine M_k, where $O_m \notin \{O_S, O_T\}$ and $O_n \neq O_S$. **Reward.** The reward comprises two components: improvement in solution quality r_{gain} and penalty for redundant solution visits $r_{penalty}$. The term r_{gain} measures the improvement over the incumbent best solution s_t^* (initialized as s_0), and is defined as: $r_{gain} = \max(C_{\max}(s_t^*) - C_{\max}(s_{t+1}), 0)$, where maximizing the cumulative reward is equivalent to minimizing the makespan $C_{\max}(s_t^*)$. The term $r_{penalty}$ discourages redundant exploration by evaluating similarity between s_t and previous states. Formally, it is computed as: $r_{penalty} = \lambda \times \frac{1}{K} \sum_{\omega \in \Omega_{\text{TopK}}} \omega$, where Ω_{TopK} denotes the top-K similarity scores between s_t and stored states, and λ is the penalty coefficient. The overall reward is defined as $r_t(s_t, a_t) = r_{gain} - r_{penalty}$, which is initially dominated by makespan improvement and gradually shifts toward solution diversity as improvements diminish.

4.2 Directed Heterogeneous Graph

Conventional disjunctive graphs lack machine nodes, hindering GNNs from accurately capturing machine states in FJSP, which in turn limits policies to effectively maximize machine utilization [37]. Inspired by the heterogeneous disjunctive graph $\mathcal{H}$ [8], we introduce a novel directed heterogeneous disjunctive graph $\overrightarrow{\mathcal{H}} = (\mathcal{O}, \mathcal{M}, \mathcal{C}, \mathcal{E})$ to represent scheduling solutions. In our graph, the operation and machine nodes are denoted as $\mathcal{O}$ and $\mathcal{M}$, while the conjunctive arc set $\mathcal{C}$ and hyper-edge set $\mathcal{E}$ represent the processing sequences on jobs and machines, respectively. Mathematically, the hyper-edge is a special edge that can join any number of vertices [52]. Since machine assignment and processing sequence are fixed in FJSP solutions, we reinterpret $\mathcal{E}$ as a directed hyper-arc set of size $|\mathcal{M}|$, i.e., $\mathcal{E} = \{E^k = (M_k, O^{k1}, O^{k2}, \ldots, O^{kn_k})\}$, where n_k is the number of operations processed on M_k. As shown in Figure 1c, each machine nodes M_k connects to its processing sequence via directed hyper-arcs E^k to explicitly encode machine status, and the unified directed arcs in $\overrightarrow{\mathcal{H}}$ enable clearer delineation and distinction of different solutions. See Appendix C for analysis.

4.3 Memory-enhanced Heterogeneous Graph Neural Network

We propose a memory-enhanced heterogeneous graph neural network (MHGNN) to extract state features from scheduling solutions. As shown in Figure 4, MHGNN encodes the topological and sequential constraint information to generate operation embeddings, which are aggregated into machine nodes to obtain machine embeddings. Meanwhile, relevant historical information is retrieved and extracted as historical action embeddings, which are then concatenated with machine and operation embeddings before being fed into the policy network to produce an action distribution. The following sections detail the embeddings design for operations, machines, and historical information.

4.3.1 Operation Node Embedding

Following [21], we focus on two key aspects of the FJSP graph: (1) the topology dynamically changes due to the transitions of MDP state, and (2) rich semantics are conveyed through two types of neighbors—job predecessors and machine predecessors—reflecting precedence constraints and processing order on machines via conjunctive arcs and hyper-arcs, respectively. Their joint representation is crucial for distinguishing solutions, forming the foundation for scheduling decisions.

To encode the topological information, we employ the Graph Isomorphism Network (GIN) [53], which is known for its strong discriminative power for non-isomorphic graphs. Specifically, given a graph $\overrightarrow{\mathcal{H}} = (\mathcal{O}, \mathcal{M}, \mathcal{C}, \mathcal{E})$, each operation $O_{ij} \in \mathcal{O}$ is encoded into a q-dimensional embedding through an L-layer GIN, where the l-th layer computes as follows:

$$\mu_{O_{ij}}^{l} = \text{MLP}^{l}\left((1+\epsilon^{l}) \cdot \mu_{O_{ij}}^{l-1} + \sum_{U \in N(O_{ij})} \mu_{U}^{l-1}\right) \tag{1}$$

where $\mu_{O_{ij}}^{l} \in \mathbb{R}^{q}$ denotes the topological embedding of operation O_{ij} at layer l, initialized with its raw features $\mathbf{x}_{o_{ij}}$, ϵ^{l} is a learnable parameter, and $N(O_{ij})$ comprises predecessor operation nodes.

To capture rich semantic relationships among operations, we apply an L-layer Graph Attention Network (GAT) with n_H attention heads, with the feature transformation at layer l computed as:

$$\tau_{O_{ij}}^{l} = \text{GAT}^{l}\left(\tau_{O_{ij}}^{l-1}, \{\tau_{U}^{l-1} | U \in N(O_{ij})\}\right) \tag{2}$$

where $\tau_{O_{ij}}^{l} \in \mathbb{R}^{q}$ denotes the semantic embedding of operation O_{ij} at layer l, initialized with raw features $\mathbf{x}_{o_{ij}}$. We obtain the operation embedding $h_{O_{ij}}^{l} \in \mathbb{R}^{2q}$ at layer l by concatenating the above two embeddingshe final-layer outputs form the set of operation node embeddings $\{h_{O_{ij}} = h_{O_{ij}}^{L} \mid \forall O_{ij} \in \mathcal{O}\}$.

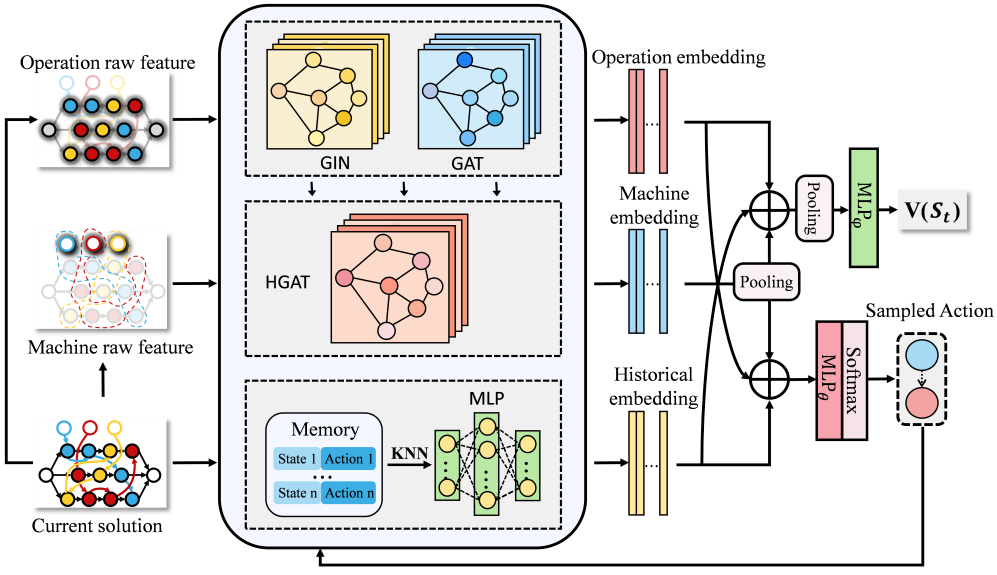


Figure 4 **The network architecture of the memory-enhanced heterogeneou GNN.**

4.3.2 Machine Node Embedding

Machine embeddings are incorporated to capture machine load by aggregating the information of their associated operation nodes, serving as an essential heuristic to guide the policy in reassigning operations from high-load machines to low-load counterparts [8]. For each machine M_k, the hyper-arc $E^{k} = (M_k, O^{k1}, O^{k2}, \ldots, O^{kn_k})$ encodes its full processing sequence. Since the importance of operations at different positions varies, we employ an L-layer heterogeneous GAT [8] to extract machine node embeddings, where nodes with different attributes are processed by distinct linear

transformations rather than a shared one when computing the attention coefficients. The machine node embedding at layer l is computed as follows:

$$h_{M_k}^l = \text{HGAT}^l\left(h_{M_k}^{l-1}, \{h_U^{l-1} | U \in N(M_k)\}\right) \tag{3}$$

where $h_{M_k}^l \in \mathbb{R}^q$ denotes machine embedding at layer l, initialized with raw features $\mathbf{y}_{M_k}$, HGAT^l is a heterogeneous graph attention layer, and $N(M_k)$ comprises operations in its processing sequence. Similarly, the graph-level machine embedding is computed as $h_M = \frac{1}{|\mathcal{M}|}\sum_{M_k \in \mathcal{M}} h_{M_k}^L$.

4.3.3 Historical Action Embedding

At step t, we store the state-action pair (s_t, a_t) in a memory module. To reduce storage and computational costs, we simplify the representation of s_t by omitting node attributes and focusing solely on the processing sequence across machines (see analysis in Appendix D). Specifically, we store the operation-machine incidence matrix $\boldsymbol{L}_t$ at each step, where the non-zero element $(\boldsymbol{L}_t)_{i,j}$ represents the processing order index of operation i on machine j, and assess the similarity between the current state $\boldsymbol{L}_t$ and each historical state $\boldsymbol{L}_{t'}$ ($t' < t$), by computing the Frobenius inner product as their similarity score $\omega_{t,t'} \in \Omega$, given by:

$$\omega_{t,t'} = \langle L_t, L_{t'} \rangle_F = \sum_{i=1}^{|\mathcal{O}|}\sum_{j=1}^{|\mathcal{M}|}(L_t)_{i,j} \cdot (L_{t'})_{i,j} \tag{4}$$

The k-nearest neighbors algorithm retrieves K actions with the highest similarity scores, forming the set $\mathcal{A}_{L_t} = \{a_1, a_2, \ldots, a_K\}$. We aim to aggregate the information in $\mathcal{A}_{L_t}$ to enhance the state embedding and encourage exploration of more diverse solutions through the design of reward. However, each historical action $a_i = [O_{m_i}, O_{n_i}, M_{k_i}]$ is denoted by discrete integer indices indicating operations and machines. Simply weighted averaging yields semantically invalid continuous values, failing to provide useful guidance. To address this, we adopt a soft voting mechanism that aggregates the three dimensions separately based on frequency-weighted similarity, as follows:

$$\tilde{O}_m = \underset{O_m \in \mathcal{A}_{L_t}^1}{\arg\max} \sum_{i=1}^{K} \omega_{t,i} \cdot \mathbb{I}(O_{m_i} = O_m) \tag{5}$$

$$\tilde{O}_n = \underset{O_n \in \mathcal{A}_{L_t}^2}{\arg\max} \sum_{i=1}^{K} \omega_{t,i} \cdot \mathbb{I}(O_{n_i} = O_n) \tag{6}$$

$$\tilde{M}_k = \underset{M_k \in \mathcal{A}_{L_t}^3}{\arg\max} \sum_{i=1}^{K} \omega_{t,i} \cdot \mathbb{I}(M_{k_i} = M_k) \tag{7}$$

where $\omega_{t,i}$ is the similarity score between s_t and the i-th most similar historical state, serving as the weight of a_i, $\mathbb{I}(\cdot)$ is the indicator function, and $\mathcal{A}_{L_t}^1$, $\mathcal{A}_{L_t}^2$ and $\mathcal{A}_{L_t}^3$ denote the unique value sets in each dimension of $\mathcal{A}_{L_t}$. The aggregated action $\tilde{a}_{L_t} = [\tilde{O}_m, \tilde{O}_n, \tilde{M}_k]$ is then mapped by a single-layer MLP to the embedding space of h_O and h_M, resulting in the historical action embedding $h_a \in \mathbb{R}^q$.

4.3.4 Decision Making

Given that machine assignments are determined in FJSP solutions (i.e., for any action $[O_m, O_n, M_k]$, O_n must be processed by M_k), the third dimension can be omitted from the action space. We enhance each operation embedding h_{Oij} by concatenating h_M and h_a, then feed them into the policy network to obtain intermediate vectors forming a matrix $F_O \in \mathbb{R}^{|\mathcal{O}| \times q}$. We multiply F_O by its transpose to obtain a score matrix F_{OO} of shape $|\mathcal{O}| \times |\mathcal{O}|$, where each element represents the priority score of an action in $\mathcal{A}_t$. Based on the Nopt2 neighborhood, we mask infeasible actions by setting their scores to $-\infty$ and apply softmax normalization to obtain the action probability distribution. By leveraging the FJSP constraints, we reduce the space complexity of $\mathcal{A}_t$ from $O(|\mathcal{O}|^2|\mathcal{M}|)$ to $O(|\mathcal{O}|^2)$, which alleviates the difficulty of learning and accelerates model convergence. A detailed analysis of this space complexity reduction is provided in Appendix E.

4.3.5 Training Algorithm

We propose an n-step PPO algorithm with a parallel greedy exploration strategy to efficiently explore the large solution space of FJSP. At each step t, the policy samples P candidate actions

Table 1 Results on small synthetic instances

	Size		Sample		Greedy		100 Iterations				200 Iterations				400 Iterations				OR-Tools[1]
			DANIEL	HGNN	DANIEL	HGNN	GD	FI	BI	MIStar	GD	FI	BI	MIStar	GD	FI	BI	MIStar	
SD1	10×5	Obj.	101.67	105.59	106.76	111.67	**100.00**	100.91	100.55	100.02	100.00	100.91	100.54	**99.88**	100.00	100.91	100.53	**99.69**	
		Gap	5.55%	9.62%	10.84%	15.94%	**3.82%**	4.77%	4.39%	3.84%	3.82%	4.77%	4.38%	**3.70%**	3.82%	4.77%	4.37%	**3.50%**	96.32(5%)
		Time[2]	0.26s	0.44s	0.16s	0.17s	8.13s	12.60s	12.09s	5.64s	16.43s	27.54s	26.21s	11.23s	32.95s	58.71s	54.73s	23.00s	
	20×5	Obj.	192.78	207.53	197.56	211.22	191.31	191.79	191.36	**190.87**	191.31	191.79	191.36	**190.79**	191.31	191.79	191.36	**190.67**	
		Gap	2.46%	10.30%	5.00%	12.26%	1.68%	1.93%	1.71%	**1.45%**	1.68%	1.93%	1.71%	**1.40%**	1.68%	1.93%	1.71%	**1.34%**	188.15(0%)
		Time	1.14s	1.05s	0.33s	0.34s	53.86s	70.73s	77.32s	15.72s	1.80m	2.46m	2.70m	33.38s	3.60m	5.20m	5.56m	1.16m	
	15×10	Obj.	153.22	160.86	161.28	166.92	151.96	152.10	151.90	**150.85**	151.96	152.10	151.90	**150.61**	151.96	152.10	151.90	**150.34**	
		Gap	6.75%	12.07%	12.37%	16.30%	5.87%	5.97%	5.83%	**5.10%**	5.87%	5.97%	5.83%	**4.93%**	5.87%	5.97%	5.83%	**4.74%**	143.53(7%)
		Time	1.90s	2.17s	0.50s	0.50s	58.62s	86.43s	90.21s	27.09s	1.97m	3.08m	3.11m	54.22s	3.93m	6.81m	6.33m	1.80m	
	20×10	Obj.	193.91	214.81	198.50	215.78	192.92	193.26	193.03	**192.71**	192.92	193.26	193.03	**192.70**	192.92	193.26	193.03	**192.68**	
		Gap	-1.06%	9.61%	1.29%	10.10%	-1.56%	-1.39%	-1.51%	**-1.67%**	-1.56%	-1.39%	-1.51%	**-1.67%**	-1.56%	-1.39%	-1.51%	**-1.68%**	195.98(0%)
		Time	2.71s	2.93s	0.68s	0.69s	2.44m	2.90m	3.09m	32.91s	4.90m	6.06m	6.64m	1.14m	9.86m	12.83m	13.87m	2.51m	
SD2	10×5	Obj.	365.26	479.37	413.73	552.80	360.17	354.22	349.66	**345.05**	360.17	354.22	349.66	**343.43**	360.17	354.22	349.66	**342.50**	
		Gap	11.96%	46.94%	26.82%	69.45%	10.40%	8.58%	7.18%	**5.77%**	10.40%	8.58%	7.18%	**5.27%**	10.40%	8.58%	7.18%	**4.98%**	326.24(96%)
		Time	0.30s	0.22s	0.16s	0.20s	6.06s	10.37s	11.74s	5.27s	12.21s	22.48s	25.65s	10.33s	24.39s	48.51s	54.57s	21.39s	
	20×5	Obj.	629.56	960.11	671.38	1047.83	626.35	622.75	619.13	**615.12**	626.35	622.75	619.13	**613.75**	626.35	622.75	619.13	**612.73**	
		Gap	4.57%	59.48%	11.52%	74.05%	4.04%	3.44%	2.84%	**2.17%**	4.04%	3.44%	2.84%	**1.95%**	4.04%	3.44%	2.84%	**1.78%**	602.04(0%)
		Time	0.78s	0.77s	0.33s	0.35s	48.48s	62.37s	72.23s	11.55s	1.61m	2.18m	2.52m	23.50s	3.30m	4.71m	5.21m	47.93s	
	15×10	Obj.	522.51	757.18	588.72	824.62	516.79	511.00	514.73	**488.22**	516.79	511.00	514.73	**476.93**	516.79	511.00	514.73	**465.71**	
		Gap	38.53%	100.75%	56.09%	118.63%	37.02%	35.48%	36.47%	**29.44%**	37.02%	35.48%	36.47%	**26.45%**	37.02%	35.48%	36.47%	**23.47%**	377.17(28%)
		Time	1.72s	1.45s	0.50s	0.57s	60.34s	89.27s	78.09s	18.09s	2.07m	3.17m	2.64m	37.42s	4.19m	6.87m	5.30m	1.31m	
	20×10	Obj.	552.38	987.57	605.37	1036.65	549.30	542.39	535.06	**530.93**	549.30	542.39	535.06	**528.92**	549.30	542.39	535.06	**525.49**	
		Gap	19.01%	112.76%	30.42%	123.34%	18.34%	16.85%	15.27%	**14.39%**	18.34%	16.85%	15.27%	**13.95%**	18.34%	16.85%	15.27%	**13.21%**	464.16(1%)
		Time	2.84s	2.91s	0.69s	0.71s	2.35m	3.06m	3.53m	25.76s	4.64m	7.07m	7.54m	53.25s	9.14m	15.33m	15.76m	1.88m	

[1] For OR-Tools, the solution and the ratio of optimally solved instances are reported.
[2] "s", "m", and "h" denote seconds, minutes, and hours, respectively.

$\{a_t^1, a_t^2, \ldots, a_t^P\}$ from A_t, evaluates their improvements in parallel, and selects the action a_t^i yielding the greatest makespan reduction for execution. This approach enables the exploration of P solutions per iteration, producing high-quality solutions in fewer iterations and significantly reducing search time. More detailed analysis and pseudo-code of our training algorithm are presented in Appendix F.

5 Experiments

In this section, we present the experimental setup, performance evaluations on synthetic and public datasets as well as the results of ablation studies.

5.1 Experimental Settings

Datasets. We evaluate our method on both synthetic data and benchmarks. The synthetic datasets SD1 [8] and SD2 [22] cover 6 scales and 100 instances per scale, and differ in distribution. For each $n \times m$ FJSP instance, the number of compatible machines $|\mathcal{M}_{ij}|$ for O_{ij} is sampled from $U(1, m)$. In SD1, the number of operations per job and the average processing time $\overline{p}_{ij}$ of O_{ij} are sampled from $U(0.8 \times m, 1.2 \times m)$ and $U(1, 20)$, respectively. The machine-specific processing time p_{ij}^k is drawn from $U(0.8 \times \overline{p}_{ij}, 1.2 \times \overline{p}_{ij})$. While in SD2, each job has m operations, and p_{ij}^k is sampled from $U(1, 99)$. Benchmarks include Hurink [54] (Edata, Rdata, Vdata; 40 instances per set over 8 scales) and Brandimarte [55] (10 instances across 7 scales). Implementation details are in Appendix G.

Baselines and Performance Metrics. We compare our method with two DRL-based construction methods, HGNN [8] and DANIEL [22]. HGNN is a heterogeneous GNN designed to encode features represented by a heterogeneous disjunctive graph, while DANIEL employs a dual-attention network on tight graph representation. Both models are retrained and tested, with results reported for sampling and greedy strategies. We also compare three hand-crafted improvement rules: greedy(GD) [46], best-improvement (BI), and first-improvement (FI) [56]. Specifically, GD selects the solution with the smallest makespan from the neighborhood, while BI and FI choose the best and first improving ones, respectively. A *restart* strategy [57] is applied in BI and FI to escape local optima by continuing the search from a new initial solution. We also compares with (near-)optimal solutions from Google OR-Tools, a powerful constraint programming solver, under a 1800 s time limit. For benchmarks, we report the results of a two-stage genetic algorithm (2SGA) [58] directly from [22].

As such incremental search methods cannot quickly access very different solutions [27, 42], their performance is sensitive to the initial solution quality. We use DANIEL to sample 100 initial solutions per instance, which are equally applied to three rule-based methods for fair comparison. *MIStar* then performs parallel iterative search, retaining the best solution. Performance is measured by the average makespan and the relative gap to the best-known solution, which is obtained via OR-Tools for synthetic datasets [22] and from [59] for public benchmarks.

Table 2 Generalization results on large synthetic instances

	Size	Sample		Greedy		100 Iterations				200 Iterations				400 Iterations				OR-Tools[1]
		DANIEL	HGNN	DANIEL	HGNN	GD	FI	BI	MIStar	GD	FI	BI	MIStar	GD	FI	BI	MIStar	
SD1	Obj.	286.77	308.55	288.61	314.71	286.04	286.19	285.95	**285.79**	286.04	286.19	285.95	**285.75**	286.04	286.19	285.95	**285.68**	
	30×10 Gap	4.41%	12.33%	5.08%	14.58%	4.14%	4.19%	4.11%	**4.05%**	4.14%	4.19%	4.11%	**4.03%**	4.14%	4.19%	4.11%	**4.01%**	274.67(6%)
	Time[2]	5.60s	6.25s	0.97s	0.97s	7.67m	10.51m	10.73m	57.03s	15.32m	22.00m	21.49m	1.83m	30.65m	47.10m	43.11m	3.77m	
	Obj.	379.71	410.76	379.28	417.87	379.04	379.05	378.92	**378.82**	379.04	379.05	378.92	**378.76**	379.04	379.05	378.92	**378.63**	
	40×10 Gap	3.76%	12.24%	3.64%	14.18%	3.57%	3.58%	3.54%	**3.51%**	3.57%	3.58%	3.54%	**3.50%**	3.57%	3.58%	3.54%	**3.46%**	365.96(3%)
	Time	9.94s	10.45s	1.30s	1.34s	18.20m	25.13m	24.04m	1.11m	36.48m	52.01m	48.46m	2.55m	1.22h	1.82h	1.63h	5.94m	
SD2	Obj.	756.52	1453.40	800.02	1536.74	753.66	748.41	739.47	**735.14**	753.66	748.41	739.47	**731.84**	753.66	748.41	739.47	**728.76**	
	30×10 Gap	9.28%	109.95%	15.57%	121.99%	8.87%	8.11%	6.82%	**6.19%**	8.87%	8.11%	6.82%	**5.72%**	8.87%	8.11%	6.82%	**5.27%**	692.26(0%)
	Time	5.76s	5.98s	0.99s	1.14s	7.65m	12.13m	12.16m	49.43s	15.04m	24.58m	25.49m	1.69m	29.50m	51.78m	52.62m	3.56m	
	Obj.	953.14	1937.96	984.55	2045.78	947.64	941.60	**930.44**	931.04	947.64	941.60	930.44	**929.13**	947.64	941.60	930.44	**925.93**	
	40×10 Gap	-4.53%	94.11%	-1.39%	104.91%	-5.08%	-5.69%	**-6.81%**	-6.75%	-5.08%	-5.69%	-6.81%	**-6.94%**	-5.08%	-5.69%	-6.81%	**-7.26%**	998.39(0%)
	Time	10.06s	11.10s	1.34s	1.36s	18.29m	25.07m	21.06m	73.29s	35.83m	54.84m	42.18m	2.52m	1.18h	1.89h	1.42h	5.40m	

[1] For OR-Tools, the solution and the ratio of optimally solved instances are reported.
[2] "s", "m", and "h" denote seconds, minutes, and hours, respectively.

Table 3 Results on public benchmarks

Method	mk			la(rdata)			la(edata)			la(vdata)		
	Obj.	Gap	Time(s)	Obj.	Gap	Time(s)	Obj.	Gap	Time(s)	Obj.	Gap	Time(s)
OR-Tools	174.20	0.99%	1447.08	935.80	0.16%	1397.43	1028.93	0.01%	899.60	919.60	-0.01%	639.17
2SGA	175.20	1.57%	57.60		-			-		812.20[1]	0.39%	51.43
DANIEL(S)[2]	180.80	4.81%	1.71	984.63	5.39%	1.99	1120.28	8.88%	2.00	934.75	1.64%	2.02
HGNN(S)	192.20	11.42%	1.72	1004.60	7.53%	1.84	1123.10	9.16%	2.44	934.23	1.58%	2.12
DANIEL(G)	184.30	6.84%	0.45	1044.85	11.84%	0.48	1176.40	14.34%	0.48	965.38	4.97%	0.48
HGNN(G)	200.00	15.94%	0.46	1035.75	10.86%	0.52	1189.33	15.59%	0.48	962.48	4.66%	0.47
GD-100[3]	178.92	3.72%	22.86	963.03	3.08%	26.09	1100.46	6.96%	4.14	924.53	0.53%	89.94
FI-100	178.85	3.68%	34.91	968.60	3.67%	36.16	1101.43	7.05%	6.77	927.33	0.83%	131.90
BI-100	178.43	3.44%	37.98	964.80	3.27%	41.25	1100.63	6.97%	5.91	927.55	0.86%	122.02
MIStar-100	**177.80** ($\pm$0.74)[4]	**3.07%**	17.50	**958.90** ($\pm$4.55)	**2.64%**	19.86	**1099.38** ($\pm$4.17)	**6.85%**	19.60	**923.55** ($\pm$1.81)	**0.42%**	20.36
GD-200	178.92	3.72%	45.73	963.03	3.08%	51.94	1100.46	6.96%	8.27	924.53	0.53%	179.99
FI-200	178.85	3.68%	74.64	968.60	3.67%	76.17	1101.43	7.05%	14.45	927.33	0.83%	272.34
BI-200	178.43	3.44%	81.03	964.65	3.25%	86.20	1100.60	6.97%	12.70	926.55	0.75%	251.11
MIStar-200	**177.70** ($\pm$0.75)	**3.01%**	36.45	**957.23** ($\pm$4.72)	**2.46%**	40.66	**1099.30** ($\pm$4.13)	**6.84%**	40.05	**922.70** ($\pm$1.71)	**0.33%**	41.83
GD-400	178.92	3.72%	91.79	963.03	3.08%	103.81	1100.46	6.96%	16.51	924.53	0.53%	360.28
FI-400	178.85	3.68%	162.16	968.60	3.67%	158.27	1101.35	7.04%	31.56	927.33	0.83%	586.32
BI-400	178.43	3.44%	169.70	964.50	3.24%	180.45	1100.33	6.94%	27.24	925.55	0.64%	521.21
MIStar-400	**177.60** ($\pm$0.63)	**2.96%**	76.35	**956.38** ($\pm$4.04)	**2.37%**	85.45	**1099.18** ($\pm$4.25)	**6.83%**	83.77	**921.85** ($\pm$1.77)	**0.24%**	87.23

[1] The makespan and gap of 2SGA on la(vdata) benchmark are computed on la1-30 instances, as reported in [58].
[2] "(S)" denotes the sampling strategy, and "(G)" denotes the greedy strategy.
[3] "-100", "-200", and "-400" denote different iteration budgets.
[4] Values are reported as the average ($\pm$ standard deviation) over 10 independent runs.

5.2 Performance on Synthetic Instances

We train our model with 100 iterations on four small sizes (10×5, 20×5, 15×10, and 20×10) from two synthetic datasets and evaluate on corresponding test instances. Table 1 reports the performance and average runtime per instance across all methods. The generalization capability of the model, evaluated on large-sized instances (30×10 and 40×10 in Table 2) and with extended iterations up to 400, is also reported. The results show that *MIStar* consistently outperforms two construction methods across various instances, with larger gains on SD2. This is because initial solutions on SD1 are near-optimal, leaving limited room for improvement, as verified by experiments on varying-quality initial solutions (see Appendix I). Compared with rule-based improvement heuristics, *MIStar* achieves better solution quality with shorter runtime and maintains consistent improvement over longer iterations, verifying the effectiveness and efficiency of its policy. See detailed analysis in Appendix H.

5.3 Performance on Public Benchmarks

We evaluate generalization on Hurink and Brandimarte benchmarks using the model trained on the 10×5 instances from SD2. While OR-Tools and 2SGA achieve near-optimal results, their computational cost limits practicality. *MIStar* maintains stable performance over both DRL-based construction and rule-base method across all benchmarks, demonstrating its effective generalization through the learned general policy. Regarding solving time, rule-based methods exhibit notable variation across Hurink dataset, with runtimes increasing from `edata` to `rdata` and `vdata` due to the growing set of assignable machines that enlarge the neighborhood. In contrast, *MIStar* maintains stable runtime across all distributions.

Table 4 Generalization Performance on Larger-Scale Instances

Instance Size	Method	Objective	Time (min)	Optimality Gap
50×15	OR-Tools	881.45	60.0	39.05%
(LB:537.2)	MIStar-200	969.7	10.6	44.60%
60×15	OR-Tools	1075.1	60.0	46.94%
(LB:570.2)	MIStar-200	1109.8	14.2	48.62%
100×10	OR-Tools	1922.65	60.0	62.22%
(LB:726.5)	MIStar-200	2138.4	15.9	66.03%
50×30	OR-Tools	Infeasible	60.0	-
(LB:N/A)	MIStar-200	1040.1	60.0	-

5.4 Results on Larger Instances

To further evaluate the generalization and scalability of our framework, we conducted experiments on even larger instances (up to 1,500 operations). For each problem size, 10 instances were randomly generated, and both *MIStar* and OR-Tools were given a one-hour time limit per instance. We report their average optimality gap across different scales, whcih is calculated based on the Lower Bound (LB) as $\left(1 - \frac{\text{LB}}{\text{Objective}}\right) \times 100\%$. For *MIStar*, the results were measured at 200 iterations using the model trained on instances of size 20×15.

The results in Table 4 highlight two critical advantages of our approach. First and most strikingly, on the highly complex 50×30 instances, OR-Tools failed to produce any feasible solution within the time limit, whereas *MIStar* consistently found high-quality solutions. This demonstrates *MIStar*'s superior robustness and scalability on truly challenging problems. Second, for the other large instances where OR-Tools did find a solution, *MIStar* achieves comparable solution quality in a small fraction of the time. For example, on the 100×10 instances, *MIStar* reaches a 66.03% optimality gap in just under 16 minutes, while OR-Tools requires a full hour to achieve a 62.22% gap. This massive speed-up, combined with the strong generalization from a model trained only on smaller instances, confirms that *MIStar* provides an effective and highly scalable solution for large-scale FJSP.

5.5 Ablation Studies

We conducted ablation experiments on the SD2 dataset to assess the memory module and parallel greedy search strategy. As evidenced in Table 7 (Appendix J), their combination improves performance: the parallel greedy search strategy significantly enhances search efficiency and mitigates the risk of early convergence to local optima. Meanwhile, the memory module optimizes policy decisions, leading to higher solution quality. We further investigated the impact of the parallel scale P on our strategy. Figure 8 (Appendix J) demonstrates that increasing P generally improves performance at the cost of longer runtimes, with the optimal value depending on the instance characteristics. Therefore, P should be properly adjusted to achieve a trade-off between runtime and solution quality.

6 Conclusion

This study proposes *MIStar*, the first DRL-based improvement heuristic framework for solving the FJSP. By formulating the iterative refinement as an MDP, we introduce a heterogeneous disjunctive graph representation tailored for FJSP solutions and employ a memory-enhanced heterogeneous graph neural network for feature extraction and thorough exploration of the solution space. Furthermore, we propose a parallel greedy search strategy that significantly reduces the number of iterations while achieving high-quality solutions. Extensive experiments on synthetic data and public benchmarks demonstrate the superior performance of *MIStar* over both traditional handcrafted improvement heuristics and DRL-based construction methods. Future work may focus on enabling the network to adaptively adjust the parallel scale P, and on extending our methodology to disruptive local moves in larger neighborhoods, such as the reconstruction of solution segments, to further enhance exploration and reduce sensitivity to initial solutions.

Acknowledgements

This work is supported by the Jilin Provincial Department of Science and Technology Project under Grant Grant20240101369JC.

References

[1] Heiner Lasi, Peter Fettke, Hans-Georg Kemper, Thomas Feld, and Michael Hoffmann. Industry 4.0. *Business & information systems engineering*, 6:239–242, 2014.

[2] Yi Wang, Hai-Shu Ma, Jing-Hui Yang, and Ke-Sheng Wang. Industry 4.0: a way from mass customization to mass personalization production. *Advances in manufacturing*, 5(4):311–320, 2017.

[3] Runwei Cheng, Mitsuo Gen, and Yasuhiro Tsujimura. A tutorial survey of job-shop scheduling problems using genetic algorithms—i. representation. *Computers & industrial engineering*, 30(4):983–997, 1996.

[4] Jian Zhang, Guofu Ding, Yisheng Zou, Shengfeng Qin, and Jianlin Fu. Review of job shop scheduling research and its new perspectives under industry 4.0. *Journal of intelligent manufacturing*, 30:1809–1830, 2019.

[5] Jin Xie, Liang Gao, Kunkun Peng, Xinyu Li, and Haoran Li. Review on flexible job shop scheduling. *IET collaborative intelligent manufacturing*, 1(3):67–77, 2019.

[6] Imran Ali Chaudhry and Abid Ali Khan. A research survey: review of flexible job shop scheduling techniques. *International Transactions in Operational Research*, 23(3):551–591, 2016.

[7] Guohui Zhang, Xinyu Shao, Peigen Li, and Liang Gao. An effective hybrid particle swarm optimization algorithm for multi-objective flexible job-shop scheduling problem. *Computers & Industrial Engineering*, 56(4):1309–1318, 2009.

[8] Wen Song, Xinyang Chen, Qiqiang Li, and Zhiguang Cao. Flexible job-shop scheduling via graph neural network and deep reinforcement learning. *IEEE Transactions on Industrial Informatics*, 19(2):1600–1610, 2022.

[9] Pedro Coelho, Ana Pinto, Samuel Moniz, and Cristovão Silva. Thirty years of flexible job-shop scheduling: a bibliometric study. *Procedia Computer Science*, 180:787–796, 2021.

[10] Philipp Wenzelburger and Frank Allgöwer. Model predictive control for flexible job shop scheduling in industry 4.0. *Applied Sciences*, 11(17):8145, 2021.

[11] Stéphane Dauzère-Pérès, Junwen Ding, Liji Shen, and Karim Tamssaouet. The flexible job shop scheduling problem: A review. *European Journal of Operational Research*, 314(2):409–432, 2024.

[12] Yoshua Bengio, Andrea Lodi, and Antoine Prouvost. Machine learning for combinatorial optimization: a methodological tour d'horizon. *European Journal of Operational Research*, 290(2):405–421, 2021.

[13] Mingzhao Wang, You Zhou, Zhiguang Cao, Yubin Xiao, Xuan Wu, Wei Pang, Yuan Jiang, Hui Yang, Peng Zhao, and Yuanshu Li. An efficient diffusion-based non-autoregressive solver for traveling salesman problem. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1*, page 1469–1480, 2025.

[14] Yubin Xiao, Di Wang, Boyang Li, Mingzhao Wang, Xuan Wu, Changliang Zhou, and You Zhou. Distilling autoregressive models to obtain high-performance non-autoregressive solvers for vehicle routing problems with faster inference speed. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 20274–20283, 2024.

[15] Yubin Xiao, Di Wang, Boyang Li, Huanhuan Chen, Wei Pang, Xuan Wu, Hao Li, Dong Xu, Yanchun Liang, and You Zhou. Reinforcement learning-based nonautoregressive solver for traveling salesman problems. *IEEE Transactions on Neural Networks and Learning Systems*, pages 1–15, 2024.

[16] Yubin Xiao, Di Wang, Xuan Wu, Yuesong Wu, Boyang Li, Wei Du, Liupu Wang, and You Zhou. Improving generalization of neural vehicle routing problem solvers through the lens of model architecture. *Neural Networks*, 187:107380, 2025.

[17] Yubin Xiao, Yuesong Wu, Rui Cao, Di Wang, Zhiguang Cao, Peng Zhao, and Yuan Jiang. Dgl: Dynamic global-local information aggregation for scalable vrp generalization with self-improvement learning. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2025.

[18] Xuan Wu, Di Wang, Chunguo Wu, Lijie Wen, Chunyan Miao, Yubin Xiao, and You Zhou. Efficient heuristics generation for solving combinatorial optimization problems using large language models. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, pages 3228–3239, 2025.

[19] Xuan Wu, Di Wang, Lijie Wen, Yubin Xiao, Chunguo Wu, Yuesong Wu, Chaoyu Yu, Douglas L. Maskell, and You Zhou. Neural combinatorial optimization algorithms for solving vehicle routing problems: A comprehensive survey with perspectives, 2024.

[20] Nina Mazyavkina, Sergey Sviridov, Sergei Ivanov, and Evgeny Burnaev. Reinforcement learning for combinatorial optimization: A survey. *Computers & Operations Research*, 134:105400, 2021.

[21] Cong Zhang, Zhiguang Cao, Wen Song, Yaoxin Wu, and Jie Zhang. Deep reinforcement learning guided improvement heuristic for job shop scheduling. In *The Twelfth International Conference on Learning Representations*, 2024.

[22] Runqing Wang, Gang Wang, Jian Sun, Fang Deng, and Jie Chen. Flexible job shop scheduling via dual attention network-based reinforcement learning. *IEEE Transactions on Neural Networks and Learning Systems*, 35(3):3091–3102, 2023.

[23] Cong Zhang, Wen Song, Zhiguang Cao, Jie Zhang, Puay Siew Tan, and Xu Chi. Learning to dispatch for job shop scheduling via deep reinforcement learning. *Advances in neural information processing systems*, 33:1621–1632, 2020.

[24] Kun Lei, Peng Guo, Wenchao Zhao, Yi Wang, Linmao Qian, Xiangyin Meng, and Liansheng Tang. A multi-action deep reinforcement learning framework for flexible job-shop scheduling problem. *Expert Systems with Applications*, 205:117796, 2022.

[25] Ruiqi Chen, Wenxin Li, and Hongbing Yang. A deep reinforcement learning framework based on an attention mechanism and disjunctive graph embedding for the job-shop scheduling problem. *IEEE Transactions on Industrial Informatics*, 19(2):1322–1331, 2022.

[26] Junyoung Park, Jaehyeong Chun, Sang Hun Kim, Youngkook Kim, and Jinkyoo Park. Learning to schedule job-shop problems: representation and policy learning using graph neural network and reinforcement learning. *International journal of production research*, 59(11):3360–3377, 2021.

[27] Nathan Grinsztajn, Daniel Furelos-Blanco, Shikha Surana, Clément Bonnet, and Tom Barrett. Winner takes it all: Training performant rl populations for combinatorial optimization. *Advances in Neural Information Processing Systems*, 36:48485–48509, 2023.

[28] Egon Balas. Machine sequencing via disjunctive graphs: an implicit enumeration algorithm. *Operations research*, 17(6):941–957, 1969.

[29] Cong Zhang, Zhiguang Cao, Yaoxin Wu, Wen Song, and Jing Sun. Learning topological representations with bidirectional graph attention network for solving job shop scheduling problem. In *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*, 2024.

[30] Ke Li, Fei Liu, Zhenkun Wang, and Qingfu Zhang. Destroy and repair using hyper-graphs for routing. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 18341–18349, 2025.

[31] Peng Zhao, You Zhou, Di Wang, Zhiguang Cao, Yubin Xiao, Xuan Wu, Yuanshu Li, Hongjia Liu, Wei Du, Yuan Jiang, and Liupu Wang. Dual operation aggregation graph neural networks for solving flexible job-shop scheduling problem with reinforcement learning. In *Proceedings of the ACM Web Conference 2025*, 2025.

[32] Peng Zhao, Zhiguang Cao, Di Wang, Wen Song, Wei Pang, You Zhou, and Yuan Jiang. Visual-enhanced multimodal framework for flexible job shop scheduling problem. In *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025.

[33] Shu Luo. Dynamic scheduling for flexible job shop with new job insertions by deep reinforcement learning. *Applied Soft Computing*, 91:106208, 2020.

[34] Yi Feng, Lu Zhang, Zhile Yang, Yuanjun Guo, and Dongsheng Yang. Flexible job shop scheduling based on deep reinforcement learning. In *2021 5th Asian Conference on Artificial Intelligence Technology (ACAIT)*, pages 660–666. IEEE, 2021.

[35] Shu Luo, Linxuan Zhang, and Yushun Fan. Dynamic multi-objective scheduling for flexible job shop by deep reinforcement learning. *Computers & Industrial Engineering*, 159:107489, 2021.

[36] Renke Liu, Rajesh Piplani, and Carlos Toro. Deep reinforcement learning for dynamic scheduling of a flexible job shop. *International Journal of Production Research*, 60(13):4049–4069, 2022.

[37] Lanjun Wan, Long Fu, Changyun Li, and Keqin Li. Flexible job shop scheduling via deep reinforcement learning with meta-path-based heterogeneous graph neural network. *Knowledge-Based Systems*, 296:111940, 2024.

[38] Zhengqi Zeng, Xiaoxia Li, and Changbo Bai. A deep reinforcement learning approach to flexible job shop scheduling. In *2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pages 884–890. IEEE, 2022.

[39] Wenquan Zhang, Fei Zhao, Yong Li, Chao Du, Xiaobing Feng, and Xuesong Mei. A novel collaborative agent reinforcement learning framework based on an attention mechanism and disjunctive graph embedding for flexible job shop scheduling problem. *Journal of Manufacturing Systems*, 74:329–345, 2024.

[40] Jonas K Falkner, Daniela Thyssens, Ahmad Bdeir, and Lars Schmidt-Thieme. Learning to control local search for combinatorial optimization. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 361–376. Springer, 2022.

[41] Constantin Waubert de Puiseau, Fabian Wolz, Merlin Montag, Jannik Peters, Hasan Tercan, and Tobias Meisen. Decision transformer for enhancing neural local search on the job shop scheduling problem. *arXiv e-prints*, pages arXiv–2409, 2024.

[42] Xinyun Chen and Yuandong Tian. Learning to perform local rewriting for combinatorial optimization. *Advances in neural information processing systems*, 32, 2019.

[43] Hao Lu, Xingwen Zhang, and Shuang Yang. A learning-based iterative method for solving vehicle routing problems. In *International conference on learning representations*, 2019.

[44] Yaoxin Wu, Wen Song, Zhiguang Cao, Jie Zhang, and Andrew Lim. Learning improvement heuristics for solving routing problems. *IEEE transactions on neural networks and learning systems*, 33(9):5057–5069, 2021.

[45] ChaoYong Zhang, PeiGen Li, ZaiLin Guan, and YunQing Rao. A tabu search algorithm with a new neighborhood structure for the job shop scheduling problem. *Computers & Operations Research*, 34(11):3229–3242, 2007.

[46] Eugeniusz Nowicki and Czeslaw Smutnicki. A fast taboo search algorithm for the job shop problem. *Management science*, 42(6):797–813, 1996.

[47] Andoni I Garmendia, Quentin Cappart, Josu Ceberio, and Alexander Mendiburu. Marco: A memory-augmented reinforcement framework for combinatorial optimization. *arXiv preprint arXiv:2408.02207*, 2024.

[48] Andoni I Garmendia, Josu Ceberio, and Alexander Mendiburu. Neural improvement heuristics for graph combinatorial optimization problems. *IEEE Transactions on Neural Networks and Learning Systems*, 2023.

[49] Zhi-Hua Zhou. *Ensemble methods: foundations and algorithms*. CRC press, 2025.

[50] Igor G Smit, Jianan Zhou, Robbert Reijnen, Yaoxin Wu, Jian Chen, Cong Zhang, Zaharah Bukhsh, Yingqian Zhang, and Wim Nuijten. Graph neural networks for job shop scheduling problems: A survey. *Computers & Operations Research*, page 106914, 2024.

[51] Monaldo Mastrolilli and Luca Maria Gambardella. Effective neighbourhood functions for the flexible job shop problem. *Journal of scheduling*, 3(1):3–20, 2000.

[52] Alain Bretto. Hypergraph theory. *An introduction. Mathematical Engineering. Cham: Springer*, 1:209–216, 2013.

[53] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? *arXiv preprint arXiv:1810.00826*, 2018.

[54] Johann Hurink, Bernd Jurisch, and Monika Thole. Tabu search for the job-shop scheduling problem with multi-purpose machines. *Operations-Research-Spektrum*, 15:205–215, 1994.

[55] Paolo Brandimarte. Routing and scheduling in a flexible job shop by tabu search. *Annals of Operations research*, 41(3):157–183, 1993.

[56] Pierre Hansen and Nenad Mladenović. First vs. best improvement: An empirical study. *Discrete Applied Mathematics*, 154(5):802–817, 2006.

[57] Helena Ramalhinho Lourenço, Olivier C Martin, and Thomas Stützle. Iterated local search: Framework and applications. In *Handbook of metaheuristics*, pages 129–168. Springer, 2018.

[58] Danial Rooyani and Fantahun M Defersha. An efficient two-stage genetic algorithm for flexible job-shop scheduling. *Ifac-Papersonline*, 52(13):2519–2524, 2019.

[59] Dennis Behnke and Martin Josef Geiger. Test instances for the flexible job shop scheduling problem with work centers. 2012.

[60] Anssi Kanervisto, Christian Scheller, and Ville Hautamäki. Action space shaping in deep reinforcement learning. In *2020 IEEE conference on games (CoG)*, pages 479–486. IEEE, 2020.

[61] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

A Comparative Analysis of Neighborhood Structures

The critical path, whose length represents the makespan $C_{\max}$, is essential in constructing feasible solutions [55]. Operations on this path are critical operations, and critical blocks are maximal sequences of adjacent critical operations processed on the same machine [45]. In local search, the neighborhood comprises feasible solutions generated by small perturbations to the current one [55]. For scheduling, these perturbations typically involve reordering operations on the critical path. The neighborhood structure directly affects the efficiency of the search algorithm, making it important to eliminate unnecessary or infeasible moves [45]. Among various neighborhood designs, the N5 neighborhood is significantly smaller than others [45]. It identifies a single critical path and defines a move by reversing either the first two or the last two operations within a critical block.

While effective for JSP [21], the N5 neighborhood cannot be applied to FJSP. Its limitation lies in only permitting reordering within one critical block, where operations are processed on the same machine. This perturbation preserves the original machine assignments, severely constraining the exploration of the solution space. More critically, unlike JSP where operations of the same job are processed on different machines, FJSP allows multiple operations of a job to be processed on the same one. Consequently, swapping operations within critical blocks in FJSP may violate job precedence constraints, resulting in infeasible solutions (Figure 5).

In contrast, we employ the Nopt2 neighborhood [51] that simultaneously modifies operation sequences and machine assignments while maintaining feasibility through time-constrained reinsertion (see paragraph 3). This structure enables broader solution space exploration and improves efficiency by using optimal insertion intervals to reduce the neighborhood size.

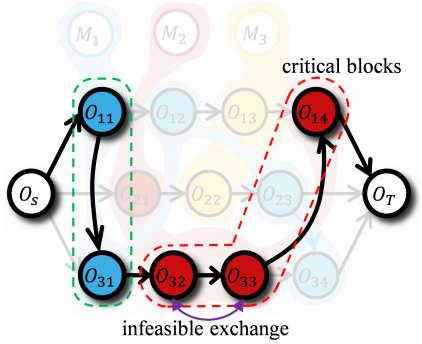


Figure 5 **Infeasible exchange caused by the N5 neighborhood in FJSP.** Dotted lines of different colors are critical blocks.

B Definition of Raw Feature Vectors

The raw features of operations and machines at each state s_t (omitting step t) are defined as follows. *Features of Operations*: Each operation node $O_{ij} \in \mathcal{O} \setminus \{O_S, O_T\}$ is represented by a 9-dimensional feature vector $\mathbf{x}_{O_{ij}}$:

1) **Minimum Processing Time**: shortest processing time among compatible machines.
2) **Average Processing Time**: mean processing time among compatible machines.
3) **Range of Processing Time**: difference between the maximum and minimum processing times among compatible machines.
4) **Machine Availability Ratio**: ratio of the number of compatible machines to total machines.
5) **Processing Time**: processing time on the assigned machine.
6) **Earliest Start Time**: earliest possible start time of the operation.
7) **Latest Start Time**: latest start time without schedule delay.
8) **Scheduling Order on Machine**: execution order on machine M_k (starting from 1).
9) **Processing Time of Job**: total processing time of parent job.

Features of Machines: Each machine node $M_k \in \mathcal{M}$ owns a 4-dimensional feature vector $\mathbf{y}_{M_k}$:

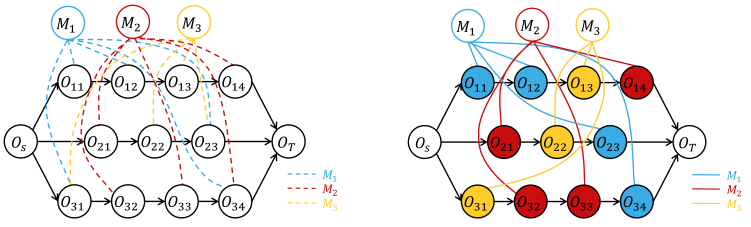
1) **Utilization Rate**: ratio of active processing time to makespan.

2) **Criticality Degree**: proportion of operations on critical path among processable operations.
3) **Load Ratio**: percentage of assigned operations versus machine capacity.
4) **Processing Efficiency**: mean processing time of assigned operations.

C Advantages of the Directed Heterogeneous Graph

In the FJSP, the global scheduling state comprises both operations and machines. However, the absence of machine nodes in traditional disjunctive graphs results in an indirect representation of machine information through numerical values [21], which limits the policy from assigning operations to available machines effectively. The heterogeneous disjunctive graph $\mathcal{H}$ [8] addresses this by introducing machine nodes (Fig. 6), where each operation connects to compatible machines via undirected edges, such that solving the FJSP is equivalent to selecting one O-M arc per operation while removing others. This reduces the graph density from $\sum_{k=1}^{|\mathcal{M}|}\binom{n_k}{2}$ to $\sum_{k=1}^{|\mathcal{M}|} n_k$ with n_k being the number of operations assignable to machine M_k and enables better injection of machine status [8].

While beneficial for construction methods, this structure provides limited advantages for improvement approaches where complete solutions contain fixed machine assignments. This leads to a total of $|\mathcal{O}|$ disjunctive arcs—more than the $|\mathcal{O}| - |\mathcal{M}|$ arcs in the conventional graph, negating the benefit of reduced graph density. Most importantly, undirected edges hinder the clear encoding of machine processing sequences, which is crucial for distinguishing between different scheduling solutions and identifying the critical path when constructing the action space [21, 51]. Instead, our heterogeneous graph $\overrightarrow{\mathcal{H}}$(Figure 1c), tailored for complete schedules, overcomes this limitation by encoding machine processing sequences through directed hyper-edges, which effectively represent the ordering of operations across different solutions without significantly increasing graph density. It adopts a unified structure with fully directed arrows, avoiding the complexity of mixed edge types and better aligning with current learning-to-search graph-based frameworks [21].



(a) (b)

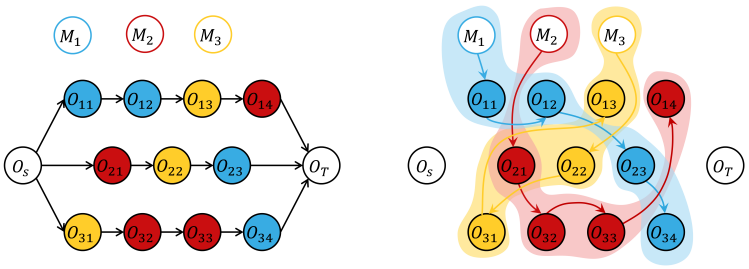
Figure 6 **Graph representations in $\mathcal{H}$ [8].** (a) FJSP instance; (b) feasible solution.

D Compact State Representation in Memory

D.1 Motivation and Compact Formulation

During the improvement, we store the state-action pair (s_t, a_t) at each step in a memory module. However, the state is represented as a complex heterogeneous graph with rich node attributes and adjacency relations, leading to excessive memory consumption if fully stored. Furthermore, evaluating similarity between such graphs typically requires GNNs to extract embeddings, introducing additional network components and computational cost.

To address this, we simplify the graph representation to better support our memory mechanism. First, since our current goal is to distinguish different scheduling solutions—rather than using node features as heuristic guidance—we omit the node attributes and retain only the graph topology. Second, for a given problem instance, solutions differ only in operation sequences on machines (i.e., hyper-arcs), while sharing the same job precedence constraints (i.e., conjunctive arcs). Based on this insight, we isolate the hyper-arcs from $\overrightarrow{\mathcal{H}}_t$ (see Fig. 7) and use them as the key feature to characterize each solution. Specifically, we use the operation-machine incidence matrix $\boldsymbol{L}_t$ as a compact representation of s_t and measure similarity via the Frobenius inner product.



(a) (b)

Figure 7 **Decomposition of FJSP solution structure.** (a) shared job precedence constraints; (b) distinct machine processing sequences.

Instead of serving as merely binary indicators of machine assignment, the non-zero element $(\boldsymbol{L}_t)_{i,j}$ represents the processing order index of operation i on its assigned machine j. This crucial design ensures that two schedules with identical machine assignments but different sequences (e.g., a good sequence vs. a poor one) are correctly distinguished. For instance, a good sequence of operations corresponding to [*1, 2, 3, 4, 5*] and a poor, reversed sequence [*5, 4, 3, 2, 1*] on the same machine will be recognized as maximally dissimilar by the Frobenius inner product. The mathematical foundation for this lies in the **Rearrangement Inequality**, which guarantees that the inner product is maximized for similarly ordered sequences and minimized for oppositely ordered ones. This allows our similarity metric to be both computationally efficient and highly sensitive to the quality of the operation sequence.

Theorem 1. *The Rearrangement Inequality states that, for two sequences* $a_1 \leq a_2 \leq \cdots \leq a_n$ *and* $b_1 \leq b_2 \leq \cdots \leq b_n$*, the inequalities*

$$a_1 b_n + a_2 b_{n-1} + \cdots + a_n b_1 \leq a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \cdots + a_n b_{\pi(n)} \leq a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$$

hold, where $\pi(1), \pi(2), \ldots, \pi(n)$ *is any permutation of* $1, 2, \ldots, n$.

D.2 Alternative Designs

We have considered and tested other expressive representations for the historical states that explicitly encode sequence information.

Multi matrices: The adjacency matrix A_t^M of operations on machines encodes the processing order between operations, while the binary operation-machine incidence matrix L_t^B indicates machine assignment. The state similarity is then calculated by computing the inner products of both matrices separately and summing the results, expressed as:

$$\omega_{t,t'} = \langle L_t^B, L_{t'}^B \rangle_F + \langle A_t^M, A_{t'}^M \rangle_F \tag{8}$$

Difference matrix: For the operation-machine incidence matrices that store processing order indices, we compute the element-wise difference between two schedules, L_t and $L_{t'}$, to obtain a difference matrix, and then use the reciprocal of its Frobenius norm to measure similarity, formulated as:

$$\omega_{t,t'} = \frac{1}{|L_t - L_{t'}|_F + \epsilon} = \frac{1}{\sqrt{\sum_{i=1}^{|\mathcal{O}|} \sum_{j=1}^{|\mathcal{M}|} |(L_t)_{i,j} - (L_{t'})_{i,j}|^2} + \epsilon} \tag{9}$$

However, these alternatives introduced higher storage or computational overhead without yielding significant performance gains. Our chosen representation thus strikes a deliberate and effective balance between representational power and efficiency.

E Complexity Analysis of Action Space

Let $F_O \in \mathbb{R}^{|\mathcal{O}| \times q}$ and $F_M \in \mathbb{R}^{|\mathcal{M}| \times q}$ denote the embedding matrices of operation and machine nodes, respectively. To assign a priority score to each possible triple-action $[O_m, O_n, M_k]$, we compute the priority matrix $F_{OOM} \in \mathbb{R}^{|\mathcal{O}| \times |\mathcal{O}| \times |\mathcal{M}|}$ through a two-step tensor operations involving F_O and F_M. As a result, the space complexity of the action space $\mathcal{A}_{original}$ is $O(|\mathcal{O}|^2 |\mathcal{M}|)$. Prior

work [60] has shown that a large number of possible actions is known to lead to over-optimistic estimates of future rewards, degrading agent performance and learning efficiency. To address this, we reshape the action space by exploiting the constraint of FJSP: each operation can only be assigned to a unique machine. Given a solution of FJSP, there exists a mapping $f_M : \mathcal{O} \to \mathcal{M}$, which allows us to omit the third dimension and represent actions as pairs $[O_m, O_n]$. The space complexity of $\mathcal{A}_{optimized}$ is reduced to $O(|\mathcal{O}|^2)$ and the optimized priority matrix is calculated as follows:

$$F_{OO} = F_O F_O^\top \in \mathbb{R}^{|\mathcal{O}|\times|\mathcal{O}|} \tag{10}$$

We can formalize the relationship between the original and optimized action spaces as follows:

Theorem 2. *The optimized action space $\mathcal{A}_{optimized}$ is functionally equivalent to $\mathcal{A}_{original}$ in terms of admissible actions under the FJSP constraints.*

Proof. For any feasible action $a = [O_m, O_n, M_k] \in \mathcal{A}_{original}$, there exists a unique $a' = [O_m, O_n] \in \mathcal{A}_{optimized}$ with $M_k = f_M(O_n)$. Conversely, any $a' \in \mathcal{A}_{optimized}$ can be mapped back to $a \in \mathcal{A}_{original}$ via f_M. Thus, this compaction preserves all feasible actions while eliminating invalid ones (i.e., actions where the operation and machine do not match). This shaping of the action space reduces learning difficulty and accelerates model convergence without compromising the agent's performance. □

Algorithm 1: Training procedure with n-step PPO using parallel search.

Input: *MIStar* with trainable parameters $\Theta = \{\delta, \theta, \varphi\}$, parallel scale P, iteration limit T, update size n, validate size v, new data generation step d, batch size B, total number of training epochs I;

Output: Trained *MIStar* with parameters $\Theta^* = \{\delta^*, \theta^*, \varphi^*\}$;

for $i = 0$ **to** $i < I$ **do**
  if $i \bmod d = 0$ **then**
    Randomly generate B instances;
  end
  Compute initial solutions $\{s_0^1, \ldots, s_0^B\}$ using DANIEL;
  for $t = 0$ **to** T **do**
    for $s_t^b \in \{s_t^1, \ldots, s_t^B\}$ **do**
      Initialize a training data buffer D^b and memory buffer H^b with size 0;
      Extract embeddings using MHGNN;
      for $p = 1$ **to** P **do**
        Sample a local move $a_t^{bp} \sim \pi_\delta(\cdot|s_t^b)$;
        Compute $C_{\max}(s_{t+1}^{bp})$ w.r.t a_t^{bp};
      end
      Identify the best action a_t^{b*} with greatest improvement ;
      Update s_t^b w.r.t a_t^{b*} and compute similarity score Ω from H^b;
      Compute reward $r_t(s_t^b, a_t^{b*}) = r_{gain} - r_{penalty}$;
      Store the data (s_t^b, a_t^{b*}, r_t) into D^b and store (s_t^b, a_t^{b*}) into H^b;
      if $t \bmod n = 0$ **then**
        Compute the generalised advantage estimates $\hat{A}_t$;
        Compute the PPO loss L, and optimize the parameters Θ for R epochs;
        Update network parameters;
      end
      Clear buffers D^b and H^b;
    end
  end
  if $i \bmod v = 0$ **then**
    Validate the policy;
  end
  $i = i + B$;
end

F The n-step PPO Algorithm with Parallel Greedy Exploration Strategy

We adopt the PPO algorithm [61] based on the actor-critic architecture as our training strategy. The actor is the policy network and the critic provides state value estimation. The network parameters are updated every n steps, which effectively addresses sparse rewards and out-of-memory issues [21]. The flexibility of FJSP leads to a large solution space, requiring numerous iterations to converge to high-quality solutions [30]. To address this challenge, we introduce a parallel greedy exploration strategy, which evaluates multiple candidate actions in parallel and greedily selects the one yielding the greatest improvement. This approach enables the exploration of multiple solutions per iteration, yielding high-quality results with fewer iterations. Moreover, by prioritizing solutions with the greatest improvement, the model is guided toward more promising paths, minimizing the interference from inefficient paths during gradient updates and accelerating convergence.

The pseudo-code of the n-step PPO algorithm is presented in Algorithm 1.

G Training Configurations

All hyperparameters were fine-tuned on the smallest-scale instances from SD2 and kept the same for all other models. Experiments were conducted on a platform equipped with an Intel i9-12900K processor with 24 cores, 64 GB memory, and an NVIDIA RTX 4090 GPU with 24 GB VRAM. Detailed specifications are provided in Table 5. Upon acceptance of this manuscript for publication, the full code will be made available on GitHub.

Table 5 Training configuration parameters

Parameter	Value
Number of MHGNN layers (L)	4
Number of Actor network layers	4
Number of Critic network layers	3
Number of attention heads in GAT (n_H)	1
Dimension of hidden layers (q)	64
GIN learnable parameter (ϵ)	0
Discount factor (γ)	1
Clipping ratio	0.2
Policy function coefficient	1
Value function coefficient	0.5
Entropy coefficient	0.01
Learning rate	5×10^{-4}
Optimizer	Adam
Memory buffer size	600
KNN retrieval count (K)	15
Similarity penalty coefficient (λ)	10
PPO update interval (n)	10
Rounds per PPO update (R)	3
Total training epochs (I)	20,000
Validation interval (v)	5
Instance generation interval (d)	20
Iterations per instance (T)	100
Batch size (B)	20
Parallel scale (P)	50

H Detailed Analysis on Synthetic Instances

Our experiments on four small-sized instances from SD2 reveal that *MIStar* generates an action space consisting of dozens to hundreds of possible moves per iteration using the Nopt2 neighborhood—up to 100 times larger than [21] for solving JSP. While this expanded action space provides greater

Table 6 Results of different initialization methods

Method	Iter.	SD2							
		10×5		20×5		15×10		20×10	
		Obj.	Gap	Obj.	Gap	Obj.	Gap	Obj.	Gap
DANIEL(S)	0	365.26	0.00%	629.56	0.00%	522.51	0.00%	552.38	0.00%
	100	345.05	5.53%	615.12	2.29%	488.22	6.56%	530.93	3.88%
	200	343.43	5.98%	613.75	2.51%	476.93	8.72%	528.92	4.25%
	400	**342.50**	6.23%	**612.73**	2.67%	**465.71**	10.87%	**525.49**	4.87%
HGNN(S)	0	479.35	0.00%	963.03	0.00%	759.54	0.00%	984.64	0.00%
	100	375.20	21.73%	745.18	22.62%	677.96	10.74%	879.48	10.68%
	200	366.35	23.57%	702.90	27.01%	626.90	17.46%	812.53	17.48%
	400	359.10	25.09%	674.93	29.92%	569.81	24.98%	737.42	25.11%
SPT	0	503.86	0.00%	799.56	0.00%	670.56	0.00%	775.03	0.00%
	100	399.25	20.76%	692.43	13.40%	578.48	13.73%	695.34	10.28%
	200	393.74	21.86%	680.62	14.88%	550.94	17.84%	673.44	13.11%
	400	389.34	22.73%	671.05	16.07%	521.96	22.16%	648.48	16.33%
MWKR	0	493.36	0.00%	949.85	0.00%	737.44	0.00%	953.11	0.00%
	100	382.09	22.55%	747.12	21.34%	664.69	9.87%	858.45	9.93%
	200	372.62	24.47%	705.15	25.76%	618.86	16.08%	799.61	16.11%
	400	364.99	26.02%	676.88	28.74%	563.56	23.58%	729.91	23.42%
FIFO	0	476.96	0.00%	931.50	0.00%	763.12	0.00%	978.03	0.00%
	100	376.29	21.11%	729.85	21.65%	669.51	12.27%	861.06	11.96%
	200	367.70	22.91%	693.35	25.57%	618.74	18.92%	799.15	18.29%
	400	360.68	24.38%	668.26	28.26%	562.59	26.28%	728.91	25.47%
RANDOM	0	585.08	0.00%	1067.38	0.00%	1062.39	0.00%	1304.70	0.00%
	100	382.24	34.67%	750.69	29.67%	733.17	30.99%	960.68	26.37%
	200	371.38	36.52%	707.33	33.73%	661.03	37.78%	865.85	33.64%
	400	365.33	**37.56%**	678.00	**36.48%**	589.53	**44.51%**	772.15	**40.82%**

optimization potential, it also introduces more suboptimal solutions, increasing the risk of local optima entrapment. Moreover, even for small-sized problems, OR-Tools could only optimally solve a small portion of instances within the time limit [22]. These observations underscore the complexity of the FJSP. Compared with rule-based improvement heuristics, *MIStar* shows superior performance under the same iteration budget, with only a marginal shortfall (0.06%) against the BI rule on the SD2 40×10 instance at 100 iterations. Notably, even equipped with *restart*, rule-based methods still tend to stagnate as iterations increase—likely due to the large and complex solution space—whereas *MIStar* continues to improve, demonstrating the effectiveness of the learned policy. Additionally, *MIStar* reduces runtime by directly outputting local moves, avoiding exhaustive evaluations across the entire neighborhood.

I Improvement over Varied Initial Solution Generation Methods

We systematically evaluate the optimization capabilities of *MIStar* using six initialization strategies: two DRL-based construction methods with sampling (DANIEL(S) and HGNN(S)), random initialization, and three classical priority dispatching rules—shortest processing time (SPT), most work remaining (MWKR), and first-in-first-out (FIFO) [55]. Experiments are conducted on the SD2 dataset with models trained on instances of the same size, to assess the performance across different initialization methods under the fixed iteration budget. Improvement is measured by the makespan gap, calculated as $\left(1 - \frac{C_{\text{best}}}{C_0}\right) \times 100\%$, where C_0 is the makespan of the initial solution and C_{best} is the incumbent optimal value.

The results are detailed in Table 6. It can be observed that the quality of initial solutions impacts the search efficiency of *MIStar*. With the same number of iterations, better initial solutions enable *MIStar* to generate higher-quality schedules, while poorer ones typically allow for larger relative improvements. Notably, after 400 iterations, solutions initialized by HGNN(S), MWKR, and FIFO outperform those generated by DANIEL(S) with 0 iteration on the 10×5 instances. This demonstrates that, with a sufficient iteration budget, *MIStar* has the potential to elevate weaker initial solutions to a performance level comparable to that of stronger ones. Moreover, even when starting from (near)-optimal solutions initialized by DANIEL(S), *MIStar* still achieves notable improvements, further validating the effectiveness of our framework.

Table 7 Performance of ablated models

Model	Iter.	SD2							
		10×5		20×5		15×10		20×10	
		Obj.	Time[1]	Obj.	Time	Obj.	Time	Obj.	Time
Baseline	100	364.27	0.76s	629.09	1.16s	519.34	1.82s	552.46	2.37s
	200	364.27	1.48s	629.09	2.44s	519.34	3.69s	552.46	4.78s
	400	364.27	2.89s	629.09	4.85s	519.34	7.25s	552.46	9.48s
w/o GIN	100	348.31	4.97s	616.71	10.72s	491.78	18.12s	548.61	24.80s
	200	347.06	10.78s	614.98	22.88s	484.36	37.58s	545.97	53.68s
	400	345.73	20.56s	613.98	45.90s	477.80	1.45m	544.31	1.91m
w/o GAT	100	350.86	5.82s	618.67	11.34s	492.76	17.85s	548.0	22.63s
	200	349.89	9.46s	617.33	22.91s	486.74	36.97s	547.97	51.02s
	400	348.33	21.41s	615.99	46.79s	480.66	1.29m	547.87	1.78m
w/o HGAT	100	350.49	5.44s	618.15	11.39s	491.97	17.95s	535.78	23.89s
	200	349.08	9.74s	616.56	23.01s	483.46	37.01s	532.83	52.47s
	400	347.17	20.10s	614.97	46.38s	473.83	1.29m	529.92	1.67m
w/o Par.	100	363.02	0.59s	627.20	0.98s	518.28	1.68s	551.39	2.41s
	200	363.02	1.22s	627.20	2.13s	518.28	4.02s	551.39	6.18s
	400	363.02	2.59s	627.20	5.05s	518.28	11.72s	551.39	17.57s
w/o Mem.	100	346.11	5.47s	615.99	13.98s	490.63	17.25s	534.31	25.47s
	200	344.76	11.04s	615.32	25.33s	486.16	35.85s	531.71	51.08s
	400	343.53	20.24s	614.80	48.91s	482.16	73.14s	528.44	1.69m
Ours	100	345.05	5.27s	615.12	11.55s	488.22	18.09s	530.93	25.76s
	200	343.43	10.33s	613.75	23.50s	476.93	37.42s	528.92	53.25s
	400	**342.50**	21.39s	**612.73**	47.93s	**465.71**	1.31m	**525.49**	1.88m

[1] "s" and "m" denote seconds and minutes, respectively.

J Results on Ablation Studies

J.1 Analysis on Components and Strategy

We conducted comprehensive ablation studies to evaluate the contributions of the key components of our framework, including both the model architecture and the search strategy. The experiments were performed on four small-size instances from the SD2 dataset.

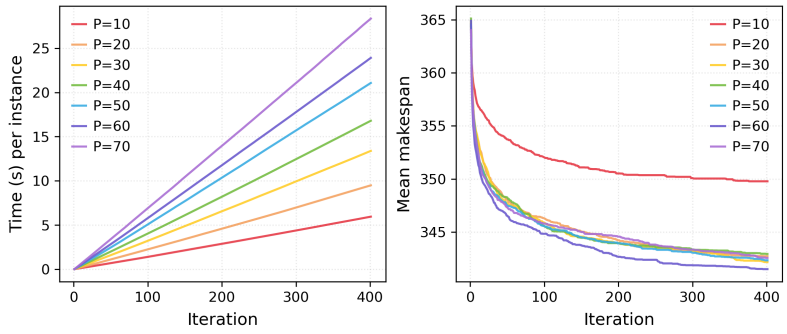
We first assessed the roles of the memory module and the parallel greedy search strategy. Three ablated variants are compared: the *Baseline* model without any enhancements, a version without the parallel greedy search strategy (*w/o Par.*), and another without the memory module (*w/o Mem.*). Results in Table 7 demonstrate the critical roles of both components. Although our parallel approach takes longer per iteration, it achieves superior solutions in fewer steps, improving search efficiency. In contrast, the model without parallelism runs faster but consistently converges to inferior local optima. Furthermore, while the memory module helps enhance the decision-making of the policy, it alone (i.e., *w/o Par.*) proves insufficient to escape local optima. This limitation may stem from the fact that the FJSP solution space is extremely large and complex, making it challenging for the policy network to achieve effective global exploration through sparse, single-step sampling. For future work, a critical direction involves dynamically constraining the search space or optimizing the memory mechanism to selectively extract salient historical features.

Our tailored representation module, which is composed of multiple specialized graph encoders, is not only theoretically motivated by the structural characteristics of FJSP graphs but is also aligned with mainstream graph learning practices [8, 21, 24, 50]. The ablation results by removing each component (i.e., *w/o GAT*, *w/o GIN* and *w/o HGAT*) confirm that each of them is essential for learning informative and discriminative state representations that support effective policy learning.

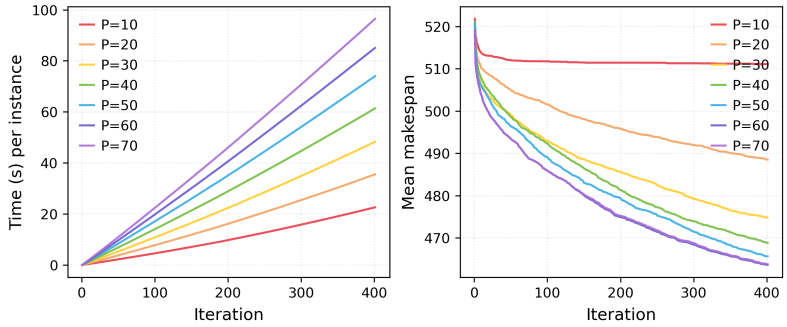
J.2 Evaluation of Search Scale

Using models trained with $P = 50$, we systematically evaluated the search strategy across different parallel scales $P \in \{10, 20, 30, 40, 50, 60, 70\}$ on SD2 instances of two sizes. The result curves are illustrated in Figure 8. It is evident that increasing P leads to longer runtimes under a fixed number of iterations. Moreover, the runtime scales linearly with the iteration count across different P values. Regarding solution quality, the 10×5 instance has a relatively small neighborhood (approximately 80 solutions), making $P = 20$ sufficient to cover most promising moves. Consequently, further increasing P does not yield noticeable improvement. On the other hand, the 15×10 instance has

around 240 neighbors, so increasing P from 10 to 60 continuously improves performance. However, the difference between $P = 60$ and $P = 70$ becomes negligible, indicating a saturation point beyond which additional evaluations offer little benefit. These observations highlight the importance of selecting an appropriate P based on the problem size and the flexibility of alternative machines to balance performance gains and computational cost. In future work, we plan to enable the network to dynamically adapt P according to instance characteristics, reducing manual tuning efforts.



(a) 10×5 instances.



(b) 15×10 instances.

Figure 8 **Runtime (left) and performance (right) curves across different parallel scale.**

2. 数据驱动决策：From Contextual Data to Newsvendor Decisions: On the Actual Performance of Data-Driven Algorithms (MS, 2026)

来源：Management Science, 2026.

https://doi.org/10.1287/mnsc.2023.02068

作者：Omar Besbes, Will Ma, Omar Mouchtaki

关键词：情境报童；加权经验风险最小化；近邻；最坏遗憾；数据质量

推荐人：段睿

推荐理由：本文以情境报童问题分析历史数据的数量和相关性如何影响决策效果，假设相近情境下的需求分布也相近。作者研究按情境相似度赋权的经验风险最小化策略，涵盖经验风险最小化、近邻与核方法，并对给定情境样本配置精确刻画最坏情形遗憾。利用报童损失的结构，原本在分布空间上的无限维优化可化为一维线搜索。这一精确分析揭示了通用上界难以展示的数据质量、样本数量与学习曲线之间的关系。

https://pubsonline.informs.org/journal/mnsc

MANAGEMENT SCIENCE
Articles in Advance, pp. 1–17
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

From Contextual Data to Newsvendor Decisions: On the Actual Performance of Data-Driven Algorithms

Omar Besbes,[a] Will Ma,[a] Omar Mouchtaki[b,*]

[a] Graduate School of Business, Columbia University, New York, New York 10027; [b] NYU Stern School of Business, New York, New York 10012
*Corresponding author

Contact: obesbes@columbia.edu, https://orcid.org/0000-0002-2982-3794 (OB); wm2428@gsb.columbia.edu, https://orcid.org/0000-0002-2420-4468 (WM); om2166@stern.nyu.edu, https://orcid.org/0000-0002-1734-9247 (OM)

Received: July 5, 2023
Revised: December 13, 2024; September 22, 2025
Accepted: November 19, 2025
Published Online in Articles in Advance: May 14, 2026

https://doi.org/10.1287/mnsc.2023.02068

Copyright: © 2026 INFORMS

Abstract. In this work, we study how the relevance/quality and quantity of past data influence performance by analyzing a contextual Newsvendor problem, in which a decision maker trades off between underage and overage costs under uncertain demand. We consider a setting in which past demands observed under "close-by" contexts come from close-by distributions and analyze the performance of data-driven algorithms through a notion of *context-dependent* worst-case expected regret. We analyze the broad class of Weighted Empirical Risk Minimization (WERM) policies which weigh past data according to their similarity in the contextual space. This class includes classical policies such as ERM, *k*-Nearest Neighbors, and kernel-based policies. Our main methodological contribution is to characterize *exactly* the worst-case regret of any WERM policy on any given configuration of contexts. To the best of our knowledge, this provides the first understanding of tight performance guarantees in any contextual decision-making problem, with past literature focusing on upper bounds via concentration inequalities. We instead take an optimization approach, and isolate a structure in the Newsvendor loss function that allows us to reduce the infinite-dimensional optimization problem over worst-case distributions to a simple line search. This in turn allows us to unveil fundamental insights that were obfuscated by previous general-purpose bounds. We characterize *actual* guaranteed performance as a function of the contexts, as well as granular insights on the learning curve of algorithms.

History: Accepted by Victor Martínez de Albéniz, operations management.
Funding: This work was supported by the Deming Center for Operations Innovation and Excellence at Columbia Business School [Doctoral Fellowship (O. Mouchtaki)].
Supplemental Material: The online appendix and data files are available at https://doi.org/10.1287/mnsc.2023.02068.

Keywords: data-driven stochastic optimization • weighted empirical risk minimization • *k*-Nearest Neighbors • contextual newsvendor • worst-case regret • data quality

1. Introduction

A fashion retailer is planning how many units to produce for a seasonal product (e.g., pairs of speckled gray gloves), in anticipation of demand, for the next selling season. It has historical sales numbers from recent years for it, as well as similar stock keeping units (SKUs) (e.g., speckled navy gloves). How should the retailer use these past data collected in contexts different from the one in which it is currently operating? Motivated by this question, this work aims to broadly understand the impact of data size and data relevance on performance in the contextual Newsvendor. The Newsvendor problem is a prototypical model of decision making in the face of uncertainty. This problem captures the common trade-off which emerges when making decisions to balance between an underage cost incurred when the resource is insufficient compared with the realized outcome and an overage cost which materializes when some resource is wasted, with this resource being the gloves that are produced and not sold in the example above. This model is widely applied across different areas such as inventory management, capacity planning, scheduling, or overbooking. Beyond the operational aspect of the model, the Newsvendor loss is of particular interest for its statistical interpretation; it generalizes the ℓ_1 loss function to allow for different weights on the error types (overestimating or underestimating) and coincides with the quantile loss used in quantile regression.

In this contextual Newsvendor problem, past data of the form $(x_i, y_i)_{i=1}^n$, where x_i denotes the context and y_i the demand observed under context x_i, are used to take an action a for a future ("out-of-sample") context x_0. The loss of the action $\ell(a, y_0)$ depends only on an unknown quantity y_0 associated with x_0. In this example, y_0 is the demand of speckled gray gloves to

materialize next winter, and the ideal production action $a = y_0$ would have loss $\ell(a, y_0) = 0$. However, two main challenges are present: (i) y_0 is not known in advance and (ii) it is drawn according to a distribution unknown to the decision maker. The observed $y_1, \ldots, y_n$ help to predict the distribution of y_0, and the contexts $x_0, x_1, \ldots, x_n$ are relevant only to the extent that they indicate how each y_i may be informative to predict the distribution of y_0. In particular, we assume a geometry (formalized later) where contexts x_i that are closer to x_0 are likely to have y_i values that are drawn from distributions "closer" to that of y_0. In this example, the context x_0 could be described by the vector (speckled, gray, 2024), and closer x_i would have more similar SKU features and come from more recent weeks.

It is generally presumed that the retailer can make a better production decision if it has more data and more contexts x_i that are close to x_0, and should place greater emphasis on the y_i values associated with these nearby x_i's when making its decision. However, despite the prevalence of contextual decision making in theory and practice, many fundamental questions are still not well understood. How do the dissimilarities between the available contexts $x_1, \ldots, x_n$ and x_0, or the quality of the historical data for the focal decision, affect performance? What levels of performance can be achieved as a function of the data at hand, and how many samples are needed to operate well? In the present paper, we explore a framework to study these questions systematically for the Newsvendor problem.

1.1. Main Contributions

To analyze how the quantity and relevance of historical data affect the performance of data-driven policies, we adopt a context-dependent notion of performance. We model demand distributions as unknown and context-specific, with the distance between distributions bounded by the dissimilarity of their contexts, a condition we refer to as the local condition (Definition 1). Policy performance is measured through expected regret, defined as the excess loss relative to the best action under the out-of-sample distribution, and we seek guarantees that hold in the worst case over distributions satisfying the local condition. By characterizing worst-case regret for different configurations of past and out-of-sample contexts, this framework captures a range of practical scenarios and makes it possible to isolate the impact of data relevance and data quantity on performance. Within this setting, we study the widely used class of Weighted Empirical Risk Minimization (ERM) policies, which encompasses ERM, k-Nearest Neighbors (k-NN), and Kernel methods.

1.1.1. Performance Characterization: Learning Without Concentration. Contrasting the typical approach of upper-bounding learning regret via concentration inequalities, we take a more precise approach of trying to quantify the *worst-case* learning regret. A priori, this optimization problem of identifying the hardest-to-learn distributions is difficult to solve, yet it is necessary for deriving insights about actual worst-case performance. Our main methodological contribution is solving this optimization problem using the special structure of the Newsvendor loss function, a method we dub "learning without concentration."

In particular, we investigate in Section 3 the special structure of the Newsvendor loss function, in conjunction with that of Weighted ERM policies and the Kolmogorov distance to derive structural properties of the optimization problem at hand. In that setting, we characterize *exactly* the structure of worst-case distributions which maximize regret for any number of samples (i.e., any data quantity) and any past context (i.e., any data relevance). Our main result, formally presented later as Theorem 2, is informally stated below:

For any sample size and any historical contexts, the problem of determining the worst-case regret of any Weighted ERM policy can be reduced from a nonconvex constrained infinite-dimensional optimization problem to a one-dimensional optimization problem. In turn, the worst-case regret can be exactly quantified.

To prove our result, we move beyond the specific class of Weighted ERM policies and adopt a more general perspective by introducing the broader class of "separable" policies. This generalization allows us to clearly identify the essential factors driving our result: specifically, how the Newsvendor loss interacts with certain properties of these policies. In particular, we show that, under the Newsvendor loss, the cumulative distribution function (cdf) of the decision prescribed by a separable policy can be expressed as a polynomial of the cumulative distribution functions of the historical samples observed. We then show that the worst-case performance for a given separable policy is achieved when the historical and out-of-sample distributions are Bernoulli distributions (Proposition 1). This allows us to reduce the initial infinite-dimensional problem to an $(n+1)$-dimensional optimization problem over the possible means of the Bernoulli distributions. Our second reduction establishes that the worst-case sequence of historical distributions admits a given structure as a function of the (unknown) out-of-sample distribution (Proposition 2). This result is obtained by exploiting the monotonic behavior of the regret of a nondecreasing separable policy with respect to the means of the Bernoulli distributions associated with historical data, allowing us to characterize the sequence of worst-case historical means. In turn, the initial problem can be reduced to a one-dimensional optimization problem over a bounded set which can be solved exactly through a line search. This completes the

reduction for the class of nondecreasing separable policies. Finally, a critical step of our approach consists in proving that the class of nondecreasing separable policies encompasses Weighted ERM (Propositions 3 and 4).

Our analysis provides the strongest results when the local condition is defined through the Kolmogorov distance (Definition 1). Importantly, our optimization-based approach is flexible and extends to other choices of local conditions. For example, in Online Appendix G we consider the Wasserstein distance and show that our method yields considerably tighter performance guarantees than state-of-the-art concentration-based approaches, even when the exact worst-case distributions cannot be characterized.

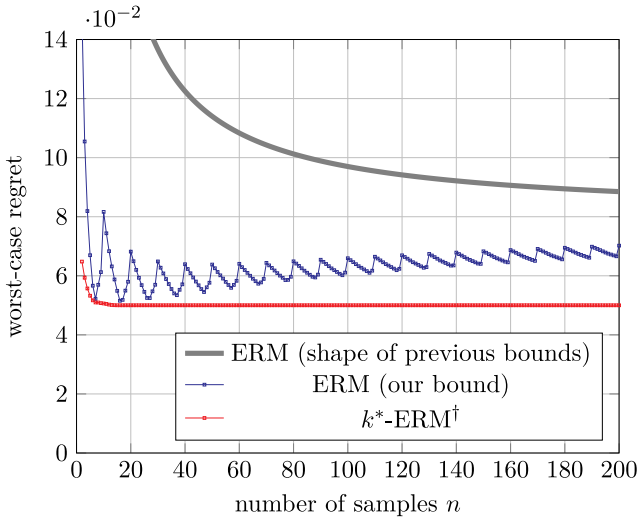
1.1.2. New Insights on Learning Behavior. Our results enable us to quantify key objects of interest and revisit the motivating questions: How does the dissimilarity between historical contexts $x_1,\dots,x_n$ and the out-of-sample context x_0 impact performance? What performance levels are achievable given the available data, and how many samples are necessary to operate effectively?

To contextualize our findings, we compare them against the prior state of the art. As articulated earlier, previous general-purpose bounds, which are applicable to general loss functions, have been the workhorse of a large portion of machine learning and sample complexity results. Over the past decades, they enabled the provision of an understanding (through upper bounds) of the impact of sample size on the regret of central policies. Despite their wide applicability, we will show that such bounds may yield overly conservative estimates, failing to accurately reflect the true worst-case performance in specific contexts.

Figure 1 summarizes our main findings in a prototypical scenario where all past contexts are identical and differ from the out-of-sample context by a fixed dissimilarity ζ. We compare our exact results for ERM with the best-known general-purpose regret bound of Mohri and Muñoz Medina (2012), based on Rademacher complexities and uniform convergence, and uncover three key insights.

First, general-purpose bounds vastly overestimate the number of samples required to achieve a given performance, whereas our analysis reduces these requirements by two to three orders of magnitude (see Table 1, discussed in detail in Section 4.1). Second, our results expand the frontier of what is achievable: performance levels previously thought unattainable by ERM, even with infinitely many samples, can in fact be achieved with only tens of samples. Third, and most importantly, the actual *shape of the learning curve*, defined as the policy's worst-case regret as a function of the sample size, is fundamentally different from what general-purpose bounds suggest. Although the latter depict a smooth, monotonically decreasing curve approaching a nonzero limit and implying that more data always help, our exact analysis shows that when $\zeta > 0$, the learning curve of ERM is oscillatory (a behavior previously observed and avoidable through randomization; see Online Appendix C). More importantly, it is *globally nonmonotonic*: performance improves initially, attains a global optimum at a finite sample size, and then deteriorates as additional samples are added. This reveals the existence of an *effective sample size*, beyond which more data harm rather than help. Motivated by this, we introduce the $k^*-\mathrm{ERM}^\dagger$ policy, which uses a subset of available samples, and we show numerically that it achieves near-optimal performance across algorithms by comparing its performance to a universal lower bound on the worst-case regret. Taken together, these findings imply that *after a few samples, data relevance, as opposed to data quantity, is the first-order limitation to improved decision making*.

Figure 1. (Color online) Illustrative Summary of Our Main Insights



Note. We note that the gray curve depicts the *shape* of "previous bounds" on the same scale as the bounds we derive; the actual previous bounds are in fact much higher.

Beyond these key insights, we illustrate in Section 4.3 the generality of our approach by evaluating the worst-case performance of various subclasses of Weighted ERM policies when sample dissimilarity degrades linearly over time. This setting models an important special case of our framework in which the outcome distributions drift over time. We observe in this case that the effective sample size is even smaller than the one obtained when all past samples have the same dissimilarity.

Finally, we examine numerically the robustness of our insights beyond the worst-case setting. In Online Appendix F, we evaluate the performance of Weighted ERM policies on instances that are non-Bernoulli and

where the distribution remains fixed as the sample size varies. Our results show that the key insights from the worst-case analysis remain broadly applicable even in these fixed-instance settings.

In summary, our paper provides new theoretical insights into data-driven decision making, highlighting the limitations of general-purpose bounds and demonstrating the practical value of tailored analyses for understanding achievable performance as a function of data relevance and data size.

1.2. Literature Review

1.2.1. Data-Driven Stochastic Optimization. Our work broadly falls under the area of data-driven stochastic optimization, where samples are used to infer an unknown out-of-sample distribution. In the absence of contexts, pioneering works (Kleywegt et al. 2002, Kim et al. 2015) analyze ERM, also referred to as Sample Average Approximation. In the presence of contexts, a wide range of statistical learning methods have been developed and analyzed (Vapnik 1999, Bousquet and Elisseeff 2002, Györfi et al. 2002). Our paper differs by taking an optimization approach and identifying exact worst-case regret for a specific decision problem.

More specifically related are papers on contextual decision problems (Kao et al. 2009, Donti et al. 2017, Elmachtoub and Grigas 2021, Bertsimas and Koduri 2022, Kannan et al. 2025). For a broader survey on data-driven decision making in revenue management, we refer the reader to the recent survey by Chen and Hu (2023). Closest to us is a line of work using "local" methods which approximate the empirical objective with Dirichlet processes (Hannah et al. 2010), k-Nearest Neighbors and Random Forest in single-stage (Bertsimas and Kallus 2020) or multistage problems (Bertsimas and McCord 2019), and Nadaraya-Watson (NW) kernels (Ban and Rudin 2019, Bertsimas and McCord 2019, Fu et al. 2021, Srivastava et al. 2021). Kallus and Mao (2023) introduce forest-based weights that are tailored to the downstream optimization problem. These papers consider methods similar in spirit to what we call Weighted ERM policies. They focus on consistency results or concentration-inequality-based upper bounds for general stochastic optimization problems. In contrast, we develop a characterization of the performance of Weighted ERM policies which holds even in settings where consistency cannot be achieved by any data-driven policy. Our methodology is also different as it allows us to quantify the impact of the geometry of past contexts on the performance of policies.

1.2.2. Newsvendor Problem. Our work also relates to the analysis of the Newsvendor problem under partial information about the demand distribution (Scarf 1958, Gallego and Moon 1993, Perakis and Roels 2008, See and Sim 2010). When there are no contexts in the Newsvendor problem, Levi et al. (2007, 2015) establish probabilistic bounds on the relative regret of ERM, and lower bounds with matching rate were derived in Cheung and Simchi-Levi (2019). Recently, Besbes and Mouchtaki (2023) characterize exactly the worst-case expected relative regret of ERM for any sample size, and derive a minimax optimal policy and associated performance. Our work significantly generalizes such analysis to the contextual setting, where historical distributions may be different and coupled through the context. We also analyze a broader class of policies to capture Weighted ERM policies. We compare in more detail our analysis with previous ones in Section 3.

Turning to contextual Newsvendor, Ban and Rudin (2019) provide finite sample bounds on the performance of ERM with a linear hypothesis class and for some Kernel methods. Qi et al. (2022) analyze robust policies, and Zhang et al. (2024) establish a closed-form solution for a distributionally robust optimization (DRO) problem under Wasserstein uncertainty and derive generalization bounds for their method. Under a "local" model similar to ours, Fu et al. (2021) derive finite bounds for robustified Kernel methods. All these works rely on concentration-based arguments; in contrast, our "learning without concentration" analysis is based on an optimization approach and further leverages the structure of the Newsvendor loss function at hand. Furthermore, these bounds are derived in settings in which context vectors are sampled from a distribution and the performance of policies is evaluated by taking an expectation over contexts. Therefore, the bounds cannot be used to derive context-dependent guarantees, which are the object of our work. Recently, Huber et al. (2019), Oroojlooyjadid et al. (2020), and Qi et al. (2023) derive approaches for contextual Newsvendor based on neural networks. These are focused on empirical performance, whereas our focus is on provable guarantees.

Other approaches have also been developed to account for heterogeneity without explicitly incorporating contextual information. Inspired by Bayesian settings, Gupta and Kallus (2022) explore the impact of data-pooling methods; Gupta and Rusmevichientong (2021) analyze the small-data large-scale setting in which the decision maker is facing several problems which are similar; and Besbes et al. (2025) analyze, for multiple loss functions including Newsvendor, a context-free model of heterogeneity. Our work generalizes their model of heterogeneity, and we provide tight finite sample guarantees whereas their results are asymptotic.

1.2.3. Learning Curves. Our work also sheds light on the behavior of learning algorithms and on their learning curves (see the survey by Mohr and van Rijn (2024)).

Data relevance is acknowledged as being an important limitation for learning, but there is no theoretical quantification of this impact. Cortes et al. (1995) empirically explore this issue, and Batini et al. (2009) and Gudivada et al. (2017) survey qualitative aspects related to relevance of data in practical applications. Furthermore, some of the "bad behaviors" of ERM unveiled in our work for the minimax setting are related to the ones surveyed in Loog and Viering (2022) for fixed distributions.

Finally, we note that our work broadly relates to quantile regression (Bhattacharya and Gangopadhyay 1990; Chaudhuri 1991; Koenker 2005, 2017), domain adaptation (Redko et al. 2020, Farahani et al. 2021), transfer learning (Pan and Yang 2010, Zhuang et al. 2020), active learning (Settles 2012), few-shot learning (Wang et al. 2020), and learning in nonstationary environments (Rakhlin et al. 2011, Mohri and Muñoz Medina 2012, Bilodeau et al. 2020, Blanchard 2022). The work most closely related to ours, that of Mohri and Muñoz Medina (2012), is discussed in Section 4.

1.3. Notation and Proofs

For any set A, $\Delta(A)$ denotes the set of probability measures on A. Furthermore, when $A \subset \mathbb{R}$, and we consider $F \in \Delta(A)$, we actually use F as the cumulative distribution function of the random variable. For any $\mu \in \mathbb{R}$, with some abuse of notation, we denote by $\mathcal{B}(\mu)$ the Bernoulli distribution with mean equal to 0 if $\mu \le 0$, 1 if $\mu \ge 1$, and μ in all other cases. For every $x \in \mathbb{R}$, we let $x^+ := \max(0, x)$ be the positive part operator. *All proofs are deferred to the Online Appendix.*

2. Problem Formulation

We consider a problem in which, given a distribution of outcome F_0 supported on the space $\mathcal{Y}$, the goal of the decision maker is to select a decision in the same space $a \in \mathcal{Y}$ to minimize the expected loss

$$L(a, F_0) := \mathbb{E}_{y\sim F_0}[\ell(a, y)],$$

where ℓ is the Newsvendor loss defined for every action $a \in \mathcal{Y}$ and outcome $y \in \mathcal{Y}$ as

$$\ell(a, y) = c_o \cdot (a - y)^+ + c_u \cdot (y - a)^+.$$

The Newsvendor loss is a form of weighted mismatch cost. One can think of the outcome as being the demand and the decision as being the inventory ordered for this demand. c_o denotes the unit cost of going "over", that is, ordering a unit of inventory that goes undemanded, whereas c_u denotes the unit cost of going "under" by missing out on a sale because of lack of inventory. We impose without loss of generality the normalization condition $c_o + c_u = 1$ and define the critical quantile $q := c_u/(c_o + c_u)$. This quantile plays a crucial role because, given a known distribution supported on $\mathcal{Y}$ with cumulative distribution F_0, one can show that a solution which minimizes the expected loss $L(\cdot, F_0)$ is the q^{th} quantile of the demand distribution defined as

$$a^*_{F_0} := \inf\{a \in \mathcal{Y} | F_0(a) \ge q\}. \tag{1}$$

2.1. Data-Generation Process

In practice, the demand distribution is not known to the decision maker. Instead, we assume that she observes a data set in which each data point is a couple $(x, y) \in \mathcal{X} \times \mathcal{Y}$ such that x is the context in which the outcome y has been observed. In this work, we assume that $\mathcal{Y}$ is a bounded interval and assume without loss of generality[1] that $\mathcal{Y} = [0, 1]$.

Consider a sequence of n historical context vectors $(x_i)_{i\in\{1,\dots,n\}} \in \mathcal{X}^n$ observed by the decision maker. For every $i \in \{1, \dots, n\}$, we denote by F_i the demand distribution conditional on the context x_i. We assume that all outcome samples are independently distributed. Hence, the vector of historical outcome observations $\boldsymbol{y} \in \mathcal{Y}^n$ is drawn from the product distribution $F_1 \times \cdots \times F_n$.

Without any assumption on the distributions $(F_i)_{i\in\{1,\dots,n\}}$ and F_0, past outcomes may be useless and completely unrelated to the future outcomes generated from the out-of-sample distribution. We now introduce the local condition, which serves to quantify relations between different distributions.

Definition 1 (Local Condition). Given a nonnegative function $d(\cdot, \cdot) : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, we say that two distributions F and F' associated with the contexts $x, x' \in \mathcal{X}$ satisfy the local condition if $\|F - F'\|_K \le d(x, x')$, where $\|\cdot\|_K$ is the Kolmogorov distance defined as $\|F - F'\|_K = \sup_{y\in\mathbb{R}} |F(y) - F'(y)|$. We call $d(\cdot, \cdot)$ the dissimilarity function.

We note that Definition 1 is a local and nonparametric structural condition which imposes that similar contexts (where similarity is defined through the function d) induce similar conditional outcome distributions measured by their Kolmogorov distance. As a consequence, we colloquially say that for a given $i \ge 1$, outcome y_i has a high relevance when F_0 and F_i satisfy the local condition and the dissimilarity $d(x_0, x_i)$ is low. In Online Appendix G, we illustrate how our approach can be generalized when the local condition is based on the Wasserstein distance.

2.2. Data-Driven Policy and Objective

Given an observed new context $x_0 \in \mathcal{X}$, and a data set containing n past context vectors in $\mathcal{X}$ with n samples (outcome realizations) in $\mathcal{Y}$, we formally define a (potentially randomized)[2] data-driven policy as a mapping π,

$$\pi : \begin{cases} \mathcal{X}^{n+1} \times \mathcal{Y}^n \to \Delta(\mathcal{Y}) \\ (\mathbf{x}, \mathbf{y}) \longmapsto \pi(\mathbf{x}, \mathbf{y}), \end{cases}$$

where $\mathbf{x} = (x_0, x_1, \ldots, x_n)$ contains the new out-of-sample context x_0 and the previously observed ones, $\mathbf{y}$ contains the n observed outcomes, and $\Delta(\mathcal{Y})$ is the space of the probability distributions supported on $\mathcal{Y}$.

A data-driven inventory policy maps past observations and the new observed context to a randomized inventory decision without the knowledge of the underlying past demand distributions or the out-of-sample one. The goal of the decision maker is to select a data-driven policy to minimize the expected regret which represents the difference between the cost incurred by the decision maker and the cost of an oracle who knows exactly the underlying demand distribution. Formally, the expected regret is defined for any decision $a \in \mathcal{Y}$ and distribution F_0 supported on $\mathcal{Y}$ as

$$R(a, F_0) = L(a, F_0) - L(a^*_{F_0}, F_0).$$

If we have a randomized action defined by a distribution $\nu \in \Delta(\mathcal{Y})$, then we overload notation to let $L(\nu, F_0) = \mathbb{E}_{a\sim\nu}[L(a, y)]$, and we let $R(\nu, F_0) = L(\nu, F_0) - L(a^*_{F_0}, F_0)$.

Our goal in this work is to understand the expected regret of data-driven policies in a fixed-design setting for arbitrary distributions satisfying the local condition (see Definition 1). To accomplish this, we first consider a natural case in which the decision maker assumes that samples observed in the same context are drawn from the same distribution. Formally, we model this setting by imposing that our dissimilarity function satisfies $d(x, x) = 0$ for every $x \in \mathcal{X}$. Then, given $\mathbf{x} = (x_0, x_1, \ldots, x_n) \in \mathcal{X}^{n+1}$, an out-of-sample context vector and a sequence of historical context vectors, the worst-case regret of a data-driven policy π is then defined as

$$\sup_{F_0 \in \Delta(\mathcal{Y})} \sup_{\substack{F_1, \ldots, F_n \in \Delta(\mathcal{Y}) \\ \|F_i - F_j\|_K \le d(x_i, x_j)\ \forall i,j \in \{0,\ldots,n\}}} \mathbb{E}_{\mathbf{y}\sim F_1 \times \cdots \times F_n}[R(\pi(\mathbf{x}, \mathbf{y}), F_0)]. \tag{2}$$

The worst-case regret in (2) reflects the assumption that samples observed in the same context are drawn from a common distribution because for every $i, j \in \{1, \ldots, n\}$ such that $x_i = x_j$, the local condition implies that $\|F_i - F_j\|_K \le d(x_i, x_j) = 0$, and thus $F_i = F_j$.

To facilitate generalizations (particularly for settings with partially observed features; see Remark 1), we next define a stronger notion of regret that allows Nature to choose potentially different distributions for each data point, even when the contexts coincide. For every $n \ge 1$ and every $\mathbf{x} = (x_0, x_1, \ldots, x_n) \in \mathcal{X}^{n+1}$, we define the worst-case regret as

$$\mathsf{Reg}_n(\pi, \mathbf{x}) = \sup_{F_0 \in \Delta(\mathcal{Y})} \sup_{\substack{F_1, \ldots, F_n \in \Delta(\mathcal{Y}) \\ \|F_0 - F_i\|_K \le d(x_0, x_i)\ \forall i \in \{1,\ldots,n\}}} \mathbb{E}_{y_1\sim F_1, \ldots, y_n \sim F_n}[R(\pi(\mathbf{x}, \mathbf{y}), F_0)]. \tag{3}$$

The formulation (3) differs from (2) by relaxing the local conditions on all pairs of distributions which do not include the out-of-sample one. In particular, (3) appears to be more conservative than (2). For example, if one wants to model a setting in which the decision maker has observed n samples in the same historical context, then (2) forces Nature to select a single historical distribution F_1 from which the n observations are drawn i.i.d., whereas (3) allows Nature to select a different distribution for each of the n samples, as long as each historical distribution is reasonably close to the out-of-sample one.

Interestingly, our main result shows that for all policies of interest considered in this paper, the two definitions ultimately yield identical worst-case regrets whenever the dissimilarity function d satisfies the triangular inequality (see Corollary 1).

Furthermore, we note that in our definition of the worst-case regret, the contexts are fixed and a policy is evaluated against all possible historical and out-of-sample distributions that satisfy the local condition. We highlight that the data-generation process we study differs from the common one in the literature in which context vectors are randomly sampled from a distribution supported on $\mathcal{X}$. In those settings, the regret guarantees obtained hold uniformly across various distributions of contexts. Hence, these approaches do not allow us to characterize the impact of the configuration of *actual* contexts observed on the performance. In contrast, our formulation considers fixed contexts and aims at understanding the regret of policies for different context configurations. Therefore, $\mathsf{Reg}_n(\pi, \mathbf{x})$ can be interpreted as the *context-dependent* robust (minimal) value associated with the data at hand under a particular policy π. This object quantifies, for a fixed sample size and a fixed configuration of the context vectors, the worst-case performance of a policy of interest. We note that when $d(x_i, x_j) = 0$ for all $i, j \in \{0, \ldots, n\}$, the local conditions imply that all distributions should be the same, and therefore, one retrieves the setting in which future and past demands are drawn from the same distribution.

Remark 1 (Partially Observed Contexts). We note that our model, through its flexibility in the dissimilarity function, captures problems in which the decision maker only observes partially the contexts. Indeed, we do not impose in (3) that samples observed in the same context have the same distribution. Moreover, we do not impose the dissimilarity function to be a distance. Therefore, we can capture partial observation of contexts by considering a dissimilarity function d of the form $d(x, x') = \tilde{d}(x, x') + \epsilon$, where $\tilde{d}$ is the dissimilarity function capturing the dissimilarity between observed contexts and ϵ is the amplitude of dissimilarity driven by unobserved factors. This special case allows to explicitly account for two types of deterioration in data relevance: unobserved factors and dissimilarity of observed contexts.

2.3. Focal Class of Policies: Weighted ERM

In this work, we anchor our analysis and discussion around the following central class of policies which minimize a weighted empirical loss.[3]

Definition 2 (Weighted Empirical Risk Minimization for Newsvendor). A Weighted ERM policy is defined by a sequence of nonnegative weights $\mathbf{w} = (w_i)_{i\in\{1,\dots,n\}}$ which could depend on the fixed contexts $\mathbf{x}$ and prescribes the action

$$\pi^{\mathbf{w}}(\mathbf{x},\mathbf{y}) = \inf\left\{ a \text{ s.t. } \frac{\sum_{i=1}^n w_i(\boldsymbol{x})\cdot \mathbb{1}\{y_i \le a\}}{\sum_{j=1}^n w_j(\boldsymbol{x})} \ge \frac{c_u}{c_u+c_o} \right\}.$$

Given a sequence of nonnegative weights $\mathbf{w} = (w_i)_{i\in\{1,\dots,n\}}$, the action prescribed by Weighted ERM is a minimizer of the weighted empirical loss $\sum_{i=1}^n w_i \cdot \ell(a,y_i)$ (see proposition 1 in Ban and Rudin (2019)). This class of policies encompasses many classical policies used in machine learning and data-driven decision making (see, e.g., Bertsimas and Kallus 2020). Notable special cases of Weighted ERM policies are the classical ERM policy (when all the weights are equal), which we denote by π^{ERM}, and the class of k-Nearest Neighbors (when the weight is one for the k-closest contexts and zero otherwise). When setting $w_i = K(d(x_i,x_0))$ for a kernel function $K:\mathbb{R}\to\mathbb{R}_+$, we retrieve policies minimizing the Nadaraya-Watson estimator of the expected loss (Nadaraya 1964, Watson 1964). Intuitively, the weights can be selected to leverage the closeness of contexts (and associated closeness of distributions imposed by the local condition).

3. Main Results: Policies and Performance Characterization

In this section, we present our main theoretical results, which characterize the performance of Weighted ERM policies, as well as more general policies. In Section 3.1, we first review a classical concentration-based approach for bounding the performance of policies. In Section 3.2, we present the alternative optimization approach we pursue in this paper. In turn, we analyze "separable policies," whose properties allow us to develop a sequence of reductions that enables us to directly characterize their performance. In Section 3.3, we then establish that Weighted ERM policies are separable policies.

3.1. Limitation of Concentration-Based Arguments

We first provide a proof sketch of a very common concentration-based argument present in the literature, when bounding the performance of a policy such as Weighted ERM. Given a family of context vectors $\mathbf{x}\in\mathcal{X}^{n+1}$ and previously observed outcomes $\mathbf{y}\in\mathcal{Y}^n$, we denote by $\hat{F}_{\mathbf{w}}$ the weighted empirical distribution defined as $\hat{F}_{\mathbf{w}}(z) = \sum_{i=1}^n w_i\cdot\mathbb{1}\{y_i\le z\}/\sum_{j=1}^n w_j$ for every $z\in\mathcal{Y}$, on which the decision of the Weighted ERM policy is based. Then the regret can be decomposed as follows:

$$\begin{aligned}
&R(\boldsymbol{\pi}^{\mathbf{w}}(\mathbf{x},\mathbf{y}),F_0)\\
&= L(\boldsymbol{\pi}^{\mathbf{w}}(\mathbf{x},\mathbf{y}),F_0) - L(a^*_{F_0},F_0)\\
&= [L(\boldsymbol{\pi}^{\mathbf{w}}(\mathbf{x},\mathbf{y}),F_0) - L(\boldsymbol{\pi}^{\mathbf{w}}(\mathbf{x},\mathbf{y}),\hat{F}_{\mathbf{w}})]\\
&\quad + [L(\boldsymbol{\pi}^{\mathbf{w}}(\mathbf{x},\mathbf{y}),\hat{F}_{\mathbf{w}}) - L(a^*_{F_0},\hat{F}_{\mathbf{w}})] + [L(a^*_{F_0},\hat{F}_{\mathbf{w}}) - L(a^*_{F_0},F_0)]\\
&\overset{(a)}{\le} 2\cdot\sup_{a\in\mathcal{Y}} |L(a,F_0) - L(a,\hat{F}_{\mathbf{w}})|,
\end{aligned}$$

where inequality (a) holds because by construction of the Weighted ERM policy the second difference is negative. Given this decomposition, most arguments aim at deriving uniform bounds (across the action space) on the difference between the loss evaluated on the distributions $\hat{F}_{\mathbf{w}}$ and F_0. This difference is controlled by introducing the weighted mixture of historical distributions $\overline{F}_{\mathbf{w}} = \sum_{i=1}^n w_i\cdot F_i/\sum_{j=1}^n w_j$, and using the triangular inequality to obtain that

$$\begin{aligned}
\sup_{a\in\mathcal{Y}} |L(a,F_0) - L(a,\hat{F}_{\mathbf{w}})| &\le \sup_{a\in\mathcal{Y}} |L(a,F_0) - L(a,\overline{F}_{\mathbf{w}})|\\
&\quad + \sup_{a\in\mathcal{Y}} |L(a,\overline{F}_{\mathbf{w}}) - L(a,\hat{F}_{\mathbf{w}})|. \quad (4)
\end{aligned}$$

The first term is then interpreted as the limiting regret with infinitely many samples, and the second term captures the loss incurred because the empirical distribution is constructed with finite samples. The first term does not depend on the realizations of $y_1,\dots,y_n$ and is generally nonzero because the weighted mixture $\overline{F}_{\mathbf{w}}$ does not faithfully capture the unknown distribution F_0; however, note that it is zero in the i.i.d. setting. This is the setting of classical statistical learning, for which we refer to Mohri et al. (2018, chapter 3 and chapter 4) and Shalev-Shwartz and Ben-David (2014, chapter 26). The second term in (4) is bounded by showing that the empirical loss concentrates around the loss under the weighted mixture. Common bounds on the second term are decreasing and converge to zero as the number of samples grows. We compare our result with bounds that use (4) in Section 4.1.

This concentration-based analysis is powerful as it can be applied to a general class of problems. However, we note that this line of argument does not account for the actual loss function at hand. Also, the bound on $R(\boldsymbol{\pi}^{\mathbf{w}}(\mathbf{x},\mathbf{y}),F_0)$ one obtains is based on a uniform bound across actions and does not take into consideration the particular action taken by the policy of interest. Finally, as explained in (4), this line of argument would separately analyze the limiting regret term and the finite sample loss term, instead of directly analyzing the finite sample regret. A natural question is then how much is lost based on such a bounding

approach. We will see in Section 4.2 that both the scale and the shape suggested by these bounds are actually an artifact of the analysis rather than reflecting the actual performance of the algorithm.

3.2. Learning Without Concentration: An Optimization Approach

In contrast to the approach outlined in Section 3.1 for upper-bounding the value of

$$\mathbb{E}_{y_1\sim F_1,\dots,y_n\sim F_n}[R(\pi(\mathbf{x},\mathbf{y}),F_0)], \tag{5}$$

we follow in this work an optimization approach to exactly evaluate the supremum value of (5) subject to constraints. Recall that for a given policy π, the object of interest $\mathsf{Reg}_n(\pi,\mathbf{x})$ defined in (3) can be phrased as the following optimization problem:

$$\sup_{F_0\in\Delta(\mathcal{Y})}\ \sup_{\substack{F_1,\dots,F_n\in\Delta(\mathcal{Y})\\ \|F_0-F_i\|_K\le d(x_0,x_i)\ \forall i}}\ \int_{\mathcal{Y}^n}(L(\pi(\mathbf{x},\mathbf{y}),F_0)-L(a^*_{F_0},F_0))\,dF_1(y_1)\dots dF_n(y_n). \tag{6}$$

We note that (6) is a DRO problem: an optimization problem where decision variables are distributions. Common approaches in the DRO literature analyze policies which use the samples to construct an uncertainty ball in which the out-of-sample distribution may lie and then solve a minimax problem over that uncertainty ball. Furthermore, the objective function in these minimax problems is usually linear in the distribution (for instance, in the case of the expected loss $L(\cdot,\cdot)$). In contrast, the problem (6) involves an objective which has an intricate dependence in all the historical distributions: they affect the decision learned by the policy, and hence appear as integrating measures and result in a nonconcave objective. Furthermore, it is a multivariate infinite-dimensional problem which involves several distributions constrained through the local conditions as opposed to a single out-of-sample distribution.

Although this is a nonconcave constrained infinite-dimensional optimization problem, we develop a sequence of reductions that allow us to solve this problem. Our methodology is conceptually related to the one recently derived by Besbes and Mouchtaki (2023) in which the authors analyze the worst-case relative regret in the i.i.d. setting. At a high level their proof relies on two critical arguments. The first step to simplify (6) consists in showing that, in the Newsvendor problem, for certain data-driven policies the objective may be transformed in the following form:

$$\int_{\mathcal{Y}^n}(L(\pi(\mathbf{x},\mathbf{y}),F_0)-L(a^*_{F_0},F_0))dF_1(y_1)\dots dF_n(y_n) = \int_{\mathcal{Y}}\Psi^\pi(F_0(y),\dots,F_n(y))dy, \tag{7}$$

where Ψ^π is a continuous function from $[0,1]^{n+1}$ to $\mathbb{R}$. The second step uses the form of the objective in (7) to show that the initial problem can be solved via pointwise optimization.

Given the contextual nature of our problem in the present paper, we depart from the i.i.d. setting, and we aim at understanding the performance of a much broader set of policies including Weighted ERM ones. We note that Besbes and Mouchtaki (2023) analyzed, in the i.i.d. setting, policies which prescribe a decision equal to an order statistic of the past samples. A natural question is whether their approach can be (i) generalized to the contextual case and (ii) directly applied to Weighted ERM policies. The answer to (ii) is negative as we show next:

Lemma 1. *There exists a Weighted ERM policy which is not an order statistic policy* (*formally introduced in Definition* B-1 *in the Online Appendix*).

Therefore, we need to extend the approach to a broader set of policies, while accounting for the contextual aspect of the problem. To that end, we will start from a general class of policies, separable policies, which satisfy (7) and show through a sequence of reductions that (6) can be simplified under this class. We then develop in Section 3.3 a series of results to understand how broad these policies are, proving that they encompass Weighted ERM.

Definition 3 (Separable Policies). We say that a data-driven policy π is a separable policy if there exists a function $P^\pi:[0,1]^n\to[0,1]$, which could depend on the fixed context $\mathbf{x}$, such that for every distribution $F_1,\dots,F_n\in\Delta(\mathcal{Y})$ and any $y\in\mathbb{R}$, we have that

$$\mathbb{P}_{\mathbf{y}\sim F_1\times\cdots\times F_n,a\sim\pi(\mathbf{x},\mathbf{y})}(a\le y)=P^\pi(F_1(y),\dots,F_n(y)).$$

We refer to P^π as the function associated with the separable policy π. Furthermore, we say that the policy π is a nondecreasing separable policy if for every $i\in\{1,\dots,n\}$ and every $h_1,\dots h_{i-1},h_{i+1},\dots,h_n\in[0,1]$, the function

$$h\longmapsto P^\pi(h_1,\dots h_{i-1},h,h_{i+1},\dots,h_n)$$

is nondecreasing.

Separable policies are ones for which the cumulative distribution of the decision at a given point $y\in\mathcal{Y}$, which is induced by the distribution of the samples used by the policy, can be expressed as a function P^π of the cumulative distributions of the samples evaluated at the same point y. Among all separable policies, a natural subclass to consider is the class of nondecreasing ones which are intuitively defined as policies for which the probability of selecting a decision less than or equal to y increases when the probability of observing a sample less than or equal to y increases. In Section 3.3 we prove that several central policies belong to the class of separable policies. Furthermore,

to provide some intuition about which policies are not separable, we prove in Proposition A-1 (see Online Appendix A) that the policy which selects the average of all samples observed is not separable.

In the remainder of this section, we analyze the worst-case regret $\mathsf{Reg}_n(\pi, \mathbf{x})$, defined in (3), of separable policies. Our next lemma formalizes the fact that for any separable policy, the objective function in Problem (6) can be simplified.

Lemma 2. *For every $n \geq 1$, any sequence of contexts $\mathbf{x} = (x_i)_{i\in\{0,\dots,n\}} \in \mathcal{X}^{n+1}$, any separable policy π, and every family of distributions $(F_i)_{i\in\{0,\dots,n\}} \in \Delta(\mathcal{Y})$, we have that*

$$\mathbb{E}_{\mathbf{y}\sim F_1\times\cdots\times F_n}[R(\pi(\mathbf{x},\mathbf{y}),F_0)] = (c_u + c_o)\cdot\int_0^1 \Psi^\pi(F_0(y),F_1(y),\dots,F_n(y))dy,$$

where Ψ^π is a mapping from $[0,1]^{n+1}$ to $\mathbb{R}$ which satisfies, for every $(z_0,\dots,z_n)\in[0,1]^{n+1}$,

$$\Psi^\pi(z_0,\dots,z_n) = P^\pi(z_1,\dots,z_n)\cdot(q - z_0) + \max\{z_0 - q, 0\}.$$

Lemma 2 shows that (7) holds for any separable policy. We next leverage this result to show that the multivariate infinite-dimensional optimization problem (6) can be reduced to a simpler one over the space of Bernoulli distributions. Recalling that $\mathcal{B}(\cdot)$ denotes the Bernoulli distribution, the following holds:

Proposition 1. *For every $n \geq 1$, any sequence of contexts $\mathbf{x} = (x_i)_{i\in\{0,\dots,n\}} \in \mathcal{X}^{n+1}$, and any separable policy π, we have*

$$\mathsf{Reg}_n(\pi,\mathbf{x}) = \sup_{\mu_0\in[0,1]} \sup_{\substack{\mu_1,\dots,\mu_n\in[0,1]\\ |\mu_i-\mu_0|\leq d(x_0,x_i)\ \forall i}} \mathbb{E}_{\mathbf{y}\sim\mathcal{B}(\mu_1)\times\cdots\times\mathcal{B}(\mu_n)} [R(\pi(\mathbf{x},\mathbf{y}),\mathcal{B}(\mu_0))].$$

Proposition 1 shows that for any separable policy, the worst-case distributions, both historical and out-of-sample, over all possible distributions supported on [0, 1] are Bernoulli distributions. Therefore, the optimization problem (3), which involves the worst case over all distributions, can be reduced to a significantly simpler one, in which the adversary only needs to decide on the means of different Bernoulli distributions (for the historical and out-of-sample distributions of outcomes).

Despite the significant reduction in complexity above, we note that optimizing over the space of Bernoulli means is still a nonconvex optimization problem with dimension $(n+1)$. Our next result enables us to further simplify the problem by characterizing the worst-case sequence of historical means $(\mu_i)_{i\geq 1}$ as a function of the out-of-sample mean μ_0.

Proposition 2. *For every $n \geq 1$, any sequence of contexts $\mathbf{x} = (x_i)_{i\in\{0,\dots,n\}} \in \mathcal{X}^{n+1}$, and any nondecreasing separable policy π, we have*

$$\sup_{\substack{\mu_1,\dots,\mu_n\in[0,1]\\ |\mu_i-\mu_0|\leq d(x_0,x_i)\ \forall i}} \mathbb{E}_{\mathbf{y}\sim\mathcal{B}(\mu_1)\times\cdots\times\mathcal{B}(\mu_n)} [R(\pi(\mathbf{x},\mathbf{y}),\mathcal{B}(\mu_0))]$$

$$= \begin{cases} \mathbb{E}_{\mathbf{y}\sim\mathcal{B}(\mu_0+d(x_0,x_1))\times\cdots\times\mathcal{B}(\mu_0+d(x_0,x_n))} [R(\pi(\mathbf{x},\mathbf{y}),\mathcal{B}(\mu_0))] \\ \qquad\qquad \text{if } \mu_0\in[0,1-q], \\ \mathbb{E}_{\mathbf{y}\sim\mathcal{B}(\mu_0-d(x_0,x_1))\times\cdots\times\mathcal{B}(\mu_0-d(x_0,x_n))} [R(\pi(\mathbf{x},\mathbf{y}),\mathcal{B}(\mu_0))] \\ \qquad\qquad \text{if } \mu_0\in[1-q,1]. \end{cases}$$

Proposition 2 establishes that for a given out-of-sample Bernoulli mean μ_0, the worst-case sequence of conditional historical distributions can be explicitly characterized—they are the furthest possible (while still satisfying the local constraint) from the out-of-sample distribution. However, the direction in which they differ depends on the economics of the problem and on the out-of-sample mean μ_0. When the mean of the out-of-sample Bernoulli distribution is below $1-q$, that is, when it is optimal to set an inventory level of zero, the worst-case sequence of distributions tends to inflate the mean as much as possible, to values of $\mu_0 + d(x_0,x_1),\dots,\mu_0 + d(x_0,x_n)$, in order to push the decision maker to carry more inventory than zero. A similar interpretation holds when the mean of the out-of-sample Bernoulli distribution is above $1-q$. The two expressions are identical if $\mu_0 = 1-q$.

In the proof of Proposition 2, we actually derive a stronger statement as we show a monotonicity property of the regret of nondecreasing separable policies as a function of the Bernoulli means. To prove this property, we need the separable policy to be nondecreasing.

By combining Proposition 1 and Proposition 2, we obtain a characterization of the worst-case regret for any nondecreasing separable policy. We present this as the following theorem, which will later be used to derive our main result about Weighted ERM policies.

Theorem 1. *For every $n \geq 1$, any sequence of contexts $\mathbf{x} = (x_i)_{i\in\{0,\dots,n\}} \in \mathcal{X}^{n+1}$, and any nondecreasing separable policy π, we have*

$$\mathsf{Reg}_n(\pi,\mathbf{x}) = \max\left\{ \sup_{\mu_0\in[0,1-q]} \mathbb{E}_{\mathbf{y}\sim\mathcal{B}(\mu_0+d(x_0,x_1))\times\cdots\times\mathcal{B}(\mu_0+d(x_0,x_n))} [R(\pi(\mathbf{x},\mathbf{y}),\mathcal{B}(\mu_0))], \sup_{\mu_0\in[1-q,1]} \mathbb{E}_{\mathbf{y}\sim\mathcal{B}(\mu_0-d(x_0,x_1))\times\cdots\times\mathcal{B}(\mu_0-d(x_0,x_n))} [R(\pi(\mathbf{x},\mathbf{y}),\mathcal{B}(\mu_0))] \right\}.$$

Theorem 1 directly follows from Propositions 1 and 2 and has two notable implications. First, it shows that computing the worst-case regret of a nondecreasing separable policy can be done efficiently as the nonconvex infinite-dimensional optimization problem (3) can actually be reduced to a one-dimensional optimization problem on a line segment [0, 1]. Furthermore,

it establishes that Bernoulli distributions form a family of hard distributions for separable policies and characterizes the worst sequence of historical distributions as a function of the out-of-sample distribution.

The next corollary relates the value of the optimization problem with multiple samples as defined in (2) and the more conservative one in (3).

Corollary 1. *If the dissimilarity d satisfies the triangular inequality, then for every $n \geq 1$, any sequence of contexts $\mathbf{x} = (x_i)_{i\in\{0,\dots,n\}} \in \mathcal{X}^{n+1}$, and any nondecreasing separable policy π, the value of the problem in (2) is equal to that of (3).*

Corollary 1 shows that when the dissimilarity function satisfies the triangular inequality, the two optimization formulations in (2) and (3) are in fact equivalent. This equivalence can be understood by examining the worst-case distributions characterized in Theorem 1. For any $i,j \in \{1,\dots,n\}$, these distributions yield worst-case means that satisfy $|\mu_i - \mu_j| = |d(x_0,x_i) - d(x_0,x_j)|$, which by the triangular inequality implies $|\mu_i - \mu_j| \leq d(x_i,x_j)$. Hence, the candidate Bernoulli distributions remain feasible even under the stricter constraints of (2). As a result, the worst-case regret in both formulations coincides. This observation highlights that, under mild structural assumptions on d, the seemingly more conservative formulation (3) yields the same worst-case regret, and our result extends to the more benign setting where samples observed in the same context must be i.i.d.

We note that Theorem 1 relies on the assumption that the local condition is defined using the Kolmogorov distance. However, our optimization-based proof technique can also yield tighter analyses for other distances. For instance, although we do not characterize the exact performance of Weighted ERM policies under the Wasserstein distance, we show in Theorem G-2 in the Online Appendix that, for any configuration of contexts, a Lagrangian relaxation combined with the proof technique of Proposition 1 reduces the original nonconvex infinite-dimensional problem to a minimax problem involving only $2n+1$ variables. Building on the ideas of Proposition 2, we further simplify this formulation in specific context configurations and for the ERM policy, leading to significantly improved guarantees compared with concentration-based bounds.

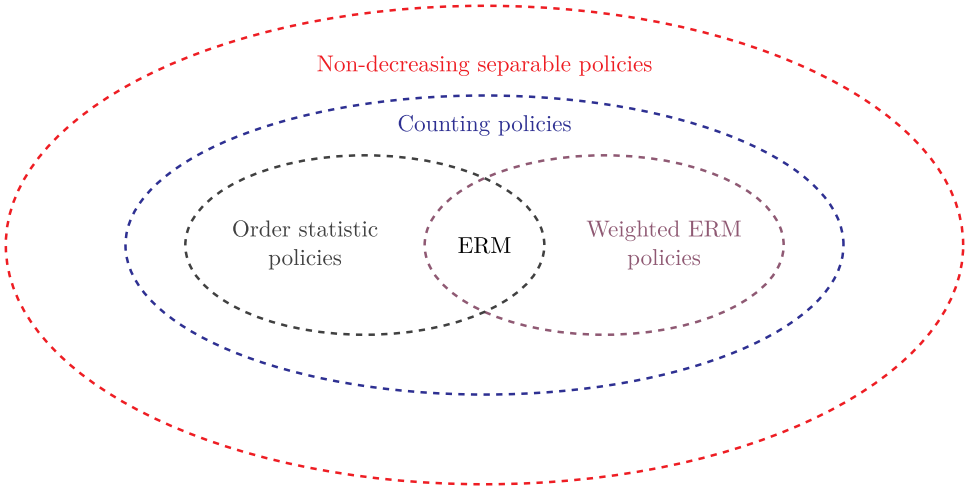
Remark 2 (Randomized Policies). It is worth noting that our main theorem also holds for any possible randomization over nondecreasing separable policies as this class of policies is closed under mixtures as formalized in Lemma A-1 in the Online Appendix. This extension to randomized policies will be valuable in various settings; we return to this in Section 4.2.

3.3. Relations Between Classes of Policies

Theorem 1 applies to any nondecreasing separable policy, but at this stage we have not yet showed that this abstract class contains any policies of interest. In particular, it is a priori nonobvious whether Weighted ERM policies are separable and nondecreasing. To show this, we first introduce the intermediate class of counting policies (Definition 4), which can be shown to be separable and nondecreasing and provide a definition that is easier to work with. We then show that Weighted ERM policies (Definition 2) and order statistic policies (Definition B-1 in the Online Appendix) are special cases of counting policies. For reference, Figure 2 illustrates the relationships between all the classes of policies we analyze in the present paper.

Definition 4 (Counting Policies). We say that a deterministic policy π is a counting policy if there exists a function $\kappa^\pi : \{0,1\}^n \to \{0,1\}$ which could depend on

Figure 2. (Color online) Classes of Policies Analyzed



Note. The figure represents the different classes of policies that we analyze and the relationships that we show in terms of containment.

the fixed contexts $\mathbf{x}$, such that, for every $y \in [0,1]$, and any $\mathbf{y} \in \mathcal{Y}^n$,

$$\mathbb{1}\{\pi(\mathbf{x},\mathbf{y}) \le y\} = \kappa^\pi(\mathbb{1}\{y_1 \le y\}, \dots, \mathbb{1}\{y_n \le y\}).$$

We refer to κ^π as the counting function associated with π.

Counting policies are intuitively defined as ones for which the decision of whether the inventory should be lower than any given value $y \in \mathcal{Y}$ is only a function of the sequence of variables $(\mathbb{1}\{y_i \le y\})_{i\in\{1,\dots,n\}}$, which indicate the past demand samples that have a value lower than y. Importantly, proving that a policy is a counting policy is simpler than showing that it is a separable policy because counting policies are defined through the decision prescribed by the policy as a function of the *realization* of past samples, whereas the definition of separable policy involves the distribution of decisions implied by the policy. Furthermore, showing that a policy is a counting policy suffices to establish that it is a nondecreasing separable policy, as formalized by the next result.

Proposition 3. *Every counting policy is a nondecreasing separable policy.*

The fact that the function κ^π is identical for every y allows us to show that counting policies must be nondecreasing. From there, we prove Proposition 3 by showing that nondecreasing counting policies induce nondecreasing separable policies.

Counting policies enable us to analyze Weighted ERM policies as we show that under the Newsvendor loss, the set of Weighted ERM policies is included in the set of counting policies.

Proposition 4. *For any sequence of nonnegative weights $\mathbf{w} = (w_i)_{i\in\{1,\dots,n\}}$, the associated Weighted ERM policy is a counting policy. Furthermore, its counting function κ^π is defined for every $\mathbf{b} \in \{0,1\}^n$ as*

$$\kappa^\pi(\mathbf{b}) = \begin{cases} 1 & \text{if } \dfrac{\sum_{i=1}^n w_i \cdot b_i}{\sum_{i=1}^n w_i} \ge \dfrac{c_u}{c_u + c_o}, \\ 0 & \text{otherwise.} \end{cases}$$

Finally, by combining Proposition 3, Proposition 4, and Theorem 1, we obtain the following theorem, which constitutes our main result. It allows us to evaluate the worst-case regret of any Weighted ERM policy.

Theorem 2. *For every $n \ge 1$, any sequence of contexts $\mathbf{x} = (x_i)_{i\in\{0,\dots,n\}} \in \mathcal{X}^{n+1}$, and any Weighted ERM policy π, we have*

$$\begin{aligned}&\mathrm{Reg}_n(\pi,\mathbf{x}) \\ &= \max\left\{ \sup_{\mu_0\in[0,1-q]} \mathbb{E}_{\mathbf{y}\sim\mathcal{B}(\mu_0+d(x_0,x_1))\times\cdots\times\mathcal{B}(\mu_0+d(x_0,x_n))}[R(\pi(\mathbf{x},\mathbf{y}),\mathcal{B}(\mu_0))], \right. \\ &\qquad \left. \sup_{\mu_0\in[1-q,1]} \mathbb{E}_{\mathbf{y}\sim\mathcal{B}(\mu_0-d(x_0,x_1))\times\cdots\times\mathcal{B}(\mu_0-d(x_0,x_n))}[R(\pi(\mathbf{x},\mathbf{y}),\mathcal{B}(\mu_0))] \right\}.\end{aligned}$$

We note that our definition of Weighted ERM policies selects the smallest action which minimizes the empirical loss, aligning with the standard definition of Weighted ERM in the Newsvendor setting. However, Theorem 2 remains valid even for policies that employ more complex tie-breaking rules when multiple actions minimize the empirical loss. We present this more general result in Online Appendix D.

4. New Insights on the Learning Behavior of Algorithms

In this section, we leverage our exact analysis of data-driven policies to derive new insights on their performance as a function of quantity and relevance of data. Although the previous section is general, we now explore different prototypical special cases of context configurations. In Sections 4.1 and 4.2, we focus on the implications of our result for a special case in which all past dissimilarities are identical, that is, $d(x_1,x_0) = \cdots = d(x_n,x_0) = \zeta$ for a fixed $\zeta \ge 0$. In such a case the data points can be treated symmetrically, and it is natural to study the (unweighted) ERM policy. In Section 4.3 we evaluate the performance of various policies when the dissimilarity deteriorates over time. In Section 4.4 we investigate the impact of misspecified dissimilarities.

4.1. Exact Sample Complexity of ERM and Achievable Performance

In this section, our goal is to understand the worst-case regret that can be achieved by ERM given a certain configuration of contexts. This question is of a very different nature from ones that can be asked in the setting in which past samples are drawn i.i.d. from the out-of-sample distribution. Indeed, when past contexts are different from the out-of-sample one, past samples are not fully indicative of the out-of-sample distribution F_0 and therefore one may not necessarily achieve a vanishing regret even with arbitrarily large sample sizes. A second question consists in understanding the number of samples required to obtain a regret guarantee lower than a given target whenever such a target can be achieved.

We will explore both questions while contrasting the answers implied by previous state-of-the-art bounds with the ones we derive. In particular, we compare our results with the work of Mohri and Muñoz Medina (2012), which, to the best of our knowledge, is the state-of-the-art bound which can be applied to our setting, as they derive a performance guarantee that holds when past outcomes are independently sampled from distributions which may differ from the out-of-sample distribution. We show in Online Appendix E that one can leverage their distribution-dependent bound to obtain the following guarantee on the regret of ERM, as a

function of the given contexts $\mathbf{x} = (x_i)_{i\in\{0,\dots,n\}}$:

$$4C_n + \frac{2\max(q,1-q)}{n}\sum_{i=1}^{n} d(x_i,x_0) + 4\max(q,1-q)\sqrt{\frac{2\pi}{n}} \cdot \left\{\Phi\left(\frac{2\max(q,1-q)}{\sqrt{n}}\right) - \Phi(0)\right\}. \quad (8)$$

In (8), Φ denotes the cumulative distribution function of the standard Gaussian and C_n is a notion of Rademacher complexity evaluated on the worst-case set of distributions identified by Theorem 1.

We present in Table 1 the number of samples n required to guarantee that ERM has a regret below some target, in the special setting where $d(x_1,x_0) = \cdots = d(x_n,x_0) = \zeta$ for a fixed $\zeta \geq 0$. We fix q to be 0.9. As a point of reference, these regret targets are given as a percentage of the "no-data regret" achieved by a decision maker who only knows the support of the demand distribution, which has been shown to equal $q\cdot(1-q)$ when this support is normalized to [0,1] (Perakis and Roels 2008).

In Table 1, we first present results for the special case in which past outcomes are drawn i.i.d. from the out-of-sample distribution ($\zeta = 0$) for reference. In this case, general-purpose bounds provide an overly pessimistic understanding of the number of samples required to achieve a given performance. Indeed, our characterization shows that the difference between the actual number of samples required and the one implied by previous bounds can be multiple orders of magnitude. We note that similar insights have been previously established by Besbes and Mouchtaki (2023), under a different performance metric of relative regret.

When ζ is positive, we observe that the discrepancy between the actual number of samples required and the one implied by previous bounds is even more acute. Indeed, previous approaches imply that, when historical distributions are different from the out-of-sample one, for example, $\zeta = 0.02$, ERM requires 734 samples (up from 279 samples in the i.i.d. case) to match the no-data regret. This suggests that ERM is a policy which would require a large number of samples to become efficient and that this sample complexity considerably deteriorates as relevance of data slightly decreases. However, our exact analysis shows that these bounds are overly conservative and do not capture the performance of ERM at all. In fact, ERM only requires three samples to achieve this same performance,[4] and the sample complexity only increases to 14 if we are targeting 25% instead of 100% of the no-data regret.

Even more notably, when the past distributions differ from the out-of-sample one ($\zeta = 0.02$), we observe that certain levels of performance are unachievable by the ERM policy even with infinitely many samples. This is a natural consequence of only accessing a past distribution that is different from the out-of-sample one on which performance will be evaluated, and even the best data-driven policy in this case has a nonvanishing regret. Table 1 shows that previous state-of-the-art bounds provide an incorrect picture of the achievable performance. Indeed, such bounds imply that, when $q = 0.9$, ERM cannot have a worst-case regret lower than 25% of the no-data regret, even with an infinite number of samples. In stark contrast, our bound shows that this performance is actually achievable and can be obtained with as few as 16 samples! The analysis above highlights that, although general-purpose upper bounds are very powerful, specialized ones are needed to fully uncover the value of data and the impact of data relevance on performance.

Table 1. Number of Samples That Ensures a Target Regret for ERM

		Regret target (as % of no-data regret)					
ζ		100%	90%	75%	50%	25%	10%
0	N^{GP}	279	338	475	1,032	3,298	24,673
	N^{exact}	3	3	4	5	14	37
0.02	N^{GP}	734	1,047	2,114	24,461	inf.	inf.
	N^{exact}	3	3	4	5	16	inf.
0.04	N^{GP}	6,228	24,493	inf.	inf.	inf.	inf.
	N^{exact}	3	4	4	6	inf.	inf.

Notes. The table reports the number of samples needed to guarantee a regret target when dissimilarities are $\zeta \in \{0, 0.02, 0.04\}$. N^{GP} (resp. N^{exact}) is the number of samples required to achieve a target regret implied by the general-purpose guarantee (8) of Mohri and Muñoz Medina (2012) (resp. our exact quantification of $\mathsf{Reg}_n(\pi^{\mathrm{ERM}},\mathbf{x})$ in Theorem 2). We report "inf." when the bound indicates that the target is unachievable even with infinitely many samples.

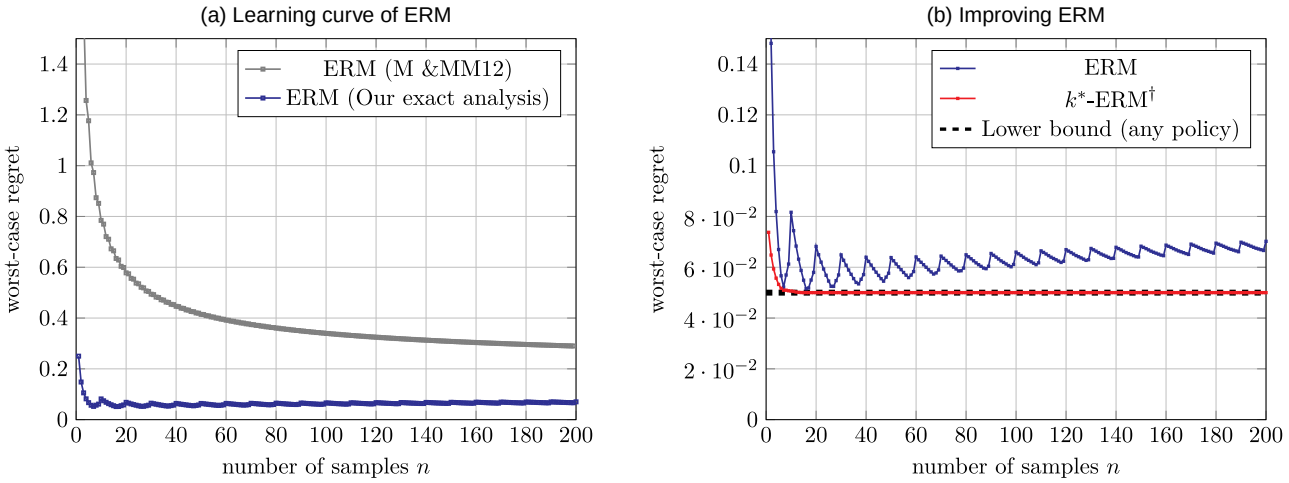
4.2. Shape of the Learning Curve of ERM and Effective Sample Size

4.2.1. Learning Curve: Unveiling New Insights. In this section we focus on the more granular question associated with the shape of how the worst-case regret evolves with the number of samples, which we call the "learning curve." We consider the following representative example throughout: dissimilarities $d(x_i,x_0)$ are $\zeta = 0.1$ for all i, and $q = 0.9$.

First, we plot in Figure 3(a) the regret guarantee of ERM as a function of n. Note that unlike Section 4.1, the regret is now expressed in absolute terms instead of as a percentage of the no-data regret.

We observe in Figure 3(a) that the scale of the curve implied by previous bounds is very different from the actual worst-case performance of ERM which is obtained by our exact analysis. This observation is consistent with the findings from Section 4.1. Moreover, this figure highlights a second shortcoming of previous bounds: they do not capture the correct

Figure 3. (Color online) Performance of ERM and Alternative Policies



Notes. (a) The figure depicts the worst-case regret of the ERM policy as implied by the bound presented by Mohri and Muñoz Medina (2012) and by ours. (b) The figure compares the performance of ERM and $k^*-\text{ERM}^\dagger$ with the lower bound achievable by any data-driven policy (see Remark 3). In these plots, $\zeta = 0.1$ and $q = 0.9$.

shape of the learning curve of ERM. Indeed, the worst-case regret implied by Mohri and Muñoz Medina (2012) is decreasing as the number of samples grows and suggests that the performance of ERM improves as we aggregate more samples. By contrast, our exact analysis unveils that the worst-case regret of ERM is nonmonotonic as a function of the sample size.

In Figure 3(b), we zoom in on the performance of ERM, displaying only the guarantee from our exact analysis, and observe two types of nonmonotonicity. First, ERM exhibits a "local" nonmonotonicity, in that the learning curve is jagged with many local minima. Such a type of behavior was already observed in the i.i.d. setting (where $\zeta = 0$) by Besbes and Mouchtaki (2023), under a different metric of regret. Figure 3(b) reveals another fundamental phenomenon, not present in the i.i.d. setting: ERM exhibits a "global" nonmonotonicity. The performance improves considerably with the first few samples, and then deteriorates after adding more samples. In other words, we observe that the learning curve of ERM admits a unique global minimum at a finite n ($n = 15$ in the example), and increasing n to ∞ tends to move away from this global minimum!

We also show in Online Appendix F that most of the insights derived through the worst-case analysis are still widely applicable when the instance is fixed across sample sizes and the demand distribution is not a Bernoulli distribution.

4.2.2. Achievable Performance and Effective Sample Size. We have just observed various phenomena associated with the performance of ERM, which lead to a natural question: How should one improve ERM in settings when historical distributions are different from the out-of-sample one?

To provide some context on the types of performance achievable in such settings, we first state a universal lower bound for any policy, even with infinite samples.

Remark 3 (Lower Bound). For $\zeta \le \min(q, 1-q)$, the regret of any data-driven policy is larger or equal to $\zeta/2$ for any sample size.

This result is implied by the proof of proposition 5 in Besbes et al. (2025), who analyze general decision problems with infinite data sizes. We investigate corrections to ERM and how close they can get to this lower bound.

The nonmonotonicities in the shape of ERM's learning curve, and in particular the observed phenomenon of global nonmonotonicity, lead us to propose the following alternate policy and subsequently define a notion of effective sample size. The alternate policy, which we refer to as $k^*-\text{ERM}^\dagger$, modifies ERM in two ways. First, we use a mixture of order statistics inspired by Besbes and Mouchtaki (2023) to avoid the local nonmonotonicity. This was sufficient to obtain a minimax policy in their setting, but it does not correct the global nonmonotonicity when $\zeta > 0$ (see Figure 4 in Online Appendix C). The second idea consists in using the optimal number of samples (denoted by k^*) even when having access to more. We formally define the $k^*-\text{ERM}^\dagger$ policy in Online Appendix C.

$k^*-\text{ERM}^\dagger$ can be evaluated by leveraging the full generality of our Theorem 1, which applies to randomized counting policies (details in Remark 2). We plot its worst-case regret as derived by our exact analysis, also in Figure 3(b), along with the lower bound on the regret of any data-driven policy. We observe that $k^*-\text{ERM}^\dagger$ considerably improves the performance

Table 2. Effective Sample Sizes for $k^*-\text{ERM}^\dagger$

	ζ					
	0.01	0.02	0.03	0.04	0.05	0.1
Effective sample size	871	380	132	95	58	15

Notes. The table reports, for different values of dissimilarity, the effective sample size, that is, the number of samples used by $k^*-\text{ERM}^\dagger$, even when having access to more samples. We note that the worst-case regret associated with these settings is computed by searching over means $\mu_0 \in \{0, 0.005, 0.01, \ldots, 1\}$ (a 201-point discretization of [0, 1]).

over ERM and resolves both types of nonmonotonicities. Quite notably, we also note that $k^*-\text{ERM}^\dagger$ is in fact near-optimal for the worst-case regret criterion (it achieves a worst-case regret within 1.001% of the lower bound) as soon as more than 15 samples are available. We therefore interpret k^*, the optimal choice of sample size for $k^*-\text{ERM}^\dagger$, as a notion of effective sample size which contains almost all the information necessary to perform well. After this point, using more samples does not allow us to improve performance by any meaningful amount, even if one were to use a different policy.

In Table 2, we investigate the effective sample size (sample size used by $k^*-\text{ERM}^\dagger$) as a function of ζ for values of ζ ranging from 0.01 to 0.1.

We note that for all the values reported in Table 2, the worst-case regret of $k^*-\text{ERM}^\dagger$ when using a sample size equal to the effective sample size is lower than 1.001% of that of the best data-driven policy using infinitely many samples. Consequently, in the presence of data which deviate from the i.i.d. setting, restricting attention to the effective sample size while using $k^*-\text{ERM}^\dagger$ enables us to match near-optimal performance across all data regimes with very few samples.

These insights sharply contrast with the behavior suggested by previous monotonic bounds. The performance of a learning policy should not be thought of as a decomposition between asymptotic error and finite sample error as described in Section 3.1; furthermore, one may achieve substantially better performance by using the effective number of samples rather than all samples.

4.3. Application to Varying Dissimilarities: Modeling Time Dependence

We conclude this section with an illustration of the generality of our theory to analyze different instances of the contextual Newsvendor. We next consider a setting which models a distribution drift over time. In this case, we set for every $i \in \{1, \ldots, n\}$ the dissimilarity $d(x_0, x_i) = i \cdot \Delta$, with $\Delta > 0$ representing the drift parameter. In this setting, one may think of y_i as the demand observed i weeks before the inventory decision, and Δ is an upper bound on the drift in demand distribution that may occur between two consecutive weeks.

We evaluate the worst-case regret for two subclasses of Weighted ERM policies with weights given by two widely used Nadaraya-Watson kernels. The first one is the class of Exponential Weighted ERM policies (EWERM) which corresponds to the kernel $K_h(z) = \exp(z/h)$. Formally, it assigns weights of the form $w_i = \gamma^i$ for a fixed $\gamma \in (0, 1]$.[5] We denote by π^γ such a policy. The second one is the class of k-NN-ERM policies which corresponds to the kernel $K_h(z) = 1\{z \le h\}$. For the instance considered in this section, it equivalently assigns weights of the form

$$w_i = \begin{cases} 1 & \text{if } i \le k \\ 0 & \text{otherwise,} \end{cases}$$

for a fixed $k \in \{1, \ldots, n\}$. We denote by π^k such a policy. We next evaluate the worst-case performance of these policies for various values of the drift parameter Δ. Given a fixed drift, we denote by $\mathbf{x}_\Delta$ the sequence of contexts satisfying the dissimilarity relation defined above.

For a fixed n we solve the following two problems,

$$\min_{\gamma \in [0,1]} \mathsf{Reg}_n(\pi^\gamma, \mathbf{x}_\Delta) \tag{9}$$

and

$$\inf_{k \in \{1, \ldots, n\}} \mathsf{Reg}_n(\pi^k, \mathbf{x}_\Delta). \tag{10}$$

Furthermore, we denote by $\gamma^*(\Delta)$ (resp. $k^*(\Delta)$) the value of γ (resp. k) which achieves the minimum value of Problem (9) (resp. (10)). The value of (9) (resp. (10)) is the minimal worst-case regret achievable by using the best parameter for each subclass of policies.

We report in Table 3 the value of these two problems and the values of the parameters achieving the minimal worst-case regret.

We note in Table 3 that when the relevance of data is heterogeneous, the number of samples used by each policy is actually even smaller than observed in Section 4.2. For instance, the value of $k^*(\Delta)$ in Table 3 can be interpreted as a notion of effective sample size comparable to the one in Table 2 (even though the

Table 3. Best Performance and Parameters for EWERM and k-NN-ERM

	EWERM		k-NN-ERM	
Δ	$\gamma^*(\Delta)$	Worst-case regret	$k^*(\Delta)$	Worst-case regret
0.0010	0.95	0.016	27	0.014
0.0025	0.91	0.023	17	0.018
0.0050	0.88	0.031	8	0.025

Note. The table reports the best choice of parameter and the achievable worst-case regret for the two classes of policies for different values of Δ ($n = 100$ and $q = 0.9$).

settings are not exactly the same). When $\Delta = 10^{-3}$, we observe that $k^*(\Delta) = 27$, implying that the least relevant data used by the policy have a dissimilarity equal to 0.027. We note that from Table 2, $k^*-\text{ERM}^\dagger$ would have used at least 200 samples when all the dissimilarities are the same with a value lower than 0.03. This difference in terms of effective sample size reveals that, in the presence of drifting distributions, the data points with low dissimilarity significantly decrease the value of data with higher dissimilarity.

We also observe that the number of samples used by k-NN-ERM is consistent with the number of samples used by EWERM. For a given $\gamma \in (0,1)$, one may interpret $1/(1-\gamma)$ as a proxy for the effective sample size of the policy π^γ. Therefore, one may ask whether $1/(1-\gamma^*(\Delta))$ is comparable to $k^*(\Delta)$. We observe in Table 3 that when $\Delta = 0.001$, $1/(1-\gamma^*(\Delta)) = 20$, which is relatively similar to $k^*(\Delta)$ as it is equal to 27. This conclusion holds across all values of Δ.

4.4. Sensitivity with Respect to the Dissimilarity

In Section 4.3 we have considered a setting with time drift and investigated how a decision maker should choose the hyper-parameter for widely used families of Weighted ERM policies (such as the k-NN-ERM policies) in order to minimize the worst-case regret. In what follows, we study the sensitivity of these results with respect to the dissimilarity.

We report in Table 4 the performance of the k-NN-ERM policy for various choices of k and Δ.

We note that the performance of the k-NN-ERM which uses $k^*(\Delta)$ by solving the minimax problem (10) is robust even when using an approximate value of Δ. For instance, if the decision maker believes that $\Delta = 0.0025$ instead of $\Delta = 0.005$ (which means that Δ is twice larger than what they assume), they would use the k-NN-ERM with a value of $k = 17$ as opposed to the recommended choice of $k = 8$. Although the error on Δ is large, the worst-case regret incurred with $k = 17$ is only 4% higher than the one with $k = 8$ when $\Delta = 0.005$. Furthermore, the worst-case regret they expect to incur goes from 0.018 to an actual worst-case regret of 0.026, which represents a 44% increase and is starkly lower than the 100% increase in Δ.

Table 4. Worst-Case Regret of k-NN-ERM

	Worst-case regret		
Δ	$k = 8$	$k = 17$	$k = 27$
0.0010	0.018	0.016	**0.014**
0.0025	0.021	**0.018**	0.020
0.0050	**0.025**	0.026	0.036

Notes. The table reports the worst-case regret for different values of k and Δ. The boldface numbers correspond to the best worst-case regret across all values of $k \in \{1,\dots,100\}$ ($n = 100$ and $q = 0.9$).

We note that beyond this sensitivity analysis, our work also raises the broader question of how to estimate dissimilarities between different contexts. Although a comprehensive treatment of this problem lies beyond the scope of the present paper, we provide and evaluate a simple estimation procedure in Online Appendix H for illustrative purposes.

5. Conclusion

In the present paper, we investigate the impact of relevance and quantity of data on performance for the prototypical contextual data-driven Newsvendor problem. We develop a new methodology to quantify exactly the worst-case regret of the broad class of Weighted ERM policies which encompasses several classical policies. Our method relies on an optimization approach we refer to as "learning without concentration" because it departs from the common concentration-based arguments previously developed in the literature.

We leverage our exact analysis to derive insights on this class of problems and show that these insights contrast with those implied by state-of-the-art upper bounds. We show that certain performances which were considered unachievable by previous bounds for ERM, even with infinitely many samples, can actually be achieved with very few samples. Furthermore, we show that, when historical distributions are different from the out-of-sample one, the worst-case regret of ERM reaches a global minimum for a finite sample size and then deteriorates when the sample size goes to ∞, whereas previous upper bounds on the worst-case regret which are decreasing implicitly prescribe accumulating more data in order to improve performance.

All in all, our analysis highlights the need to develop problem-specific bounds in order to capture the *actual* shape and scale of the learning curve. Indeed, concentration-based analysis is powerful as it can be applied to a general class of problems. However, the shape and scale suggested by these bounds are an artifact of the analysis rather than the actual performance of the algorithm.

We note that our analysis is tailored to the Newsvendor loss, and it remains an open question to what extent such exact characterizations can be extended to other decision-making problems. Several exciting research avenues remain regarding the generalizability of our method. A first natural direction is to use our tractable characterization of the worst-case performance to design new algorithms, for example, by optimizing over the weights of the Weighted ERM policy. A second direction is to extend the analysis beyond the Kolmogorov distance to alternative notions of distributional proximity; for instance, in Online Appendix G we illustrate how a Lagrangian relaxation can yield bounds under a Wasserstein distance, and

further work could establish "learning without concentration" guarantees more broadly. A third avenue is to study distributions with additional structure, such as moment or shape constraints, which may yield bounds even closer to the realized performance of algorithms. Finally, it would be valuable to explore the applicability of our approach to entirely different problem classes beyond Newsvendor, and to investigate whether (and how) analogous sharp worst-case characterizations can be obtained in these more general decision-making settings.

Acknowledgments

The authors thank the editor, the associate editor, as well as three anonymous referees for valuable suggestions which led to significant improvements in the final version of the paper. The authors are also grateful to David Cheikhi, Vishal Gupta, Michael Huang, Guillaume Roels, Ali Sadighian, and Meng Qi for their helpful discussions and valuable insights.

Endnotes

[1] The analysis directly generalizes to support $[0, M]$ for any $M > 0$ with appropriate modifications. When $\mathcal{Y}$ is unbounded, no policy can achieve a finite worst-case expected regret (defined in (3)).

[2] For simplicity of notation, when π is deterministic, we will denote by $\pi(\boldsymbol{x}, \boldsymbol{y})$ the action a played by $\pi(\boldsymbol{x}, \boldsymbol{y})$ as opposed to the point-mass distribution which puts all mass at a.

[3] We note that our framework and some of our results extend to a superset of these policies; see Section 3.3.

[4] We remark here that ERM does not use knowledge of the support of demand, whereas the minimax no-data policy does. This explains why it takes a few samples to match its performance.

[5] We note that when the dissimilarity satisfies $d(x_0, x_i) = i \cdot \Delta$, one can reparametrize the exponential kernel as follows: $K_h(d(x_0, x_i)) = \exp\left(\frac{d(x_0, x_i)}{h}\right) = \exp\left(\frac{\Delta}{h} \cdot i\right) = \gamma^i$.

References

Ban G-Y, Rudin C (2019) The big data newsvendor: Practical insights from machine learning. *Oper. Res.* 67(1):90–108.

Batini C, Cappiello C, Francalanci C, Maurino A (2009) Methodologies for data quality assessment and improvement. *ACM Comput. Surveys* 41(3):1–52.

Bertsimas D, Kallus N (2020) From predictive to prescriptive analytics. *Management Sci.* 66(3):1025–1044.

Bertsimas D, Koduri N (2022) Data-driven optimization: A reproducing kernel Hilbert space approach. *Oper. Res.* 70(1):454–471.

Bertsimas D, McCord C (2019) From predictions to prescriptions in multistage optimization problems. Preprint, submitted April 26, https://arxiv.org/abs/1904.11637.

Besbes O, Mouchtaki O (2023) How big should your data really be? Data-driven newsvendor: Learning one sample at a time. *Management Sci.* 69(10):5848–5865.

Besbes O, Ma W, Mouchtaki O (2025) Beyond IID: Data-driven decision making in heterogeneous environments. *Management Sci.* 71(12):10538–10555.

Bhattacharya PK, Gangopadhyay AK (1990) Kernel and nearest-neighbor estimation of a conditional quantile. *Ann. Statist.* 18(3): 1400–1415.

Bilodeau B, Negrea J, Roy DM (2020) Relaxing the i.i.d. assumption: Adaptively minimax optimal regret via root-entropic regularization. Preprint, submitted July 13, https://arxiv.org/abs/2007.06552.

Blanchard M (2022) Universal online learning: An optimistically universal learning rule. Loh P-L, Raginsky M, eds. *Proc. 35th Conf. Learn. Theory*, Proceedings of Machine Learning Research (PMLR, New York), 1077–1125.

Bousquet O, Elisseeff A (2002) Stability and generalization. *J. Machine Learn. Res.* 2:499–526.

Chaudhuri P (1991) Nonparametric estimates of regression quantiles and their local Bahadur representation. *Ann. Statist.* 19(2):760–777.

Chen N, Hu M (2023) Frontiers in service science: Data-driven revenue management: The interplay of data, model, and decisions. *Service Sci.* 15(2):79–91.

Cheung WC, Simchi-Levi D (2019) Sampling-based approximation schemes for capacitated stochastic inventory control models. *Math. Oper. Res.* 44(2):668–692.

Cortes C, Jackel LD, Chiang W-P (1995) Limits on learning machine accuracy imposed by data quality. Tesauro G, Touretzky D, Leen T, eds. *Adv. Neural Inform. Processing Systems*, vol. 7 (MIT Press, Cambridge, MA), 239–246.

Donti P, Amos B, Kolter JZ (2017) Task-based end-to-end model learning in stochastic optimization. Guyon I, Von Luxburg U, Bengio S, Wallach H, Fergus R, Vishwanathan S, Garnett R, eds. *Adv. Neural Inform. Processing Systems 30 (NIPS 2017)* (Curran Associates, Red Hook, NY), 5484–5494.

Elmachtoub AN, Grigas P (2021) Smart "predict, then optimize." *Management Sci.* 68(1):9–26.

Farahani A, Voghoei S, Rasheed K, Arabnia HR (2021) A brief review of domain adaptation. Stahlbock R, Weiss GM, Abou-Nasr M, Yang C-Y, Arabnia HR, Deligiannidis L, eds. *Adv. Data Sci. Inform. Engrg. Proc. ICDATA 2020 IKE 2020* (Springer, Cham, Switzerland), 877–894.

Fu M, Li X, Zhang L (2021) Data-driven feature-based newsvendor: A distributionally robust approach. Preprint, submitted July 13, http://dx.doi.org/10.2139/ssrn.3885663.

Gallego G, Moon I (1993) The distribution free newsboy problem: Review and extensions. *J. Oper. Res. Soc.* 44(8):825–834.

Gudivada V, Apon A, Ding J (2017) Data quality considerations for big data and machine learning: Going beyond data cleaning and transformations. *Internat. J. Adv. Software* 10(1):1–20.

Gupta V, Kallus N (2022) Data pooling in stochastic optimization. *Management Sci.* 68(3):1595–1615.

Gupta V, Rusmevichientong P (2021) Small-data, large-scale linear optimization with uncertain objectives. *Management Sci.* 67(1): 220–241.

Györfi L, Kohler M, Krzyzak A, Walk H (2002) *A Distribution-Free Theory of Nonparametric Regression*, vol. 1 (Springer, New York).

Hannah L, Powell W, Blei D (2010) Nonparametric density estimation for stochastic optimization with an observable state variable. Lafferty J, Williams C, Shawe-Taylor J, Zemel R, Culotta A, eds. *Adv. Neural Inform. Processing Systems*, vol. 23 (Curran Associates, Red Hook, NY), 820–828.

Huber J, Müller S, Fleischmann M, Stuckenschmidt H (2019) A data-driven newsvendor problem: From data to decision. *Eur. J. Oper. Res.* 278(3):904–915.

Kallus N, Mao X (2023) Stochastic optimization forests. *Management Sci.* 69(4):1975–1994.

Kannan R, Bayraksan G, Luedtke JR (2025) Technical note—Data-driven sample average approximation with covariate information. *Oper. Res.* 73(6):3245–3259.

Kao Y-H, Roy B, Yan X (2009) Directed regression. Bengio Y, Schuurmans D, Lafferty J, Williams C, Culotta A, eds. *Adv. Neural Inform. Processing Systems*, vol. 22 (Curran Associates, Red Hook, NY), 889–897.

Kim S, Pasupathy R, Henderson SG (2015) A guide to sample average approximation. Fu M, ed. *Handbook of Simulation Optimization* (Springer, New York), 207–243.

Kleywegt AJ, Shapiro A, Homem-de Mello T (2002) The sample average approximation method for stochastic discrete optimization. *SIAM J. Optim.* 12(2):479–502.

Koenker R (2005) *Quantile Regression* (Cambridge University Press, Cambridge, UK).

Koenker R (2017) Quantile regression: 40 years on. *Annual Rev. Econom.* 9(1):155–176.

Levi R, Perakis G, Uichanco J (2015) The data-driven newsvendor problem: New bounds and insights. *Oper. Res.* 63(6): 1294–1306.

Levi R, Pál M, Roundy RO, Shmoys DB (2007) Approximation algorithms for stochastic inventory control models. *Math. Oper. Res.* 32(2):284–302.

Loog M, Viering T (2022) A survey of learning curves with bad behavior: Or how more data need not lead to better performance. Preprint, submitted November 25, https://arxiv.org/abs/2211.14061.

Mohr F, van Rijn JN (2024) Learning curves for decision making in supervised machine learning: A survey. *Machine Learn.* 113(11): 8371–8425.

Mohri M, Muñoz Medina A (2012) New analysis and algorithm for learning with drifting distributions. Bshouty NH, Stoltz G, Vayatis N, Zeugmann T, eds. *Algorithmic Learn. Theory ALT 2012* (Springer, Berlin), 124–138.

Mohri M, Rostamizadeh A, Talwalkar A (2018) *Foundations of Machine Learning* (MIT Press, Cambridge, MA).

Nadaraya EA (1964) On estimating regression. *Theory Probab. Appl.* 9(1):141–142.

Oroojlooyjadid A, Snyder LV, Takáč M (2020) Applying deep learning to the newsvendor problem. *IISE Trans.* 52(4):444–463.

Pan SJ, Yang Q (2010) A survey on transfer learning. *IEEE Trans. Knowledge Data Engrg.* 22(10):1345–1359.

Perakis G, Roels G (2008) Regret in the newsvendor model with partial information. *Oper. Res.* 56(1):188–203.

Qi M, Cao Y, Shen Z-J (2022) Distributionally robust conditional quantile prediction with fixed design. *Management Sci.* 68(3): 1639–1658.

Qi M, Shi Y, Qi Y, Ma C, Yuan R, Wu D, Shen Z-J (2023) A practical end-to-end inventory management model with deep learning. *Management Sci.* 69(2):759–773.

Rakhlin A, Sridharan K, Tewari A (2011) Online learning: Stochastic and constrained adversaries. Preprint, submitted April 27, https://arxiv.org/abs/1104.5070.

Redko I, Morvant E, Habrard A, Sebban M, Bennani Y (2020) A survey on domain adaptation theory: Learning bounds and theoretical guarantees. Preprint, submitted April 24, https://arxiv.org/abs/2004.11829.

Scarf H (1958) A min-max solution of an inventory problem. Arrow KJ, Karlin S, Scarf H, eds. *Studies in the Mathematical Theory of Inventory and Production* (Stanford University Press, Stanford, CA), 201–209.

See C-T, Sim M (2010) Robust approximation to multiperiod inventory management. *Oper. Res.* 58(3):583–594.

Settles B (2012) *Active Learning* (Morgan & Claypool, San Rafael, CA).

Shalev-Shwartz S, Ben-David S (2014) *Understanding Machine Learning: From Theory to Algorithms* (Cambridge University Press, Cambridge, UK).

Srivastava PR, Wang Y, Hanasusanto GA, Ho CP (2021) On data-driven prescriptive analytics with side information: A regularized Nadaraya-Watson approach. Preprint, submitted October 10, https://arxiv.org/abs/2110.04855.

Vapnik VN (1999) An overview of statistical learning theory. *IEEE Trans. Neural Networks* 10(5):988–999.

Wang Y, Yao Q, Kwok JT, Ni LM (2020) Generalizing from a few examples: A survey on few-shot learning. *ACM Comput. Surveys* 53(3):1–34.

Watson GS (1964) Smooth regression analysis. *Sankhyā Indian J. Statist. Ser. A* 26(4):359–372.

Zhang L, Yang J, Gao R (2024) Optimal robust policy for feature-based newsvendor. *Management Sci.* 70(4):2315–2329.

Zhuang F, Qi Z, Duan K, Xi D, Zhu Y, Zhu H, Xiong H, He Q (2020) A comprehensive survey on transfer learning. *Proc. IEEE* 109(1):43–76.

Copyright of Management Science (INFORMS) is the property of INFORMS: Institute for Operations Research & the Management Sciences and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.

3. 库存学习：Learning in Lost-Sales Inventory Systems with Stochastic Lead Times and Random Supplies (MS)

来源：Management Science.

https://doi.org/10.2139/ssrn.4671416

作者：Xin Chen, Jiameng Lyu, Shilin Yuan, Yuan Zhou

关键词：库存管理；缺货失销；随机提前期；随机供给；在线学习；遗憾分析

推荐人：金柯

推荐理由：本文研究同时存在随机提前期与随机供给数量的缺货失销库存系统，决策者事先不知道需求与供给分布，且只能观察删失数据。作者提出具有理论保证的学习算法，以最佳恒定订货策略为基准，在给定提前期参数下获得随时间规模近似平方根增长的累计遗憾保证。方法的核心是变换凸性框架，并利用系统结构建立耦合与集中结果来估计长期成本。数值实验验证了算法效果，相关分析工具也可用于其他具有变换凸性结构的运营问题。

https://pubsonline.informs.org/journal/mnsc

MANAGEMENT SCIENCE
Articles in Advance, pp. 1–18
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Learning in Lost-Sales Inventory Systems with Stochastic Lead Times and Random Supplies

Xin Chen,[a] **Jiameng Lyu,**[b,*] **Shilin Yuan,**[c,*] **Yuan Zhou**[d,e]

[a] Naveen Jindal School of Management, The University of Texas at Dallas, Richardson, Texas 75080; [b] Department of Management Science, School of Management, Fudan University, Shanghai 200433, China; [c] School of Management, Huazhong University of Science and Technology, Wuhan 430074, China; [d] Yau Mathematical Sciences Center & Department of Mathematical Sciences, Tsinghua University, Beijing 100084, China; [e] Beijing Institute of Mathematical Sciences and Applications, Beijing 101408, China

*Corresponding authors

xin.chen@utdallas.edu (X. Chen), jiamenglyu@fudan.edu.cn (J. Lyu), shilinyuan-research@outlook.com (S. Yuan), yuan-zhou@tsinghua.edu.cn (Y. Zhou)

https://orcid.org/0000-0002-5168-4823 (X. Chen), https://orcid.org/0000-0002-4688-5276 (J. Lyu), https://orcid.org/0009-0002-7892-0344 (S. Yuan), https://orcid.org/0009-0008-1706-6539 (Y. Zhou)

Received: December 21, 2023
Revised: May 24, 2025; March 23, 2026
Accepted: March 30, 2026
Published Online in Articles in Advance: September 3, 2026

https://doi.org/10.1287/mnsc.2023.04203

Copyright: © 2026 INFORMS

Abstract. Supply uncertainty, characterized by stochastic lead times and random supply quantities, has attracted increasing attention from academia, industries, and governments, particularly in the aftermath of the COVID-19 pandemic. In this paper, we consider the problem of managing lost-sales inventory systems with general supply uncertainty: stochastic lead times and random supplies. Unlike the previous studies, we assume the decision maker has no prior information on the stochastic demand and supply. We propose the first provably effective learning algorithm for inventory management problems with censored demand and supply data under general supply uncertainty. Then, we establish a cumulative regret of $\tilde{O}(L+\mathrm{poly}(\overline{L})\cdot\sqrt{T})$ for this learning algorithm compared with the best constant-order policy, where $\overline{L}$ is the upper bound of the random part, and L is the deterministic part of the stochastic lead times. We also conduct numerical experiments to demonstrate the effectiveness of our algorithm. Our approach lies in developing a new framework for transformed convexity. Furthermore, we address the unique challenges of our problem through new techniques, for example, estimating the long-run cost by establishing coupling and concentration results utilizing the system structures. These techniques are also of independent interest. Beyond our problem, our framework provides broad implications for other operations management (OM) problems exhibiting transformed convexity.

History: Accepted by J. George Shanthikumar, data science.
Funding: J. Lyu and Y. Zhou are supported by the National Natural Science Foundation of China [Grant 52494974]. S. Yuan is supported by the National Natural Science Foundation of China [Grants 72588101, 72425008].
Supplemental Material: The online appendix and data files are available at https://doi.org/10.1287/mnsc.2023.04203.

Keywords: inventory management • lost sales • supply uncertainty • stochastic lead times • random supplies • online learning • regret analysis

1. Introduction

In recent years, especially during and after the COVID-19 pandemic, the structure of supply chains has emerged to be increasingly complex, which makes the problem of supply uncertainty more prominent worldwide and attracts increasing attention from academia, industries, and governments (Bu et al. 2020, Amazon 2023, White House 2023, Chen et al. 2024). Supply uncertainty predominantly encompasses two categories of uncertainties: uncertainty in the lead time and uncertainty in the supply quantity (Snyder and Shen 2019). The uncertainty in lead time, which is characterized by stochastic lead times, can arise because of factors such as stockouts at the supplier, delays in manufacturing or transit, and so on. In such instances, the lead time needs to be regarded as a stochastic variable instead of a constant. Likewise, the uncertainty in the supply quantity, which is referred to as the random supplies and characterized by the random supply function,[1] can be attributed to the probabilistic characteristics encompassing capacity availability, yield, and the total orders from all firms involved. Because of its great importance, there is a large body of literature focusing on the supply uncertainty problem, which either aims to characterize the structure of the optimal solution (Ciarallo et al. 1994, Wang and Gerchak 1996, Dada et al. 2007, Chen et al. 2018, Feng and Shanthikumar 2018, Chen and Gao 2019) or focus on finding asymptotically

optimal heuristic policies (Xin and Goldberg 2016, Bu et al. 2020, Bai et al. 2026).

The abovementioned existing literature studying supply uncertainty mostly assumes that full information about the supply chain is known by the firm manager. However, in practice, this is rarely the case. Specifically, on the demand side, the demand distribution is usually unknown to the firm manager. On the supply side, it is also difficult for the firm manager to acquire the supply information such as the supply distribution, random supply function, and distributions of the stochastic lead times because the structure of supply chains has become increasingly complex nowadays.

Motivated by the practical needs discussed above, in this paper, we aim to propose a provably effective learning algorithm for inventory systems with unknown demand distribution and supply information under the general supply uncertainty: stochastic lead times and random supplies. Specifically, we consider a single-product periodic-review inventory model with lost sales in which an order placed at a period will be delivered after a stochastic lead time and the delivered amount is a function of the order quantity and a random component. We assume that the stochastic lead times can be written as a deterministic part L plus a random part with support set $\{0,1,\ldots,\overline{L}\}$, which is independent of the random component in the supply function.[2] Under the assumption that demand distribution, lead time distribution, and the format and the distribution of the random component of the supply function are all unknown, we establish the $\tilde{\mathcal{O}}(L+\text{poly}(\overline{L})\cdot\sqrt{T})$[3] compared with an optimal constant-order policy under complete information.

As far as we know, our paper is the first one to study inventory management problems with unknown demand and supply information under the general supply uncertainty: stochastic lead times and random supplies. In literature, few works study the learning problem of supply uncertainty with unknown supply information. A recent work (Chen et al. 2024) falls into the learning realm, which proposes a learning algorithm for inventory systems with constant lead times and random supplies. However, Chen et al. (2024) only achieve part of the above goal and still assume some supply information is known, whereas we study the general uncertainty and do not assume any supply information is known. Specifically, first, they do not consider the supply uncertainty in lead time and assume constant lead times. In contrast, our paper includes the more general stochastic lead times setting. Addressing stochastic lead times represents a significant step toward incorporating reality into learning models in operations and bridging the gap with practice. Second, although Chen et al. (2024) assume both the demand and supply distributions are unknown, their algorithm requires the accurate formulation of the random supply function, including the function type and parameters to design the algorithm, which might not be practical in many scenarios.[4] Instead, our paper successfully relaxes their restrictive assumption on known random supply functions (i.e., we do not require any information about the formulation of the supply function). Because the constant lead times setting is a special case of the stochastic lead times setting with $\overline{L}=0$, it is easy to note that our algorithm could achieve a cumulative regret of $\tilde{\mathcal{O}}(L+\sqrt{T})$ in the constant lead times setting. The detailed comparisons are presented in Table 1.

Similar to Chen et al. (2024), we use the best constant-order policy as our benchmark. This is well justified by the good performance of constant-order policies when lead times are large. Indeed, for the infinite horizon counterpart of our model under the assumption of full knowledge of distributional information, Xin and Goldberg (2016) and Bai et al. (2026) demonstrated that the performance gap between the best constant-order policy and the true optimal policy decays exponentially fast in the deterministic part L for the deterministic lead time case and the stochastic lead time case, respectively. Our stochastic lead time model follows what's proposed in Bai et al. (2026), involving a large deterministic component and a random component, which captures a variety of business settings. For example, for the construction industry, the delivery times for prestressed concrete products and steel pipe and tube were six to seven months and four to five months, respectively, during and after the COVID-19 pandemic.[5] Similarly, the pandemic has caused a significant increase in lead times for semiconductor chips and electronic components. Lead times for semiconductor chips have increased from about 13 weeks to above 20 weeks, whereas lead times for electronic components have gone from 16 weeks to more than 52 weeks.

Whereas constant-order policies offer operational simplicity, learning and optimizing the best constant-order policy in lost-sales inventory systems with stochastic lead times and random supplies presents several key challenges. First, the cost function induced by random supplies is inherently nonconvex, breaking a key assumption underlying many existing algorithms that rely heavily on convexity. Second, the stability (Lipschitz continuity) of the long-run average cost is essential for learning algorithm design and regret analysis, yet proving such stability is nontrivial because of the absence of a closed-form expression. Third, accurately estimating the long-run average cost is challenging because of the persistent influence of each decision over time and the intricate dynamics introduced by stochastic lead times.

1.1. Our Contributions

In the following, we present an overview of our contributions.

Table 1. Comparison Between Our Results and Chen et al. (2024)

Source	Stochastic lead times	Assumption on lead times	Random supplies	Assumption on known random supply function	Cumulative regret
Chen et al. (2024)	✗	Constant	✓	Required	$\tilde{\mathcal{O}}(L+\sqrt{T})$
Our result	✗	Constant	✓	Not required	$\tilde{\mathcal{O}}(L+\sqrt{T})$
Our result	✓	Stochastic	✓	Not required	$\tilde{\mathcal{O}}(L+\text{poly}(\overline{L})\cdot\sqrt{T})$

Notes. $\overline{L}$ is the upper bound of the random part, and L is the deterministic part of the stochastic lead times. Constant lead times setting is a special case of the stochastic lead times setting with $\overline{L}=0$.

1.1.1. Developing a New Framework for Transformed Convexity. We begin by uncovering a key structural property: the long-run average cost function in our inventory problem, although nonconvex in its original form, becomes convex after a suitable variable transformation. We term this property *transformed convexity* (see Assumption 3). This insight motivates a general framework for *optimizing transformed-convex functions with bandit feedback*. To formalize this, we first analyze a simplified setting where the objective function satisfies transformed convexity and unbiased, independent and identically distributed (i.i.d.) bandit feedback is available. We establish minimax lower bounds of $\Omega(\nu\sqrt{T})$ for cumulative regret and $\Omega(\nu/\sqrt{T})$ for simple regret (see Equation (3) for definition), where ν is the condition number capturing the distortion induced by the transformation.[6] We then show that a quadrisection search algorithm achieves optimal cumulative and simple regret, whereas a trisection search algorithm achieves optimal simple regret. These results (see Section 3) form the foundation of our general framework.

1.1.2. Addressing Key Challenges in Our Inventory Systems. Building on this foundation, we tackle the significantly more complex setting of lost-sales inventory control with unknown demand and supply functions under stochastic lead times and random supplies for the first time in the literature. Our paper represents a significant step toward incorporating reality into learning models in inventory management and bridging the gap with practice. We establish the Lipschitz continuity of the long-run average cost and explicitly bound the Lipschitz constant by leveraging convexity and system properties. In contrast to the simplified setting, estimating the long-run average cost here is particularly difficult because of the intricate dynamics of the inventory system. To address this, we first establish a coupling result (Lemma 3) showing that applying the same order quantity over a sequence of periods drives the distribution of on-hand inventory close to its limiting distribution. Next, we augment the on-hand inventory with the stochastic lead time process to construct a Markov chain and prove its V-geometric ergodicity[7] through careful analysis of the inventory process structure. Finally, we apply a McDiarmid-type concentration inequality for V-geometrically ergodic Markov chains to establish the desired estimation accuracy (see Corollary 1).

1.1.3. Providing Broad Implications for Other OM Problems. Our framework addresses a general problem: *optimizing transformed-convex functions with bandit feedback*. Beyond inventory management with constant-order policies, this perspective connects to classic operations management (OM) problems such as newsvendor models with random supply, dynamic pricing with unknown nonparametric demand function, and arrival/service rate control in queueing systems (see Section 4). Moreover, the stability results for the inventory system under constant-order policies offer insight into parameter sensitivity for other G/G/1 queueing systems. Finally, to the best of our knowledge, this is the first work to utilize V-geometric ergodicity to establish concentration bounds for inventory problems. Our analysis provides a novel foundation for studying concentration results in complex stochastic systems.

1.2. Related Works

In this section, we introduce the following streams of literature related to our paper and discuss how our paper is appropriately placed in contemporary literature by giving comparisons with closely related existing works.

1.2.1. Classic Research on Inventory Systems with Supply Uncertainty. There is a large body of literature that assumes the demand distribution and the stochastic lead times distribution are known in advance by decision makers. These works, which aim to illustrate the structure of optimal policies or develop tractable approximate optimal policies, can be divided into two streams. Some works consider the i.i.d. stochastic lead times assumption, which might cause orders to cross in time (Song and Zipkin 1996b, 2009; Stolyar and Wang 2022). However, as pointed out in Hadley and Whitin (1963), because in most practical cases, orders cannot cross in time, the i.i.d. stochastic lead times assumption is not proper to be considered in these cases. The seminal work by Kaplan (1970) proposed an assumption to model the no-crossover phenomena, which is known

as *Kaplan assumption* (see Section 2 for a detailed definition). Later, lots of works emerged to adopt this assumption and develop approximately optimal policies (to name a few, Nahmias 1979, Song and Zipkin 1996a, Janakiraman and Roundy 2004, Zipkin 2008, Bai et al. 2026).

Inventory systems with random supply (unreliable suppliers) are also well studied in the classic research, and lots of random supply models were proposed to model the randomness of the supply, in which there are two well-studied classic models, the random yield model (Yano and Lee 1995, Bollapragada and Morton 1999, Huh and Nagarajan 2010) and the random capacity model (Ciarallo et al. 1994; Wang and Gerchak 1996; Feng 2010; Chen et al. 2015, 2018; Chen and Gao 2019), which model the random nature of the yield and the capacity of suppliers. In addition, Dada et al. (2007) proposed a highly nonlinear function class that greatly expands the candidate random supply functions. Cachon (2003) introduced a supply function that models a situation where the supplier distributes total output among multiple firms in proportion to their order quantities. Existing works on inventory systems involving random supply mainly focus on developing heuristics and demonstrating their performance under the assumption that the demand distribution and the supply distribution are known in advance. For example, Bu et al. (2020) demonstrated that the optimal constant-order policy is asymptotically optimal with large lead times in lost-sales inventory systems with positive deterministic lead times and random supply. Bai et al. (2026) established the asymptotic optimality of the optimal constant-order policy in a more complex inventory control problem, which further allows the lead times to be stochastic. Our work is the first paper to study the learning problem for inventory systems with both stochastic lead times and random supply.

1.2.2. Learning in Lost-Sales Inventory Systems with Positive Deterministic Lead Time. In recent years, several papers have delved into the study of lost-sales inventory systems with positive lead times in cases where the demand distribution is unknown to the decision makers. Because the optimal policy is computationally intractable, existing papers all considered adopting asymptotic heuristics as the benchmark. Huh et al. (2009a), Zhang et al. (2020), and Agrawal and Jia (2022) are a series of works that select the best base-stock policy as the benchmark, which has been shown to be asymptotically optimal in Huh et al. (2009b). Huh et al. (2009a) proposed a stochastic gradient descent (SGD) based algorithm and achieved a regret of $\tilde{\mathcal{O}}(\exp(L)\cdot T^{2/3})$, where L is the positive lead time. Zhang et al. (2020) improved the regret to $\tilde{\mathcal{O}}(\exp(L)\cdot\sqrt{T})$ with a different SGD-based algorithm. With a convex bandit optimization-based algorithm, Agrawal and Jia (2022) successfully eased the exponential dependency on L and established a regret of $\tilde{\mathcal{O}}(L\cdot\sqrt{T})$. Lyu et al. (2024) developed upper confidence bound (UCB)–type learning algorithms to learn the best capped base-stock policy and best base-stock policy. Chen et al. (2024) studied the learning problem with both positive lead times and random supply. Taking the optimal constant-order policy as the benchmark, which has been shown to be asymptotically optimal in Bu et al. (2020), they established a regret of $\tilde{\mathcal{O}}(L+\sqrt{T})$. Because the work in Chen et al. (2024) is closely related to our paper, we conduct a comprehensive comparison and in-depth discussion in the introduction section and other places. Here, we provide a concise summary of this comparison.

Our approach differs fundamentally from theirs and offers several key advantages. First, whereas their method requires exact specification of the random supply function's form and parameters, which is often impractical, our framework operates without any supply model formulation, ensuring robustness against more specification errors. Second, we significantly generalize previous results by handling stochastic lead times rather than just constant delays, overcoming substantial technical challenges. We are not aware of other works that incorporate lead time uncertainty into learning models in inventory management. Third, our framework provides broader implications for other OM problems with transformed convexity, such as newsvendor models with random supply and dynamic pricing with unknown nonparametric demand function. Finally, our numerical results consistently demonstrate superior performance across diverse scenarios. Notably, in situations involving model misspecification, the numerical performance of our approach significantly outperforms theirs.

1.2.3. Other Related Works on Learning for Inventory Systems. In addition to the abovementioned works closely related to our work, there are other papers studying inventory systems with demand learning in different settings. The following is an incomplete list, and interested readers can refer to the review by Gao and Zhang (2022b) for a detailed discussion.

Some works assume that the demand data can be fully observed, in which case the sample average approximation (SAA) method is commonly used to learn the distribution based on the observed samples of the demand (Levi et al. 2015; Lin et al. 2022; Qin et al. 2022, 2023; Keskin et al. 2026). For learning in lost-sales inventory systems with zero lead time, the seminal work by Huh and Rusmevichientong (2009) proposed an SGD-based algorithm in a single-product setting. Shi et al. (2016) extended the SGD-based algorithm to a multiproduct single constraint inventory system. Taking a new technical perspective, Lyu et al. (2024)

proposed a novel minibatch-SGD-based metapolicy that works for a wide range of inventory systems with myopic optimal policy, including multiproduct and multiconstraint systems. Chen et al. (2025b) proposed three first-order stochastic optimization algorithms for minimizing compositions of convex functions and random functions and studied their sample complexity. Their framework can be applied to the newsvendor problem with zero lead time and known random supply function formulation. In addition, Yuan et al. (2021) explored inventory systems with demand learning and fixed ordering costs, Zhang et al. (2018) studied the periodic-review perishable inventory systems with fixed lifetimes, Chen et al. (2020) investigated the learning problem for backlogging inventory systems with random production capacity and zero lead time, and Gong and Simchi-Levi (2023) studied online inventory models with unknown cyclic demand distributions. Gao and Zhang (2022a) proposed an efficient learning framework tailored for multiproduct inventory systems featuring customer choice considerations. Feng et al. (2026) developed the transfer learning and cross-learning method to learn across related inventory systems; Feng et al. (2025) studied the contextual newsvendor problem and proposed several data-integrated methods by utilizing structural properties. Both works are based on the operational data analytics (ODA) framework proposed in Feng and Shanthikumar (2023).

1.3. Organization

The remainder of this paper is organized as follows. In Section 2, we present the problem formulations, including the definition of stochastic lead times and random supply functions, the performance measure, and two important structural results of the transformed long-run average cost function. In Section 3, we present our general framework for optimizing transformed-convex functions with bandit feedback. In Section 4, we apply our general framework to three important operations management tasks to illustrate the broad applicability of our framework. In Section 5, we address key challenges in our focused inventory systems and apply our general framework. In Section 6, we extend the algorithm design and analysis to Markov-modulated environments and the indivisible product setting. To demonstrate the empirical performance of our policy, we conduct several numerical experiments and present the results in Section 7. Finally, we summarize our paper in Section 8.

2. Problem Formulation

We consider a periodic-review lost-sales inventory system managed by a firm under censored demand, stochastic lead times, and random supplies. In each period over T-period selling season, the demand D_t is sampled from a common distribution D and is independent over time. The lead time of the order in period t is stochastic and denoted by a random variable L_t, which means that the order at period t will arrive at period $t+L_t$. We assume the lead times are equal to a deterministic part L plus a random part with support set $\{0,1,\dots,\overline{L}\}$. Note that we do not assume L_t is independent over time.

We introduce the random supply variables Z_t and random supply function $s(q,Z):\mathbb{R}^2\to\mathbb{R}^+$ to model the supply uncertainty. The random supply variables Z_t are independent over time and identically distributed with common distribution Z. The random supply function maps the order quantity and the realization of the random supply variable to the quantity that actually arrived.

Initially, the inventory level is $I_1=0$. During each period $t\in\{1,2,3,\dots\}$, we assume the following sequence of events.

1. At the beginning of period t, the firm decides an order quantity o_t (based on observed data in history, which will be clarified soon) and places the order o_t. For convenience, we assume that $o_\tau=0$ for all $\tau\leqslant 0$.

2. The orders placed at time set $\{i:\ i+L_i=t\}$ arrive. Following the convention in Bu et al. (2020), we also assume the random supply variables are realized when received; that is, $Z_i,\ i\in\{i:\ i+L_i=t\}$ are realized at time t.[8] We assume Z_i cannot be observed and only the supply quantity (censored supply data) $s(o_i,Z_i)$ can be observed. The total inventory received is

$$X_t:=\sum_{\{i:\ i+L_i=t\}} s(o_i,Z_i).$$

3. The demand D_t is realized and satisfied to the maximal extent by the on-hand inventory. We assume the unsatisfied demand is lost and only the sale quantity (censored demand) $\min\{I_t+X_t,D_t\}$ can be observed. The leftover inventory (which is also the inventory level at the beginning of the next period) is

$$I_{t+1}=(I_t+X_t-D_t)^+. \tag{1}$$

We assume the on-hand inventory incurs a holding cost of h per unit and unsatisfied demand incurs a lost-sales cost of b per unit.[9] For any nonanticipative ordering policy $\pi=(\pi_t)_{t\geqslant 1}$ with $\pi_t:\mathcal{H}_t\longmapsto o_t$, where $\mathcal{H}_t$ is the historically observed data at the beginning of period t (inventory levels $\{I_\tau\}_{\tau\leqslant t}$, order quantities $\{o_\tau\}_{\tau<t}$, received quantities $\{X_\tau\}_{\tau<t}$, and censored demands $\{\min\{I_\tau+X_\tau,D_\tau\}\}_{\tau<t}$), the total cost incurred at period t can be denoted by

$$C_t^\pi=h\cdot(I_t+X_t-D_t)^+ +b\cdot(D_t-I_t-X_t)^+.$$

The long-run average inventory cost incurred by policy π is defined as

$$C_\infty^\pi=\limsup_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[C_t^\pi].$$

Now, we turn to the discussion of the regret. Because the optimal policy of this inventory problem is quite

complicated, selecting it as the benchmark of learning will make the objective of minimizing the regret computationally intractable. Fortunately, a simple heuristic *constant-order policy*, which orders the same quantity in all periods, has been shown by Bai et al. (2026) to be asymptotically optimal as the lead times increase. Considering the lead times are large in a variety of applications, in this paper, we use the best constant-order policy as the benchmark.

Denoting the constant-order policy that places the order q in all periods as π_q and assuming $q^* = \arg\min_{q\in\mathbb{R}^+} C_\infty^{\pi_q}$, we define the *cumulative regret* as follows:

$$\mathcal{R}(T) = \sum_{t=1}^{T} \mathbb{E}[C_t^{\pi}] - T \cdot C_\infty^{\pi_{q^*}}. \tag{2}$$

In addition to the cumulative regret, *simple regret* is another popular performance measure that evaluates the performance gap between the final recommendation and the optimal solution. For our problem, we define the simple regret as

$$\mathcal{S}(T) := \mathbb{E}[C_\infty^{\pi_{\hat{q}_T}}] - C_\infty^{\pi_{q^*}}, \tag{3}$$

where $\hat{q}_T$ represents the final recommended order quantity after T periods.

Under a constant policy, the on-hand inventory process $\{I_t\}_{t=1}^{\infty}$ can be regarded as the waiting time process of a G/G/1 queue. To make sure that under any candidate constant-order policy $\pi_q, q \in [0, \overline{q}]$, the traffic intensity is less than one and the on-hand inventory process will not explode, we impose the following basic assumption of Chen et al. (2024) throughout this paper.

Assumption 1. *The optimal constant-order quantity q^* is upper bounded by $\overline{q}$, which is known to the firm. Moreover, $\overline{q} < q'$, where $q' = \inf\{q : \mathbb{E}[s(q,Z)] = \mathbb{E}[D]\}$.*

In this assumption, we assume $\overline{q}$ is known to the firm. In Sections 6.3 and Section EC.6.3 in the Online Appendix, we will discuss how to relax this assumption.

2.1. Random Supply Function

For any $q \in [0, q']$, we first define the expected supply function $\mu(q) := \mathbb{E}[s(q,Z)]$. Let $\overline{\mu} := \mathbb{E}[s(\overline{q},Z)]$ and $q(\mu) = \inf\{q \in \mathbb{R} : \mu(q) = \mu\}$. Then, for any $\mu \geq 0$, we define the transformed random supply function as

$$\hat{s}(\mu, Z) := s(q(\mu), Z).$$

For algorithm design and regret analysis, we make the following assumptions on the random supply function.

Assumption 2. *There exists a constant δ such that the random supply function satisifies the following four conditions.*

1. *For any realization of Z and $q \in [0, q(\overline{\mu} + \delta/2)]$, $s(q,Z)$ is nondecreasing w.r.t. q, and there exists a positive constant $\overline{D}$ and $\overline{s}$ such that $D \leqslant \overline{D}$ and $s(q,Z) \leqslant \overline{s}$. Without loss of generality, we assume $\overline{s} \leqslant \overline{D}/2$.*
2. *$\mu(q)$ is strictly increasing in $q \in [0, q']$ and $\mathbb{E}[D] - \overline{\mu} \geqslant \delta > 0$, where, we futher require that δ is a positive constant and without loss of generality, we assume $\delta \leqslant \overline{D}^2/\overline{L}\overline{s}$.*
3. *The transformed random supply function is stochastically linear in midpoint; that is, $\{\hat{s}(\mu, Z) : \mu \in [0, \overline{\mu} + \delta/2]\} \in$ SL(mp).*
4. *There exist two positive constants l_q and L_q such that for any $q_1, q_2 \in [0, q(\overline{\mu} + \delta/2)]$, it holds that $l_q|q_1 - q_2| \leqslant |\mathbb{E}[s(q_1, Z)] - \mathbb{E}[s(q_2, Z)]| \leqslant L_q|q_1 - q_2|$.*

Assumption 2.1 is basic and mild assumptions on the boundedness of D, $s(q,Z)$. Assumption 2.2 is a mild assumption on $\mu(q)$ to avoid the on-hand inventory approaching infinity, which is common in literature (e.g., Bu et al. 2020, Chen et al. 2024). In Assumption 2.3, $\{\hat{s}(\mu, Z) : \mu \in [0, \overline{\mu} + \delta/2]\} \in$ SL(mp) refers to that for any convex function $\phi(\cdot)$, $\mathbb{E}[\phi(\hat{s}(\mu, Z)]$ is convex w.r.t. μ. This property is critical for the establishment of the convexity of the transformed long-run average cost function, and it is verified by Feng and Shanthikumar (2018) that many common random supply functions (e.g., examples listed below) satisfy this assumption. In Section EC.3 in the Online Appendix, we present the detailed definition of SL(mp) developed by Shaked and Shanthikumar (2007) and Feng and Shanthikumar (2018) and discuss how to use supply data to verify this assumption. Assumption 2.4 is a regular assumption on the function $\mu(q)$, which ensures that an interval of q after the map $\mu(q)$ is nondegenerate and is essential for the regret analysis of our quadrisection algorithm.

The following are four examples of random supply functions, which are classic and common in the literature (see e.g., Dada et al. 2007, Feng and Shanthikumar 2018, Bu et al. 2020, Chen et al. 2024). Note the first is the random yield model and the second is the random capacity model, and we refer interested readers to Dada et al. (2007) for a detailed discussion of the motivation and origin of these random supply functions. In Section EC.3.2 in the Online Appendix, we verify all these random supply functions satisfy Assumption 2 under some mild conditions on Z.

1. $s(q,Z) = q \cdot Z$.
2. $s(q,Z) = \min\{q, Z\}$.
3. $s(q,Z) = qZ/(q + \alpha Z^{\rho})$ for $\rho \leqslant 1$ and $\alpha > 0$.
4. $s(q,Z) = kq/(q + Z)$ for some $k > 0$.

2.2. Stochastic Lead Times

For convenience of presentation, this paper assumes the lead time process follows the classic *Kaplan model* (Kaplan 1970), and extensions to more general lead time models will be presented in Section 6.1. We first introduce a sequence of i.i.d. random variables $\{\rho_t, t \geqslant 1\}$ with common distribution ρ and support set

$\{L, L+1, \ldots, L+\overline{L}\}$, where L is a nonnegative integer. The stochastic lead times process $\{L_t, t \geqslant 1\}$ is defined based on the sequence $\{\rho_1, \rho_2, \ldots, \rho_T\}$ as follows: the event $\{\rho_t = n\}$ means that in period t, all orders that have been outstanding for at least n periods would arrive at period t. Therefore, the event $\{L_t = n\} = \{\rho_t > 0, \rho_{t+1} > 1, \ldots, \rho_{t+n-1} > n-1, \rho_{t+n} \leqslant n\}$, and we know $\mathbb{P}[L_t = n] = \mathbb{P}[\rho_t > 0] \cdot \mathbb{P}[\rho_{t+1} > 1] \cdots \mathbb{P}[\rho_{t+n-1} > n-1] \cdot \mathbb{P}[\rho_{t+n} \leqslant n]$.

It is easy to note that under the Kaplan model, the orders do not cross in time, and $\{L_t\}_{t=1}^{T}$ are identically (and dependently) distributed and can be decomposed as the deterministic part L plus the random part with support set $\{0, 1, \ldots, \overline{L}\}$.

Next, we consider the total order quantity $X_t^{\pi_q}$ arriving at t under the constant-order policy π_q and stochastic lead times. The following lemma shows that although $X_t^{\pi_q}$ has a complex form depending on lead time models, it is identically distributed, and its expectation is simply $\mathbb{E}[s(q, Z)]$.

Proposition 1. *If the multidimensional process* $\{(L_t, L_{t+1}, \ldots, L_{t+L+\overline{L}})\}_{t=1}^{\infty}$ *is stationary, the total arriving quantities* $\{X_t^{\pi_q} : t \geqslant L+\overline{L}+1\}$ *under constant-order policy* π_q *are identically distributed. We denote the common distribution by* X^{π_q}, *and we have* $\mathbb{E}[X^{\pi_q}] = \mathbb{E}[s(q, Z)]$.

The above conclusion is stated for general lead time processes, and it is easy to see that under the Kaplan model, $\{(L_t, L_{t+1}, \ldots, L_{t+L+\overline{L}})\}_{t=1}^{\infty}$ is decided by i.i.d. sequence $\{\rho_t, t \geqslant \}$ and is stationary.

Under policy π_q, the on-hand inventory updating formulation Equation (1) can be written as

$$I_{t+1}^{\pi_q} = (I_t^{\pi_q} + X_t^{\pi_q} - D_t)^+. \tag{4}$$

Because when $t \geqslant L+\overline{L}+1$ and both $\{D_t : t \geqslant L+\overline{L}+1\}$ and $\{X_t^{\pi_q} : t \geqslant L+\overline{L}+1\}$ are identically distributed, it is easy to note that the above updating dynamics (Equation (4)) are the same as those of the waiting time process of a G/G/1 queue[10] with interarrival times $\{D_t : t \geqslant L+\overline{L}+1\}$ and service time $\{X_t^{\pi_q} : t \geqslant L+\overline{L}+1\}$, which helps us study the limiting distribution of on-hand inventory.

2.3. Limiting Distribution of On-Hand Inventory

In this subsection, we apply the conclusions of G/G/1 queues to study the limiting distribution of on-hand inventory. By Proposition 1 and Assumption 2, for $t \geqslant L+\overline{L}+1$, we have

$$\mathbb{E}[X_t^{\pi_q}] = \mathbb{E}[s(q, Z)] < \mathbb{E}[D].$$

By the well-known stability and waiting time distribution convergence results of G/G/1 queue (Loynes 1962), we obtain that $\{I_t^{\pi_q} : t \geqslant 1\}$ converge in distribution to $I_\infty^{\pi_q}$

Thus, the long-run average inventory cost incurred by policy π_q is

$$C_\infty^{\pi_q} = h \cdot \mathbb{E}[I_\infty^{\pi_q} + X^{\pi_q} - D]^+ + b \cdot \mathbb{E}[D - X^{\pi_q} - I_\infty^{\pi_q}]^+. \tag{5}$$

By the facts that

$$I_\infty^{\pi_q} \overset{d}{=} (I_\infty^{\pi_q} + X^{\pi_q} - D)^+ \quad \text{and}$$
$$I_\infty^{\pi_q} + X^{\pi_q} - D = (I_\infty^{\pi_q} + X^{\pi_q} - D)^+ - (D - X^{\pi_q} - I_\infty^{\pi_q})^+,$$

it is easy to note that

$$\mathbb{E}[(I_\infty^{\pi_q} + X^{\pi_q} - D_t)^+] = \mathbb{E}[I_\infty^{\pi_q}] \quad \text{and}$$
$$\mathbb{E}[D - s(q, Z) - I_\infty^{\pi_q}]^+ = \mathbb{E}[D] - \mathbb{E}[X^{\pi_q}].$$

Invoking the above two equations into Equation (5), we can rewrite $C_\infty^{\pi_q}$ as

$$C_\infty^{\pi_q} = h \cdot \mathbb{E}[I_\infty^{\pi_q}] + b \cdot \mathbb{E}[D] - b \cdot \mathbb{E}[X^{\pi_q}].$$

Because $\mathbb{E}[D]$ is independent of the adopted policy, we could define the *pseudocost* by subtracting $b \cdot \mathbb{E}[D]$ from $C_\infty^{\pi_q}$ as follows:

$$\overline{C}_\infty^{\pi_q} = h \cdot \mathbb{E}[I_\infty^{\pi_q}] - b \cdot \mathbb{E}[X^{\pi_q}].$$

Let $\overline{C}_t^{\pi} = C_t^{\pi} - b \cdot D_t$. With a slight abuse of notation, we also use $\overline{C}_t^{\pi}$ for the observable pseudo cost $h \cdot I_t$-$b \cdot D_t$. It holds that

$$\begin{aligned}\mathcal{R}(T) &= \sum_{t=1}^{T} \mathbb{E}[\overline{C}_t^{\pi} + b \cdot D_t] - T(\overline{C}_\infty^{\pi_{q^*}} + b \cdot \mathbb{E}[D]) \\ &= \sum_{t=1}^{T} \mathbb{E}[\overline{C}_t^{\pi}] - T \cdot \overline{C}_\infty^{\pi_{q^*}}.\end{aligned} \tag{6}$$

From the above simple derivation, we know that the cumulative regret of the true cost (Equation (2)) is equal to the cumulative regret of the pseudocost (the equivalence of the simple regret is similar). In the remainder of this paper, we focus on the regret of the pseudocost. The transformation of the cost enables us to obtain the bandit feedback of the pseudocost function. Thus, by designing a learning algorithm that only requires bandit feedback, we can effectively circumvent the censoring issues. With a slight abuse of terminology, we still refer to $\overline{C}_\infty^{\pi_q}$ as the long-run average cost and denote $\overline{C}_\infty^{\pi_q}$ as $C(q)$.

2.4. Structural Properties of Long-Run Average Cost Function $C(q)$

Because of the complex structure of the random supply function, the long-run average cost function $C(q)$ lacks convexity. Fortunately, with the variable transformation $\mu(q) := \mathbb{E}[s(q, Z)]$, we could establish the convexity of $C(q(\mu)) = h \cdot \mathbb{E}[I_\infty^{\pi_{q(\mu)}}] - b \cdot \mathbb{E}[X^{\pi_{q(\mu)}}]$, where $q(\mu) = \inf\{q \in \mathbb{R} : \mu(q) = \mu\}$. We refer to $C(q(\mu))$ as the *transformed long-run average cost function* and denote it as $\mathrm{TC}(\mu)$.

The following lemma shows the convexity of $\mathrm{TC}(\mu)$ with respect to μ, which is critically important for both algorithm design and regret analysis.

Lemma 1. *The transformed long-run average cost function* $\mathrm{TC}(\mu)$ *is convex in* $[0,\overline{\mu}]$.

The convexity behavior under this type of variable transformation can be traced back to the seminal work of Feng and Shanthikumar (2018). To the best of our knowledge, we are the first to utilize this type of convexity of $C(q)$ in the learning algorithm design and corresponding regret analysis.

Because it is difficult for us to test the performance of all candidate order quantities in a continuous set, the stability of the long-run average cost function with respect to the order quantity, which is reflected by its Lipschitz continuity, becomes crucially important for the design and analysis of online learning algorithms.

In the following, we establish the desired Lipschitz continuity by explicitly constructing the Lipschitz constant, where we are most interested in the dependence of the Lipschitz constant on $\overline{L}$. Chen et al. (2024) demonstrate the stability result of the long-run average cost function through the convexity property of its transformed version in the constant lead times setting. However, as their proof is essentially an existence proof, it seems difficult for their proof method to figure out the Lipschitz constant explicitly, including the dependence on $\overline{L}$, in the stochastic lead times setting.

Lemma 2. $C(q)$ *is* L_C*-Lipschitz continuous on* $[0,\overline{q}]$*; that is, for any* $q_1,q_2\in[0,\overline{q}]$*, we have*

$$|C(q_1)-C(q_2)|\leqslant L_C|q_1-q_2|, \tag{7}$$

where $L_C=L_q\cdot\mathcal{O}(\overline{L})$*, and* L_q *is the Lipschitz constant of* $\mathbb{E}[s(q,Z)]$ *in Assumption 2.*

At a higher level, we initially establish the Lipschitz continuity of $\mathrm{TC}(\mu)$ with respect to μ by leveraging its convexity properties. We then relate the Lipschitz constant to $\mathbb{E}[I_\infty^{\pi_{q_{(\overline{\mu}+\delta/2)}}}]$ and derive its upper bound by exploiting the system properties, where the dependence on $1/\delta$ emerges. Then, derive the Lipschitz continuity of $C(q)$ by noting $|\mu(q_1)-\mu(q_2)|=|\mathbb{E}[s(q_1,Z)]-\mathbb{E}[s(q_2,Z)]|\leqslant L_q|q_1-q_2|$ (Assumption 2). Please refer to Section EC.3.5 in the Online Appendix.

3. General Framework: Bandit Optimization with Transformed Convexity

In Section 2.4, we established two desirable structural properties of the long-run average cost function $C(q)$, Lipschitz continuity and convexity under variable transformation $\mu(q)=\mathbb{E}[s(q,Z)]$. When this transformation is *known*, convex bandit optimization can be conducted in the μ-space and subsequently mapped back to the class of constant-order policies. However, $\mu(q)$ is *unknown* because both the supply function $s(q,\cdot)$ and the distribution of Z are unknown. Consequently, how to optimize $C(q)$ under these unknown conditions using only bandit feedback remains unsolved in the literature.

Beyond the optimization of constant-order policies, a variety of OM problems exhibit similar structural properties. Examples include newsvendor problem with random supply and dynamic pricing with unknown nonparametric demand function (see Section 4 for a detailed description of these problems). We denote the objective function in these problems as $f(x):[a,b]\longmapsto\mathbb{R}$ and assume it to be L_f-Lipschitz. Below, we formalize the key (structural property) assumption on $f(x)$:

Assumption 3 (Transformed Convexity). *There exists a strictly monotonic function* $g(x)$ *and a convex function* $h(\cdot)$ *such that* $f(x)=h(g(x))$*. Besides, there exist positive constants* l_g,L_g *such that for any* $x_1,x_2\in[a,b]$*,* $l_g|x_1-x_2|\leqslant|g(x_1)-g(x_2)|\leqslant L_g|x_1-x_2|$.

The assumption captures the hidden convexity of a function under an unknown variable transformation. We refer to this property as *transformed convexity*. We define $\nu=L_g/l_g\geqslant 1$ to be the *conditional number* that measures the distortion factor of transformation $g(x)$. Noting that convex functions satisfy the above definition with $g(\cdot)$ being the identity map and $\nu=1$, transformed convexity is thus a generalization of standard convexity.

To better illustrate our method for this optimization problem, we consider minimizing a function $f(x)$ satisfying Assumption 3 with bandit feedback under the following basic dynamics. In later sections, we will demonstrate that the algorithm design and analysis in this section can be extended to solve our focused problem in Section 2. An algorithm and environment interact over T periods. In period t, the algorithm chooses x_t and observes the noisy bandit feedback $y_t=f(x_t)+\epsilon_t$, where $\{\epsilon_t\}_{t=1}^T$ is a sequence of centered conditional σ-sub-Gaussian random variables. Specifically, $\mathbb{E}[\epsilon_t|\mathcal{F}_t]=0$ and $\mathbb{E}[\exp(\epsilon_t^2/\sigma^2)|\mathcal{F}_t]\leqslant 2$ with filtration $\mathcal{F}_t=\{x_1,\epsilon_1,\ldots,x_{t-1},\epsilon_{t-1},x_t\}$.

Let $\pi=(\pi_t)_{t=1}^T$ be an admissible policy where $\pi_t:\mathcal{F}_{t-1}\cup\epsilon_{t-1}\longmapsto[a,b]$ is measurable. After T periods, the algorithm π can output a final decision $\hat{x}_T$ based on all historical information. Let Π be the set of all admissible policies. We evaluate π using cumulative regret $\mathcal{R}(T)$ defined as

$$\begin{aligned}\mathcal{R}(T)&:=\mathbb{E}\left[\sum_{t=1}^T f(x_t^\pi)\right]-Tf(x^*)\ \text{ and}\\ \mathcal{S}(T)&:=\mathbb{E}[f(\hat{x}_T^\pi)]-f(x^*).\end{aligned} \tag{8}$$

When $f(\cdot)$ is convex, Shamir (2013) proved that the optimal cumulative regret and simple regret are $\Theta(\sqrt{T})$

and $\Theta(1/\sqrt{T})$, respectively. In the following theorem, we show that in our problem, the lower bounds for both cumulative regret and simple regret depend on the condition number of the transformation $\nu = L_g/l_g \geqslant 1$, which distinguishes our setting from the convex case.

We define function class $\mathcal{F}_\nu$ to be the set of all 1-Lipschitz functions $f(x) = h(g(x))$ such that $h(x)$ is 1-Lipschitz and $g(x)$ has conditional number of ν. Then, we can prove the following lower bound.

Theorem 1. *Suppose that* $[a,b] = [-1,1]$ *and* $\{\epsilon_t, t \geqslant 1\}$ *are i.i.d. with normal distribution* $\mathcal{N}(0,1)$. *For any* T *and any fixed* $C \geq 1$ *and any* $1 \leq \nu \leqslant C$, *we have the following lower bound for cumulative regret,*

$$\inf_{\pi\in\Pi}\sup_{f\in\mathcal{F}_\nu}\left\{\mathbb{E}\left[\sum_{t=1}^{T} f(x_t^\pi)\right] - Tf(x_f^*)\right\} \geqslant \frac{\nu\sqrt{T}}{16C},$$

and the following lower bound for simple regret,

$$\inf_{\pi\in\Pi}\sup_{f\in\mathcal{F}_\nu}\{\mathbb{E}[f(\hat{x}_T^\pi)] - f(x_f^*)\} \geqslant \frac{\nu}{16C\sqrt{T}},$$

where x_f^* *is the optimal solution of* $f(x)$.

We want to point out that although $f(x) = h(g(x))$ and L_f may be affected by the conditional number ν of $g(x)$, the above linear dependence on ν is directly because of ν itself but not the Lipschitz constant L_f (intermediate result of ν). As we will see in the following Theorem 2 in Section 3.1, the regret has a logarithmic dependence on L_f and linear dependence on ν. Therefore, it is the value of ν that significantly affects the convergence of the algorithm, but not L_f.

3.1. Optimality of Quadrisection Search Algorithm

Algorithm 1 (Quadrisection Search)

1: Initialization: Input T and $N_1(\cdot)$. Let the initial active interval $[l_1, r_1] = [a,b]$.
2: for epochs $\tau = 1,2,\ldots,$ do
3: Let $w_\tau := r_\tau - l_\tau$. Set $x_l^\tau := l_\tau + w_\tau/4$, $x_c^\tau := l_\tau + w_\tau/2$, and $x_r^\tau := l_\tau + 3w_\tau/4$.
4: for round $i = 1,2,\ldots,$ do
5: Let $\gamma_i = 2^{-i}$. Apply $x \in \{x_l^\tau, x_c^\tau, x_r^\tau\}$ each for $N_1(\gamma_i)$ periods. Terminate when the total periods reach T. Collect observations and join these periods into $\mathcal{T}_{\tau,i,x}$, and compute $\mathrm{UB}_{\gamma_i}(x)$, $\mathrm{LB}_{\gamma_i}(x)$ as follows:

$$\mathrm{UB}_{\gamma_i}(x) = \frac{1}{|\mathcal{T}_{\tau,i,x}|}\sum_{t\in\mathcal{T}_{\tau,i,x}} y_t + \gamma_i \quad \text{and}$$

$$\mathrm{LB}_{\gamma_i}(x) = \frac{1}{|\mathcal{T}_{\tau,i,x}|}\sum_{t\in\mathcal{T}_{\tau,i,x}} y_t - \gamma_i.$$

6: if $\mathrm{LB}_{\gamma_i}(x_l^\tau) \geqslant \min\{\mathrm{UB}_{\gamma_i}(x_c^\tau), \mathrm{UB}_{\gamma_i}(x_r^\tau)\} + \gamma_i$ ▷ Case 1
7: then $l_{\tau+1} := x_l^\tau$ and $r_{\tau+1} = r_\tau$. Go to the next epoch $\tau+1$.
8: else if $\mathrm{LB}_{\gamma_i}(x_r^\tau) \geqslant \min\{\mathrm{UB}_{\gamma_i}(x_c^\tau), \mathrm{UB}_{\gamma_i}(x_l^\tau)\} + \gamma_i$ ▷ Case 2
9: then $l_{\tau+1} := l_\tau$ and $r_{\tau+1} = x_r^\tau$. Go to the next epoch $\tau+1$.
10: else Go to the next round $i+1$. ▷ Case 3
11: end for
12: end for

The quadrisection search algorithm (Algorithm 1) is a classic optimization algorithm for stochastic convex optimization with bandit feedback developed by Agarwal et al. (2013).

3.1.1. Algorithm Overview.

In each epoch τ, the decision maker will select three checkpoints $x_l^\tau, x_c^\tau, x_r^\tau$ that quarter the active region of x. Each epoch is further subdivided into rounds. In each round i, for each $x \in \{x_l^\tau, x_c^\tau, x_r^\tau\}$, the decision maker will first set order quantity as x for $\mathcal{O}(1/\gamma_i^2)$ consecutive periods, and then the decision maker will collect the noisy cost to compute the high-probability $\mathcal{O}(\gamma_i)$-radius confidence interval $[\mathrm{LB}_{\gamma_i}(x), \mathrm{UB}_{\gamma_i}(x)]$ of $f(x)$. Finally, if the $\mathcal{O}(\gamma_i)$-radius confidence intervals suffice to resolve the difference between checkpoints, the decision maker will discard a portion of the active region and proceed to the next epoch (case 1 and case 2). Taking case 1 as an example, $\mathrm{LB}_{\gamma_i}(x_l^\tau) \geqslant \min\{\mathrm{UB}_{\gamma_i}(x_c^\tau), \mathrm{UB}_{\gamma_i}(x_r^\tau)\} + \gamma_i$ shows that $f(x_l^\tau) \geqslant f(x_r^\tau)$ with high probability; thus, we discard the portion (l_τ, x_l^τ). If not, the decision maker will halve γ_i and proceed to the next round (case 3).

Agarwal et al. (2013) established the algorithm's optimality in cumulative regret for convex objectives. However, our problem involves a nonconvex objective. In the following theorem, we demonstrate that this classic algorithm, which was originally designed for convex functions, can also achieve optimal cumulative and simple regret for the transformed convex objectives without knowledge of the variable transformation.

Theorem 2. *Suppose that* $f(\cdot)$ *satisfies Assumption 3. Running Algorithm 1 with* $N_1(x) = (2\sigma^2 \log T)/x^2$ *leads to*

$$\mathcal{R}(T) \leqslant 128\sigma(1+\nu)T_{4/3}\left((b-a)L_f\sqrt{T}/2\right)\sqrt{T} + L_f(b-a),$$

$$\mathcal{S}(T) \leqslant 12(1+\nu)\sqrt{\frac{2\sigma^2{}_{4/3}\left((b-a)L_f\sqrt{T}/2\right)\log T}{3T}} + \frac{L_f(b-a)}{T},$$

where the simple regret bound is obtained by outputting the midpoint checkpoint of the longest epoch prior to period T.[11]

The above findings not only provide an optimal algorithm for solving the problem at hand but also open the avenue for applying this classic algorithm to a broader range of OM problems.

The key point in the cumulative regret analysis lies in bounding the regret during *unsuccessful* rounds, where the threshold γ_i fails to resolve the difference between checkpoints. Because we cannot separate the checkpoints, we know the objective function values at these points are close, satisfying $\max\{f(x_l^\tau), f(x_c^\tau), f(x_r^\tau)\} - \min\{f(x_l^\tau), f(x_c^\tau), f(x_r^\tau)\} \leqslant \mathcal{O}(\gamma_i)$.

For clarity, we focus on the case where $x^* \leqslant x_c^\tau$ (the case $x^* > x_c^\tau$ follows analogously) to demonstrate the application of transformed convexity and the emergence of the condition number ν. Recall the variable transformation $f(x) = h(g(x))$, where $h(\cdot)$ is convex. By the monotonicity of $g(x)$, we have $g(x^*) \leqslant g(x_c^\tau)$. The increasing slope property of convex functions yields

$$\frac{h(g(x_c^\tau)) - h(g(x^*))}{g(x_c^\tau) - g(x^*)} \leqslant \frac{h(g(x_r^\tau)) - h(g(x_c^\tau))}{g(x_r^\tau) - g(x_c^\tau)}.$$

Furthermore, the Lipschitz property of $g(\cdot)$ implies

$$\frac{g(x_c^\tau) - g(x^*)}{g(x_r^\tau) - g(x_c^\tau)} \leqslant \frac{L_q \cdot w_\tau/2}{l_q \cdot w_\tau/4}.$$

Combining these with $f(x_r^\tau) - f(x_c^\tau) \leqslant \mathcal{O}(\gamma_i)$, we can bound the regret in this round as

$$f(x_c^\tau) - f(x^*) \leqslant (f(x_r^\tau) - f(x_c^\tau)) \frac{g(x_c^\tau) - g(x^*)}{g(x_r^\tau) - g(x_c^\tau)} \leqslant O(\nu\gamma_i),$$

where $\nu = L_q/l_q$ denotes the condition number.

3.2. Trisection Search Algorithm

The trisection search algorithm represents another popular optimization approach (Avriel and Wilde 1966). Unlike quadrisection, each epoch τ of trisection selects checkpoints x_l^τ, x_r^τ that trisect the active region. However, this method cannot achieve optimal cumulative regret for general convex functions (please refer to Section EC.4.1 in the Online Appendix for a hard instance). Figure 1 illustrates this difference: whereas quadrisection maintains proximity between optimal values and checkpoint values in unsuccessful rounds (Figure 1(a)), trisection may have optimal values far from checkpoint values even when $f(x_l^\tau)$ and $f(x_r^\tau)$ are equal (Figure 1(b)). This fundamental limitation prevents effective cumulative regret bounding in such rounds.

Although the optimal cumulative regret cannot be achieved, we present a simpler version of the trisection search algorithm (Algorithm 2), which can achieve the optimal simple regret as follows.

Algorithm 2 (Trisection Search)

1: **Initialization:** Input T, and N. Let the initial active interval $[l_1, r_1] = [a, b]$.
2: **for** epochs $\tau = 1, 2, \ldots,$ **do**
3: Let $w_\tau := r_\tau - l_\tau$. Set $x_l^\tau := l_\tau + w_\tau/3$, and $x_r^\tau := l_\tau + 2w_\tau/3$.
4: Apply $x \in \{x_l^\tau, x_r^\tau\}$ each for N periods. Terminate when the total periods reach T. Collect observations and join these periods into $\mathcal{T}_{\tau,x}$, and compute $\mathrm{A}(x) = \frac{1}{|\mathcal{T}_{\tau,x}|}\sum_{t\in\mathcal{T}_{\tau,x}} y_t$.
5: **if** $\mathrm{A}(x_l^\tau) > \mathrm{A}(x_r^\tau)$ **then** $l_{\tau+1} := x_l^\tau$ and $r_{\tau+1} = r_\tau$; **else** $l_{\tau+1} := l_\tau$ and $r_{\tau+1} = x_r^\tau$.
6: **end for**

Theorem 3. *Suppose that $f(\cdot)$ satisfies Assumption 3. Running Algorithm 2 with $N = T/(2\log_{3/2}T)$ leads to*

$$\mathcal{S}(T) \leqslant 4\nu\left(\log_{3/2}T\right)\sqrt{\frac{\sigma^2 \log T \log_{3/2}T}{T}} + \frac{2(b-a)L_f}{T},$$

where the simple regret bound is obtained by outputting any points at the end of the last epoch.

4. Illustrative Applications: Pricing, Newsvendor, and Queueing

In this section, we highlight the broad applicability of our framework by presenting three illustrative applications involving three canonical problems in operations management: dynamic pricing, the newsvendor problem with random supplies, and arrival/service rate control in queueing systems.

4.1. Dynamic Pricing with Unknown Nonparametric Demand

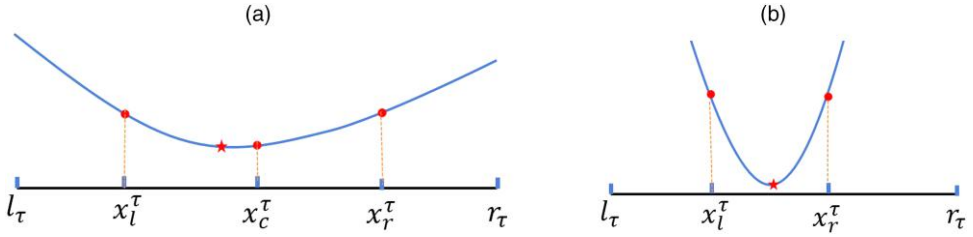
Whereas previous works have explored dynamic pricing under unknown demand, they largely fall into two categories: (i) parametric models where the functional form of $D(p|\theta)$ is known while the parameter θ is unknown, or (ii) nonparametric models that require direct strong concavity in p. Our framework can address the nonparametric dynamic pricing problem using only bandit feedback and very weak transformed concavity in p.

4.1.1. Model and Bandit Feedback. In each period of the dynamic pricing model, the seller posts a price $p \in [a, b]$ and observes a random demand $D(p, \xi)$, where ξ is an unobserved shock, and the demand function is unknown. The one-period revenue is $r(p) = p\,D(p, \xi)$. To fit the minimization framework, define the cost as negative revenue:

$$y_t := f(p_t) + \epsilon_t := -p_t D(p_t, \xi_t).$$

The goal is to minimize $f(p) := -\mathbb{E}[pD(p, \xi)]$. If $p \in [a, b]$ is bounded and demand is bounded almost surely, say, $0 \leqslant D(p, \xi) \leqslant D_{\max}$, then the observation $y_t = -p_t D(p_t, \xi_t)$ is bounded in $[-bD_{\max}, -a \cdot 0]$, implying conditional sub-Gaussian noise. Moreover, if $R(d)$ is L_R-Lipschitz in d (a mild regularity condition implied, e.g., by bounded demand and smoothness of λ^{-1}), then $h(d) = -R(d)$ is L_R-Lipschitz, and hence, $f(p) = h(\lambda(p))$ is Lipschitz with $L_f \leqslant L_R L_\lambda$.

Figure 1. (Color online) Illustration of Quadrisection and Trisection in the Unsuccessful Round



4.1.2. Transformed Convexity. In dynamic pricing, $f(p)$ is generally not convex in p, as this property fails even for standard models such as exponential demand ($D(p) = e^{-\alpha p}$). Instead, the standard structural assumption relies on the reparametrization $d = \lambda(p)$, under which the expected revenue is concave in d, as follows.

Assumption 4. *There exists a strictly monotone map $d = \lambda(p)$ such that the following hold.*

1. *Function $R(d) := \mathbb{E}[\lambda^{-1}(d)\, D(\lambda^{-1}(d), \xi)]$ is concave in d over $d \in [\lambda(a), \lambda(b)]$.*
2. *There exist $0 < l_\lambda \leqslant L_\lambda < \infty$ such that for all $p_1, p_2 \in [a, b]$,*

$$l_\lambda |p_1 - p_2| \leqslant |\lambda(p_1) - \lambda(p_2)| \leqslant L_\lambda |p_1 - p_2|.$$

This assumption is significantly more general and robust than direct concavity in p. For instance, many common demand functions (e.g., logit, exponential) satisfy Assumption 4 that revenue is concave in the demand rate d, even if it is not concave in p.

4.1.3. Applying Our Framework. Under Assumption 4, let

$$g(p) := \lambda(p), \qquad h(d) := -R(d).$$

Then, $h(\cdot)$ is convex (as the negative of a concave function), and

$$f(p) = -\mathbb{E}[pD(p, \xi)] = -R(\lambda(p)) = h(g(p)),$$

which matches Assumption 3. Assumption 4 also includes the bi-Lipschitz condition for $g(p) = \lambda(p)$; that is, Assumption 3 holds with condition number $\nu = L_g/l_g = L_\lambda/l_\lambda$. Therefore, we can apply our framework to this problem and have the theoretical guarantees in Section 3.

4.2. Newsvendor Problems with Random Supplies

In this section, we apply our framework to address the data-driven newsvendor problem with random supplies under the *unknown* supply functions. This application serves as a simplified version of the problem formulated in Section 2. We present it here as an illustrative warm-up to demonstrate our general framework.

4.2.1. Model and Bandit Feedback. In each period, the decision maker chooses an order quantity $q \in [0, \overline{q}]$. The realized supply is $s(q, Z)$, where both $s(\cdot, \cdot)$ and the distribution of Z are unknown. Demand D is random and bounded (Assumption 2). Similar to the derivations in the Section 2, by dropping the constant term independent of q, the problem reduces to minimizing the following objective:

$$f(q) = (h + b)\, \mathbb{E}[(s(q, Z) - D)^+] - b\, \mathbb{E}[s(q, Z)].$$

Even under censored demand, the decision maker can obtain the bandit feedback $y_t = (h + b)(s(q_t, Z_t) - D_t)^+ - bs(q_t, Z_t)$ from the sales and supply data, and we have $y_t = f(q_t) + \epsilon_t$.

4.2.2. Transformed Convexity. We assume that the random supply $s(q, Z)$ satisfies Assumption 2. As in Section 2.1, we define the expected supply $\mu(q) := \mathbb{E}[s(q, Z)]$, the inverse map $q(\mu) = \inf\{q \in \mathbb{R} : \mu(q) = \mu\}$, and the transformed random supply $\hat{s}(\mu, Z) := s(q(\mu), Z)$. By construction, $\mathbb{E}[\hat{s}(\mu, Z)] = \mu$. We can express f as a function of μ:

$$f(q) = (h + b)\, \mathbb{E}[(\hat{s}(\mu(q), Z) - D)^+] - b\, \mu(q) =: H(\mu(q)),$$

where, for $\mu \in [0, \overline{\mu}]$,

$$H(\mu) := (h + b)\, \mathbb{E}[(\hat{s}(\mu, Z) - D)^+] - b\, \mu.$$

Therefore, $f(q) = H(\mu(q))$ already takes the compositional form in Assumption 3 with $g(q) = \mu(q)$. By the third term of Assumption 2, we have that $\{\hat{s}(\mu, Z) : \mu \in [0, \overline{\mu}]\} \in \mathrm{SL(mp)}$. Applying the SL(mp) implication and taking expectation over D, we have $\mu \longmapsto \mathbb{E}[(\hat{s}(\mu, Z) - D)^+]$ is convex on $[0, \overline{\mu}]$. Hence, $H(\cdot)$ is convex.

4.2.3. Applying Our Framework. Under Assumption 2, it is straightforward to verify the Lipschitz continuity of the objective function. Specifically, $\mu(\cdot)$ satisfies the bi-Lipschitz condition, and $f(q) = H(\mu(q))$ is Lipschitz continuous with constant $L_f \leqslant O(h + b)L_q$. Consequently, Assumption 3 holds with $g(q) = \mu(q)$. Because all requirements are satisfied, we can readily apply our framework to this problem to obtain the theoretical guarantees presented in Section 3.

4.3. Arrival/Service Rate Control in Queueing Systems

In this section, we illustrate how our framework applies to queueing systems. Arrival rate and service rate are two important features in many service systems.

- First, the decision maker can adjust the price charged to customers to control the arrival rate.
- Second, the decision maker can control the service rate via operational expenditure: increasing staffing, processing effort, or other service-related investment can improve the service rate.

This leads naturally to the optimization problem of price and service expenditure. Existing learning literature assumes direct convexity in price and service rate (Chen et al. 2024, 2026b). Recently, Chen et al. (2025a) investigated the hidden convexity in queueing systems. This motivates us to consider leveraging hidden convexity and our framework to solve optimization problems in queueing.

4.3.1. Model Formulation. Consider a single-server queue. We assume that $\mu > \lambda$ for stability. Let $Q_\infty(\lambda, \mu)$ be the steady-state queue length under rates (λ, μ). The long-run objective consists of a congestion term and the costs required to induce the chosen arrival and service rates as

$$\mathbb{E}[Q_\infty(\lambda, \mu)] + C_a(\lambda) + C_s(\mu),$$

where $C_a(\cdot)$ denotes the cost (or negative revenue) associated with arrival rate λ, and $C_s(\cdot)$ denotes the cost associated with service rate μ.

For example, in the $M/M/1$ queue, the queue length $\mathbb{E}[Q_\infty(\lambda, \mu)] = \lambda/(\mu - \lambda)$, which is convex in λ given μ fixed and is convex in μ given λ is fixed. Similar convexity also appears in more general queues (see Chen et al. (2025a) for more examples). Therefore, we consider applying our framework to the arrival/service rate control problem as follows.

4.3.2. Pricing-Based Arrival Rate Control. Suppose the service rate μ is fixed and the decision maker controls only the arrival rate through price. Let p denote the price charged to customers, and the induced arrival rate is $\lambda = \Lambda(p)$. The optimization problem becomes

$$\min_{p \in [\underline{p}, \overline{p}]} \mathbb{E}[Q_\infty(\Lambda(p), \mu)] + C_a(\Lambda(p)),$$

where the uncontrolled term and cost are omitted.

The key assumption, analogous to those used throughout the paper, is that the mapping $\Lambda(p)$ is monotonic and bi-Lipschitz and that $\mathbb{E}[Q_\infty(\lambda, \mu)] + C_a(\lambda)$ is convex in λ. Therefore, the pricing problem is a one-dimensional hidden-convexity problem and can be addressed by our algorithm.

4.3.3. Cost-Based Service Rate Control. Suppose the arrival rate λ is fixed and the decision maker controls the service rate through service expenditure. Let s denote the service expenditure (or capacity investment), and the induced service rate is $\mu = M(s)$. The problem becomes

$$\min_{s \in [\underline{s}, \overline{s}]} \mathbb{E}[Q_\infty(\lambda, M(s))] + C_s(M(s)).$$

We assume that the mapping $M(s)$ is monotonic and bi-Lipschitz and that $\mathbb{E}[Q_\infty(\lambda, \mu)] + C_s(\mu)$ is convex in μ. Under these assumptions, the problem becomes another instance of our framework.

5. Core Application: Inventory Control Under General Supply Uncertainty

With the general framework established and its broad utility demonstrated, we now return to the primary motivation of this study: the management of lost-sales inventory systems under stochastic lead times and random supplies.

In Section 3, we established the optimality of quadrisection search for transformed-convex bandit optimization under idealized interaction dynamics. In that setting, the availability of unbiased, i.i.d., and sub-Gaussian feedback provided a favorable environment for constructing accurate objective function estimates. However, the inventory system of interest here presents significantly more challenges. Although our objective function $C(q)$ satisfies Lipschitz continuity and transformed convexity, the complexity of inventory control dynamics under stochastic lead times and random supplies presents unique challenges for constructing accurate estimations and establishing nonasymptotic concentration results.

5.1. Challenge I: Biased Feedback

Note that $C(q)$ represents the long-run average cost under the stationary distribution. However, in any finite time horizon, the observed cost data comes from transient distributions that have not yet converged to this limiting distribution. Consequently, these finite-period cost observations provide biased estimates of $C(q)$. On the algorithm side, to address this issue, we adapt Algorithm 1. Specifically, for each $x \in x_l^\tau, x_c^\tau, x_r^\tau$, we apply the constant-order policy for $N_0 + N_1(\gamma_i)$ periods but only use the final $N_1(\gamma_i)$ observations for estimation. Here, N_0 is a fixed constant independent of γ_i, and $N_1(\gamma_i) = \mathcal{O}(1/\gamma_i^2)$. The initial N_0 periods serve as a warm-up phase to diminish bias in the cost data, whereas the subsequent $N_1(\gamma_i)$ periods are used for constructing the estimates. On the analytic side, we establish a coupling result (Lemma 3), which proves that after applying the same constant-order policy over consecutive periods, the distribution of on-hand inventory approaches the limiting inventory distribution.

5.1.1. Challenge II: Complexity of Stochastic Process Because of the stochastic lead times, there exists an enduring impact of instantaneous decisions across periods. This causes the cost data across periods to be neither independent nor Markovian. Whereas we can augment the cost data with the stochastic lead times process to form a Markov chain, the complex nature of stochastic lead times makes constructing concentration results for this augmented Markov chain highly nontrivial. In our problem, the Markov chain is not uniformly ergodic (rapidly mixing), as Huh et al. (2009a), Lyu et al. (2024), and concentration inequality for uniform ergodic Markov chains cannot be applied. Through careful analysis and utilization of the inventory process structure (see Section EC.5 in the Online Appendix), we establish the chain's V-geometric ergodicity and derive concentration bounds via a McDiarmid-type inequality for Markov chains (Lemma 1). Combining the concentration results with the above coupling result yields the confidence interval for our estimator for $C(q)$.

After addressing these challenges and running Algorithm 1 with the aforementioned adaptation, we establish the following theorem.

Theorem 4. *Suppose that Assumption 1 and Assumption 2 hold. Let* $\nu = L_q/l_q$. *Setting* $N_0 = \tilde{\Theta}(L+\overline{L})$, $N_1(x) := \tilde{\Theta}(1/x^2)$, *we have that*

$$\mathcal{R}(T) \leqslant \tilde{\mathcal{O}}(L + \mathrm{poly}(\overline{L})\nu\sqrt{T}) \quad \text{and}$$
$$\mathcal{S}(\mathrm{T}) \leqslant \tilde{\mathcal{O}}\left(\frac{L+\mathrm{poly}(\overline{L})\nu}{\sqrt{T}}\right)$$

Running the trisection search algorithm (Algorithm 2), we could have the following theorem.

Theorem 5. *Suppose that Assumption 1 and Assumption 2 hold. Let* $\nu = L_q/l_q$. *Setting* $N_0 = \tilde{\Theta}(L+\overline{L})$, $N := \Theta(T/\log T)$, *we have that*

$$\mathcal{S}(T) \leqslant \tilde{\mathcal{O}}\left(\frac{L+\mathrm{poly}(\overline{L})\nu}{\sqrt{T}}\right).$$

To prove the above theorems, we need to establish the coupling and concentration results, and then we invoke these results into the proof scheme of Theorem 2 and Theorem 3 to obtain the final proof.

Note that the first term in the cumulative regret definition in Equation (6) is $\sum_{t=1}^{T}\mathbb{E}[\overline{C}_t^{\pi}]$ rather than $\sum_{t=1}^{T}\mathbb{E}[C(q_t)]$, as defined in the general framework (see Equation (8)). We can bound this gap by $\tilde{\mathcal{O}}(L+\overline{L}+\sqrt{T})$, which does not affect the main results. Specifically, the number of rounds across all epochs is bounded by $\mathcal{O}(\log^2 T)$. Furthermore, in each round, the length of the N_0 segment can be bounded by $\mathcal{O}(L+\overline{L})$. Additionally, the concentration results (Corollary 1) yield for $N_1(\gamma_i)$ estimation periods in each round the gap $\sum_{t=1}^{N_1(\gamma_i)}\mathbb{E}[\overline{C}_t - C(q)] \leqslant \mathcal{O}(\sqrt{|N_1(\gamma_i)|})$. Thus, in each round, this gap could be bounded by $\mathcal{O}(L+\overline{L}+\sqrt{|N_1(\gamma_i)|})$ Combining this with the number of rounds, by the Cauchy–Schwarz inequality, we obtain

$$\sum_{t=1}^{T}\mathbb{E}[\overline{C}_t^{\pi}] - \sum_{t=1}^{T}\mathbb{E}[C(q_t)] \leqslant \mathcal{O}(L+\overline{L}+\sqrt{T}).$$

Thus, by integrating the structural properties, the confidence interval of our estimator, and the general guarantees of quadrisection and trisection search (Theorem 2 and Theorem 3), we can prove the above theorems, which are the theoretical guarantee of these two algorithms for our inventory problem. See Section EC.1 in the Online Appendix for the proof roadmap and relationship of the lemmas and theorems.

5.2. Coupling and Concentration Results

In this section, we first establish an important coupling lemma, which demonstrates a coupling result between the actual inventory level and the limiting distribution; then, assuming the actual inventory level follows the limiting distribution, we construct the concentration lemmas by proposing several novel techniques; finally, combining these lemmas, we obtain the confidence interval of our estimators (see Corollary 1).

For any $q \in [0, \overline{q}]$, to obtain a stochastic process that follows the same distribution of $I_{\infty}^{\pi_q}$, we define a stochastic process $(\tilde{I}_t^{\pi_q})_{t\geqslant 1}$ according to the following dynamics:

$$\tilde{I}_1^{\pi_q} \overset{d}{=} I_{\infty}^{\pi_q} \quad \text{and} \quad \tilde{I}_{t+1}^{\pi_q} = (\tilde{I}_t^{\pi_q} + X_t - D_t)^+.$$

Recalling that the I_t is the actual inventory-level stochastic process, we have the following coupling lemma.

Lemma 3. *With probability* $1 - 1/2T^2$, *from any fixed period* $k \in [T]$, *after placing a constant-order* $q \in [0, \overline{q}]$ *for* $L+\overline{L}+\kappa_1$ *consecutive periods,* $(I_t)_{t\geqslant k+L+\overline{L}+\kappa_1}$ *will couple with the stochastic process* $(\tilde{I}_t^{\pi_q})_{t\geqslant k+L+\overline{L}+\kappa_1}$; *that is,*

$$\mathbb{P}[I_{t+L+\overline{L}+\kappa_1} = \tilde{I}_{t+L+\overline{L}+\kappa_1}^{\pi_q},\ \forall t \geqslant k] \geqslant 1 - \frac{1}{2T^2},$$

where $\kappa_1 = 4\overline{D}^2\ln(6C_1T^2)/\delta^2 + 4\overline{L}\overline{s}/\delta$ *and* $C_1 = \exp\left(\frac{\delta^2}{8\overline{D}^2}\right)/\left(\exp\left(\frac{\delta^2}{8\overline{D}^2}\right) - 1\right)$.

We note that Chen et al. (2024) established a similar coupling result in the constant lead times setting. The proof of Chen et al. (2024) heavily relies on Lundberg's inequality to bound the tail probability of the limiting inventory, which involves a problem-specific parameter relating to the nontrivial solution $z > 0$ of the equation $\mathbb{E}[\exp(z\cdot(X^{\pi_q} - D))] = 1$. The dependence of such a parameter on $\overline{L}$ seems difficult to figure out in the stochastic lead times setting. Thus, to prove the polynomial dependence of the regret on $\overline{L}$, we propose a novel

method to establish the coupling between the actual inventory level stochastic process I_t and the stochastic process $\tilde{I}_t^{\pi_q}$, with all parameters explicitly established.

Proof Sketch. For any fixed $k \in [T]$, we define $k' = k + L + \overline{L}$ and denote the target event $\{I_{t+\kappa_1} = \tilde{I}_{t+\kappa_1}^{\pi_q},\ \forall t \geqslant k'\}$ as $\mathcal{A}$. We then construct three auxiliary events, $\mathcal{A}_1 := \{I_{k'} \leqslant a\}$, $\mathcal{A}_2 := \{\tilde{I}_{k'}^{\pi_q} \leqslant a\}$, and $\mathcal{A}_3 := \{\sum_{j=k'}^{k'+\kappa_1}(D_j - X_j^{\pi_q}) \geqslant a\}$, where $\mathcal{A}_1$ refers to the event that the on-hand inventory of period k' is upper bounded by a positive constant a, $\mathcal{A}_2$ refers to the event that the inventory of the limiting distribution of period k' is upper bounded by a, and $\mathcal{A}_3$ refers to the event that through a consecutive time of periods the inventory consumption is lower bound by some a. Through this construction, it holds that $\mathcal{A}_1 \cap \mathcal{A}_2 \cap \mathcal{A}_3 \subset \mathcal{A}$. Then, we carefully determine a and prove that all three auxiliary events happen with a high probability. □

Note that $\overline{C}_t = h \cdot I_t - b \cdot X_t$ and $\overline{C}_\infty^{\pi_q} = h \cdot \mathbb{E}[I_\infty^{\pi_q}] - b \cdot \mathbb{E}[X^{\pi_q}]$. To obtain the concentration result of $\overline{C}_t$, in this section, we establish the concentration results of X_t and I_t individually.

Carefully utilizing the definition of X_t by Azuma–Hoeffding inequality, we could establish the concentration result of X_t as follows.

Lemma 4. *Let $N_1 = N_1(\gamma_i)$ as specified in Section* EC.5 *and $t_0 \in [T]$ is a fixed time step. If $\mathbb{E}[X_j] = \mathbb{E}[X^{\pi_q}] = \mathbb{E}[s(q, Z)]$, $t_0 \leqslant j \leqslant N_1 + t_0$, where X_j is the total order quantity arriving at j, it holds that*

$$\mathbb{P}\left[\left|\frac{1}{N_1}\sum_{j=t_0+1}^{t_0+N_1} X_j - \mathbb{E}[X^{\pi_q}]\right| \geqslant \frac{\gamma_i}{b}\right] \leqslant \frac{1}{4T^2}.$$

The following lemma states the concentration result of I_t.

Lemma 5. *Let $N_1 = N_1(\gamma_i)$ as specified in Section* EC.5 *and $t_0 \in [T]$ is a fixed time step. It holds that*

$$\mathbb{P}\left[\left|\frac{1}{N_1}\sum_{j=t_0+1}^{t_0+N_1} \tilde{I}_j^{\pi_q} - \mathbb{E}[\tilde{I}_\infty^{\pi_q}]\right| \geqslant \frac{\gamma_i}{h}\right] \leqslant \frac{1}{4T^2}.$$

Compared with the concentration of X_t, the proof of Lemma 5 is much more involved. To prove Lemma 5, we first introduce a projection function to handle the unboundedness of the on-hand inventory and decompose the objective into three parts. Then, we augment the on-hand inventory with the stochastic lead times process to get a Markov chain. Finally, we prove the chain has the desired properties for using the McDiarmid-type inequality and complete the proof.

Recalling the coupling result between the actual inventory level stochastic process I_t and the stochastic process $\tilde{I}_t^{\pi_q}$ established in Lemma 3, by Lemma 4 and Lemma 5, we could get the following desired confidence interval of our estimator $\frac{1}{|\mathcal{T}_{\tau,i,q}|}\sum_{t\in\mathcal{T}_{\tau,i,q}} \overline{C}_t$. This confidence interval is necessary for the application of Algorithms 1 and 2.

Corollary 1. *For any fixed τ, i, $q \in \{q_l^\tau, q_c^\tau, q_r^\tau\}$, it holds that*

$$\mathbb{P}\left[\left|\frac{1}{|\mathcal{T}_{\tau,i,q}|}\sum_{t\in\mathcal{T}_{\tau,i,q}} \overline{C}_t - \overline{C}_\infty^{\pi_q}\right| \geqslant 2\gamma_i\right] \leqslant \frac{1}{T^2}.$$

6. Extensions

The algorithm design and analysis in this paper are developed based on several assumptions. Specifically, we assume that the lead time process follows the Kaplan model, decisions are allowed to be continuous, and we know a prior upper bound about demand. This section demonstrates that our algorithm design and analysis can be extended to a more general Markov-modulated environment. Furthermore, the core design ideas of the algorithm can be adapted to indivisible products, and the assumption regarding the upper bound on demand can be relaxed.

6.1. Markov-Modulated Environment

In this section, we extend the algorithm design and analysis to Markov-modulated environments. To be specific, we assume that there exists a finite-state, stationary, ergodic Markov chain $\{w_t\}_{t=1}^\infty$ called nature, and we consider systems with one of the following features.

- *Markov-modulated demand.* Each state w of the Markov chain $\{w_t\}_{t=1}^\infty$ is related to a distribution D^w. The demand D_t has a distribution of D^{w_t} and is independent of the history.
- *Markov-modulated lead time process I.* Each state w is related to a distribution ρ^w supported on $\{L, L+1, \ldots, L+\overline{L}\}$, and the lead time process is decided by $\{\rho_t\}_{t=1}^\infty$ similar to the Kaplan model, where ρ_t has a distribution of ρ^{w_t} and is independent of the history.
- *Markov-modulated lead time process II.* Each state w is related to a distribution L^w supported on $\{L, L+1, \ldots, L+\overline{L}\}$. Lead time L_t has a distribution of L^{w_t} and is independent of the history. If the Markov chain has only one state, this model is i.i.d. lead time processes.

Such an assumption is widely adopted in literature when modeling fluctuating demand and supply (Song and Zipkin 1996a, Bai et al. 2026, Chen et al. 2026a), and the underlying Markov chain is used to model the fluctuating environment that could determine the demand and lead time process. Note that Markov demand or lead times are a special case of the above Markov-modulated setting. For lost-sales inventory systems with the above features, constant-order policies have been proven to be asymptotically optimal (Bai et al.

2026). Therefore, we still choose the optimal constant-order policy as a benchmark and can generalize the current algorithm design and analysis to the above context.

In Section EC.6.1 of the Online Appendix, we detail the algorithmic adaptations and assumptions on Markov-modulated environments, formal regret guarantee, and high-level proof ideas.

6.2. Indivisible Product

An implicit yet critical assumption in the design and analysis of previous Algorithm 1 is that order quantities can be noninteger values. However, in certain scenarios, products may be indivisible, implying both demand and order quantities are integers. Therefore, previous algorithms are not suitable for this context, and we address this extension in this section. For systems with indivisible products, bracket policies are proved to be asymptotically optimal only in the deterministic supply function setting (Bai et al. 2026). Therefore, we consider $s(q,Z)=q$ and choose the optimal bracket policy as the benchmark and consider learning optimal bracket policies.

In Section EC.6.2 of the Online Appendix, we detail a detailed definition of model, benchmark, and feedback, the algorithmic adaptations and assumptions on Markov-modulated environments, formal regret guarantee, and high-level proof ideas.

6.3. Relaxation of Prior Information on $\overline{q}$

In Assumption 1 of Section 2, we assume there exists an upper bound $\overline{q}$ on the optimal solution q^*. Similar prior knowledge about the optimal solution is quite common in the existing literature, such as Huh et al. (2009a) and Zhang et al. (2020). However, under the constant-order policies, the inventory system behaves similarly to a queueing system, and it is likely to be unstable if the order parameter q is too large. Therefore, we also assume the upper bound $\overline{q}$ cannot be too large, and $\overline{q}<q'$, where q' is given by $\mathbb{E}[s(q',Z)]=\mathbb{E}[D]$. These two requirements are somewhat in conflict. Similar issues also appear in the existing literature involving constant-order policies (Chen et al. 2024) or queueing systems (Chen et al. 2024, 2026b). Therefore, in Section EC.6.3 in the Online Appendix, we discuss how to find a satisfactory upper bound $\overline{q}$ spending very few periods. We show that the estimation step only takes $\mathcal{O}(\log T)$ periods and would not affect the order of regret, thus relaxing Assumption 1. This idea may also apply to other data-driven OM problems, such as learning in queueing systems.

At a high level, by incorporating this brief exploration-based estimation stage, we eliminate the need for restrictive prior knowledge regarding the supply-demand stability limit. This two-stage approach, first identifying a "safe" search region and then optimizing within it, provides a robust framework for managing complex inventory systems under high uncertainty without the risk of system instability or explosion. For a more comprehensive discussion and detailed technical descriptions, please refer to Section EC.6.3 of the Online Appendix.

7. Numerical Experiments

In this section, we conduct extensive numerical experiments to study the performance of our algorithms. Because of the limited space, we only provide a brief summary of the experiments and conclusions, and the detailed experiment design and results are deferred to the appendix.

Our numerical experiments encompass the following aspects.

- In Section EC.7.1 in the Online Appendix, we compare our algorithm with the main algorithm of Chen et al. (2024), a simulation-based successive elimination algorithm, and we call it SE for short. The results show that our algorithm achieves better cumulative regret for problems with deterministic lead times, and we think it is because our algorithm better takes advantage of the structural properties of the cost function, such as transformed convexity.
- In Section EC.7.2, we investigate the effect of model misspecifications on quadrisection search (QS) and SE algorithms. Because our algorithms do not need detailed information about supply models for optimization, they are not sensitive to misspecifications and are more suitable for scenarios where supply model information is unknown.
- In Section EC.7.3, we consider the stochastic lead times setting. We compare our QS algorithm with SE algorithm under cumulative regret and trisection search (TS) under simple regret.
- In Section EC.7.4, we discuss how to use the data obtained during the algorithm's execution to provide more information about the demand and supply of the inventory system.
- In Section EC.7.5, to improve the empirical performance of QS in short horizons, we implement a counterfactual data augmentation variant, QS-AUG, which reuses historical demand, supply, and lead time observations collected during the search to improve the cost estimation at newly tested points. We observe that QS-AUG consistently outperforms the baseline QS under both cumulative regret and simple regret. These results indicate that, while preserving the core quadrisection search structure, improving data efficiency through counterfactual sample reuse can lead to significant practical performance gains. They also suggest that, in algorithm design, attention should not be limited to comparing theoretical convergence rates because many algorithms often share similar convergence orders. Instead, how to more

effectively exploit and reuse existing data may play a more decisive role in practical performance.

8. Conclusion

In this paper, we consider managing the lost-sales inventory systems with general supply uncertainty: stochastic lead times and random supplies without assuming prior information on the demand and supply. The main contribution of our paper is the first provably effective learning algorithm for inventory management problems under these conditions, with a regret of $\tilde{\mathcal{O}}(L+\text{poly}(\overline{L})\cdot\sqrt{T})$ compared with the best constant-order policy. Our approach lies in developing a new framework for transformed convexity. Furthermore, we address the unique challenges of our problem through new techniques, some of which are of independent interest.

In this paper, we develop a one-dimensional bandit optimization framework under transformed convexity because of the one-dimensional nature of our decision variable. Extending this framework to a multidimensional setting represents a promising direction for future research. This extension may require the application of alternative optimization techniques, such as the Kiefer–Wolfowitz stochastic approximation, while leveraging the similar idea of utilizing transformed convexity in our paper to establish the regret guarantees. For other future directions, one can consider characterizing the tightness of the bound on $\overline{L}$. This technical problem presents significant challenges and may rely on the underlying stochastic lead time model. Beyond our problem, our framework has broader implications for OM problems exhibiting transformed convexity, such as the newsvendor problem with random supply and dynamic pricing. One can consider adopting our framework to other problems with transformed convexity. One can also consider the learning problems for a class of MDP systems studied in Bai et al. (2023) in which constant-control policies are asymptotically optimal when lead times increase.

Acknowledgments

The authors are grateful to the department editor, the associate editor, and the referees for their insightful comments and constructive suggestions throughout the review process. Authors are listed in alphabetical order with implied co-first authorship.

Endnotes

[1] The random supply function $s(q,Z)$ maps the order quantity q and a realization of the random supply variable Z, whose distribution is referred to as the supply distribution, to the quantity that actually arrived.

[2] The randomness of the supply function can be attributed to the manufacturing process of suppliers, whereas the uncertainty of lead times is mainly because of exogenous reasons such as transportation delays and port congestion. Thus, the independent assumption is reasonable.

[3] $\text{poly}(\overline{L})$ refers to the polynomial dependence on $\overline{L}$.

[4] In practice, this additional assumption about the accurate formulation of the random supply function might face the following two issues: (1) using a specific random supply function to model the random supply phenomena may cause the problem of model misspecification, and (2) it is also difficult for the firm manager to know the parameters of some random supply functions.

[5] See Wang (2021).

[6] We define $\nu = L_g/l_g \geqslant 1$ as the condition number of the transformation $g(x)$, where l_g and L_g are Lipschitz constants satisfying $l_g|x_1-x_2| \leqslant |g(x_1)-g(x_2)| \leqslant L_g|x_1-x_2|$ for all $x_1, x_2 \in [a,b]$.

[7] V-geometric ergodicity means that a Markov chain converges to its stationary distribution at a geometric rate, with the convergence speed weighted by a function $V(\cdot)$ that reflects how far the starting state is from the limiting distribution. Therefore, V-geometric ergodicity is quite different from uniform ergodicity, where the convergence to the limiting distribution is uniform over the state space.

[8] This assumption is practical. For example, in semiconductor manufacturing, the random supply variable could denote the percentage of chips that meet the nominal specification in all productions (Hsu and Bassok 1999), which is realized only after the firm's inspection.

[9] In this paper, we adopt the assumption of linear ordering cost. Furthermore, the cost can be assumed to be zero without loss of generality (Janakiraman and Roundy 2004).

[10] In this paper, the interarrival time and service time of a G/G/1 queue can be general stochastic processes over period $t \geqslant 1$, but the two processes are independent.

[11] In our problem setting, the direct implication from cumulative regret to simple regret does not hold because of the absence of convexity.

References

Agrawal S, Jia R (2022) Learning in structured MDPs with convex cost functions: Improved regret bounds for inventory management. *Oper. Res.* 70(3):1646–1664.

Agarwal A, Foster DP, Hsu D, Kakade SM, Rakhlin A (2013) Stochastic convex optimization with bandit feedback. *SIAM J. Optim.* 23(1):213–240.

Amazon (2023) Build supply chain resilience in the face of uncertainty. *Amazon Bus*. Accessed August 11, 2026, https://business.amazon.com/en/discover-more/blog/build-supply-chain-resilience-in-the-face-of-uncertainty.

Avriel M, Wilde DJ (1966) Optimality proof for the symmetric Fibonacci search technique. *Fibonacci Quart.* 4(3):265–269.

Bai X, Chen X, Li M, Stolyar A (2026) Asymptotic optimality of semi-open-loop policies in lost-sales inventory models with stochastic lead times. *Management Sci.* Forthcoming.

Bai X, Chen X, Li M, Stolyar A (2023) Asymptotic optimality of semi-open-loop policies in Markov decision processes with large lead times. *Oper. Res.* 71(6):2061–2077.

Bollapragada S, Morton TE (1999) Myopic heuristics for the random yield problem. *Oper. Res.* 47(5):713–722.

Bu J, Gong X, Yao D (2020) Constant-order policies for lost-sales inventory models with random supply functions: Asymptotics and heuristic. *Oper. Res.* 68(4):1063–1073.

Cachon GP (2003) Supply chain coordination with contracts. *Handbooks in Operations Research and Management Science*, vol. 11 (North-Holland, Amsterdam), 227–339.

Chen X, Gao X (2019) Stochastic optimization with decisions truncated by positively dependent random variables. *Oper. Res.* 67(5):1321–1327.

Chen X, Gao X, Hu Z (2015) A new approach to two-location joint inventory and transshipment control via $l^\sharp$-convexity. *Oper. Res. Lett.* 43(1):65–68.

Chen X, Gao X, Pang Z (2018) Preservation of structural properties in optimization with decisions truncated by random variables and its applications. *Oper. Res.* 66(2):340–357.

Chen X, Hu Y, Zhao M (2026a) Landscape of policy optimization for finite horizon MDPs with general state and action. Preprint, submitted September 25, https://arxiv.org/abs/2409.17138.

Chen X, Hong G, Liu Y (2026b) Online learning and optimization for queues with unknown arrival rate and service distribution. *Oper. Res.* 74(4):2053–2074.

Chen X, Liu Y, Hong G (2024) An online learning approach to dynamic pricing and capacity sizing in service systems. *Oper. Res.* 72(6):2677–2697.

Chen W, Shi C, Duenyas I (2020) Optimal learning algorithms for stochastic inventory systems with random capacities. *Production Oper. Management* 29(7):1624–1649.

Chen X, Xin L, Zhao M (2025a) Hidden convexity in queueing models. Preprint, submitted November 6, https://arxiv.org/abs/2511.03955.

Chen X, He N, Hu Y, Ye Z (2025b) Efficient algorithms for a class of stochastic hidden convex optimization and Its applications in network revenue management. *Oper. Res.* 73(2):704–719.

Chen B, Jiang J, Zhang J, Zhou Z (2024) Learning to order for inventory systems with lost sales and uncertain supplies. *Management Sci.* 70(12):8631–8646.

Ciarallo FW, Akella R, Morton TE (1994) A periodic review, production planning model with uncertain capacity and uncertain demand—Optimality of extended myopic policies. *Management Sci.* 40(3):320–332.

Dada M, Petruzzi NC, Schwarz LB (2007) A newsvendor's procurement problem when suppliers are unreliable. *Manufacturing Service Oper. Management* 9(1):9–32.

Feng Q (2010) Integrating dynamic pricing and replenishment decisions under supply capacity uncertainty. *Management Sci.* 56(12):2154–2172.

Feng Q, Shanthikumar JG (2018) Supply and demand functions in inventory models. *Oper. Res.* 66(1):77–91.

Feng Q, Shanthikumar JG (2023) The framework of parametric and nonparametric operational data analytics. *Production Oper. Management* 32(9):2685–2703.

Feng Q, Li L, Shanthikumar JG (2026) Transfer learning, cross learning and co-learning across newsvendor systems with operational data analytics (ODA). *Management Sci.*, ePub ahead of print May 7, https://doi.org/10.1287/mnsc.2023.03688.

Feng Q, Shanthikumar JG, Wu J (2025) Contextual data-integrated newsvendor solution with operational data analytics (ODA). *Management Sci.* 71(11):9384–9403.

Gao X, Zhang H (2022a) An efficient learning framework for multiproduct inventory systems with customer choices. *Production Oper. Management* 31(6):2492–2516.

Gao X, Zhang H (2022b) Inventory control with censored demand. Chen X, Jasin S, Shi C, eds. *The Elements of Joint Learning and Optimization in Operations Management*, Springer Series in Supply Chain Management, vol. 18 (Springer, Cham, Switzerland), 273–303.

Gong XY, Simchi-Levi D (2023) Bandits atop reinforcement learning: Tackling online inventory models with cyclic demands. *Management Sci.* 70(9):6139–6157.

Hadley G, Whitin TM (1963) *Analysis of Inventory Systems* (Prentice-Hall, Englewood Cliffs, NJ).

Hsu A, Bassok Y (1999) Random yield and random demand in a production system with downward substitution. *Oper. Res.* 47(2):277–290.

Huh WT, Nagarajan M (2010) Linear inflation rules for the random yield problem: Analysis and computations. *Oper. Res.* 58(1):244–251.

Huh WT, Rusmevichientong P (2009) A nonparametric asymptotic analysis of inventory planning with censored demand. *Math. Oper. Res.* 34(1):103–123.

Huh WT, Janakiraman G, Muckstadt JA, Rusmevichientong P (2009a) An adaptive algorithm for finding the optimal base-stock policy in lost sales inventory systems with censored demand. *Math. Oper. Res.* 34(2):397–416.

Huh WT, Janakiraman G, Muckstadt JA, Rusmevichientong P (2009b) Asymptotic optimality of order-up-to policies in lost sales inventory systems. *Management Sci.* 55(3):404–420.

Janakiraman G, Roundy RO (2004) Lost-sales problems with stochastic lead times: Convexity results for base-stock policies. *Oper. Res.* 52(5):795–803.

Kaplan RS (1970) A dynamic inventory model with stochastic lead times. *Management Sci.* 16(7):491–507.

Keskin NB, Min X, Song JSJ (2026) The nonstationary newsvendor: Data-driven nonparametric learning. *Management Sci.* Forthcoming.

Levi R, Perakis G, Uichanco J (2015) The data-driven newsvendor problem: New bounds and insights. *Oper. Res.* 63(6):1294–1306.

Lin M, Huh WT, Krishnan H, Uichanco J (2022) Data-driven newsvendor problem: Performance of the sample average approximation. *Oper. Res.* 70(4):1996–2012.

Loynes RM (1962) The stability of a queue with non-independent inter-arrival and service times. *Math. Proc. Cambridge Philos. Soc.* 58(3):497–520.

Lyu C, Zhang H, Xin L (2024) UCB-type learning algorithms with Kaplan-Meier estimator for lost-sales inventory models with lead times. *Oper. Res.* 72(4):1317–1332.

Lyu J, Xie J, Yuan S, Zhou Y (2024) A minibatch stochastic gradient descent-based learning metapolicy for inventory systems with myopic optimal policy. *Management Sci.* 71(7):5572–5588.

Nahmias S (1979) Simple approximations for a variety of dynamic leadtime lost-sales inventory models. *Oper. Res.* 27(5):904–924.

Qin H, Simchi-Levi D, Wang L (2022) Data-driven approximation schemes for joint pricing and inventory control models. *Management Sci.* 68(9):6591–6609.

Qin H, Simchi-Levi D, Zhu R (2023) Sailing through the dark: Provably sample-efficient inventory control. Preprint, submitted December 8, https://doi.org/10.2139/ssrn.4652347.

Shaked M, Shanthikumar JG (2007) *Stochastic Orders*, Springer Series in Statistics (Springer, New York).

Shamir O (2013) On the complexity of bandit and derivative-free stochastic convex optimization. *Proc. 26th Annual Conf. Learn. Theory* (PMLR, New York), 3–24.

Shi C, Chen W, Duenyas I (2016) Nonparametric data-driven algorithms for multiproduct inventory systems with censored demand. *Oper. Res.* 64(2):362–370.

Snyder LV, Shen ZJM (2019) *Fundamentals of Supply Chain Theory* (John Wiley & Sons, Hoboken, NJ).

Song JS, Zipkin PH (1996a) Inventory control with information about supply conditions. *Management Sci.* 42(10):1409–1419.

Song JS, Zipkin PH (1996b) The joint effect of leadtime variance and lot size in a parallel processing environment. *Management Sci.* 42(9):1352–1363.

Song JS, Zipkin P (2009) Inventories with multiple supply sources and networks of queues with overflow bypasses. *Management Sci.* 55(3):362–372.

Stolyar AL, Wang Q (2022) Exploiting random lead times for significant inventory cost savings. *Oper. Res.* 70(4):2496–2516.

Wang X (2021) Post-COVID lead times. *Cumming Group* (October 18), https://cumming-group.com/en/news-insights/blogs/post-covid-lead-times/.

Wang Y, Gerchak Y (1996) Periodic review production models with variable capacity, random yield, and uncertain demand. *Management Sci.* 42(1):130–137.

White House (2023) Fact sheet: President Biden announces new actions to strengthen America's supply chains, lower costs for families, and secure key sectors. Fact sheet, The White House, Washington, DC.

Xin L, Goldberg DA (2016) Optimality gap of constant-order policies decays exponentially in the lead time for lost sales models. *Oper. Res.* 64(6):1556–1565.

Yano CA, Lee HL (1995) Lot sizing with random yields: A review. *Oper. Res.* 43(2):311–334.

Yuan H, Luo Q, Shi C (2021) Marrying stochastic gradient descent with bandits: Learning algorithms for inventory systems with fixed costs. *Management Sci.* 67(10):6089–6115.

Zhang H, Chao X, Shi C (2018) Perishable inventory systems: Convexity results for base-stock policies and learning algorithms under censored demand. *Oper. Res.* 66(5): 1276–1286.

Zhang H, Chao X, Shi C (2020) Closing the gap: A learning algorithm for lost-sales inventory systems with lead times. *Management Sci.* 66(5):1962–1980.

Zipkin P (2008) On the structure of lost-sales inventory models. *Oper. Res.* 56(4):937–944.

Copyright of Management Science (INFORMS) is the property of INFORMS: Institute for Operations Research & the Management Sciences and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.

4. 学习分支：Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly (OR, 2026)

来源：Operations Research. Vol. 74, No. 4, 2026, pp. 1811–1833.
https://doi.org/10.1287/opre.2023.0615

作者：Zhengzhong You, Yu Yang, Xinshang Wang, Wotao Yin

关键词：学习分支；分支定价切割；列生成；车辆路径；精确算法

推荐人：林家祥

推荐理由：分支定价切割算法中的变量不断变化，分支又常引入稠密约束，使通用整数规划中的分支学习方法难以直接应用。本文提出两阶段学习分支策略：第一阶段利用低成本特征筛选候选，第二阶段通过部分测试减少启发式测试负担。作者建立分支决策耗时与质量之间的权衡模型，并据此动态决定测试的候选数量。数值研究显示，该策略在容量受限和带时间窗车辆路径问题上显著缩短求解时间，且保持精确求解框架。

https://pubsonline.informs.org/journal/opre

OPERATIONS RESEARCH
Vol. 74, No. 4, July–August 2026, pp. 1811–1833
ISSN 0030-364X (print), ISSN 1526-5463 (online)

Contextual Areas

Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly

Zhengzhong You,[a,b] Yu Yang,[a,b,*] Xinshang Wang,[c] Wotao Yin[c]

[a] Department of Industrial and Systems Engineering, University of Florida, Gainesville, Florida 32611; [b] Center for Applied Optimization, University of Florida, Gainesville, Florida 32611; [c] Alibaba Group US, Bellevue, Washington 98004

*Corresponding author

Contact: you.z@ufl.edu, https://orcid.org/0009-0007-3480-5578 (ZY); yu.yang@ise.ufl.edu, https://orcid.org/0000-0002-0502-7603 (YY); xinshang.w@alibaba-inc.com, https://orcid.org/0000-0003-4683-7167 (XW); wotao.yin@alibaba-inc.com, https://orcid.org/0000-0001-6697-9731 (WY)

Received: November 8, 2023
Revised: January 5, 2025; May 11, 2025
Accepted: January 14, 2026
Published Online in Articles in Advance: February 24, 2026

Area of Review: Transportation

https://doi.org/10.1287/opre.2023.0615

Copyright: © 2026 INFORMS

Abstract. Branching is one of the most important components in branch-price-and-cut (BPC) algorithms for solving vehicle routing problems (VRPs) exactly. However, learning to branch is much more challenging in BPC than in branch-and-cut algorithms that are used for solving general mixed integer programs because branching, in this case, is generally performed by adding a dense constraint to the restricted master problem (RMP), and meanwhile, the variables in the RMP change constantly. To address such challenges, we propose the first effective *learning-to-branch framework* in BPC algorithms, leading to a novel *two-stage learning-based branching* (2LBB) strategy. This serves as an innovative learning-based enhancement for the cutting-edge three-phase branching strategy for column-generation-based algorithms. In the 2LBB, the first stage focuses on narrowing down the list of promising candidates using computationally cheap features thereby lessening dependence on linear programming testing. The second stage, meanwhile, diminishes the burden on heuristic testing through an innovative partial testing approach. Moreover, we propose a novel theoretical model characterizing the fundamental tradeoff between time spent making a single branching decision and the resulting branching quality. A formula, derived from the model, for dynamically adjusting the number of candidates to select for the second stage achieves consistently superior performance to ones obtained from trial-and-error tuning. The derivation easily generalizes to most branching strategies requiring time-consuming score computation. Through an extensive numerical study, we demonstrate that a dynamic version of the 2LBB, denoted by 2LBB-dy, achieves approximately 45% and 50% time reduction, respectively, compared with the state-of-the-art (SOTA) hand-crafted branching strategy in solving the capacitated vehicle routing problem (CVRP) and vehicle routing problem with time windows. In addition, our RouteOpt (an exact VRP solver available at https://github.com/Zhengzhong-You/RouteOpt), when equipped with the 2LBB-dy, achieves a 47% time reduction compared with the SOTA VRPSolver for the CVRP.

Funding: This work was supported by the National Science Foundation [Grant CMMI-2309667] and the Alibaba Group US [Grant AGR00022274].

Supplemental Review: The empirical results in this paper were replicated. The code, data, and files required to reproduce the results were reviewed and are available at https://doi.org/10.1287/opre.2023.0615.

Supplemental Material: This article includes an online appendix, computer code and data supporting the study's findings, and replication files. All supplemental materials, including the code, data, and files required to reproduce the results, are available at https://doi.org/10.1287/opre.2023.0615.

Keywords: learning to branch • column generation • exact methods • vehicle routing

1. Introduction

The vehicle routing problem (VRP) is a well-known discrete optimization problem that has been extensively studied in the operations research and management science community since its inception in 1959 (Dantzig and Ramser 1959). It seeks routes of the minimum cost (usually measured by the total travel distance) to serve a set of consumers, subject to restrictions such as vehicle capacity in the capacitated VRP (CVRP) and time windows in the VRP with time windows (VRPTW). Optimizing the routes can also decrease waiting time and improve customer satisfaction, leading to wide applications in the industry such as e-commerce (Vinsensius et al. 2020), healthcare

(Heching et al. 2019), and emergency response (Zheng 2018), where the timely delivery of products and services is crucial.

Nonetheless, VRPs are generally challenging to solve optimally because of their combinatorial nature or, more intrinsically, the so-called NP-hardness. Approaches to solving VRPs exactly can generally be categorized into two types: *branch-and-cut* (B&C) algorithms solving compact (in terms of variables) formulations such as vehicle or commodity flow-based formulations (Bektaş et al. 2011), and *branch-price-and-cut* (BPC) algorithms solving the set-partitioning formulation (SPF; Costa et al. 2019). Branching is an indispensable component of these methods.

It is well known that the choice of branching variable significantly impacts the effectiveness of B&C algorithms for solving general mixed integer programs (MIPs; Achterberg and Wunderling 2013). Expectedly, branching remains one of the most important decisions in BPC algorithms for solving VRPs. In particular, our numerical experiments show that selecting the branching variable randomly increases computation time by more than eight times compared with the three-phase branching (3PB) strategy employed in the state-of-the-art VRP solver, the VRPSolver developed in Pessoa et al. (2020). Moreover, in contrast to pricing and cutting planes, which are usually customized based on specific problem structures for acceleration (Lysgaard et al. 2004, Lübbecke and Desrosiers 2005), branching does not rely on such information as much and tends to bear better generalization across a wide range of VRP variants. Therefore, the study of branching strategy holds significant potential in enhancing BPC methods for various vehicle routing–related problems.

To enhance the readability of this paper, the meanings of some key abbreviations are summarized in Table 1, sorted in order of decreasing frequency of use.

1.1. Motivation

Strong branching (SB; Applegate et al. 1995) is believed to be the most reliable way to obtain a small branch-and-bound tree (BBT; Costa et al. 2019). It considers a set of variables required to be integral but currently taking fractional values, hereafter referred to as *candidates*, computes the gains from branching on each of them, and selects one with the highest gains to branch on (see Section 3.2 for more details). Despite its high efficacy in producing small-sized BBTs, SB can be extremely time-consuming because column generation (CG) is required to solve the child linear programming (LP) problems for all the candidates. To alleviate the computational burden, a hierarchical branching scheme is often adopted. In particular, the VRPSolver employs a 3PB strategy modified from Pecin et al. (2017b).

The 3PB, recognized as state-of-the-art (SOTA) branching strategy in BPC algorithms for solving VRPs, progressively narrows branching candidates through three phases: initial screening using pseudocost and fractionality, subsequent LP testing shortening the list, and finally heuristic testing selecting the candidate with the highest heuristic score. By gradually applying more computationally intensive evaluations to fewer candidates, 3PB effectively balances computational cost and decision quality, making it the leading branching strategy for BPC algorithms prior to this work. Its well-structured phased approach inspires the design of our two-stage learning-based branching (2LBB) and lays the foundation for making learning-accelerated branching strategies practical.

Although considered highly efficient, 3PB has two major weaknesses. Firstly, the screening process's *complete* reliance on pseudocost and fractionality before LP testing can potentially miss good candidates. Secondly, it requires performing *complete* time-consuming testing, whereas our newly proposed partial testing demonstrates that 80% of the time spent on complete heuristic testing is unnecessary. Overall, this research seeks to overcome these weaknesses to significantly reduce the computational effort while maintaining, if not improving, the branching decision quality.

Recently, considerable research effort has been dedicated to leveraging advanced machine learning (ML) techniques to mimic SB in the context of solving general MIPs and has successfully demonstrated that learned strategies can beat best-performing ones currently employed in solvers such as SCIP and CPLEX (Khalil et al. 2016, Alvarez et al. 2017, Yang et al. 2022). These ML models are trained with readily available historical and current-node information to solve compact formulations (i.e., formulations with a nonexponential and usually fixed number of variables). Once

Table 1. List of Abbreviations and Meanings

Abbreviation	Meaning	Abbreviation	Meaning
3PB	Three-phase branching	2LBB	Two-stage learning-based branching
BPC	Branch-price-and-cut	CG	Column generation
SB	Strong branching	BKF	Best k formula
BBT	Branch-and-bound tree	RMP	Restricted master problem
BBN	Branch-and-bound node	B&C	Branch-and-cut
SPF	Set-partitioning formulation	SOTA	State-of-the-art

trained, the ML models can select high-quality branching variables with little extra computation cost, achieving significant acceleration overall.

Nonetheless, models designed for branching selection in general MIPs mostly rely on three categories of features: static ones (e.g., initial LP matrix coefficients), dynamic ones from the current branch-and-bound node (BBN), and dynamic history-dependent ones (Alvarez et al. 2017, Yang et al. 2022). These approaches fail for BPC algorithms solving noncompact formulations, for example, SPF. To the best of our knowledge, no successful attempt to improve branching in BPC via ML has been accomplished before this study. The unfortunate ineffectiveness of existing learning-to-branch paradigms and the considerable difficulty in making them work for BPC are mainly because of the following three reasons.

First, to balance the resultant BBT, branching is typically not performed on a nominal variable in the restricted master problem (RMP) but rather achieved by adding a dense constraint (see more explanation in Section 3.2), after which, repetitive CG is needed to compute the exact LP value of a child BBN. For clarity, we refer to the LP at the current BBN as the *parent LP*, the one at a child BBN obtained from the parent LP by adding the branching constraint as the *starting child LP*, and the one resulting from solving the starting child LP with CG as the *ending child LP*. This branching constraint, together with the variables that have yet to be generated in the repetitive CG, causes the ending child LP to significantly deviate from the parent LP. Worse still, such deviation is not known until the starting child LP is solved with CG, making it extremely difficult, if not impossible, to accurately predict the LP value changes based on conventional features of the current BBN. Second, generated by CG iteratively, variables are added to the RMP over time. As a result, minimal historical information about the recently generated variables is available to aid the branching variable selection, which, unfortunately, is essential for existing ML models in the literature to make accurate predictions. Third, the accumulated effect of the first reason leads to acute changes to the RMP, where the current RMP can be almost completely different from the initial RMP at the root. Thus, the standard static features lose predictive power. In summary, these three reasons bring about unique obstacles to using dynamic current-node, dynamic history-dependent, and static features, accordingly, for learning to branch in BPC algorithms.

To overcome these challenges, new features that capture the underlying problem structure, rather than focusing solely on the RMP, are needed. Although these new features can be useful, their effectiveness depends on appropriate learning methods and workflows. We believe the newly proposed two-stage scheme addresses these needs. In the first stage, graph-based features that are computationally cheap are used to narrow down the promising candidates where the learning target is the ranking of SB scores, thus reducing the dependence on LP testing. In the second stage, more computationally intensive features derived from LP testing are employed. At this stage, the learning target shifts from SB score ranking to the ratio of optimal value changes between the starting and ending child LPs relative to that of the parent LP. By applying a novel partial testing (see Section 4.4) that integrates predictions from the second stage ML model with LP scores, we efficiently determine which starting child LP should be further tested, enabling us to identify the best candidate with much less computational effort.

Moreover, SB, 3PB, and 2LBB necessitate testing multiple branching candidates before finalizing the decision. However, determining the optimal number of candidates to evaluate for minimizing overall solution time remains challenging as there is no established method to balance the tradeoff between the time spent on making a single branching decision and the resulting decision quality. To shed light on this fundamental tradeoff, we propose a novel theoretical model for deriving *Best K Formula* (BKF), which dynamically decides the best number of candidates to test based on the solution's progress. By embedding the distribution of local branching improvements, the BKF optimizes this tradeoff, leading to additional improvement in computational efficiency. It is important to emphasize that 2LBB is specifically designed for algorithms that incorporate CG. In contrast, the BKF is broadly applicable to branching strategies involving computationally intensive testing, including both BPC algorithms (the focus of this study) and B&C algorithms that do not require CG, as supported by our experiments in Section EC.8.4 of the e-companion.

In this paper, we develop a novel learning-to-branch framework with a theory-based dynamic adjustment scheme that can overcome the weaknesses of the current state-of-the-art branching strategy in BPC, namely 3PB. Although this study focuses on VRPs, we believe that most ideas introduced herein are easily transferable to BPC for addressing operational challenges in supply chain operations, healthcare, and the airline industry, for example, inventory routing problems (Engineer et al. 2012, Desaulniers et al. 2016), healthcare planning problems (Trautsamwieser and Hirsch 2014, Qiu et al. 2022), and crew rostering problems (Quesnel et al. 2020, Breugem et al. 2022).

1.2. Major Contributions and Outline

This paper concerns improving the branching decisions in BPC algorithms for solving VRPs exactly. It makes the following major contributions.

- *We propose the first effective learning-to-branch framework in BPC algorithms*. The first stage relies on

graph-based features that are computationally cheap to narrow down the promising candidates, whereas the second stage finalizes the decision with substantially less computation effort by utilizing features obtained from LP testing. This design ensures that the framework makes branching decisions fast and effectively selects high-quality candidates to generate a small-sized BBT. Furthermore, by integrating richer information at each stage and defining appropriate learning tasks, it advances both efficiency and effectiveness beyond 3PB, making it a SOTA branching method for BPC algorithms.

- *We establish a novel theoretical model that gives rise to a formula for dynamically adjusting a key parameter in 2LBB and 3PB.* The model characterizes the tradeoff between the testing time and branching decision quality, deciding the number of candidates to evaluate. It also generalizes to branching strategies reliant on computationally intensive scores and explains the performance differences between 2LBB and 3PB for CVRP and VRPTW instances.
- *We integrate the 2LBB method into RouteOpt and perform extensive numerical experiments to support the design of the two-stage learning workflow*. Our 2LBB reduces the average computation times by more than 45% compared with the SOTA 3PB method and generalizes well across varying instance sizes by consistently lowering CPU time and BBT size. When embedded into RouteOpt together with other effective techniques (see Section EC.5 of the e-companion), 2LBB achieves a 47% time reduction on benchmark instances compared with the VRPSolver and allows RouteOpt to prove optimality for an open CVRP instance.

The rest of the paper is organized as follows. In Section 2, we review related literature on branching in BPC algorithms and existing learning-to-branch methods. In Section 3, we include some preliminaries to facilitate readability. We describe our learning-to-branch framework in Section 4 and detail the learning methodology employed in the framework in Section 4.5. A theoretical model providing insights into the underlying tradeoffs is presented in Section 5. The effectiveness of the proposed framework is demonstrated via an extensive numerical study in Section 6. Lastly, in Section 7, we make some concluding remarks and identify future research directions. Details of selected proofs and numerical results are included in the e-companion.

2. Literature Review

Numerous VRP variants have been extensively investigated in the past 6.5 decades, leading to many effective exact algorithms. Laporte (2009) provides a thorough review of this topic. Currently, the best-performing ones are BPC algorithms, solving CVRP instances with up to 856 customers (Lima et al. 2014, Pessoa et al. 2020). This paper focuses on learning to branch in BPC algorithms. We first review branching strategies employed in BPC algorithms from two perspectives: branching variable type and selection. Then, we discuss some most related ML attempts.

2.1. Branching Variable Type

Branching on arcs (or edges in the undirected case) is typically the default approach in B&C methods for solving the two-index vehicle flow formulation (Laporte and Nobert 1983). This prevalence is because the variables in such a flow formulation explicitly correspond to arcs/edges, and such branching selection leads to the most common branching on a single nominal variable in the formulation. Desrosiers et al. (1984) introduced branching on arcs in a BPC method for solving the VRPTW, which has become the primary choice in BPC algorithms (Irnich et al. 2010, Pessoa et al. 2020). Costa et al. (2019) underscored the popularity and highlighted its simplicity, ease of implementation, and robustness. In light of these considerations, we also adopt branching on edges in our learning-to-branch framework.

Despite the widespread application of branching on edges, Augerat et al. (1995) observed that this approach, for the CVRP, poses a relatively localized influence that might result in undesirably small changes in the dual bound after branching. Thus, they proposed to branch on sets, which entails selecting a set of vertices. If the total flow, induced by the current solution, passing through this set is fractional, then branching is performed on all edges incident to this vertex set. This branching strategy can be viewed as an aggregated version of branching on edges, thus intuitively causing a broader impact. It has been applied in some BPC (Fukasawa et al. 2006, Jepsen et al. 2008, Pecin et al. 2017a, Yang et al. 2023) and B&C (Naddef and Rinaldi 2002, Baldacci et al. 2004, Lysgaard et al. 2004) algorithms. In a recent study, Silva and Uchoa (2022) proposed cluster branching, which can be viewed as a special case of branching on sets. It first divides the vertices into multiple clusters based on geographical location and then defines two sets of variables, that is, intra- and intercluster variables. Their empirical study demonstrated that this strategy could substantially reduce the CPU time for instances with clustered customer locations. Notably, it proved optimality for multiple CVRP and VRPTW open instances.

2.2. Branching Variable Selection

Once the type of variables to branch on is determined, the question reduces to how to make high-quality selections efficiently, which is the major focus of studies on branching strategies. The most straightforward approach is to select randomly without referring to any additional information, which, not surprisingly, leads to a large-sized BBT and thus undesirable ineffectiveness. A slightly more effective alternative is the

most fractional branching, which selects the variable with the largest fractionality (Augerat et al. 1995). The pseudocost branching, proposed by Bénichou et al. (1971) for solving general MIPs, has been widely adopted because of its high effectiveness and computational efficiency. It utilizes historical information to compute pseudocosts, which requires proper initialization. Applegate et al. (1995) introduced SB for solving the traveling salesman problem, which has been widely observed to produce small-sized BBTs. However, SB is generally not practical because of the time-consuming SB score computation. Linderoth and Savelsbergh (1999) combined the strengths of pseudocost branching and SB by using SB to initialize pseudocosts and improved the performance.

Pecin et al. (2017b) proposed to combine the most fractional branching, pseudocost branching, and SB into a 3PB in the context of solving VRPs by BPC algorithms. This approach comprises an initial screening phase and two estimation phases: one by solving LPs without CG and the other with heuristic CG. Such a phased strategy significantly reduces the number of branching candidates to consider, which, in turn, allows one to afford moderate extra computation while still selecting a branching variable of close-to-best quality with high chance. The 3PB, together with limited memory subset row cuts (Pecin et al. 2017c) and enumeration (Baldacci et al. 2008), helped to optimally solve open benchmark instances with up to 199 customers. The 3PB has since been adopted by the VRPSolver, further underscoring its efficacy and applicability.

2.3. ML for Solving VRPs

ML has proven effective in enhancing various existing algorithms for solving large-scale discrete optimization problems, including VRPs. The application of ML to approaching VRPs can be roughly categorized into two types. The first type seeks to directly construct heuristic solutions (Khalil et al. 2017, Nazari et al. 2018, Costa et al. 2020) or predict the feasibility of a given problem (Mai and Navet 2021, van der Hagen et al. 2024). However, these heuristic-based ML methods, especially graph neural network–based methods, struggle with generalization, scale poorly with instance size, and lack quality guarantees. Their performance degrades significantly as the number of nodes increases, both in terms of solution quality and inference time (Cappart et al. 2023). In contrast, our ML models are significantly less complex, and therefore, they demonstrate strong generalization capability while maintaining the optimality guarantee.

The second type focuses on accelerating some specific components in a sophisticated exact algorithm. Morabit et al. (2021) applied graph neural networks to pick columns likely to improve the objective value of the current RMP. Such selection significantly reduced the size of the RMP, and thus, a 30% decrease in computing time for solving the root node was achieved. In another closely related work (Morabit et al. 2022), the authors applied supervised learning (logistic regression, random forest, and neural network) to identify arcs that are most likely to be used in an optimal solution. The unselected arcs were subsequently discarded, and the resulting residual graph was used for heuristic pricing. Readers can refer to Tahir et al. (2021) and Zhang et al. (2022) for additional works falling into this category.

Despite much success achieved in applying ML to enhance branching decisions for solving general MIPs (Khalil et al. 2016, Alvarez et al. 2017, Gasse et al. 2019, Yang et al. 2022), to the best of our knowledge, there is no known work on learning to branch in BPC algorithms for solving VRPs. A slightly related work is Pereira et al. (2022), where the authors used ML to assist in solving the crew pairing problem. Note that, instead of trying to solve it exactly, they aimed to compute a feasible solution via a branch-and-price heuristic. Building on the reviewed pioneering efforts, our paper seeks to fill this research gap.

3. Preliminaries

Before jumping into the intricacies of the proposed framework, to facilitate readability, we present necessary preliminaries about a general BPC method for solving VRPs. We first review the SPF with a focus on the associated dynamics in the solution process. Subsequently, we discuss the branching component, underscoring the differences between its usage in BPC and in B&C for solving MIPs. Lastly, we elaborate on an effective branching strategy widely adopted in BPC algorithms, which addresses the high computational overhead of SB while keeping the BBT size relatively small.

3.1. SPF

The standard SPF for VRPs, including the classic CVRP and VRPTW, takes the following general form:

$$\begin{aligned} \min \ & \sum_{r\in\Omega} c_r x_r \\ \text{s.t.} \ & \sum_{r\in\Omega} a_{ir} x_r = 1, && \forall i \in N, \\ & \sum_{r\in\Omega} w_{jr} x_r \le w_j, && \forall j \in Q, \\ & x_r \in \{0,1\}, && \forall r \in \Omega, \end{aligned} \tag{1}$$

where N, Q, and Ω denote the sets of customers to serve, the index set for side constraints, and the set of vehicle routes, respectively. The coefficient c_r represents the cost of route r. Coefficient a_{ir} is a binary indicator, with $a_{ir} = 1$ if route r visits customer i, and 0

otherwise. The binary decision variable x_r equals one if route r is selected in the solution and zero otherwise. It is important to note that it suffices to include only elementary routes in the SPF, that is, the customers of the route can only be visited exactly once. However, to expedite the CG process, *ng*-route relaxation (Baldacci et al. 2011) is often applied, which allows nonelementary routes. In this case, a_{ir} extends to a nonnegative integer representing the number of times route r visits customer i. Inequalities $\sum_{r\in\Omega} w_{jr}x_r \le w_j$, $\forall j \in Q$, represent the additional side constraints. For example, in the multidepot VRP (Contardo and Martinelli 2014), constraints $\sum_{r\in\Omega_j} x_r \le m_j$, $\forall j \in D$, enforce the restriction on the fleet size, where D is the set of depots, m_j is the fleet size of depot j, and $\Omega_j \subseteq \Omega$ denotes the set of routes that start and end at depot $j \in D$.

Initially, only a small subset of routes $\Omega' \subseteq \Omega$ are included. Formulation (1) with Ω replaced by $\Omega' \cup \Omega''$ is referred to as the RMP, where Ω'' is the set of variables generated via CG. To solve the LP relaxation of the RMP, columns (variables) with negative reduced costs are generated iteratively via CG, also known as pricing. If the optimal solution to the RMP is fractional, branching is applied to help achieve integrality. We remark that the new variables complicate branching selection because historical data on their behaviors is lacking. Additionally, cutting planes increase the dimension of the dual problem and may even alter the structure of the pricing subproblem (Pecin et al. 2017b). Branching further complicates the situation by introducing branching constraints that accumulate as the tree grows deeper. Essentially, the BPC solution process involves alternating rounds of column and row generation, resulting in highly dynamic changes (Figure 1) to the RMP, which challenges existing learning-to-branch paradigms.

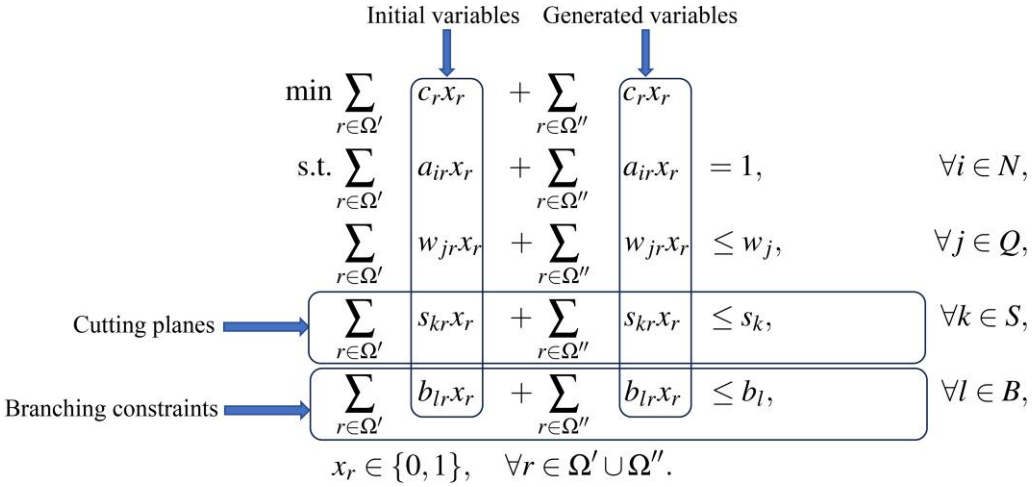
3.2. Branching Component

Although branching on a nominal variable in MIPs is a popular choice, it is not the case in the BPC context because imposing $x_r = 0$ is not straightforward for CG and is often ineffective as it creates extremely unbalanced BBTs. In contrast, branching on an edge in the underlying graph requires no change in CG and leads to a more balanced tree, making it a mainstream choice in BPC algorithms (Costa et al. 2019). Branching on a given edge can be generally achieved by adding a dense constraint of the form $\sum_{r\in\Omega} b_r x_r \le 0$ or $\sum_{r\in\Omega} b_r x_r \ge 1$, where b_r counts the number of times route r traverses the edge. It is worth mentioning that such branching constraints also enforce restrictions on variables that have yet to be generated.

SB considers a set of branching candidates and performs tests by branching on each of them separately. Let z denote the optimal value of the parent LP. For a given candidate, let z_l and z_r be the LP values at the left and right child BBNs, respectively, when branched on. Then the SB score s of this particular branching choice is computed by the product rule (Achterberg 2007): $s = \max\{z_l - z, \epsilon\} \cdot \max\{z_r - z, \epsilon\}$, where $\epsilon > 0$ is typically set to 10^{-6}. SB then selects to branch on the one with the largest score. The calculation of pseudocost is similar to that of s by using historical information. Let $(\hat{x}_r)_{r\in\Omega'\cup\Omega''}$ be the solution to the parent LP. For a given edge e, let $\hat{x}(e)$ be the total flow on e computed by $\hat{x}(e) = \sum_{r\in\Omega'\cup\Omega''} a_{ir}\hat{x}_r$. Then the edge e can be considered as a branching candidate if $\hat{x}(e)$ is fractional. Let $\overline{\Delta z_l}$ and $\overline{\Delta z_r}$ denote the average value of $(z_l - z)$ and $(z_r - z)$ collected from historical branching, respectively. The pseudocost is computed as $\max\{(\hat{x}(e) - \lfloor\hat{x}(e)\rfloor)\cdot\overline{\Delta z_l}, \epsilon\} \cdot \max\{(\lceil\hat{x}(e)\rceil - \hat{x}(e))\cdot\overline{\Delta z_r}, \epsilon\}$. In this study, we adopt this way for our pseudocost calculation.

However, calculating such z_l and z_r values for each candidate can still need a large number of CG

Figure 1. (Color online) Dynamic Changes of the RMP in the BPC Solution Process



iterations, making this process time-consuming. To make SB faster, one can limit the number of candidates to test and apply heuristic CG in place of the much slower exact CG to solve the starting child LPs or even bypass CG completely. Without exact CG, the SB score computed is not precise, potentially compromising the quality of the branching decision. For the clarity of presentation, we thereafter refer to the procedure of branching on each candidate and solving the starting child LPs without any CG to compute an SB score as *LP testing* and refer to this SB score as *LP score*. The counterparts are called *exact testing* and *heuristic testing* when solving with exact CG and heuristic CG, respectively. The resultant SB scores are accordingly called *heuristic score* and *exact score*. Moreover, in this paper, unless stated otherwise, the term pseudocost always refers to the pseudocost computed using historical LP scores. Pseudocosts derived from heuristic and exact scores are termed *heuristic pseudocost* and *exact pseudocost*, respectively.

3.3. Three Phase Branching Selection

The 3PB is considered the state-of-the-art branching strategy in BPC algorithms for solving VRPs, which combines the aforementioned two ideas for enhancing SB. In particular, it runs progressively time-consuming testing on a gradually reduced candidate list and totally avoids performing exact CG. Basically, it balances the tradeoffs between the time consumed for branching decision making and the decision quality. As suggested by its name, the 3PB (Figure 2) comprises three phases—initial screening, LP testing, and heuristic testing—which gradually shorten the candidate list until a final decision is reached.

In the initial screening, the complete set of branching candidates, denoted by $\mathcal{C}$, is first divided into two groups: one with initialized pseudocosts and the other without. For the former group, the top $\hat{\theta}_p$ candidates, denoted by $\widehat{\mathcal{C}}_p$, are selected based on pseudocosts, whereas for the other group, the top $\hat{\theta}_f$ candidates with the highest fractionality, denoted by $\widehat{\mathcal{C}}_f$, are chosen. The parameters $\hat{\theta}_p$ and $\hat{\theta}_f$ are typically set equal and let $\hat{\theta}_{pf} := \hat{\theta}_p + \hat{\theta}_f$. The $\hat{\theta}_{pf}$ candidates in $\widehat{\mathcal{C}}_p \cup \widehat{\mathcal{C}}_f$ then go through LP testing and the top $\tilde{\theta}$ candidates ranked by LP scores constitutes the set $\widetilde{\mathcal{C}}$. Lastly, heuristic testing is performed on candidates in $\widetilde{\mathcal{C}}$, and the one with the largest heuristic score is the final decision.

4. Learning-to-Branch Framework

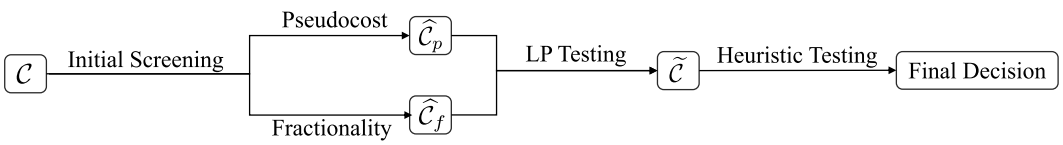
Similar to Khalil et al. (2016), Alvarez et al. (2017), and Yang et al. (2022), and the various related studies cited therein, our goal is to apply ML to mimic SB in a computationally efficient manner. Unfortunately, learning-to-branch paradigms used in Yang et al. (2022) that have been proven to work well in the context of solving MIPs fail to achieve any prediction accuracy above 20% according to our preliminary experiments, confirming that learning to branch in BPC algorithms is much more difficult. Such difficulty stems from causes explained in Section 1.1, which are addressed by our novel learning-to-branch framework.

4.1. High-Level Idea

As explained in Section 1.1, 3PB lacks the critical information necessary for efficiently making good branching decisions, and the three mentioned challenges prevent us from leveraging proven useful features from the literature for learning to optimize MIPs. In this section, we first identify new features to overcome the second and third challenges—the absence of historical information about recently generated (nominal) variables and the ineffectiveness of standard static features caused by the accumulated changes in the RMP. For the second challenge, we propose substituting the missing information with features of implicit variables. Because branching occurs on an edge in the graph—corresponding to a static variable in an equivalent compact formulation that remains unaffected by the CG process—historical data on these implicit variables can be collected throughout the solution process. To address the third challenge, our framework uses features derived from the graph (e.g., the depot and customer locations and the induced proximities), which offer consistent predictive power and require one-time computation, as a substitute for static RMP-derived features.

The first challenge—the unknown beforehand deviation of an ending child LP from its parent LP—is particularly difficult to overcome. Even with the features mentioned above, achieving accurate predictions for the deviation remains challenging. Performing exact CG, as is done in exact testing is the only reliable way to obtain this uncertain deviation. However, exact testing is often time-consuming, even when evaluating just two candidates. Our second stage aims to limit its use to necessary cases. Specifically, when the

Figure 2. Workflow of the 3PB



deviation between an ending child LP and its starting LP is expected to be minimal, exact testing for that starting child LP can be skipped without risk. However, directly predicting this deviation is as complex as predicting the exact SB score since knowing the deviation would allow for deriving the SB score. Drawing from the success of *ranking* methods in learning-to-branch tasks for MIPs, our second-stage model focuses on predicting the rank of the *deviation ratio*, ρ, for each starting child LP. Let $s := z_1 - z$ and $\hat{s} := z_2 - z$, where z, z_1, and z_2 are the optimal values of the corresponding parent, starting, and ending child LPs, respectively. This deviation ratio, defined as $\rho := \frac{s}{\hat{s}} \geq 1$, reflects the similarity between the starting and ending child LPs and thus serves as a measure of the necessity for exact testing. By ranking the ρ values for all candidates—each with two starting child LPs (one from the left branch and the other from the right)—we can selectively limit exact testing to only those starting child LPs with high predicted deviation.

4.2. Workflow

Our framework builds on the cutting-edge handcrafted 3PB detailed in Section 3.3. Recall that 3PB has two primary weaknesses. The first is the less effective initial screening, which leads to solving a large number of (often in the hundreds) starting child LPs without CG in the LP testing phase. The second is that it needs to perform several heuristic testing to determine the best candidate from those identified by LP testing. Although the number of heuristic testing may not be very large, each testing involves multiple iterations of heuristic CG and solving intermediate LPs, making it potentially over ten times slower than an LP testing.

To overcome these two weaknesses, our framework employs two ML models: one before the LP testing phase (*M1*) and one after LP testing (*M2*), as illustrated in Figure 3. *M1* further selects promising candidates from the set given by initial screening, thereby reducing the need to solve hundreds of LPs without CG in the LP testing phase. *M2* aids in selecting a subset of starting child LPs for testing while ensuring decisions match those from testing *all* starting child LPs. We term such testing as *partial testing*.

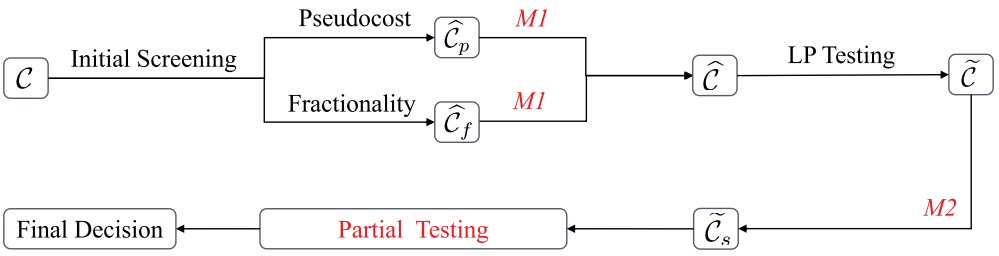
Specifically, in the first stage, *M1 mimics the behavior of performing exact testing on candidates that have passed the initial screening*, with the learning goal of predicting the ones with the highest exact scores. In the second stage, *M2 predicts whether the starting child LP value (z_1) accurately approximates the ending child LP value (z_2) on the candidates that have passed LP testing*. Generally speaking, starting child LPs expected to have a large deviation ratio will be subjected to partial testing for a more precise estimate of z_2. Details on the second stage are provided in Section 4.4.

4.3. First-Stage Selection

We propose a phased design with two stages of ML (Figure 3) that define two sequential learning tasks. The first task utilizes easily computable features to filter out a large proportion of unpromising candidates from $\widehat{\mathcal{C}}_p$ and $\widehat{\mathcal{C}}_f$, producing a refined candidate set, $\widehat{\mathcal{C}}$. Notably, *M1* is applied separately to $\widehat{\mathcal{C}}_p$ and $\widehat{\mathcal{C}}_f$ to obtain $\widehat{\mathcal{C}}$, ensuring that some candidates from $\widehat{\mathcal{C}}_f$, if any, are retained. Without this treatment, our experiments indicate that *M1* tends to choose candidates almost exclusively from $\widehat{\mathcal{C}}_p$ when $\widehat{\mathcal{C}}_p \cup \widehat{\mathcal{C}}_f$ altogether is fed into *M1*, leading to unsatisfactory performance.

The reason is mainly twofold. First, the candidates in $\widehat{\mathcal{C}}_p$, that is, those with pseudocosts initialized, are of higher quality, in an average sense, than those in $\widehat{\mathcal{C}}_f$ because the pseudocost of a variable is initialized when it has successfully passed the initial screening. In contrast, the pseudocosts of candidates in $\widehat{\mathcal{C}}_f$ have not been initialized thus far, indicating that they failed such screening. Second, the pseudocosts get initialized fast as the BBT grows, causing $\widehat{\mathcal{C}}_f$ to vanish quickly. As a result, $\widehat{\mathcal{C}}_p$ constitute a vast majority of the training data, so *M1* naturally prioritizes members in $\widehat{\mathcal{C}}_p$. Nonetheless, some candidates from $\widehat{\mathcal{C}}_f$ can still be good branching choices, especially for branching decisions near the root of the BBT, where most of the pseudocosts have not yet been initialized. It is essential not to miss such candidates considering the decisions close to the root are generally of great importance (Pecin et al. 2017b, Pessoa et al. 2020).

Figure 3. (Color online) Workflow of Our Learning-to-Branch Framework



4.4. Second-Stage Selection

The key idea of the second-stage selection is to utilize the predicted ranking of ρ (output from *M2*) to avoid unnecessary testing. More specifically, this selection process relies on Conditions (2)–(4), proven to be sufficient in Proposition 1. It is based on the straightforward intuition: for two candidates $e_i, e_j \in \widetilde{\mathcal{C}}$, if the current testing score, not necessarily the exact score, of e_i exceeds that of e_j, and the deviation ratio of e_i is no larger than that of e_j, then e_j can be removed from $\widetilde{\mathcal{C}}$, as its exact score cannot surpass that of e_i. Further details are provided in Section 4.4.2.

It is rare, however, that these three conditions are satisfied for all candidate pairs (e_*, e_j) at the beginning, where $e_* \in \widetilde{\mathcal{C}}$ has the highest exact score. To address this, Algorithm 1 conducts exact testing on a subset of the starting child LPs in a dynamic manner to ensure these conditions hold across all pairs (e_*, e_j). Consequently, all such e_j will be removed from $\widetilde{\mathcal{C}}$, leaving only e_*, thus achieving the selection goal.

4.4.1. Assumptions on the Behavior of *M2*.

Let l_i and r_i denote the left and right starting child LPs, respectively, generated by adding the branching constraint associated with the ith candidate $e_i \in \widetilde{\mathcal{C}}$ and define $\widetilde{\mathcal{C}}_s := \bigcup_{i=1}^{|\widetilde{\mathcal{C}}|} \{l_i, r_i\}$ as the set of all starting child LPs. Given the input features, *M2* outputs a rank by the ρ values for each starting child LP in $\widetilde{\mathcal{C}}_s$.

We formalize some assumptions based on the observed behavior of *M2*, which relate the deviation ratio, ρ, to the output ranking of *M2*, denoted by d.

Assumption 1. *d is a function of the deviation ratio, that is, $d = h(\rho)$, where $h : [1, \infty) \to \mathbb{Z}_+ \cup \{-1\}$.*

Assumption 2. *h is a monotonically increasing function such that $h(\rho) = -1$ if and only if $\rho = 1$.*

Assumption 3. *$h(\rho_1) = h(\rho_2) \le \underline{d}$ implies that $\max\left\{\frac{\rho_1}{\rho_2}, \frac{\rho_2}{\rho_1}\right\} \le \delta$, where $\underline{d}$ and δ are predefined parameters.*

If *M2* achieves perfect accuracy, the assumptions can be justified as follows: (1) ρ should be the sole factor influencing the prediction d, thereby ensuring the existence of h; (2) d should increase monotonically as deviation ratio ρ increases, aligning with the learning target of *M2*; and (3) based on our observation, a small d indicates that s is a strong approximation of $\hat{s}$, and thus the corresponding ρ_1 and ρ_2 should be close to one when $h(\rho_1) = h(\rho_2) \le \underline{d}$. Although the same d value does not guarantee identical ρ values, a minor difference is expected because both ρ_1 and ρ_2 are close to one in this case. Therefore, it is reasonable to assume that $\max\left\{\frac{\rho_1}{\rho_2}, \frac{\rho_2}{\rho_1}\right\}$ is bounded by a constant δ.

4.4.2. Conditions to Ensure a Larger Exact Score.

The conditions introduced here are fundamental to the proposed partial testing procedure. They establish sufficient criteria to prove that e_i has a larger exact score than e_j, even without knowing their exact scores. To proceed, we first clarify the notation. Let S_k denote the difference between the optimal value of the *current* child LP $p_k \in \widetilde{\mathcal{C}}_s$ and its parent LP value, and D_k represents the *current* deviation ratio. Note that current signifies dynamic determination based on the state of testing. If a child LP $p_k \in \widetilde{\mathcal{C}}_s$ has not yet been tested, S_k is set to s_k, and D_k is set to d_k. Otherwise, S_k is updated to $\hat{s}_k$, and D_k is updated to $h(1) = -1$ to indicate that p_k has been tested. For clarity, when no superscript is attached, we do not distinguish the left or right child LP. When necessary, we employ the superscript l for the left child and r for the right; for example, S_i^l and S_i^r denote the S values of the corresponding left and right child LPs associated with branching candidate e_i. The current score of e_i is represented as $S_i^{l\times r} := S_i^l \times S_i^r$, and its exact score is denoted by $\hat{s}_i^{l\times r} := \hat{s}_i^l \times \hat{s}_i^r$.

Definition. We define e_i as a *better* branching candidate than e_j if the following conditions hold for the candidate pair (e_i, e_j):

$$S_i^{l\times r} > S_j^{l\times r} \cdot \delta^{\phi_1+\phi_2}, \tag{2}$$

$$(D_i^{\max} < D_j^{\max} \text{ and } \phi_1 = 0) \text{ or } (D_i^{\max} = D_j^{\max} \le \underline{d} \text{ and } \phi_1 = \mathbb{1}_{D_i^{\max} \ne -1}), \tag{3}$$

$$(D_i^{\min} < D_j^{\min} \text{ and } \phi_2 = 0) \text{ or } (D_i^{\min} = D_j^{\min} \le \underline{d} \text{ and } \phi_2 = \mathbb{1}_{D_i^{\min} \ne -1}), \tag{4}$$

where $D_i^{\max} := \max\{D_i^l, D_i^r\}$, $D_i^{\min} := \min\{D_i^l, D_i^r\}$, $D_j^{\max} := \max\{D_j^l, D_j^r\}$, and $D_j^{\min} := \min\{D_j^l, D_j^r\}$.

Condition (2) specifies that the score $S_i^{l\times r}$ for e_i must be greater than the score for e_j, scaled by a factor $\delta^{\phi_1+\phi_2}$. Conditions (3) and (4) further require that the $D_i^{\max}$ and $D_i^{\min}$ values of e_i are either strictly less than those of e_j or, if equal, must not exceed the predefined threshold $\underline{d}$. Together, these conditions ensure that e_i can be regarded as a better branching candidate than e_j, because they provide sufficient evidence that e_i achieves a larger exact score than e_j, as demonstrated in Proposition 1.

Proposition 1. *Conditions (2), (3), and (4) are sufficient to prove that e_i has a larger exact score than e_j; that is, $\hat{s}_i^{l\times r} > \hat{s}_j^{l\times r}$.*

By verifying the conditions outlined in Proposition 1, it is possible to replicate the selection made by SB without requiring explicit computation of the exact scores, thereby significantly reducing computational complexity. Given a pair of candidates, the one with a larger current score $S^{l\times r}$ is referred to as the *Leader*, and the other is the *Competitor*. The candidate $e_* \in \widetilde{\mathcal{C}}$ that is

better than the others is called the *Winner*, and the goal of our selection process is to identify it with the minimum necessary testing.

4.4.3. Partial Testing Procedure. In the ideal scenario, these conditions would hold for all candidate pairs (e_*, e_j) from the very beginning, making e_* the winner, thereby eliminating the need for any additional testing. However, as indicated by our preliminary experiments, this rarely occurs in practice. Candidates with larger LP scores often tend to have larger ρ values, preventing the three conditions from holding simultaneously. To eventually satisfy these conditions, exact testing on some starting child LPs needs to be performed to update the $S^{l\times r}$ and D values.

The challenge lies in strategically selecting a subset of $\widetilde{\mathcal{C}}_s$ for testing to meet the conditions as efficiently as possible. For a given pair (e_i, e_j) with $S_i^{l\times r} > S_j^{l\times r}$, the strategy is to prioritize e_i, and perform testing on its starting LPs, that is, l_i and r_i. If, after testing, the updated $S_i^{l\times r}$ remains larger, then the conditions are satisfied, eliminating the testing need for e_j. On the other hand, starting with l_j (or r_j) may lower both $S_j^{l\times r}$ and D_j^l (or D_j^r), and testing for e_i is still required.

Algorithm 1 outlines the aforementioned partial testing procedure. The procedure begins by initializing S_k and D_k to s_k and d_k, respectively, for all starting child LPs $p_k \in \widetilde{\mathcal{C}}_s$. In the subsequent sorting step, the candidates are sorted based on their current scores $S^{l\times r}$. The top two candidates, e_i and e_j, are selected to check if they satisfy Conditions (2), (3), and (4). If these conditions are satisfied, e_j will be removed from $\widetilde{\mathcal{C}}$. If not, exact CG will be performed on p_k, which is chosen as either the left starting child LP l_i or the right r_i depending on the relative values of D_i^l and D_i^r. Following this, S_k will be updated to $\hat{s}_k$, and D_k will be updated to -1, after which the sorting step will be repeated.

Algorithm 1 (Partial Testing Procedure)

Step 1. Sorting: If $|\widetilde{\mathcal{C}}| = 1$, **Stop**.
Sort candidates in $\widetilde{\mathcal{C}}$ in descending order by $S^{l\times r}$, identify top candidates e_i, e_j.

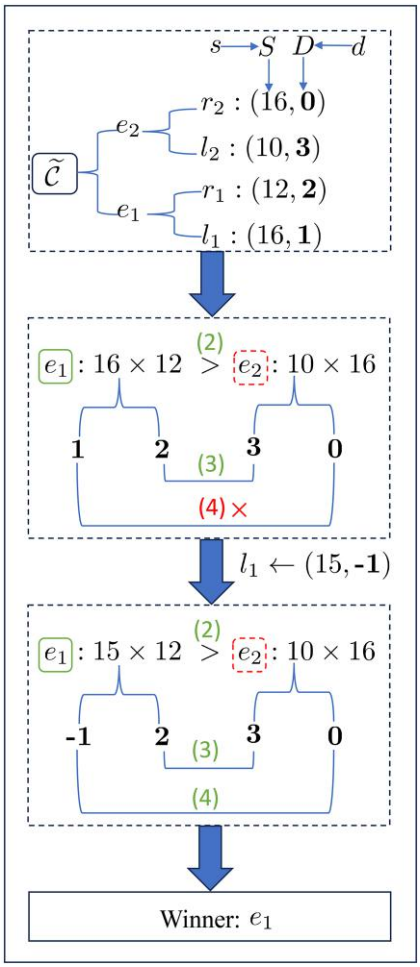
Step 2. Checking: Set $p_k \leftarrow$ NULL.
If Condition (3) failed: if $D_i^l \geq D_i^r$, set $p_k \leftarrow l_i$, otherwise set $p_k \leftarrow r_i$.
If Condition (4) failed: if $D_i^l \leq D_i^r$, set $p_k \leftarrow l_i$, otherwise set $p_k \leftarrow r_i$.

Step 3. Testing:
If p_k equals l_i: Compute $\hat{s}_i^l$, and update $S_i^l \leftarrow \hat{s}_i^l$, $D_i^l \leftarrow -1$, $S_i^{l\times r} \leftarrow S_i^l \times S_i^r$;
If p_k equals r_i: Compute $\hat{s}_i^r$, and update $S_i^r \leftarrow \hat{s}_i^r$, $D_i^r \leftarrow -1$, $S_i^{l\times r} \leftarrow S_i^l \times S_i^r$;
Otherwise: Eliminate e_j from $\widetilde{\mathcal{C}}$.
Go to Step 1.

An illustration example is provided in Figure 4, where $\underline{d} = 3$, and the set of candidates $\widetilde{\mathcal{C}} = \{e_1, e_2\}$, with initial values $(S_1^l, D_1^l) = (16, \mathbf{1})$, $(S_1^r, D_1^r) = (12, \mathbf{2})$, $(S_2^l, D_2^l) = (10, \mathbf{3})$, and $(S_2^r, D_2^r) = (16, \mathbf{0})$. In Figure 4, the square around e_1 indicates that e_1 is the leader, and the dash square around e_2 signifies that e_2 is the competitor. Conditions (2), (3), and (4), are indexed as (2), (3), and (4), respectively, with the green color indicating a satisfied status and red for an unsatisfied status.

During the sorting step, e_1 is determined to be the leader because $S_1^{l\times r} = 16 \times 12 = 192$ is larger than $S_2^{l\times r} = 10 \times 16 = 160$. Consequently, e_2 is designated as the competitor. The algorithm proceeds to the checking step (Step 2), where Conditions (3) and (4) are evaluated. Because Condition (4) is not satisfied, l_1 is chosen to compute $\hat{s}_1^l$, which updates S_1^l from 16 to 15 and sets D_1^l to -1. The algorithm then returns to the sorting step (Step 1). Despite the update, e_1 remains the leader because 15×12 is still greater than 10×16. The checking step is repeated, and this time both Conditions (3) and (4) are satisfied, so no further testing is required,

Figure 4. (Color online) Example of the Partial Testing Process in the Proposed Framework



and e_2 is removed from $\widetilde{\mathcal{C}}$. This leaves e_1 as the only candidate in $\widetilde{\mathcal{C}}$ and thus the winner. Proposition 2 confirms that, after a finite number of iterations, the winner can be identified by Algorithm 1.

Proposition 2. *Algorithm 1 terminates in a finite number of steps and identifies the winner e_* that is better than any other candidate $e_j \in \widetilde{\mathcal{C}}$.*

4.4.4. Acceleration Technique. When $|\widetilde{\mathcal{C}}|$ is large, Algorithm 1 can still require significant computation to identify e_*. We propose to address it by reusing information obtained from testing that has already been conducted. Given Assumptions 2 and 3, Proposition 3 estimates the exact score by approximating ρ based on other known deviation ratio values.

Proposition 3. *Given a set of pairs $\mathcal{S} := \{(\rho_\ell, h(\rho_\ell))\}_\ell \cup \{(M,M),(1,-1)\}$ with M sufficiently large and a candidate pair (e_i, e_j). If $D_i = -1$, set $\rho_1 = 1$, otherwise, let $\rho_1 = \min_{(\rho_k, h(\rho_k))\in\mathcal{S}}\{\rho_k : h(\rho_k) > D_i\}$. Define $\rho_2 := \max_{(\rho_k, h(\rho_k))\in\mathcal{S}}\{\rho_k : h(\rho_k) < D_j\}$. It holds that e_i has a larger exact score than e_j, that is, $\widehat{s}_i^{l\times r} > \widehat{s}_j^{l\times r}$, if*

$$S_i^{l\times r} > S_j^{l\times r}\cdot\delta_a \tag{5}$$

is satisfied, where $\delta_a = \min\{a(D_i^l, D_j^l)\cdot a(D_i^r, D_j^r), a(D_i^l, D_j^r)\cdot a(D_i^r, D_j^l)\}$, and

$$a(D_1,D_2) = \begin{cases} \min\left\{\delta, \frac{\rho_1}{\rho_2}\right\}, & \text{if } D_1 = D_2 \le \underline{d}, \\ \min\left\{1, \frac{\rho_1}{\rho_2}\right\}, & \text{if } D_1 < D_2, \\ \frac{\rho_1}{\rho_2}, & \text{if } D_1 > D_2 \text{ or } D_1 = D_2 > \underline{d}. \end{cases}$$

Condition (5) is checked at the beginning of Step 2 in Algorithm 1. If it is satisfied, Step 2 terminates directly with p_k set to NULL and it moves directly to Step 3. In practice, testing four starting child LPs is often sufficient for Condition (5) to conclude that no further testing is necessary. This significantly accelerates the entire selection process, especially when $|\widetilde{\mathcal{C}}|$ is 10 or larger.

Remarks. Although the heuristic testing in 3PB and the proposed partial testing procedure in our 2LBB expedite the selection through different mechanisms, the best numerical performance is achieved when these approaches are combined. Specifically, we replace the exact CG with heuristic CG in Algorithm 1 in our implementation when performing the testing. This is motivated by the observation that heuristic CG is significantly faster than exact CG, and more importantly, the optimal value obtained from heuristic CG serves as a reliable way to approximate that of the ending child LP, leading to the same selection outcome as exact CG in most cases. Although Propositions 1 and 3 may no longer guarantee sufficient conditions for certifying a larger exact score, the reliability of these approximations ensures that such cases are rare in practice.

4.5. Learning Methodology

The two ML models, *M1* and *M2*, in our ML module aim to predict the correct ranking with different learning goals. For *M1*, the goal is to rank candidates based on their exact scores, whereas for *M2*, the goal is to rank candidates based on the deviation ratio. *Learning to rank*, compared with more common supervised learning methods such as regression or classification, tends to achieve higher accuracy in our task because the learning objective is relatively easier.

The input for a ranking task typically consists of multiple lists, each containing several items. For the kth list with n_k items, let $\vec{f}_i^k \in \mathbb{R}^d$ denote the d-dimensional feature vector of item i, and $r_i^k \in \{1,\dots,n_k\}$ denote its rank in descending order based on exact scores or the deviation ratio. Each list can be expressed as (x^k, y^k), where $x^k = \{\vec{f}_1^k,\dots,\vec{f}_{n_k}^k\}$ and $y^k = \{r_1^k,\dots,r_{n_k}^k\}$, forming training data for learning-to-branch tasks. However, directly using ranks as labels is unnecessary because candidates with similar scores typically have comparable performance. To reduce overfitting and enhance generalization, we recompute labels $\bar{y}_i^k$ following the approach in Yang et al. (2022):

$$\bar{y}_i^k = W - \max\left\{0, \left\lceil \frac{\tilde{s}^k - s_i^k}{\tilde{s}^k - s_{(n_k)}^k}\cdot W \right\rceil\right\}, \quad \text{where}$$
$$\tilde{s}^k = \alpha\cdot s_{(1)}^k, \quad \alpha\in(0,1). \tag{6}$$

Candidates with scores above the threshold $\tilde{s}^k$ (commonly set with $\alpha = 0.8$) receive the maximum label W, indicating equally optimal choices for branching.

We characterize each branching candidate using three types of features: $\mathcal{F}_E$, $\mathcal{F}_I$, and $\mathcal{F}_D$. The $\mathcal{F}_E$ features provide detailed LP-related information, including dual solution values and changes in LP objective values obtained from solving child LPs without using CG, which are computationally expensive. To the best of our knowledge, the $\mathcal{F}_E$ features are completely new in the literature. In contrast, $\mathcal{F}_I$ and $\mathcal{F}_D$ are inexpensive to compute. The $\mathcal{F}_I$ features capture the fundamental properties of candidates that are independent of previous LP results, whereas the $\mathcal{F}_D$ features are derived from previously computed $\mathcal{F}_E$ values. Together, these features support a two-stage learning approach (*M1* and *M2*), which effectively selects and ranks branching candidates. For detailed descriptions of feature definitions, computations, data collection, and training procedures, please refer to Section EC.3 of the e-companion.

5. Theoretical Model for Determining the Number of Candidates

Branching strategies, such as the SB, 3PB, and 2LBB, require performing computationally intensive testing on a set of branching candidates to achieve small-sized BBTs. Nonetheless, determining the size of the set to test so that the overall solution time is minimized remains challenging because of a lack of a systematic way to balance the tradeoff between the time consumed for making a single branching decision and the decision quality. The size is typically decided by a trial-and-error numerical approach and remains unchanged once decided. In Section EC.7.2 of the e-companion, we numerically decide a good static $\hat{\theta}$ for the 2LBB as a baseline. In this section, we propose a novel theoretical model that theoretically decides the best size based on the solution progress, which is the first to embed a distribution of the local branching improvements into a tree size estimation. More precisely, we derive the so-called *best K formula* (BKF) to dynamically adjust the size to achieve the minimum total computational time under our assumptions.

The BKF is broadly applicable to branching strategies that employ one-phase testing, such as SB, or multiphase testing where either only one phase's duration can vary (e.g., 3PB) or one phase dominates the overall computation time (e.g., 2LBB). We refer to such a phase as the *key testing phase*. In 3PB and 2LBB, LP testing generally serves as the key phase. Notably, in 2LBB, this phase often accounts for more than 80% of the total branching time. Therefore, BKF determines $\hat{\theta}_{pf}$ in 3PB and $\hat{\theta}$ in 2LBB. For multiphase strategies, the applicability of BKF hinges on the assumption that the total branching variable selection time is approximately equal to the time spent on the key testing phase, plus $\tilde{c}$ representing the combined duration of other phases. If $\tilde{c}$ is negligible compared with the key phase, it can be treated as a constant. Unless stated otherwise, all proofs related to these derivations are included in Section EC.2 of the e-companion.

5.1. Derivation of the BKF

Given a set of n branching candidates, we seek to decide the number, k, of candidates randomly chosen from this set to go through the key testing phase such that the minimum possible time for solving the problem is achieved. Our theoretical model builds on the following assumptions.

Assumption 4. *The BBT is a binary tree and thus* $n_{\text{inner}} = n_{\text{leaf}} - 1$, *where* n_{inner} *is the number of inner BBNs, and* n_{leaf} *is the number of leaf BBNs. The time for solving each inner BBN consists of those consumed by branching, pricing, and adding cuts. For leaf BBNs, branching is not needed and the time only comprises the other two parts.*

Assumption 5. *It takes a constant time* $2t$ *to perform testing for a single branching candidate (two branches), a constant time* $\tilde{c}$ *for other phases in making a branching decision, and a constant time* c' *for pricing and adding cuts at a BBN. Moreover,* $c \geq t$, *where* $c := c' + \frac{\tilde{c}}{2}$.

Assumption 6. *The tree size estimation follows Le Bodic and Nemhauser (2017). For a given branching candidate, the corresponding dual bound improvements at the left and right child BBNs, denoted by* o_1 *and* o_2, *respectively, remain constant with* $o_1 \geq o_2 \geq 1$. *We define testing score* r *to be the geometric mean of* o_1 *and* o_2; *that is,* $r := \sqrt{o_1 \cdot o_2}$. *The value of* r *varies for different branching candidates.*

Assumption 7. *The testing score for the* n *branching candidates take distinct values from* $\{\frac{j}{n} \cdot R : j = 1, \ldots, n\}$, *where* R *is maximum testing score. The value of* r_i *for candidate* i *is not known until it undergoes the testing.*

Let $N(G)$ denote the size of the subtree rooted at the current BBN with a local optimality gap of G. In particular, for each leaf BBN, $N(0) = 1$. According to Le Bodic and Nemhauser (2017), $N(G)$ can be approximated by $N(G) = \varphi^{G-F} N(F)$, where F is no larger than G, and φ is the unique solution to the equation $x^{o_1} - x^{o_1 - o_2} - 1 = 0$. Given that φ does not admit a closed form, analyzing the properties of equations involving φ is complex. Therefore, we propose to approximate φ by $\varphi_g := 2^{\frac{1}{r}}$. Theorem 1 characterizes the quality of such approximation.

Theorem 1. *The approximation error* $\left|\frac{\varphi_g - \varphi}{\varphi_g}\right| \leq \frac{2^{-\frac{1}{o_1 v}} \left|2^{\frac{1}{v}} - 2^{-v + \frac{1}{v}} - 1\right|}{o_1 v^2}$, *where* $v := \sqrt{\frac{o_2}{o_1}} \in (0, 1]$. *Moreover, the error is monotonically decreasing in terms of* v *when* $v \geq \frac{1}{3}$.

Theorem 1 bounds the approximation error. In particular, when $o_1 = 10$ and $v = \sqrt{\frac{1}{2}}$, which is typical for VRP instances, the error is no greater than 0.006. It is worth mentioning that this approximation only needs to be applied to the best one from the set of candidates considered and a good candidate usually has a value of o_2 close to that of o_1, making v close to one and thus the error bound close to zero. Consequently, φ_g serves as a good approximation, and the tree size $N(G)$ is estimated to be $N(F) \cdot 2^{\frac{G-F}{r}}$ in our subsequent analysis.

The total computation time for the subtree is calculated as $T(k) = (c' + \tilde{c} + 2kt) n_{\text{inner}} + c' \cdot n_{\text{leaf}} = c' N(G) + (\frac{\tilde{c}}{2} + kt)(N(G) + 1) \approx (c' + \frac{\tilde{c}}{2} + kt) N(G) = (c + kt) N(G)$. When the calculation is performed at the root, $T(k)$ estimates the total time it takes to solve the problem. Combining the above two equations leads to

$$T(k) \approx (c + kt) N(F) \cdot 2^{\frac{G-F}{r}}. \tag{7}$$

In BPC algorithms, enumeration tends to succeed when the local gap is small enough, helping close a BBN fast. Therefore, the subtree rooted at the current

BBN, in this case, approximately has a constant size. In other words, when F is close to zero, $N(F)$ can be considered constant that is independent of k or r. If $N(F)$ can be estimated accurately, such information should be exploited. Otherwise, we can always use the case $F = 0$ with $N(0) = 1$ in Equation (7).

Intuitively speaking, increasing the number of candidates to test is likely to result in a higher testing score because of the increased probability of identifying the best candidate. The monotone relationship between r and k poses challenges in determining the value of k that minimizes $T(k)$. To address this, we first build a quantified connection between r and k. Let $j_1, \dots, j_k$ be the k randomly selected from the n candidates. After testing, the candidate with the highest score among the k tested ones is selected. Thus, in view of Assumption 7, the final testing score $r = \max\{r_{j_1}, \dots, r_{j_k}\}$ is a random variable. We proceed by examining the following two cases.

(i) Pure random selection. This case applies when there is no prior knowledge about the n candidates such as in the SB. The expected value of $T(k)$ is computed by

$$\begin{aligned}\mathbb{E}[T(k)] &= \mathbb{E}\left[(c+kt)N(F)\cdot 2^{\frac{G-F}{r}}\right] \\ &= (c+kt)N(F)\sum_{i=k}^{n}\frac{\binom{i-1}{k-1}}{\binom{n}{k}}2^{\frac{G-F}{\frac{i}{n}\cdot R}}. \end{aligned} \tag{8}$$

(ii) Conditional random selection. The selection is conditioned on that only candidates with a testing score larger than $\frac{(n-m)R}{n}$ will be selected:

$$\begin{aligned}T_k := \mathbb{E}[T(k)|m] &= \mathbb{E}\left[(c+kt)N(F)\cdot 2^{\frac{G-F}{r}}|m\right] \\ &= (c+kt)N(F)\sum_{i=k}^{m}\frac{\binom{i-1}{k-1}}{\binom{m}{k}}2^{\frac{G-F}{\frac{i+n-m}{n}\cdot R}}. \end{aligned} \tag{9}$$

Case (ii) applies to the usage of *M1* because *M1* possesses some basic prediction power enabling it to effectively exclude $(n-m)$ unpromising ones. The output $k = \hat{\theta}$ candidates are considered as being randomly selected from the best m ones. When $m = n$, case (ii) reduces to (i), so case (i) can be considered as a special case. Therefore, we focus on case (ii) in the following discussion.

Proposition 4. *Given two branching strategies with a respective constant number of testing k_1 and k_2 $(k_1 > k_2)$ and respective m values of m_1 and m_2, the ratio of their expected solving time for the same instance $\tau := \frac{\mathbb{E}[T(k_1)|m_1]}{\mathbb{E}[T(k_2)|m_2]}$ decreases as ω increases, where $\omega := \frac{c}{t}$.*

The above proposition can be readily proved as follows. For the same instance, c and t remain unchanged across different strategies. In view of Equation (9), it follows that τ is proportional to $\frac{\omega+k_1}{\omega+k_2}$. Because $k_1 > k_2$ and $\frac{\omega+k_1}{\omega+k_2} = 1 + \frac{k_1-k_2}{\omega+k_2}$, we have that τ decreases with ω. Proposition 4 suggests that for instances with a larger ω, overestimating the parameter k is less detrimental when $\tau > 1$. This insight can provide guidance on tuning k for instances with distinct structures. Additionally, upon successful enumeration at a BBN, pricing becomes much faster; that is, ω significantly decreases. In this case, a larger k is less desirable, and thus k should be reduced. Such a change in ω will be further studied numerically in Section 6.

Proposition 5. *Given two branching strategies selecting the same number of candidates, that is, $k_1 = k_2 = k$, but with $m_1 > m_2$, it follows that $\tau > 1$.*

Proposition 5 indicates that for two strategies using the same number of testing but differ in their m values, the one with a larger m will have a longer computational time than the other. Intuitively, a smaller m suggests a higher selection quality, leading to a more effective branching scheme. This correlation between m and τ is further supported by the comparison of the two case studies in Sections 6.1 and 6.2.

Let $k^* \in \arg\min_{1\le k\le m} T_k$; that is, when k is set to k^*, the expected computational time is minimized. When m is relatively small, k^* can be obtained by computing T_k for $k = 1, \dots, m$, and select the smallest one. However, for large m, such computations become prohibitively expensive. In the next section, we provide an efficient way to compute k^*.

5.2. Asymptotic Study

In this section, we analyze the asymptotic behavior of the total computational time T_k as $n \to \infty$, which gives rise to a practical way of dynamically adjusting the key parameter $\hat{\theta}$ for *M1*. In this analysis, *M1* is considered to have a fixed prediction power in the sense that its m value grows proportionally with n. In other words, $\alpha := \frac{m}{n}$ is a constant with $\alpha \in (0,1)$. We demonstrate that k^* can be computed without explicitly computing T_k for every k in the range $\{1, \dots, m\}$. More specifically, we narrow down the possible values of k^* to just several options. Additionally, we establish that as $n \to \infty, T_k$ possesses a definite integral representation. This integral form facilitates a more efficient computation of T_k. Let $\mathbb{E}[r|m]$ be the conditional expectation of r given m. Lemma 1 gives a closed form for computing it.

Lemma 1. *The conditional expectation of r can be computed by $\mathbb{E}[r|m] = \sum_k^m \frac{\binom{i-1}{k-1}}{\binom{m}{k}}\frac{i+n-m}{n}R = \frac{(n+1)k+n-m}{n(k+1)}R$.*

Define $\underline{T}_k := (c+kt)N(F)2^{\frac{G-F}{\mathbb{E}[r|m]}}$. In view of the convexity of the function $f(x) = 2^{1/x}$ and Jensen's inequality, $\underline{T}_k$ lower bounds T_k. For any given $k \in \{1, \dots, m\}$,

Theorem 2 establishes the existence of limits for T_k and $\underline{T}_k$ as $n \to \infty$, which enables us to focus on $\underline{T}_k$ when analyzing the asymptotic behavior of T_k.

Theorem 2. *For $k \in \{1,\dots,m\}$, the limits $\widehat{T}_k := \lim_{n\to\infty} T_k$ and $\widehat{\underline{T}}_k := \lim_{n\to\infty} \underline{T}_k$ exist. Moreover, $\widehat{T}_k = (c+kt)kN(F)\int_0^1 x^{k-1}B^{\frac{1}{\alpha x+1-\alpha}}\mathrm{d}x$ and $\widehat{\underline{T}}_k = (c+kt)N(F)B^{\frac{1}{1-\alpha/(k+1)}}$, where $B := 2^\beta$ and $\beta := \frac{G-F}{R}$.*

Theorem 3. *The sequence $\widehat{U}_k := (c+kt)kN(F)B^{\frac{1}{1-\alpha}}(\frac{\alpha}{1-\alpha}\ln(B))^{-k}\gamma\left(k, \frac{\alpha}{1-\alpha}\ln(B)\right)$ upper bounds $\widehat{T}_k$, where $\gamma(s,x) := \int_0^x t^{s-1}e^{-t}\mathrm{d}t$ is the lower incomplete gamma function.*

For any given $\hat{k} \in \{1,\dots,m\}$, we define $\Delta_{\hat{k}} := \{1 \le k \le m : \widehat{\underline{T}}_k > \widehat{U}_{\hat{k}}\}$. According to Theorems 2 and 3, $\widehat{T}_k$ is lower bounded by $\widehat{\underline{T}}_k$ and upper bounded by $\widehat{U}_k$. Consequently, any $k \in \Delta_{\hat{k}}$ cannot achieve the smallest $\widehat{T}_k$. Furthermore, the following Theorem 4 describes the monotonicity property of $\widehat{\underline{T}}_k$ and suggests a convenient way to determine $\Delta_{\hat{k}}$ without computing $\widehat{\underline{T}}_k$ for every $k \in \{1,\dots,m\}$.

Theorem 4. *$\widehat{\underline{T}}_k$ is monotonically decreasing for $k \in \{1,\dots,\underline{k}^*\}$ and monotonically increasing for $k \in \{\underline{k}^*,\dots,m\}$, where $\underline{k}^* = \max\{\arg\min_{1\le k\le m} \widehat{\underline{T}}_k\}$.*

Combining the lower and upper bounds of $\widehat{T}_k$ with its monotonicity, we can narrow down the range of k^* to a small number of options as shown in Theorem 5. In Proposition 6, we further provide an explicit expression for $\underline{k}^*$, enabling us to decide the value of k^* with even less computation.

Theorem 5. *Let k_l' be the largest $k < \underline{k}^*$ and k_r' be the smallest $k > \underline{k}^*$, respectively, such that $\widehat{\underline{T}}_k \ge \widehat{U}_{\underline{k}^*}$. It then follows that k^* belongs to the set $\{k_l'+1,\dots,k_r'-1\}$.*

In view of the monotonicity of $\widehat{\underline{T}}_k$, both k_l' and k_r' can be efficiently computed.

Corollary 1. *When $\widehat{\underline{T}}_{\underline{k}^*}+1 \ge \widehat{U}_{\underline{k}^*}$ and $\widehat{\underline{T}}_{\underline{k}^*}-1 \ge \widehat{U}_{\underline{k}^*}$, it holds that $k^* = \underline{k}^*$.*

Corollary 1 provides a sufficient condition for $k^* = \underline{k}^*$, which is a special case of Theorem 5 with $k_l' = \underline{k}^* - 1$ and $k_r' = \underline{k}^* + 1$. In this case, there is no need to compute any $\widehat{T}_k$. It is worth mentioning that the lower incomplete gamma function can be computed very efficiently (in microseconds) by many existing numerical libraries (e.g., Boost C++ library), rendering the evaluation of $\widehat{U}_k$ significantly faster than that of $\widehat{T}_k$.

Corollary 2. *Let k_l'' be the largest $k < \underline{k}^*$ and k_r'' be the smallest $k > \underline{k}^*$, respectively, such that $\widehat{\underline{T}}_k \ge \widehat{T}_{\underline{k}^*}$, it follows that $k_l'' \ge k_l'$ and $k_r'' \le k_r'$. Moreover, $k^* \in \{k_l''+1,\dots,k_r''-1\}$.*

Corollary 2 can be proved straightforwardly. According to Theorem 3, we have $\widehat{T}_{\underline{k}^*} \le \widehat{U}_{\underline{k}^*}$. In view of the monotonicity of the $\widehat{\underline{T}}_k$ and the definitions of k_l'', k_r'', k_l', and k_r', we have $k_l'' \ge k_l'$ and $k_r'' \le k_r'$. Therefore, Corollary 2 further narrows the range of k^*. For $k \le k_l''$, it follows that $\widehat{T}_k \ge \widehat{\underline{T}}_k \ge \widehat{\underline{T}}_{k_l''} \ge \widehat{T}_{\underline{k}^*}$, which leads to $k^* \ge k_l'' + 1$. For $k \ge k_r''$, we also have the same inequality, proving the other direction; that is, $k^* \le k_r'' - 1$. Therefore, k^* belongs to the set $\{k_l''+1,\dots,k_r''-1\}$. Let

$$k_1 = \sqrt{(\alpha\beta\ln\sqrt{2})(\alpha\beta\ln\sqrt{2}+2\alpha+2\omega-2)} + \alpha\beta\ln\sqrt{2}+\alpha-1. \tag{10}$$

Proposition 6. *If $k_1 > 1$, then $\underline{k}^* \in \arg\min_{k\in\{\lfloor k_1\rfloor,\lceil k_1\rceil\}} \widehat{\underline{T}}_k$.*

Proposition 6 computes $\underline{k}^*$ by evaluating the value of k_1 and suggests that when n is sufficiently large, $\underline{k}^*$ tends to be a constant.

The procedure for efficiently determining the value of k^* when n is large is outlined as follows.

1. Compute k_1 by Equation (10). If $k_1 \le 1$, then $\underline{k}^* = 1$, otherwise, pick $\underline{k}^* \in \arg\min_{k\in\{\lfloor k_1\rfloor,\lceil k_1\rceil\}} \widehat{\underline{T}}_k$.

2. Check if $\widehat{\underline{T}}_{\underline{k}^*}+1 \ge \widehat{U}_{\underline{k}^*}$ and $\widehat{\underline{T}}_{\underline{k}^*}-1 \ge \widehat{U}_{\underline{k}^*}$ hold. If so, then set $k^* = \underline{k}^*$. Otherwise, compute k_l'', k_r'' and pick $k^* \in \arg\min_{k\in\{k_l''+1,\dots,k_r''-1\}} \widehat{T}_k$.

6. Numerical Study

Our extensive numerical study is performed using RouteOpt, an exact solver for the CVRP and VRPTW, achieving state-of-the-art performance. Some implementation details of RouteOpt are included in Section EC.5 of the e-companion. Although the default configuration of RouteOpt, which activates all optimization techniques, is most suitable for practical use, the simultaneous deployment of cutting planes, arc elimination, and enumeration at all BBNs can significantly interfere with the branching process. To clearly demonstrate the effectiveness of our methodology, we also evaluate a second configuration where cutting planes and arc elimination are applied only at the root node, with enumeration completely disabled. Restricting cutting planes to the root node during branching studies is standard practice (Khalil et al. 2016, Alvarez et al. 2017, Yang et al. 2022), and methods such as arc elimination and enumeration are not typically used in standard B&C algorithms. For convenience, we denote experiments conducted with the default (first) and second configurations of RouteOpt as Settings I and II, respectively (Table 2). In Sections 6.1 and 6.2, we

Table 2. Summary of the Configurations Used in the Two Experimental Settings

Setting	Cutting plane	Arc elimination	Enumeration
I	All BBNs	All BBNs	Activated
II	Root node	Root node	Disabled

present two case studies on the CVRP and VRPTW, respectively, that demonstrate the effectiveness of the 2LBB derived from the proposed framework. Finally, Section 6.3 is dedicated to our attempt to solve open CVRP instances. Preliminary experiments performed to determine and justify the design of some key components of our learning-to-branch framework are presented in Section EC.7 of the e-companion.

For the instances tested in this study, we adopt a systematic naming convention to categorize and clarify the characteristics of each batch. The naming format is *problem-size-replication-type*, where *problem* specifies the type of problem (e.g., VRPTW or CVRP); *size* denotes the number of customers, with *V* indicating a varied size within the batch; *replication* represents the number of instances in the batch; and *type* identifies the category of instances, including *0, 1, X*, and *H2*. Specifically, *0* and *1* correspond to newly generated instances, where *0* is used for training data generation and *1* for performance evaluation. The *X* category refers to the X series benchmarks for CVRP (Uchoa et al. 2017), whereas *H2* denotes type 2 instances for VRPTW (Homberger and Gehring 2005). We used instance batches CVRP-120-90-1, CVRP-150-90-1, CVRP-180-90-1, and CVRP-V-42-X for CVRP and VRPTW-160-60-1, VRPTW-180-60-1, VRPTW-200-60-1, and VRPTW-200-22-H2 for VRPTW to demonstrate the improvement achieved by the learned branching strategy when integrated into RouteOpt. Further details on the generation, attributes of these instances, and the rationale behind their selection can be found in Section EC.4 of the e-companion.

In our experiments, two sets of computing resources were used: HiPerGator and Dell workstations. HiPerGator is a supercomputing cluster at the University of Florida. We used the hpg-default partition running Red Hat Enterprise Linux 8.8, where each node is equipped with two AMD EPYC 770264-core processors @2.0 GHz and 1003 GB RAM. On Hipergator, the code was compiled by g++ 12.2.0. Our group has access to 1080 CPU cores and 8640 GB RAM, which allows us to run hundreds of jobs simultaneously. Training data generation was efficiently conducted using this computing resource. For experiments necessitating precise time comparisons, we used four identical Dell workstations. Each workstation runs on Ubuntu 20.04 LTS and is equipped with the 12th Gen Intel(R) Core(TM) i9-12900K CPU @3.2 GHz and 128 GB RAM. On the workstations, g++ version 9.4.0 was used to compile the code. For all experiments in this section, Gurobi 10.0.0 (Gurobi Optimization, LLC 2023) was used as the LP and MIP solver inside our RouteOpt, except in Section 6.1.4, where CPLEX 22.1.0 (IBM Corporation 2023) was employed to ensure a fair comparison with VRPSolver. All code was run using a single thread. The source code of RouteOpt, together with all the ML code and CVRP and VRPTW instances, has been made publicly available on GitHub.

6.1. Case Study I: CVRP

In this case study, we use the CVRP to demonstrate the performance of our framework. We integrate the trained ML module into RouteOpt and evaluate the performance improvement by solving instances in batches CVRP-120-90-1, CVRP-150-90-1, CVRP-180-90-1, and CVRP-V-42-X, aiming to highlight the superiority of 2LBB over the benchmark 3PB and emphasize its generalization capability. Based on preliminary experiments in Section EC.7 of the e-companion, we set $\hat{\theta}_{pf} = 100$, $\hat{\theta} = 10$, and $\tilde{\theta} = 3$ for 2LBB, with the only difference for 3PB being $\tilde{\theta} = 2$. For the dynamic version, we assign $\eta = 0.72$, whereas α is set to 0.60 for 3PB-dy and 0.37 for 2LBB-dy, respectively. For the parameters of the second-stage selection, we set $\underline{d} = 2$ and $\delta = 1$. To assess the performance difference due solely to branching, experiments on batches CVRP-120-90-1, CVRP-150-90-1, and CVRP-180-90-1 are conducted under experimental Setting II, using the optimal value u^* or $u^0 := (1 + 0.1\%)u^*$ as the upper bound. To further demonstrate its generalization across different settings and compatibility with other techniques, CVRP-V-42-X is tested under Setting I. Details on training the ML models are included in Section EC.9.1 of the e-companion.

6.1.1. Superiority of the 2LBB. We compare the performance of RouteOpt using the 2LBB and the benchmark 3PB for instance batches CVRP-150-90-1. Additionally, we include two corresponding dynamic adjustments, that is, 3PB-dy and 2LBB-dy, which leverage the BKF to dynamically determine the number of time-consuming testing conducted. We report the time for solving the root node (Root/s), the root gap (Root gap/%), LP testing time (LP/s), heuristic testing time (Heuristic/s), the overall solution time (CPU/s), and the number of nodes explored (No. of nodes). Tables 3 and 4 present the geometric means of these metrics when two upper bounds (UBs) are used, that is, u^* and u^0, respectively. Detailed results for each instance are included in Sections EC.10.1 and EC.10.2 of the e-companion.

We observe that when u^* is used as the UB, employing the learning models is beneficial, regardless of whether BKF is applied. Specifically, when BKF is not used, the time reduction (2LBB versus 3PB) can reach 36%, whereas with BKF, the time reduction (2LBB-dy versus 3PB-dy) is approximately 27%. This lower ratio suggests that BKF may play a more significant role in the 3PB strategy. Indeed, in lower-level branching decisions, where limited historical information is available, the performance of the initial screening—on which 3PB heavily relies—is significantly deteriorated. However,

Table 3. Comparison of 3PB, 2LBB, 3PB-dy, and 2LBB-dy When u^* Serves as the UB

Version	Root/s	Root gap/%	LP/s	Heuristic/s	CPU/s	No. of nodes
3PB	24.5	0.13	218.1	139.8	570.2	297.9
2LBB			166.2	29.2	362.5	222.1
3PB-dy			152.0	108.9	436.8	245.5
2LBB-dy			102.9	21.8	317.2	270.1

when incorporating *M1*, 2LBB is less affected by the lack of historical information. Consequently, adopting a strategy of testing more at lower levels and less at higher levels can be expected to enhance the performance of 3PB more effectively.

From the perspective of the node metric, we observe less node reduction (25%) than time reduction (36%) when comparing 2LBB with 3PB. The additional time reduction arises from the second stage, where fewer heuristic testing are required while maintaining a similar tree size, as shown in Section EC.7.3 of the e-companion. This time saving also causes 2LBB to require less time on average to solve a BBN (excluding LP testing), that is, a smaller c, which further contributes to different behavior observed for BKF. Specifically, 3PB-dy processes 18% fewer nodes than 3PB, whereas 2LBB-dy processes 22% more nodes than 2LBB. Note that in BKF, a smaller c often leads to fewer testing being performed and thus a larger tree size. Despite these differences in tree size, both 3PB and 2LBB benefit from BKF, achieving time reductions of 23% and 13%, respectively. Overall, the 2LBB enhanced with BKF achieves the best performance, with a 44% time reduction compared with the baseline 3PB.

For the case where u^0 is provided as the UB, similar to the conclusion drawn from Table 3, 2LBB still outperforms 3PB regardless of whether BKF is applied. However, the time reduction of 2LBB versus 3PB decreases to 31%. This reduction also applies to the versions using BKF. Further investigation reveals that, although 2LBB finds better solutions or identifies the optimal solution more quickly than 3PB, the ratio of the time spent finding the solution to the overall solution time is lower for 3PB. Specifically, as shown in Figure 5, 3PB typically finds a better UB after spending 60% of the overall time, whereas 2LBB requires 73%. When a better solution is found, a large number of nodes can be pruned, accelerating the remaining solution process. We believe this explains why the time reduction diminishes. Another interesting observation is that 2LBB-dy explores 7% fewer nodes than 2LBB, which is the opposite of what we observed in Table 3. This is because, when given a larger UB, the tree size is larger, making BKF tend to run more LP testing to achieve the best possible tradeoff between testing efforts and tree size.

6.1.2. Generalization Capability of the 2LBB.

Generalization refers to the capability of a trained ML model to effectively adapt to new, previously unseen data drawn from the same distribution as the one used during model training. Generalization affects the practicability of a trained model. We investigate the generalization of our 2LBB by evaluating its performance in solving instances of different sizes. More specifically, we collect the training data by solving instance batch CVRP-150-270-0 and apply the resulting 2LBB to solve batches CVRP-120-90-1, CVRP-150-90-1, and CVRP-180-90-1. As illustrated by the ratios of tree size and CPU time with respect to the 3PB in Figure 6, the effectiveness of the 2LBB approach is consistent across instances with 120, 150, and 180 customers. These results confirm that our methodology generalizes well. Especially, the node ratio decreases as the size increases, indicating that 2LBB is potentially more powerful in solving larger instances.

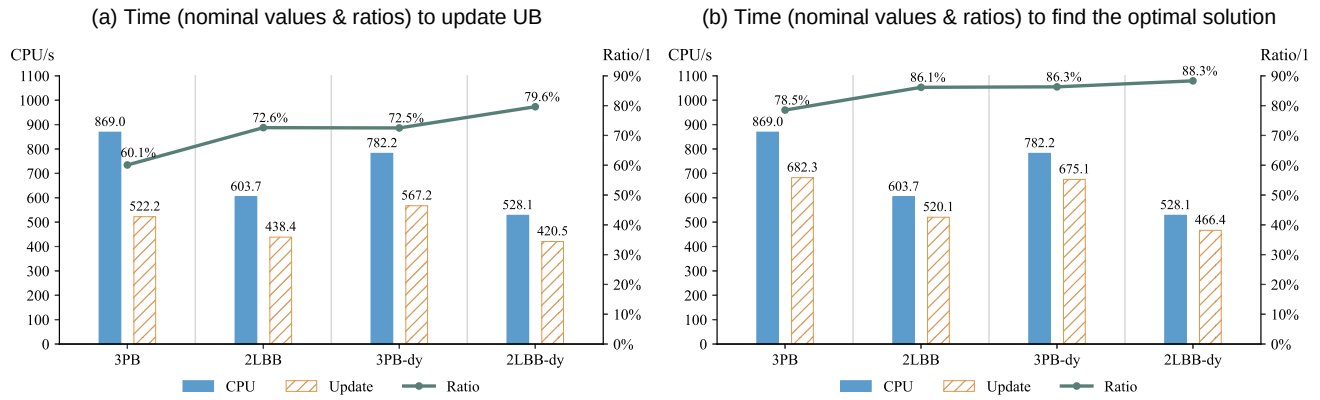
6.1.3. Performance on Established CVRP Benchmark Instances.

In this section, we report the computational results for the established CVRP benchmark instances, that is, batch CVRP-V-42-X. Most of these instances remain very challenging to prove optimality within several days, even with the best-known value as the initial UB. Some instances may still remain unsolved. Only reporting the results of instances that can be solved within a few hours is not enough to draw rigorous conclusions. Therefore, to include as many instances as possible, we decided to solve the more difficult instances (34 of 42 instances) by setting

Table 4. Comparison of 3PB, 2LBB, 3PB-dy, and 2LBB-dy When u^0 Serves as the UB

Version	Root/s	Root gap/%	LP/s	Heuristic/s	CPU/s	No. of nodes
3PB	24.2	0.24	356.3	238.1	869.0	463.8
2LBB			317.8	71.9	603.7	398.3
3PB-dy			390.2	181.3	782.2	347.6
2LBB-dy			269.2	57.6	528.1	372.1

Figure 5. (Color online) Comparison on Time (Nominal Values and Ratios) for Updating the UB and Finding the Optimal Solution



an initial *cutoff* that is geometrically 0.14% lower than the best-known value, allowing the solver to stop within approximately two hours. It is important to note that the specified cutoff might not always represent a valid UB. The solver could terminate without finding any feasible solution. In this case, we conclude the given initial cutoff is a valid lower bound (LB). However, if at least one feasible solution is found, then the output solution is guaranteed to be optimal. For clarity, we consistently use the term *solve* to encompass both outcomes.

The key reason for not comparing the LB within the same time limit, as often done in other papers, is that BKF is not designed to maximize the LB under tight running time constraints but rather focuses on minimizing the total solution time, even though this necessitates spending significantly more time at the early stages of branching. In the dynamic version, fewer nodes are explored if the solver stops prematurely, potentially resulting in a worse LB and leading to the incorrect conclusion that the dynamic version underperforms. However, the dynamic version may significantly outperform the static version in later stages, because the initial intensive efforts lead to fewer nodes needing to be solved, each also requiring less testing. Ultimately, the dynamic version requires less time for the complete solution process. Additionally, it is both inconsistent and challenging to estimate total time reduction by comparing LB improvements.

Table 5 shows that 2LBB-dy still achieves the best performance among all four versions of RouteOpt, specifically achieving a 15% time reduction compared with 2LBB and extending to a 38% time reduction for 3PB. Additionally, because cutting planes are enabled, which increases the average time excluding LP testing for solving a BBN, the dynamic version opts for more LP testing, which results in a node reduction of 31% and 16% for 3PB-dy versus 3PB and 2LBB-dy versus 2LBB, respectively. What is noteworthy is that 3PB-dy achieves a 28% time reduction compared with 3PB, demonstrating again the potential of the generalized and independent learning mechanism of BKF, which can significantly enhance its impact in learning-free branching strategies including the 3PB.

From the detailed results in Section EC.10.3 of the e-companion, we observed that for instances requiring fewer branchings—specifically, fewer than 64 using 2LBB—the time reduction of 2LBB-dy compared with

Figure 6. (Color online) Generalization Capability of 2LBB Across Different Instance Sizes

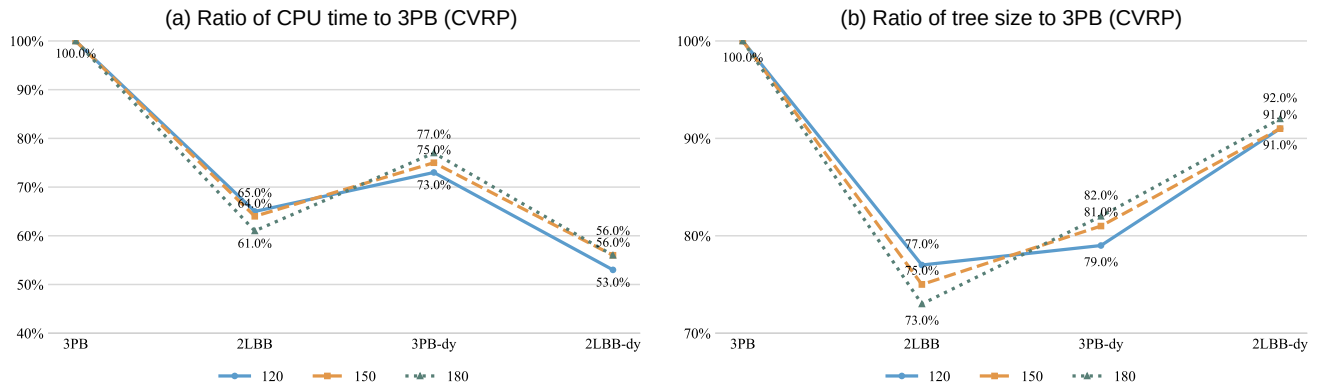


Table 5. Comparison of 3PB, 2LBB, 3PB-dy, and 2LBB-dy on Batch CVRP-V-42-X

Version	Root/s	Root gap/%	LP/s	Heuristic/s	CPU/s	No. of nodes
3PB	366.8	0.11	389.2	329.2	4,324.3	100.3
2LBB			321.9	91.2	3,184.3	74.9
3PB-dy			410.6	256.4	3,120.7	69.3
2LBB-dy			329.0	86.1	2,691.5	62.7

2LBB and 3PB is 9% and 32%, respectively. Conversely, for instances with more than 64 branchings, we observe time reductions of 22% and 43%, respectively. This larger time reduction for instances requiring extensive branching indicates that when more branchings are needed, a static $\hat{\theta}$ is less effective. Consequently, the application of BKF in solving more difficult instances gives rise to even more pronounced acceleration.

We also noticed that the time reduction brought by BKF decreases for CVRP as the average route length increases. Specifically, when using 8 as the threshold to categorize instances into two groups—short routes (<8) and long routes (≥8)—we observed that for the short group, the time reduction of 2LBB-dy compared with 2LBB is 22%, whereas it significantly decreases to 10% for the long group. A similar observation can be made for 3PB-dy. The reason is that instances with longer routes incur significantly higher pricing time for each node, resulting in a smaller ratio of branching selection time versus overall time. Even though BKF helps greatly reduce computational time for later branching selection, the overall time reduction remains relatively minor. Additionally, the long group instances have smaller BBT sizes than the short group, which further exacerbates this effect.

6.1.4. Comparison with the VRPSolver. The VRPSolver, developed in Pessoa et al. (2020), is recognized as the best-performing tool for solving the CVRP exactly. We compare its performance with the RouteOpt version equipped with 2LBB-dy, which achieves the best results among the four tested versions. For testing, we select the CVRP-V-42-X instance batch. To ensure a fair comparison, we use the same cutoff, the same LP and MIP solver, CPLEX (version 22.1.0 in our experiments; IBM Corporation 2023) with identical parameters. VRPSolver with extension version v0.5.24 and BaPCod version v0.74, both using default parameters, are employed to solve the instances.

From Table 6, we observe that 2LBB-dy processes the root node much faster, largely because of RouteOpt's less aggressive cutting plane addition compared with VRPSolver, which we refer to as the branching-focused strategy, in contrast to VRPSolver's cutting-focused strategy. We adopt the branching-focused strategy for two main reasons. First, in 2LBB-dy, good branching decisions can be made without excessive computational efforts, making it more efficient to focus on branching rather than performing numerous rounds of cutting plane addition that offer minimal improvement to the lower bound. Second, RouteOpt's current heuristics for separating rank-1 cuts are not as effective as VRPSolver's. RouteOpt typically has a larger root gap, approximately 0.03% larger in this case.

Despite such a larger root gap, RouteOpt still reduces the overall solution time by 47% compared with VRPSolver. RouteOpt requires significantly less time at each node for both the pricing and cutting phases thanks to the less aggressive cutting strategy and the 2LBB-dy scheme that efficiently makes good branching decisions. This substantial time reduction per node compensates for the significantly increased number of nodes explored. We emphasize that this acceleration is not simply the result of a direct switch to a branching-focused strategy. Rather, it is the effectiveness brought by 2LBB-dy that ensures each branching decision requires less computational effort without little sacrifice in quality, ultimately leading to the success of our RouteOpt.

6.2. Case Study II: VRPTW

We further demonstrate the effectiveness of 2LBB and BKF by applying them to various VRPTW instance batches: VRPTW-180-60-1, VRPTW-160-60-1, VRPTW-200-60-1, and VRPTW-200-22-H2. The critical parameters used, including those for the dynamic version (α and η), are consistent with those in the CVRP case study, with the exception of VRPTW-200-22-H2. For

Table 6. Comparison of VRPSolver and 2LBB-dy on Batch CVRP-V-42-X

Version	Root/s	Root gap/%	LP/s	Heuristic/s	CPU/s	No. of nodes
VRPSolver	1,178.8	0.08	336.1	116.4	5,773.3	21.8
2LBB-dy	429.1	0.11	316.6	146.2	3,055.1	64.0

Table 7. Comparison of 3PB, 2LBB, 3PB-dy, and 2LBB-dy with u^* Serving as the UB

Version	Root/s	Root gap/%	LP/s	Heuristic/s	CPU/s	No. of nodes
3PB	24.0	0.17	175.3	124.0	414.0	176.0
2LBB			141.9	24.1	254.1	138.8
3PB-dy			119.2	92.2	317.0	157.6
2LBB-dy			79.1	17.1	201.5	164.5

this instance batch, we observed a decay in the effectiveness of *M1* and selected a larger $\alpha = 0.45$ to address it. Additionally, we chose a smaller $\eta = 0.5$ in response to the decreased testing scores observed as branching progresses. All these instances are tested under Setting II, even for VRPTW-200-22-H2, because most of them require very few branchings or are very hard to solve even after hours of effort using Setting I. Therefore, no comparison with VRPSolver is made in this case study, because we do not enable RouteOpt to utilize all of its techniques. Readers can refer to Section EC.9.2 of the e-companion for training details.

6.2.1. Superior of the 2LBB. In this section, we compare the performance of four versions of RouteOpt for instance batch VRPTW-180-60-1 and report the same metrics used in the CVRP case study. Detailed results for each instance are provided in Sections EC.10.5 and 10.6 of the e-companion.

As shown in Table 7, the results are consistent with those from the CVRP case study. Specifically, the time reduction for 2LBB compared with 3PB is 39%, and for 2LBB-dy compared with 3PB-dy, it is 36%. 2LBB-dy exhibits the best performance, achieving a 51% time reduction compared with 3PB. Furthermore, by incorporating BKF, 3PB-dy reduces computational time by 23% compared with 3PB. This once again demonstrates the effectiveness of BKF over a static scheme, even in a learning-free branching strategy.

From Table 8, a similar reduction in time reduction is observed. Specifically, the time reduction of 2LBB compared with 3PB diminishes to 34%, and for 2LBB-dy compared with 3PB, the reduction is 48%. As previously explained in the CVRP case study, this diminishing effect can be attributed to the relatively early updating of the UB or the early finding of the optimal solution, as illustrated in Figure 7. Moreover, because the UB is larger, BKF tends to prioritize decreasing the tree size. Consequently, the node ratios for both 3PB-dy versus 3PB and 2LBB-dy versus 2LBB have decreased compared with the results when the optimal value u^* is used as the UB.

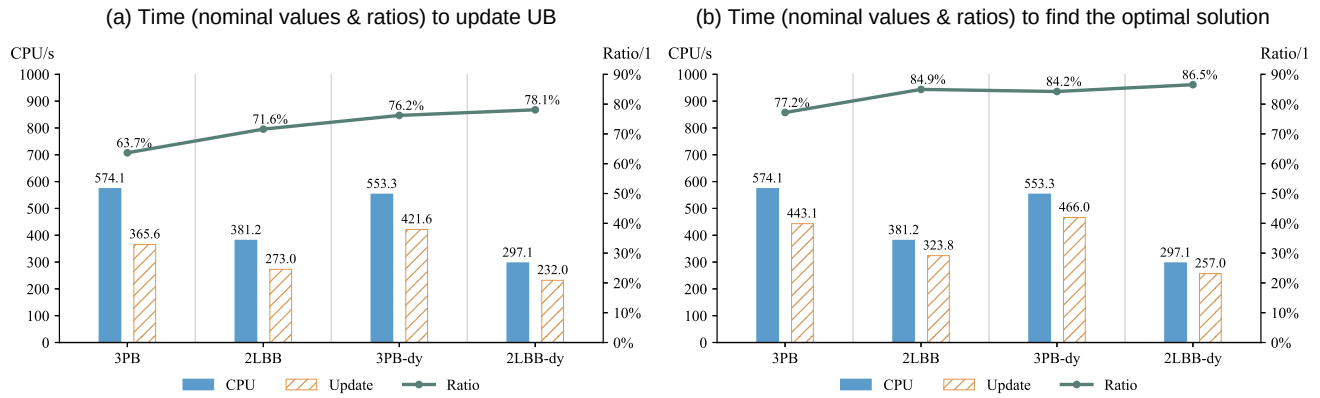
6.2.2. Generalization Capability of the 2LBB. In this section, we collect training data by solving instance batch VRPTW-180-600-0 and subsequently apply the trained *M1* and *M2* to solve instance batches VRPTW-160-60-1, VRPTW-180-60-1, and VRPTW-200-60-1. As illustrated by the ratios of tree size and CPU time with respect to 3PB in Figure 8, the effectiveness of the 2LBB approach is consistent across instances with 160, 180, and 200 customers, achieving an approximate 40% time reduction. This reduction increases to 50% after incorporating BKF. Even without introducing the learning models to RoutOpt, applying just BKF to 3PB results in a 10%–26% time reduction, with the reduction increasing as the instance size grows. This again indicates that BKF becomes even more effective when dealing with larger instances.

6.2.3. Performance on Established VRPTW Benchmark Instances. We report the computational results for the established VRPTW benchmark instances: batch VRPTW-200-22-H2. Our preliminary results suggest that these instances tend to favor an aggressive cutting strategy that achieves a very small root gap and thus requires minimal need for branching. This characteristic, unfortunately, is not ideal for demonstrating the effectiveness of our learning-to-branch scheme. To increase the reliance on branching decisions, we continue employing Setting II in this set of experiments. Furthermore, we stop adding cuts when the current gap between the LB and the cutoff is within 0.50%. This adjustment necessitates significantly more branching decisions, thereby suitable for demonstrating the differences among the four branching strategies. The cutoff is chosen to be 0.64% (on a geometric mean basis) smaller than the optimal value

Table 8. Comparison of 3PB, 2LBB, 3PB-dy, and 2LBB-dy When u^0 Serving as the UB

Version	Root/s	Root gap/%	LP/s	Heuristic/s	CPU/s	No. of nodes
3PB	21.8	0.28	258.0	187.8	574.1	245.9
2LBB			237.3	52.0	381.2	218.1
3PB-dy			269.4	164.9	553.3	220.1
2LBB-dy			158.0	39.8	297.1	219.9

Figure 7. (Color online) Comparison on Time (Nominal Values and Ratios) for Updating the UB and Finding the Optimal Solution



for 8 of the 22 instances to ensure the solver stops within a four-hour limit when using 2LBB-dy. For the remaining 14 relatively easy instances, the optimal value serves as the cutoff.

As shown in Table 9, instance batch VRPTW-200-22-H2 exhibits significantly different characteristics compared with other instances. Specifically, for the 3PB version, where each branching decision performs LP testing for 10 candidates and heuristic testing for 2 candidates, the time per LP testing calculates to approximately $\frac{253.4}{10b} \approx \frac{25.3}{b}$ seconds, and the time per heuristic test is $\frac{1284.8}{2b} \approx \frac{642.4}{b}$ seconds, where b is the number of branchings. This indicates that heuristic testing is more than 25 times as time-consuming as LP testing, suggesting that applying BKF to the heuristic phase could offer substantial time reduction. After incorporating BKF into the heuristic phase as described in Section EC.6.1.2 of the e-companion, the dynamic versions show significant improvements. Comparing 3PB-dy to 3PB, a time reduction of 29% and a reduction in the number of nodes by 32% were achieved. Similarly, when comparing 2LBB-dy with 2LBB, a time reduction of 26% and a reduction in the number of nodes by 44% were achieved. Furthermore, when comparing 2LBB-dy to 3PB, a CPU time reduction of 51% was achieved.

6.3. Open CVRP Instances

In this section, we investigate the performance of RoutOpt across four versions to tackle open instances for which optimal solutions are unknown. We picked five instances from the CVRPLIB X series: X-n294-k50, X-n313-k71, X-n336-k84, X-n344-k43, and X-n384-k52. These instances seem relatively easier to solve because they possess short average route lengths. While preparing this paper, we became aware that X-n344-k43 was solved optimally on September 14, 2023, by the VRPSolver with the cluster branching strategy in Silva and Uchoa (2022). Because no details about their results were reported, the computational effort they spent to solve it is unclear. We decided to still report the relevant results for completeness. It is worth mentioning that using one thread can take an excessive amount of time (several months) to complete the computation, which necessitates taking advantage of the rich computational resources we can access.

Figure 8. (Color online) Generalization Capability of 2LBB Across Different Instance Sizes

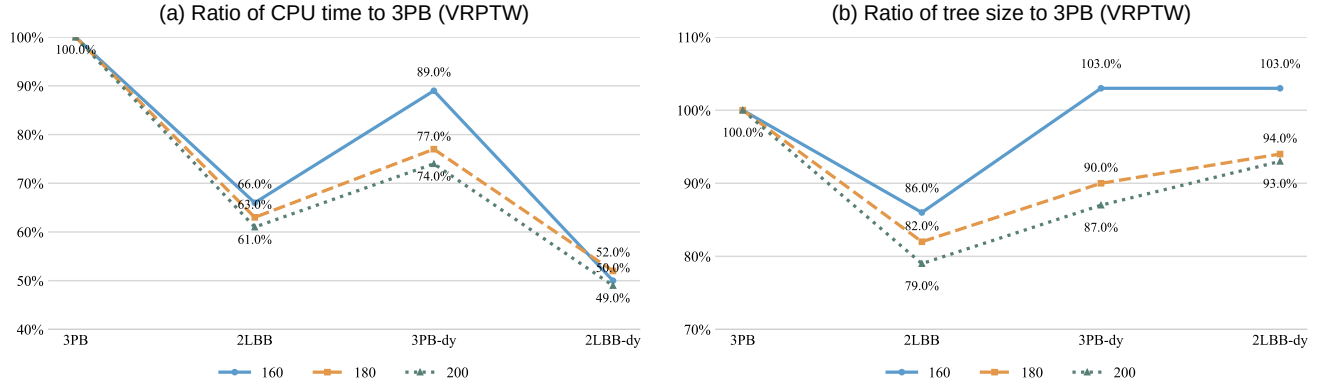


Table 9. Comparison of 3PB, 2LBB, 3PB-dy, and 2LBB-dy on Batch VRPTW-200-22-H2

Version	Root/s	Root gap/%	LP/s	Heuristic/s	CPU/s	No. of nodes
3PB	169.5	0.38	253.4	1,284.8	2,959.2	114.7
2LBB			245.4	324.7	1,944.8	122.7
3PB-dy			253.7	743.9	2,112.5	78.2
2LBB-dy			147.8	351.8	1,442.6	68.6

More specifically, we leverage the supercomputing resources on HiPerGator following the two-step procedure elaborated below. In the solution process, enumeration starts to succeed at some BBNs when the gap becomes small, and such BBNs are referred to as *enumerated BBNs*. We write the information of each enumerated BBN into a separate file and prune it from the BBT. This process continues until there are no open BBNs in the BBT and is referred to as the first step. This step is much more time-consuming, taking around a week because no parallelism is implemented. In the second step, each enumerated BBN is solved as a standalone job using one thread and 16 GB of memory, which usually finishes within three hours. HiPerGator enables us to process 500 jobs simultaneously, and the second step typically takes less than a day.

The 3PB is configured to handle such extremely difficult instances with $\hat{\theta}_{pf} = 30$ and $\tilde{\theta} = 3$. For a fair comparison, $\hat{\theta}_{pf} = 100$, $\hat{\theta} = 30$, and $\tilde{\theta} = 3$ are used for 2LBB. We managed to prove optimality for two instances: X-n294-k50 and X-n344-k43. Table 10 lists the time for the root node (Root/s), the lower bound for the root node (LB), and the upper bound as input (UB). In addition to the same time metrics, we add one more column indicating the number of enumerated BBNs (No. of E-nodes).

We observe that the 2LBB consistently outperforms the 3PB for both instances. Specifically, for X-n294-k50, the time reduction is 21%, and for X-n344-k43, the time reduction is 36%. As expected, both dynamic versions achieve better performance than their static counterparts, especially in the comparison of 3PB-dy and 3PB for instance X-n294-k50. Although 3PB devotes much more time to branching selection, the number of nodes is 1.4 times that of 3PB-dy, showing the effectiveness of the dynamic version. It is worth noting that, for instance X-n294-k50, 3PB-dy achieves better performance, a time reduction of 8%, than 2LBB-dy. Considering that the difference in time spent on LP testing is minimal, the remaining factor could be inaccuracies in the second stage selection procedure. We remark that in this experiment, the selection procedure uses the same $\underline{d}$ and δ throughout the paper, which may not be an ideal choice for instances with more than 20,000 BBNs. To elaborate, a large $\underline{d}$ and a small δ may cause the second stage to skip necessary testing, leading to suboptimal branching decisions. This negative impact can be amplified for instances with a large BBT size. It also suggests that a dynamic version adjusting this parameter should be beneficial.

7. Concluding Remarks

We propose the first effective learning framework that significantly enhances the SOTA 3PB strategy for exact BPC algorithms, along with a novel theoretical model that establishes the foundations for balancing branching decision time and quality. The 2LBB-dy, derived from this framework, demonstrates exceptional performance, achieving approximately 45% and 50% time reduction over the SOTA 3PB for CVRP and VRPTW, respectively, and enabling our RouteOpt to significantly outperform the best existing competitor, VRPSolver. Through a comprehensive numerical study, we further demonstrate that the learned strategy generalizes well to large unseen problems with similar structures and thus is especially suitable for scenarios where similar problems need to be solved on a regular basis, such as last-mile delivery in a given region.

Table 10. Comparison of 3PB, 2LBB, 3PB-dy, and 2LBB-dy on Instances X-n294-k50 and X-n344-k43

Instance	Version	Root/s	LB	UB	LP/h	Heuristic/h	CPU/h	No. of nodes	No. of E-nodes
X-n294-k50	3PB	79	46,928	47,161	28.3	25.4	170.5	23,355	11,678
	2LBB				24.0	10.0	135.1	20,123	10,062
	3PB-dy				9.0	15.7	111.6	16,353	8,176
	2LBB-dy				9.4	8.3	121.6	21,297	10,649
X-n344-k43	3PB	172	41,922	42,050	1.5	2.6	13.7	891	444
	2LBB				1.2	0.6	8.8	731	363
	3PB-dy				1.0	1.5	9.6	707	349
	2LBB-dy				0.7	0.5	7.8	687	342

2LBB-dy can be readily extended to problems beyond VRP, such as job scheduling and network optimization, especially those depending on column generation. Although the specific features used to train the ML models may differ across problem domains, a key insight from this work is the importance of focusing on features that capture the problem's intrinsic structure, such as the distribution of node locations and demands in the CVRP, rather than solely relying on the nominal mathematical formulation. That said, the effectiveness of 2LBB depends on the predictive accuracy of *M1* and *M2*. In more complex problems than CVRP or VRPTW, such as inventory routing problems (Engineer et al. 2012, Desaulniers et al. 2016), healthcare planning problems (Trautsamwieser and Hirsch 2014, Qiu et al. 2022), and crew rostering problems (Quesnel et al. 2020, Breugem et al. 2022), hierarchical branching is often employed, involving multiple types of branching variables. Using a single model to rank candidates across all variable types may reduce prediction accuracy and degrade 2LBB-dy's performance. A more practical solution is to train separate models for each variable type, rank candidates within their respective groups, and then use an additional stage to select the overall best candidate for branching.

Another prospective research direction concerns adapting the current method to other branching variable types or multiple variables at a time (Yang et al. 2021). For example, the recently proposed clustering branching (Silva and Uchoa 2022) has helped to solve over 20 open instances, suggesting its potential to tackle instances with certain structures. Nevertheless, the clustering branching gives rise to an exponentially large number of branching candidates (possible clusters), posing an evident challenge in the learning process. Moreover, incorporating new features characterizing such cluster structures is anticipated to be nontrivial. Finally, despite the wide applicability of our theoretical model, some branching variable types might deviate far from Assumption (g), suggesting a need for alternative assumptions for such cases. Most ideas proposed in deriving the BKF are expected to still apply therein.

Acknowledgments

The authors thank the area editor, associate editor, and three anonymous referees for providing valuable comments and suggestions.

References

Achterberg T (2007) Constraint integer programming. PhD thesis, Technische Universitat, Berlin.

Achterberg T, Wunderling R (2013) Mixed integer programming: Analyzing 12 years of progress. *Facets of Combinatorial Optimization* (Springer, Berlin), 449–481.

Alvarez AM, Louveaux Q, Wehenkel L (2017) A machine learning-based approximation of strong branching. *INFORMS J. Comput.* 29(1):185–195.

Applegate D, Bixby R, Chvátal V, Cook W (1995) Finding cuts in the tsp (a preliminary report). Technical report, Center for Research in Parallel Computing, Rice University, Houston.

Augerat P, Naddef D, Belenguer J, Benavent E, Corberan A, Rinaldi G (1995) Computational results with a branch and cut code for the capacitated vehicle routing problem. Technical Report No. 949-M, Université Joseph Fourier, Grenoble, France.

Baldacci R, Christofides N, Mingozzi A (2008) An exact algorithm for the vehicle routing problem based on the set partitioning formulation with additional cuts. *Math. Programming* 115(2):351–385.

Baldacci R, Hadjiconstantinou E, Mingozzi A (2004) An exact algorithm for the capacitated vehicle routing problem based on a two-commodity network flow formulation. *Oper. Res.* 52(5): 723–738.

Baldacci R, Mingozzi A, Roberti R (2011) New route relaxation and pricing strategies for the vehicle routing problem. *Oper. Res.* 59(5):1269–1283.

Bektaş T, Erdoğan G, Røpke S (2011) Formulations and branch-and-cut algorithms for the generalized vehicle routing problem. *Transportation Sci.* 45(3):299–316.

Bénichou M, Gauthier JM, Girodet P, Hentges G, Ribière G, Vincent O (1971) Experiments in mixed-integer linear programming. *Math. Programming* 1(1):76–94.

Breugem T, Dollevoet T, Huisman D (2022) Is equality always desirable? Analyzing the trade-off between fairness and attractiveness in crew rostering. *Management Sci.* 68(4):2619–2641.

Cappart Q, Chételat D, Khalil EB, Lodi A, Morris C, Veličković P (2023) Combinatorial optimization and reasoning with graph neural networks. *J. Machine Learn. Res.* 24(130):1–61.

Contardo C, Martinelli R (2014) A new exact algorithm for the multi-depot vehicle routing problem under capacity and route length constraints. *Discrete Optim.* 12:129–146.

Costa L, Contardo C, Desaulniers G (2019) Exact branch-price-and-cut algorithms for vehicle routing. *Transportation Sci.* 53(4): 946–985.

Costa PR, Rhuggenaath J, Zhang Y, Akcay A (2020) Learning 2-opt heuristics for the traveling salesman problem via deep reinforcement learning. Pan SJ, Sugiyama M, eds. *Proc. 12th Asian Conf. Machine Learn.* vol. 129 (PMLR, New York), 465–480.

Dantzig GB, Ramser JH (1959) The truck dispatching problem. *Management Sci.* 6(1):80–91.

Desaulniers G, Rakke JG, Coelho LC (2016) A branch-price-and-cut algorithm for the inventory-routing problem. *Transportation Sci.* 50(3):1060–1076.

Desrosiers J, Soumis F, Desrochers M (1984) Routing with time windows by column generation. *Networks* 14(4):545–565.

Engineer FG, Furman KC, Nemhauser GL, Savelsbergh MW, Song JH (2012) A branch-price-and-cut algorithm for single-product maritime inventory routing. *Oper. Res.* 60(1):106–122.

Fukasawa R, Longo H, Lysgaard J, Aragão M, Reis M, Uchoa E, Werneck RF (2006) Robust branch-and-cut-and-price for the capacitated vehicle routing problem. *Math. Programming* 106(3): 491–511.

Gasse M, Chételat D, Ferroni N, Charlin L, Lodi A (2019) Exact combinatorial optimization with graph convolutional neural networks. Wallach H, Larochelle H, Beygelzimer A, d'Alché-Buc F, Fox E, Garnett R, eds. *Adv. Neural Inform. Processing Systems* (Curran Associates Inc., Red Hook, NY), 15554–15566.

Gurobi Optimization, LLC (2023) Gurobi optimizer reference manual. Accessed February 13, 2026, https://www.gurobi.com/documentation/10.0/refman/index.html.

Heching A, Hooker JN, Kimura R (2019) A logic-based benders approach to home healthcare delivery. *Transportation Sci.* 53(2): 510–522.

Homberger J, Gehring H (2005) A two-phase hybrid metaheuristic for the vehicle routing problem with time windows. *Eur. J. Oper. Res.* 162(1):220–238.

IBM Corporation (2023) IBM ILOG CPLEX optimization studio CPLEX user's manual. https://www.ibm.com/docs/en/icos/22.1.0?topic=manuals-cplex-users-manual.

Irnich S, Desaulniers G, Desrosiers J, Hadjar A (2010) Path-reduced costs for eliminating arcs in routing and scheduling. *INFORMS J. Comput.* 22(2):297–313.

Jepsen M, Petersen B, Spoorendonk S, Pisinger D (2008) Subset-row inequalities applied to the vehicle-routing problem with time windows. *Oper. Res.* 56(2):497–511.

Khalil E, Dai H, Zhang Y, Dilkina B, Song L (2017) Learning combinatorial optimization algorithms over graphs. Guyon I, Von Luxburg U, Bengio S, Wallach H, Fergus R, Vishwanathan SVN, Garnett R, eds. *Adv. Neural Inform. Processing Systems*, vol. 30, (Curran Associates, Inc., Red Hook, NY), 6348–6358.

Khalil E, Le Bodic P, Song L, Nemhauser G, Dilkina B (2016) Learning to branch in mixed integer programming. *Proc. AAAI Conf. Artificial Intelligence* 30.

Laporte G (2009) Fifty years of vehicle routing. *Transportation Sci.* 43(4):408–416.

Laporte G, Nobert Y (1983) A branch and bound algorithm for the capacitated vehicle routing problem. *Oper. Res. Spektrum* 5(2):77–85.

Le Bodic P, Nemhauser G (2017) An abstract model for branching and its application to mixed integer programming. *Math. Programming* 166(1–2):369–405.

Lima I, Uchoa E, Pecin D, Pessoa A, Poggi M, Vidal T, Subramanian A, et al. (2014) CVRPLIB: Capacitated vehicle routing problem library. http://vrp.galgos.inf.puc-rio.br/index.php/en/updates.

Linderoth JT, Savelsbergh MW (1999) A computational study of search strategies for mixed integer programming. *INFORMS J. Comput.* 11(2):173–187.

Lübbecke ME, Desrosiers J (2005) Selected topics in column generation. *Oper. Res.* 53(6):1007–1023.

Lysgaard J, Letchford AN, Eglese RW (2004) A new branch-and-cut algorithm for the capacitated vehicle routing problem. *Math. Programming* 100(2):423–445.

Mai TL, Navet N (2021) Deep learning to predict the feasibility of priority-based ethernet network configurations. *ACM Trans. Cyber-Physical Systems* 5(4):1–26.

Morabit M, Desaulniers G, Lodi A (2021) Machine-learning–based column selection for column generation. *Transportation Sci.* 55(4):815–831.

Morabit M, Desaulniers G, Lodi A (2022) Machine-learning-based arc selection for constrained shortest path problems in column generation. *INFORMS J. Optim.* 5(2):191–210.

Naddef D, Rinaldi G (2002) Branch-and-cut algorithms for the capacitated VRP. Toth P, Vigo D, eds. *The Vehicle Routing Problem* (SIAM, Philadelphia), 53–84.

Nazari M, Oroojlooy A, Snyder L, Takác M (2018) Reinforcement learning for solving the vehicle routing problem. Bengio S, Wallach H, Larochelle H, Grauman K, Cesa-Bianchi N, Garnett R, eds. *Advances in Neural Information Processing Systems*, vol. 31 (Curran Associates Inc., Red Hook, NY), 9839–9849.

Pecin D, Contardo C, Desaulniers G, Uchoa E (2017a) New enhancements for the exact solution of the vehicle routing problem with time windows. *INFORMS J. Comput.* 29(3):489–502.

Pecin D, Pessoa A, Poggi M, Uchoa E (2017b) Improved branch-cut-and-price for capacitated vehicle routing. *Math. Programming Comput.* 9(1):61–100.

Pecin D, Pessoa A, Poggi M, Uchoa E, Santos H (2017c) Limited memory rank-1 cuts for vehicle routing problems. *Oper. Res. Lett.* 45(3):206–209.

Pereira P, Courtade E, Aloise D, Quesnel F, Soumis F, Yaakoubi Y (2022) Learning to branch for the crew pairing problem. Les Cahiers du GERAD. Technical Report, GERAD, HEC Montréal.

Pessoa A, Sadykov R, Uchoa E, Vanderbeck F (2020) A generic exact solver for vehicle routing and related problems. *Math. Programming* 183(1):483–523.

Qiu H, Wang D, Yin Y, Cheng TE, Wang Y (2022) An exact solution method for home health care scheduling with synchronized services. *Naval Res. Logist.* 69(5):715–733.

Quesnel F, Desaulniers G, Soumis F (2020) Improving air crew rostering by considering crew preferences in the crew pairing problem. *Transportation Sci.* 54(1):97–114.

Silva JMP, Uchoa E (2024) Cluster branching for vehicle routing problems. *INFORMS J. Comput.* 36(6):1774–1791.

Tahir A, Quesnel F, Desaulniers G, El Hallaoui I, Yaakoubi Y (2021) An improved integral column generation algorithm using machine learning for aircrew pairing. *Transportation Sci.* 55(6):1411–1429.

Trautsamwieser A, Hirsch P (2014) A branch-price-and-cut approach for solving the medium-term home health care planning problem. *Networks* 64(3):143–159.

Uchoa E, Pecin D, Pessoa A, Poggi M, Vidal T, Subramanian A (2017) New benchmark instances for the capacitated vehicle routing problem. *Eur. J. Oper. Res.* 257(3):845–858.

van der Hagen L, Agatz N, Spliet R, Visser TR, Kok L (2024) Machine learning–based feasibility checks for dynamic time slot management. *Transportation Sci.* 58(1):94–109.

Vinsensius A, Wang Y, Chew EP, Lee LH (2020) Dynamic incentive mechanism for delivery slot management in e-commerce attended home delivery. *Transportation Sci.* 54(3):567–587.

Yang Y, Boland N, Savelsbergh M (2021) Multivariable branching: A 0-1 knapsack problem case study. *INFORMS J. Comput.* 33(4):1354–1367.

Yang Y, Boland N, Dilkina B, Savelsbergh M (2022) Learning generalized strong branching for set covering, set packing, and 0–1 knapsack problems. *Eur. J. Oper. Res.* 301(3):828–840.

Yang Y, Yan C, Cao Y, Roberti R (2023) Planning robust drone-truck delivery routes under road traffic uncertainty. *Eur. J. Oper. Res.* 309(3):1145–1160.

Zhang X, Chen L, Gendreau M, Langevin A (2022) Learning-based branch-and-price algorithms for the vehicle routing problem with time windows and two-dimensional loading constraints. *INFORMS J. Comput.* 34(3):1419–1436.

Zheng YJ (2018) Emergency train scheduling on Chinese high-speed railways. *Transportation Sci.* 52(5):1077–1091.

Zhengzhong You is an independent researcher. His research interests include branch-price-and-cut algorithms, column generation, and learning-augmented methods for large-scale integer and combinatorial optimization, with applications to exact vehicle routing. He is a lead developer of RouteOpt, an open-source, modular exact solver for vehicle routing problems.

Yu Yang is an assistant professor in the Department of Industrial and Systems Engineering at the University of Florida (UF) and assistant director of the UF Center for Applied Optimization. His research focuses on machine learning and artificial intelligence for combinatorial optimization, large-scale optimization theory, and their applications in supply chain management and sustainable energy system design. His team developed RouteOpt.

Xinshang Wang is a researcher at Alibaba Group (U.S.). His research focuses on large-scale integer and stochastic optimization, with an emphasis on learning-augmented algorithms for revenue management, assortment and pricing, and operational decision making in logistics and supply chain systems.

Wotao Yin is an applied mathematician, scientist, and engineer and is the director of the Decision Intelligence Lab at the Alibaba DAMO Academy. His research includes computational optimization, operator splitting methods, and their applications in image processing and machine learning. He has received many prestigious awards, including the Morningside Gold Medal, the NSF CAREER Award, the Alfred P. Sloan Research Fellowship, and the INFORMS Egon Balas Prize.

Copyright of Operations Research is the property of INFORMS: Institute for Operations Research & the Management Sciences and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.

5. 排序与选择：Data-Driven Ranking and Selection Under Input Uncertainty (OR, 2024)

来源：Operations Research. Vol. 72, No. 2, 2024, pp. 781–795.
https://doi.org/10.1287/opre.2022.2375

作者：Di Wu, Yuhao Wang, Enlu Zhou

关键词：排序与选择；仿真优化；输入不确定性；流式数据；序贯消除

推荐人：刘元辛

推荐理由：本文研究输入分布未知、数据按批次逐步到达的仿真排序与选择问题，目标是在尽量少收集批次的情况下可靠识别最优方案。作者提出移动平均估计量，汇总在不同输入分布估计下产生的仿真输出，再基于精确和渐近置信带构造序贯消除程序，并推广至无差异区间情形。数值实验显示，这些程序能控制由输入不确定性引起的错误；合理调整旧仿真结果的丢弃比例，还可进一步提升效率。

https://pubsonline.informs.org/journal/opre

OPERATIONS RESEARCH
Vol. 72, No. 2, March–April 2024, pp. 781–795
ISSN 0030-364X (print), ISSN 1526-5463 (online)

Methods

Data-Driven Ranking and Selection Under Input Uncertainty

Di Wu,[a] Yuhao Wang,[b] Enlu Zhou[b,*]

[a] Amazon Web Services, Seattle, Washington 98109; [b] School of Industrial and Systems Engineering, Georgia Institute of Technology, Atlanta, Georgia 30332

*Corresponding author

Contact: woody10074026@gmail.com, https://orcid.org/0000-0003-3911-0955 (DW); yuhaowang@isye.gatech.edu, https://orcid.org/0000-0002-9863-273X (YW); enlu.zhou@isye.gatech.edu, https://orcid.org/0000-0001-5399-6508 (EZ)

Received: March 29, 2021
Revised: July 20, 2022; April 10, 2022
Accepted: August 21, 2022
Published Online in Articles in Advance: October 18, 2022

Area of Review: Simulation

https://doi.org/10.1287/opre.2022.2375

Copyright: © 2022 INFORMS

Abstract. We consider a simulation-based ranking and selection (R&S) problem with input uncertainty, in which unknown input distributions can be estimated using input data arriving in batches of varying sizes over time. Each time a batch arrives, additional simulations can be run using updated input distribution estimates. The goal is to confidently identify the best design after collecting as few batches as possible. We first introduce a moving average estimator for aggregating simulation outputs generated under heterogeneous input distributions. Then, based on a sequential elimination framework, we devise two major R&S procedures by establishing exact and asymptotic confidence bands for the estimator. We also extend our procedures to the indifference zone setting, which helps save simulation effort for practical usage. Numerical results show the effectiveness and necessity of our procedures in controlling error from input uncertainty. Moreover, the efficiency can be further boosted through optimizing the "drop rate" parameter, which is the proportion of past simulation outputs to discard, of the moving average estimator.

Funding: The authors gratefully acknowledge support by the National Science Foundation Division of Civil, Mechanical and Manufacturing Innovation [Grant CMMI-1453934] and Division of Mathematical Sciences [Grant DMS2053489] and the Air Force Office of Scientific Research [Grants FA9550-19-1-0283 and FA9550-22-1-0244].

Supplemental Material: The electronic companion is available at https://doi.org/10.1287/opre.2022.2375.

Keywords: ranking and selection • optimization via simulation • input uncertainty • fixed confidence • streaming data • online estimation • indifference zone

1. Introduction

Stochastic simulation is used widely for evaluating and optimizing complex systems arising in manufacturing, transportation, and many other domains. Building a stochastic simulation model requires accounting for external random factors (e.g., lead time, traveling time, demand load) that affect the system's performance. A common practice is to model such randomness by probability distributions, from which random samples are generated to simulate real-world scenarios. These distributions are referred to as "input distributions."

In practice, input distributions need to be estimated from finite historical data known as "input data," for which the estimation error results in *input uncertainty* (IU). Aside from this extrinsic uncertainty, the simulation output is also subject to *stochastic uncertainty* (SU), which is the intrinsic uncertainty induced by random samples generated from input distributions. However, these two types of uncertainty differ in nature: whereas SU can be reduced by investing simulation effort, IU is determined by the availability of input data, which is largely beyond a modeler's control. Ignoring IU can be risky when simulation results are used to inform decisions (see, e.g., Zhou and Xie 2015).

In many application problems, input data are collected frequently to reduce the IU of the simulation model. The simulation model, which is often time-consuming to run, is then used to compare different designs/strategies to find the best one with high confidence. Such a procedure is often referred to as ranking and selection (R&S). Data often come on a fast time scale (such as daily), but system designs cannot be changed so frequently; otherwise, it not only incurs excessive cost and labor, but also might cause instability to the system. Hence, simulation becomes extremely useful to test and compare potential designs/strategies before they are implemented in the real system. The following application examples illustrate such a problem setting:

1. Start a ride-sharing service in an airport: In some cities, ride sharing used to be regulated (disallowed) in the airport. When such regulation is lifted, the platform faces a cold-start problem for pricing and matching strategies because of the lack of historical data at the

airport (such as driver and rider arrival rates, pricing and earnings elasticities), which are the inputs to a typical simulation system that the platform uses to choose the pricing and matching strategies. Such simulation can take hours per replication to capture the day/week effect in the market condition. As new data come in daily, the simulation input is updated, and new simulations are run to compare potential pricing and matching strategies until one best strategy is identified and then rolled out to the platform.

2. Supply chain optimization: A multinational retailer makes use of a complex supply chain simulation model to evaluate and compare a few potential inventory policies (including the current policy in use). The model's input distributions capture the uncertainty in production lead time, transit lead time, demand, and so on. As data accrues daily over the selling season, the retailer continuously updates the input models and runs simulations until it identifies the best inventory policy with high confidence and then implements the policy on the real system.

3. Budget R&D capital: A company conducts testing on a set of products with the goal of finding the product with the highest expected net gain within the shortest possible time period. Testing each product is time costly and subject to stochastic error, so multiple replications of testing are needed. As outside investments reveal over time, the company uses this data to update its estimate of the market condition of each product with reduced uncertainty and runs new tests until it narrows down to one best product and then releases it to the market.

These examples motivate us to consider a fixed confidence R&S problem in which new input data arrives over time in batches of possibly varying and random sizes. In each time stage, the number of simulation replications that can be run is limited because of the expensive simulation cost and the time length of each stage. On the one hand, we can use the new data batch to update the input distribution of the simulation model so as to reduce IU. Because of the limited simulation replications at each time stage, it is necessary to aggregate the simulation outputs across different time stages to reduce SU in the performance estimate. On the other hand, simulations are run under a different (updated) input distribution at each stage, and hence, the simulation outputs are correlated and differently distributed across different time stages. This creates a major challenge in designing an R&S procedure for this setting because most classical R&S requires an independent and identically distributed (i.i.d.) condition on the simulation outputs. Further complications lie in how to (i) update the input models and (ii) aggregate simulation outputs generated under heterogenous input models. Whereas there are well-established methods for these two tasks, whether they fit into the R&S framework needs investigation.

To address these challenges in the problem setting, we propose a moving average estimator for system performance to aggregate the simulation outputs across time stages. Intuitively, the moving average estimator drops the obsolete simulation outputs, keeping the more recent outputs that are generated under closer input distributions to the latest one. A parameter, called the drop rate, is used to balance the trade-off between bias (because of keeping the old simulation outputs) and variance (because of the limited number of simulation outputs) in the performance estimate. We then build on the sequential elimination (SE) framework, which was first developed by Even-Dar et al. (2002; 2006), to perform R&S using the moving average estimator. Specifically, by computing the confidence bands to account for both the IU and SU of the moving average estimates, the SE procedures eliminate one (statistically) inferior design each time. We summarize the contributions of this paper as follows:

1. To our knowledge, this paper, along with our earlier conference paper, Wu and Zhou (2019), are the first to consider streaming input data in R&S problems and design a data-driven approach. The simulation outputs generated from different stages are neither independent nor identically distributed, which is a major theoretical challenge that makes our approach drastically different from the classic R&S procedures.

2. We introduce a moving average estimator to aggregate simulation outputs generated under heterogenous input models and substantially extend a sequential elimination framework to handle IU with streaming input data.

3. We design sequential elimination procedures, called SEIU and SEIU-MCB, based on confidence bands established using two different approaches. The SEIU approach relies on exact confidence bands but tends to be conservative; the SEIU-MCB approach relies on asymptotically valid but tighter confidence bands. Specifically, the latter leverages results from "multiple comparison with the best" (MCB) (Chang and Hsu 1992) and an asymptotic normality result that we establish for characterizing the trade-off between IU and SU under streaming data, which is of independent interest and can be viewed as a multistage generalization of a well-known result in Cheng and Holland (1997). We also extend the two SE procedures to the indifference zone setting, named SEIU(IZ) and SEIU-MCB(IZ), which further boosts the procedures for practical usage.

4. The necessity and effectiveness of the procedures are demonstrated numerically through a simple quadratic problem and a more sophisticated production inventory problem. Furthermore, we show that SEIU-MCB can be accelerated by optimizing the drop rate parameter that shows up in the moving average estimator.

We should remark that the aggregation of simulation outputs under different input distributions via the likelihood ratio method is studied by Feng and Staum (2015; 2017), Eckman and Feng (2018), and Feng and Song (2019) under the name "green simulation" and recently by Liu and Zhou (2020) for simulation optimization. Such methods have the potential to be extended to the setting in this paper. We leave the adaptation of these methods to data-driven R&S as an open problem.

1.1. Literature Review

The mixed effect of IU and SU on simulation output is studied in a variety of contexts. In terms of quantifying the impact of IU on a single system design's simulation output, the earliest method at least dates back to Barton and Schruben (1993), followed by many other works, including but not limited to Cheng and Holland (1997), Chick (2001), Zouaoui and Wilson (2003; 2004), Barton et al. (2014), Xie et al. (2014; 2016), Lam and Qian (2022), Lin et al. (2015), Lam and Zhou (2017), Feng and Song (2019), Song and Nelson (2015), and Zhu et al. (2020).

These works assume a fixed batch of input data, but recently Zhou and Liu (2018) and Liu and Zhou (2019) considered streaming input data—the same setting as this paper. However, they take the likelihood ratio method to estimate the performance and a sampling-based method to quantify IU, whereas we develop a moving average estimator and analytically compute confidence bands to quantify IU. We refer the reader to Corlu et al. (2020) for a recent review on the topic of input uncertainty.

There is an abundant and fast-growing R&S literature. Although a comprehensive review is out of this paper's scope, in what follows, we make our best effort to present our work through a broader perspective.

In simulation optimization or ordinal optimization (Ho et al. 1992), R&S arises in the context of identifying the best system design (or a subset of good designs) through noisy simulation outputs. Simulation-based R&S is predominantly studied under the assumption that SU is the only source of uncertainty. Research in this field can roughly be categorized into two problem settings: fixed budget and fixed confidence. In the fixed budget setting, the total number of simulation runs is constrained, and the goal is to maximize the probability of correct selection (PCS) of the best design. In this paper, we focus on the fixed confidence setting, in which the goal is to attain a prespecified PCS using as little simulation effort as possible.

The study on fixed confidence R&S is primarily focused on the IZ formulation, in which the goal is to select the best system design with a target probability when the top two designs differ by at least some value δ in expected performance.

Since Bechhofer (1954), the IZ formulation has been studied actively by the statistics community. In simulation literature, the milestone work that adapts IZ to simulation-based R&S is Kim and Nelson (2001), in which the proposed KN procedure not only outperforms classic statistical procedures, but also allows the use of common random numbers (CRN), a variance reduction technique, for further enhancement. The KN procedure is extended to steady-state simulation in Goldsman et al. (2002) and Kim and Nelson (2006) and also inspires numerous subsequent works on fully sequential procedures, including Batur and Kim (2006), Hong and Nelson (2005; 2007), Pichitlamken et al. (2006), Hong (2006), and so on. More recently, Frazier (2014) proposes an IZ procedure that is Bayesian in spirit but provides a guarantee on the frequentist PCS; Luo et al. (2015) designs procedures for large-scale R&S problems in parallel computing environments; and Fan et al. (2016) develops procedures that do not require an IZ parameter.

The fixed confidence R&S problem is also studied extensively in the multiarmed bandits literature under the name of best arm identification (BAI). Except for a minor difference in performance criterion, R&S and BAI are essentially the same mathematical problem. Nevertheless, the research in the two fields diverges in several aspects with the most notable difference that BAI focuses more on characterizing complexity results and designing algorithms that match some worst case or problem-specific lower bounds, such as the seminal works of Even-Dar et al. (2002) and Mannor and Tsitsiklis (2004) followed by Gabillon et al. (2012), Karnin et al. (2013), Jamieson and Nowak (2014), and Kaufmann et al. (2016) and culminating in Garivier and Kaufmann (2016).

Aside from these frequentist results, Russo (2016) proposes some simple algorithms from a Bayesian viewpoint.

Compared with the studies on classic R&S and BAI, which have gradually matured over the past few decades, research on R&S under IU is only starting to gain momentum. One stream of work takes a distributionally robust optimization approach by assuming that the true input distribution is contained in a finite set of known distributions (i.e., ambiguity set) with a goal of selecting the design with the best worst case performance over the ambiguity set, such as Gao et al. (2017), Xiao and Gao (2018), Xiao et al. (2020), Fan et al. (2020), and Wu and Zhou (2017). Another line of research aims to screen out as many inferior designs as possible in the presence of IU, such as Corlu and Biller (2013, 2015) and Song and Nelson (2019). Our work has a close connection with Song and Nelson (2019) in that we both assume parametric input distributions and incorporate the MCB framework from Chang and Hsu (1992). However, despite these similarities, our settings

of streaming input data are fundamentally different. In Song and Nelson (2019), the performance estimate only consists of i.i.d. samples simulated under the same input distribution, whereas in this paper, we sequentially update input distributions over time and aggregate past simulation outputs from heterogeneous input distributions. Therefore, the methodologies therein cannot be easily extended to our problem.

All the papers on R&S under IU listed here share one assumption in common: input data are a static data set that does not grow over time. Streaming data have only been considered recently for R&S by Wu and Zhou (2019), which is a preliminary conference version of this paper. Song and Shanbhag (2019) and Liu et al. (2021; 2022) study continuous simulation optimization with streaming data.

The rest of the paper is organized as follows. We formulate the problem in Section 2 and extend the sequential elimination framework in Section 3. We derive the exact and asymptotically valid confidence bands in the SE framework and present the corresponding procedures SEIU and SEIU-MCB in Sections 4 and 5, respectively. We extend the procedures to the IZ setting in Section 6. Finally, we numerically demonstrate the procedures' performance in Section 7 and conclude in Section 8.

2. Problem Formulation

2.1. Review on Fixed Confidence R&S Without IU

We begin with a brief review on the classic fixed confidence R&S problem and lay down some basic notations. Suppose that we are given a set of designs $\mathcal{I} = \{1, \dots, K\}$, and the goal is to find the design with the highest expected performance. A design $i \in \mathcal{I}$ is evaluated through repeated simulation runs, and during each run, we first generate a random sample ξ from a distribution P_i and then compute the output by evaluating a deterministic function of i and ξ.

The input distributions $\{P_i\}$ are used as input to the simulation model to capture various sources of real-world randomness. In classic R&S literature, $\{P_i\}$ are assumed to be known, and the simulation outputs for any design i, denoted by $\{X_{i,r}\}_r$, are i.i.d. Let $\mu_i := \mathbb{E}_{P_i}[X_{i,1}]$ be the true mean performance of design i. Then, μ_i can be estimated by averaging the simulation outputs $\{X_{i,r}\}_r$. Because of finite-sample error, the probability of selecting the best design (PCS) is often adopted to quantify the confidence of selection. In the fixed confidence setting, there is no constraint on the simulation budget, but the PCS is required to exceed a prespecified level. We refer to procedures that guarantee to attain the PCS target as *statistically valid*. The core of the fixed confidence R&S problem is to find statistically valid procedures that terminate after as few simulation runs as possible.

2.2. Fixed Confidence R&S with Streaming Input Data

A limitation of the classic R&S framework is that true input distributions are rarely known in practice; instead, they must be estimated using finite data, which incurs IU that propagates to simulation output. Throughout the paper, we assume that all the designs share the same set of input distributions. Furthermore, to facilitate the characterization of IU, we impose the following structure on the input distributions.

Assumption 1. *The set of input distributions belongs to a known parametric family, which contains S mutually independent distributions with unknown parameters $\{\boldsymbol{\theta}_1^c, \boldsymbol{\theta}_2^c, \dots, \boldsymbol{\theta}_S^c\}$, where $\boldsymbol{\theta}_s^c \in \mathbb{R}^{d_s}$.*

The parametric assumption is common in the literature of R&S. Let $d := \sum_{s=1}^S d_s$ be the total dimension of parameters and $\boldsymbol{\theta} := [\boldsymbol{\theta}_1^\top, \dots, \boldsymbol{\theta}_S^\top]^\top$ be a vector in $\mathbb{R}^d$ concatenating all parameters. The input distributions then form a product measure $P_{\boldsymbol{\theta}} = \prod_{s=1}^S P_{\boldsymbol{\theta}_s}$. Also, let $\boldsymbol{\theta}^c \in \mathbb{R}^d$ denote the true input parameter. Under Assumption 1, the expected performance of design i becomes a function of θ and is, therefore, denoted $\mu_i(\boldsymbol{\theta})$.

The negative impact of IU on R&S can be seen as follows. For simplicity, assume throughout the paper that $b \in \mathcal{I}$ is the unique true best design, that is, $\mu_b(\boldsymbol{\theta}^c) > \max_{i \neq b} \mu_i(\boldsymbol{\theta}^c)$. When θ^c is unknown and is replaced by a finite-sample estimate $\hat{\boldsymbol{\theta}}$, the best design under $\hat{\boldsymbol{\theta}}$ may be different from b, in which case one cannot correctly identify the true best design even using infinite simulation runs. If input data are given as a static data set that does not grow over time, then our best hope is to gauge the impact of IU and provide a more conservative statistical guarantee (e.g., a lower PCS) because of limited resolution in performance estimation.

Assuming a static input data set is reasonable in applications in which gathering data is expensive, labor-intensive, or time-consuming. Nevertheless, with the advancement in big data technology, additional data can sometimes be collected efficiently and economically. This motivates us to consider an R&S problem with streaming input data, which we elaborate on as follows.

Suppose that, for each input distribution $P_{\boldsymbol{\theta}_s^c}$, the input data consists of i.i.d. samples denoted by $\{\zeta_{s,1}, \zeta_{s,2}, \dots\}$. The input data arrives in batches sequentially in time, and the sample size of batch t is $n_s(t) \geq 1$. Denote by $N_s(t) := \sum_{\ell=1}^t n_s(\ell)$ the total number of data samples up to batch t and let $N_s(0) = 0$. We allow $n_s(t)$ to vary in s and t in order to have the most flexible model for multiple streams of input data with time-varying batch sizes.

The arrival process of input data, thus, divides the simulation process into multiple time stages. At the beginning of stage t, a new batch of input data arrives, which, together with historical data, is used to compute an updated parameter estimate $\hat{\boldsymbol{\theta}}_t$. After that, incremental simulations are run by drawing i.i.d. samples from the updated input distributions $P_{\hat{\boldsymbol{\theta}}_t}$. Similarly, let $m(t) \geq 1$ be the number of incremental simulation runs for each design at stage t and let $M(t) := \sum_{\ell=1}^{t} m(\ell)$ be the total number of runs up until stage t. The simulation batch size $m(t)$ can vary across different stages because the interarrival times between two adjacent data batches could be different. However, here, we require the same batch size of simulation runs for each design because we make a pairwise comparison, which uses the CRN strategy.

Figure 1 illustrates how simulation outputs are generated for a single design, for which we highlight the following observations. First, the parameter estimates $\{\hat{\boldsymbol{\theta}}_t\}$ are correlated because they are computed using the same stream of input data. Second, because $\{\hat{\boldsymbol{\theta}}_t\}$ are random variables that generally take different values, the simulation outputs X are not identically distributed across different stages. Third, conditioned on $\{\hat{\boldsymbol{\theta}}_t\}$, the outputs X are i.i.d. within the same stage and independent across stages, but unconditional independence no longer holds because X are jointly affected by the correlation among $\{\hat{\boldsymbol{\theta}}_t\}$. In sum, the simulation outputs are neither independent nor identically distributed, which sets the problem apart from most of the R&S studies.

The online estimation of an individual design's performance under streaming input data is an important problem in its own right. In this paper, however, we are concerned with designing R&S procedures that can achieve a prespecified PCS upon termination. To make the problem well-defined, we first need to address the following questions:

i. What is the estimator of the input parameter θ^c?

ii. How do we aggregate non-i.i.d. simulation outputs to estimate $\mu_i(\boldsymbol{\theta}^c)$?

Many existing methods can be applied to these two problems, but most of them do not suit our purpose. The key is to find an estimator that allows a tractable decomposition of the correlation among different stages' simulation outputs. This motivates us to make the following assumption.

Assumption 2. *For each input parameter $\boldsymbol{\theta}_s^c$, there exists a function D_s such that*

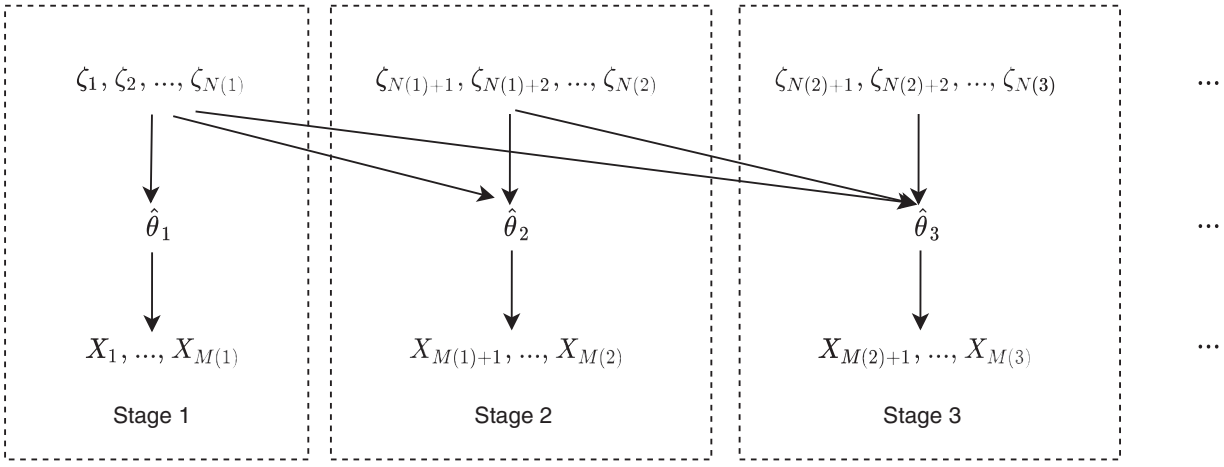
$$\hat{\boldsymbol{\theta}}_{s,t} := \frac{1}{N_s(t)} \sum_{j=1}^{N_s(t)} D_s(\zeta_{s,j}) \tag{1}$$

is an unbiased estimator of $\boldsymbol{\theta}_s^c$.

Assumption 2 can often be satisfied through reparameterization. Specifically, for a parametric family of distributions with k unknown parameters, the parameters (or some transformation of the parameters, which can be transformed back to the parameters) can often be estimated by the first k moments with $D_s(x)$ being the functions $x, x^2, \ldots, x^k$. For example, if we reparameterize the normal distribution by the first two moments, then $D_s(\zeta) = [\zeta, \zeta^2]^\top$ satisfies the desired property. Let $D_{s,j}$ be shorthand for $D_s(\zeta_{s,j})$. Then, $\{D_{s,j}\}$ are i.i.d. transformed input data samples with $\mathbb{E}[D_{s,1}] = \boldsymbol{\theta}^c$. As we see in Section 5, the additive form of $\hat{\boldsymbol{\theta}}_{s,t}$ plays an important role in designing R&S procedures.

The other question is how to estimate $\mu_i(\boldsymbol{\theta}^c)$ using samples generated under different input distributions.

Figure 1. Illustration of Simulating a Single Design with Streaming Input Data, Where ζ Denotes Input Data, $\hat{\theta}$ Is the Input Parameter Estimate, and X Denotes Simulation Output



To simplify indexing, let $\{X_{i,r}(\hat{\boldsymbol{\theta}}_t)\}_{r=1}^{m(t)}$ denote the batch of simulation outputs generated at stage t under the input distribution $P_{\hat{\boldsymbol{\theta}}_t}$. We consider the following moving average estimator:

$$\widehat{\mu}_{i,t} := [M(t) - M(t_\eta)]^{-1} \sum_{\ell=t_\eta+1}^{t} \sum_{r=1}^{m(\ell)} X_{i,r}(\hat{\boldsymbol{\theta}}_\ell), \tag{2}$$

where $\eta \in [0,1)$ is called the drop rate, which controls the amount of effective samples used for computation, and $t_\eta := \lfloor \eta t \rfloor$ is the number of stages "discarded." In words, the estimator in (2) only averages the latest $(1-\eta)$ portion of all the simulation output, hence the name "moving average." The idea of throwing away some earlier samples is motivated by the following two extreme cases:

i. $\eta = 0$: Keeping all the outputs helps reduce SU but also retains all the biases $\{\mu_i(\hat{\boldsymbol{\theta}}_t) - \mu_i(\boldsymbol{\theta}^c)\}$ that accumulate over time.

ii. $\eta = 1$: This roughly corresponds to keeping only the latest output $X_{i,m(t)}(\hat{\boldsymbol{\theta}}_t)$, which reduces the bias but also prohibits the estimator from converging to $\mu_i(\boldsymbol{\theta}^c)$.

In Section 5, we explicitly characterize this trade-off between IU and SU through η and show that optimzing η can lead to a lower asymptotic variance than in both of the two extreme cases.

We now formally state the problem as follows. We aim to find the best design among K alternatives via simulation. The designs share the same input distribution with an unknown parameter θ^c, which can be estimated via (1) using input data arriving in batches. The arrival process divides time into multiple stages, and during each stage, we can update the input parameter estimate $\hat{\boldsymbol{\theta}}$ and run incremental simulations under $P_{\hat{\boldsymbol{\theta}}}$. The performance of each design is estimated via a moving average estimator in (2). Given $\alpha \in (0,1)$, the goal is to design a procedure that terminates after a number of stages and outputs the true best design with probability $\geq 1-\alpha$.

On a side note, classic R&S procedures may still be applied to our problem. For instance, we can collect a few batches of input data and apply KN as if the input distribution estimates are accurate. However, statistical validity is no longer guaranteed under the influence of IU. Whereas, in theory, this issue can be overcome with sufficient data, classic procedures have no way of telling when to safely stop collecting data.

3. Sequential Elimination Framework

In fixed confidence R&S, an important idea underlying many sequential procedures is to construct a "continuation region" in which a design can be confidently labeled as suboptimal if its performance estimate exits the region. One example is the triangular region in the KN procedure (see Kim and Nelson 2001). In our problem setting, most of the existing methods are inapplicable because they rely on key assumptions such as i.i.d. and normality. To construct a continuation region in the presence of IU, we resort to an SE framework in Even-Dar et al. (2002, 2006).

In short, an SE procedure sequentially eliminates inferior designs until there is only one left, and the surviving design is output as the estimated best design. Let $\delta_{ij}(\boldsymbol{\theta}) := \mu_i(\boldsymbol{\theta}) - \mu_j(\boldsymbol{\theta})$ and $\hat{\delta}_{ij,t} = \widehat{\mu}_{i,t} - \widehat{\mu}_{j,t}$. A requirement for an SE procedure is a collection of the confidence bands for $\delta_{ij}(\boldsymbol{\theta}^c)$,

$$\mathbb{P}\left\{\bigcap_{t=1}^{\infty}\bigcap_{i<j}\left\{|\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)| \leq w_{ij,t}\right\}\right\} \geq 1-\alpha, \tag{3}$$

where $w_{ij,t} \to 0$ as $t \to \infty$. The elimination happens to design i if there exists a design j such that

$$\hat{\delta}_{ij,t} + w_{ij,t} = \widehat{\mu}_{i,t} - \widehat{\mu}_{j,t} + w_{ij,t} < 0. \tag{4}$$

That is if the upper confidence of δ_{ij} is below zero. To see why (4) achieves statistical validity, notice that, if the confidence bands achieve perfect coverage, that is, $\delta_{ij}(\boldsymbol{\theta}^c) \in [\hat{\delta}_{ij,t} - w_{ij,t}, \hat{\delta}_{ij,t} + w_{ij,t}]$ for all i, j, and t, then for any design $i \neq b$ (recall that b is the unique best design),

$$\widehat{\mu}_{b,t} - \widehat{\mu}_{i,t} + w_{bi,t} \geq \mu_b(\boldsymbol{\theta}^c) - \mu_i(\boldsymbol{\theta}^c) > 0,$$

meaning that design b is never eliminated. Meanwhile, because $w_{ij,t} \to 0$ as $t \to \infty$, the procedure terminates in finite time almost surely, and the final output must be design b.

In applying the SE procedure, the core question is how to construct the confidence bands $\{w_{ij,t}\}$. Notice that the false selection happens at any stage when the true optimal design is eliminated. Therefore, we can write the probability of false selection (PFS) as

$$\begin{aligned}
\text{PFS} &= \mathbb{P}(\text{The optimal design } b \text{ is eliminated at some } t) \\
&= \mathbb{P}\left(\bigcup_{t=1}^{\infty}\{\text{The optimal design } b \text{ is eliminated at } t\}\right) \\
&= \sum_{t=1}^{\infty}\mathbb{P}(\text{The optimal design } b \text{ is eliminated at } t) \\
&\leq \sum_{t=1}^{\infty}\mathbb{P}\left(\bigcup_{i<j}\{|\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)| > w_{ij,t}\}\right),
\end{aligned} \tag{5}$$

where the second equality follows from the fact that the events {The optimal design b is eliminated at t}, $t = 1, 2, \ldots$ are disjoint, and the last inequality follows

from

$$\{\text{The optimal design } b \text{ is eliminated at } t\} \subseteq \bigcup_{i<j}\{|\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)| > w_{ij,t}\}.$$

Hence, if we can find $w_{ij,t}$ such that

$$\mathbb{P}\left(\bigcup_{i<j}\{|\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)| > w_{ij,t}\}\right) \le \frac{6\alpha}{\pi^2 t^2},$$

then we have

$$\sum_{t=1}^{\infty}\mathbb{P}\left(\bigcup_{i<j}\{|\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)| > w_{ij,t}\}\right) \le \sum_{t=1}^{\infty}\frac{6\alpha}{\pi^2 t^2} = \alpha. \tag{6}$$

We obtain that PCS is at least $1-\alpha$.

Notice that we can also use a nonpairwise framework that derives the confidence bands $w_{i,t}$ for each μ_i. Nonetheless, the pairwise framework allows for the use of CRN, which often sharpens the comparison between designs. Furthermore, the common input distribution may introduce additional positive correlation that further reduces variance. All things considered, it is promising to achieve $w_{ij,t} < w_{i,t} + w_{j,t}$ and, therefore, faster termination. Therefore, we only focus on the pairwise framework.

The problem is reduced to finding tight confidence bands for $\hat{\delta}_{ij,t}$. In the classic IU-free setting in which $\hat{\delta}_{ij,t}$ are averages of i.i.d. samples, many concentration bounds (e.g., Hoeffding's inequality, Gaussian tail bounds) are readily available. Unfortunately, these bounds do not apply directly to the case with IU. In the upcoming section, we derive IU-compatible confidence bands, on top of which we build SE procedures.

4. SEIU Procedures

We propose procedures based on confidence bands derived through an exact approach. These confidence bands allow us to build idealized SE procedures, which assume full knowledge on several key parameters. The idealized procedures are shown to be statistically valid and are equipped with upper bounds on their expected runtime. We also briefly discuss how to estimate the unknown parameters in practice to end this section.

4.1. Derivation of Exact Confidence Bands

We recall a few notations and concepts before deriving the confidence bands. Denote by $\Theta \subseteq \mathbb{R}^d$ the space in which θ lives. The stream of transformed input data for the sth input distribution, denoted by $\{D_{s,t}\}_t$, are i.i.d. random vectors in $\mathbb{R}^{d_s}$ whose sample mean is an unbiased estimator of $\boldsymbol{\theta}_s^c$. Also recall that a random variable X with mean $\mu = \mathbb{E}[X]$ is called *sub-Gaussian* if there exists $\sigma > 0$ such that $\mathbb{E}[e^{s(X-\mu)}] \le \exp(\sigma^2 s^2/2)$ for all $s \in \mathbb{R}$, in which case we write $X \sim subG(\sigma^2)$, and σ^2 is called a *variance proxy*. In the following assumption, $D_{s,t,j}$ denotes the jth component of the random vector $D_{s,t}$.

Assumption 3.

i. *Θ is a compact subset of $\mathbb{R}^d$.*

ii. *For any s, t, and j, $D_{s,t,j} \sim subG(v_s^2)$ for some $v_s > 0$ and $\{D_{s,t,j}\}$ are i.i.d.*

iii. *For any $\boldsymbol{\theta} \in \Theta$ and $i \in \mathcal{I}$, $X_i(\boldsymbol{\theta}) \sim subG(\bar{\sigma}_i^2)$ for some $\bar{\sigma}_i > 0$ independent of θ.*

iv. *For any design i, the performance function μ_i is Lipschitz continuous with respect to input parameter θ. More specifically, there exists $\bar{L}_i > 0$ such that*

$$|\mu_i(\boldsymbol{\theta}_1) - \mu_i(\boldsymbol{\theta}_2)| \le \bar{L}_i \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_1, \quad \forall \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \Theta,$$

where $\|\cdot\|_1$ denotes the ℓ^1-norm.

In Assumption 3, the major condition is the compactness of Θ, which can be satisfied in most real-world applications in which input distributions are supported on a bounded set (e.g., number of visitors on a given day, customer waiting time, etc.). With compactness, $\bar{\sigma}_i^2$ can often be taken as the maximum variance proxy over Θ, and the Lipschitz continuity of μ_i becomes a local property that usually holds in practice. Also, the boundedness of input data and simulation outputs implies their sub-Gaussianity. Furthermore, Assumption 3 implies that there exists $\bar{\sigma}_{ij} > 0$ such that $X_i(\boldsymbol{\theta}) - X_j(\boldsymbol{\theta}) \sim subG(\bar{\sigma}_{ij}^2)$ for all $\boldsymbol{\theta} \in \Theta$ and there exists $\bar{L}_{ij} > 0$ such that $\delta_{ij}(\cdot) = \mu_i(\cdot) - \mu_j(\cdot)$ is $\bar{L}_{ij}$-Lipschitz continuous on Θ.

We now motivate exact confidence bands for the moving average estimator $\hat{\delta}_{i,t}$. The goal is to find bounds $B_t(\cdot)$ such that, for any fixed $x > 0$, we have

$$\mathbb{P}\{|\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)| > x\} \le B_t(x),$$

where $B_t(x) \to 0$ as $t \to \infty$. To gain intuition for an IU-compatible bound, consider a simplified setting in which $\boldsymbol{\theta}^c \in \mathbb{R}$ and there is no SU, that is,

$$\hat{\delta}_{ij,t} = \frac{1}{t}\sum_{\ell=1}^{t}\delta_{ij}(\hat{\boldsymbol{\theta}}_\ell). \tag{7}$$

In this case, error only comes from the biases $\{\delta_{ij}(\hat{\boldsymbol{\theta}}_\ell) - \delta ij(\boldsymbol{\theta}^c)\}$. Then, for any $x > 0$,

$$\mathbb{P}\{|\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)| > x\} \le \mathbb{P}\left\{\frac{1}{t}\sum_{\ell=1}^{t}|\delta_{ij}(\hat{\boldsymbol{\theta}}_\ell) - \delta_{ij}(\boldsymbol{\theta}^c)| > x\right\} \le \mathbb{P}\left\{\sum_{\ell=1}^{t}|\hat{\boldsymbol{\theta}}_\ell - \boldsymbol{\theta}^c| > tx/\bar{L}_{ij}\right\} \tag{8}$$

by the Lipschitz continuity of δ_{ij}. To further decompose the last term in (8), notice that, for any nonnegative sequence $\{\omega_\ell\}$ with $\sum_{\ell=1}^t \omega_\ell = 1$, we have

$$\mathbb{P}\left\{\sum_{\ell=1}^t |\hat{\boldsymbol{\theta}}_\ell - \boldsymbol{\theta}^c| > tx/\bar{L}_{ij}\right\} \le \mathbb{P}\left\{\bigcup_{\ell=1}^t \left\{|\hat{\boldsymbol{\theta}}_\ell - \boldsymbol{\theta}^c| > \omega_\ell tx/\bar{L}_{ij}\right\}\right\} \le \sum_{\ell=1}^t \mathbb{P}\left\{|\hat{\boldsymbol{\theta}}_\ell - \boldsymbol{\theta}^c| > \omega_\ell tx/\bar{L}_{ij}\right\}.$$

The question is whether we can choose $\{\omega_\ell\}$ such that the resulting bound converges to zero as $t \to \infty$. Note that taking equal weights $\omega_\ell = 1/t$ is not helpful because the resulting bound is bounded from below by a positive constant:

$$\sum_{\ell=1}^t \mathbb{P}\left\{|\hat{\boldsymbol{\theta}}_\ell - \boldsymbol{\theta}^c| > x/\bar{L}_{ij}\right\} \ge \mathbb{P}\left\{|\hat{\boldsymbol{\theta}}_1 - \boldsymbol{\theta}^c| > x/\bar{L}_{ij}\right\} = \mathbb{P}\left\{|D_1 - \boldsymbol{\theta}^c| > x/\bar{L}_{ij}\right\}.$$

Instead, an intuitive way to tighten this bound is to allow larger deviations (assign a greater ω_ℓ) for earlier stages and smaller deviations for later stages. This idea is made concrete by the following proposition.

Proposition 1. *Let Assumption 3 hold. Then, for any design i and any $x, y > 0$,*

$$\mathbb{P}\left\{|\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)| > x + y\right\} \le 2\exp\left\{-\frac{[M(t) - M(t_\eta)]x^2}{2\bar{\sigma}_{ij}^2}\right\} + 2(t - t_\eta)\Sigma_{s=1}^S d_s \exp\left\{-\frac{[M(t) - M(t_\eta)]^2 y^2}{2d^2\bar{L}_{ij}^2 v_s^2 \gamma_{s,\eta}^2(t)}\right\}, \tag{9}$$

where $\gamma_{s,\eta}(t) := \sum_{\ell=t_\eta+1}^t m(\ell)/\sqrt{N_s(\ell)}$.

Remark 1. The term $\gamma_{s,\eta}$ loosely characterizes the rate at which the IU-induced error accumulates over time: the $1/\sqrt{N_s(\ell)}$ term reflects how fast IU (i.e., estimation error in $\hat{\boldsymbol{\theta}}_\ell$) diminishes as input data grows, but the error is multiplied by a factor of $m(\ell)$ because, during each stage, we perform $m(\ell)$ simulation runs under the inaccurate input distribution $P_{\hat{\boldsymbol{\theta}}_\ell}$.

Remark 2. In the proof of Proposition 1, the choice of $\omega_\ell \propto m(\ell)/\sqrt{N_s(\ell)}$ echoes our interpretation of $\gamma_{s,\eta}$, and it reweighs IU at different stages to yield a useful upper bound. It is also worth noting that, despite the multiplicative factor $(t - t_\eta)$ in (9), the bound still converges to zero exponentially fast as $t \to \infty$.

Proposition 1 provides IU-compatible bounds for a pair of designs. When it comes to designing SE procedures, however, it is more advantageous to use the following simultaneous bounds for all pairs of designs, which exploits the fact that all designs share the same input distribution and is, therefore, tighter than directly applying Bonferroni's inequality to (9).

Lemma 1. *Let Assumption 3 hold. Then, for any $x_{ij}, y_{ij} > 0$,*

$$\mathbb{P}\left\{\bigcup_{i<j}\left\{|\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)| > x_{ij} + y_{ij}\right\}\right\} \le \sum_{i<j} \exp\left\{-\frac{[M(t) - M(t_\eta)]x_{ij}^2}{2\bar{\sigma}_{ij}{}^2}\right\} + 2(t - t_\eta)\Sigma_{s=1}^S d_s \max_{i<j}\left\{\exp\left\{-\frac{[M(t) - M(t_\eta)]^2 y_{ij}^2}{2v_s^2 d^2 \bar{L}_{ij}^2 \gamma_{s,\eta}^2(t)}\right\}\right\}, \tag{10}$$

where $\gamma_{s,\eta}(t) := \sum_{\ell=t_\eta+1}^t m(\ell)/\sqrt{N_s(\ell)}$.

4.2. The SEIU Procedures

Based on Lemma 1, we design the SEIU procedure based on exact confidence band derivation, in which SEIU stands for sequential elimination under input uncertainty. The procedure relies on the key parameters $v, \bar{\sigma}_{ij}$, and $\bar{L}_{ij}$, which are assumed to be known for the moment.

Procedure: SEIU

- Initialization. $\alpha \in (0,1), \eta \in (0,1), \mathcal{S} = \mathcal{I}, t = 1$.
- Step 1. At stage t, update the estimate of θ^c using new input data and run additional simulations. For each pair of designs $i > j$ in $\mathcal{S}$, compute $\hat{\delta}_{ij,t} = \widehat{\mu}_{i,t} - \widehat{\mu}_{j,t}$ using (2).
- Step 2. Compute the confidence bands $w_{ij,t} = u_{ij,t} + v_{ij,t}$ for each pair of designs $i < j$ using

$$u_{ij,t} = 2\bar{\sigma}_{ij}\sqrt{\frac{\ln\left(\sqrt{\frac{K(K-1)\pi^2}{3\alpha}}t\right)}{M(t) - M(t_\eta)}}, \tag{11}$$

$$v_{ij,t} = \max_{1\le s\le S}\left\{\frac{dv_s\bar{L}_{ij}\gamma_{s,\eta}(t)}{M(t) - M(t_\eta)}\sqrt{2\ln\left(\frac{2Sd_s\pi^2(1-\eta)t^3}{3\alpha}\right)}\right\}, \tag{12}$$

where $\gamma_{s,\eta}(t) := \sum_{\ell=t_\eta+1}^t m(\ell)/\sqrt{N_s(\ell)}$.

- Step 3. For each design $i \in \mathcal{I}$, if

$$\min_{i>j}\left\{\hat{\delta}_{ij,t} + w_{ij,t}\right\} < 0, \tag{13}$$

then set $\mathcal{S} \leftarrow \mathcal{S} \setminus \{i\}$. Go to Output if $|\mathcal{S}| = 1$; otherwise, set $t \leftarrow t + 1$ and go to step 1.
- Output. Output the only design in $\mathcal{S}$.

In the SEIU procedure presented here, $u_{i,t}$ and $v_{i,t}$ are chosen to ensure that PCS $\geq 1-\alpha$. Specifically, by substituting x and y with u and v in Lemma 1, we have

$$\mathbb{P}\left\{\bigcup_{i<j}\left\{|\hat{\delta}_{ij,t}-\delta_{ij}(\boldsymbol{\theta}^c)|>u_{ij,t}+v_{ij,t}\right\}\right\}\leq\frac{6\alpha}{\pi^2t^2},$$

that is, the probability of any design's estimate at any stage exiting its confidence band does not exceed $\frac{6\alpha}{\pi^2t^2}$, and as a result, the probability of false selection does not exceed α as shown in (5) and (6). To run SEIU, a user first needs to specify the target PCS level $1-\alpha$ and a drop rate η for the moving average estimator $\widehat{\mu}_{i,t}$. Then, the procedure collects input data and runs simulations alternately and removes design(s) from the survival set $\mathcal{S}$ whenever the elimination criterion (13) is met. Finally, the last surviving design is output as the selected best.

The following theorem shows the statistical validity of the procedure SEIU and characterizes the number of stages needed for the procedure to terminate, denoted as $\tilde{\tau}$, assuming full knowledge of $\nu_s, \bar{\sigma}_{ij}$ and $\bar{L}_{ij}$.

Theorem 1. *Let Assumption 3 hold, and suppose that ν_s, $\bar{\Sigma}_{ij}$, and $\bar{L}_{ij}$ are known. Then, SEIU selects the best design with probability at least $1-\alpha$. Furthermore,*

$$\mathbb{E}[\tilde{\tau}]\leq\max_{i\neq b}\min_{j\neq i}\tilde{\tau}_{ij}+\alpha, \tag{14}$$

where

$$\tilde{\tau}_{ij}=\inf\{t|w_{ij,t}<\delta_{ij}(\boldsymbol{\theta}^c)\}. \tag{15}$$

The intuition behind the bound on $\mathbb{E}[\tilde{\tau}]$ is that, if all confidence bands achieve perfect coverage, then they eventually become narrow enough for us to distinguish between any two designs. Hence, $\tilde{\tau}_{ij}$ is the worst case time it takes for a design j to eliminate design i, which is infinity if j is inferior to i. By taking the minimum of $\tilde{\tau}_{ij}$ over all designs j other than i, we obtain the longest survival time for i. Thus, the total runtime does not exceed the maximum survival time of all suboptimal designs. Although $\tilde{\tau}_{ij}$ does not have a simple closed form, it clearly depends on two factors: (i) the widths of the confidence bands $\{w_{ij,t}\}$ and (ii) the difference between means, that is, $\delta_{ij}(\boldsymbol{\theta}^c)$.

As in many elimination-based procedures, SEIU can be prestopped at any stage to output a subset $\mathcal{S}$ that contains the true best with probability at least $1-\alpha$. Moreover, the following corollary provides a selection at any stage with an IZ-type guarantee.

Corollary 1. *Let Assumption 3 hold and suppose that $\nu_s, \bar{\sigma}_{ij}$, and $\bar{L}_{ij}$ are known. Following the SEIU procedure, at any stage t, with the remaining set of designs $\mathcal{S}$, the design with the largest moving average mean, that is, the design $i^*=\arg\max_{i\in\mathcal{S}_t}\widehat{\mu}_{i,t}$, is ϵ_t-optimal with probability $1-\alpha$, that is, with probability $1-\alpha$, $\mu_{i^*}(\boldsymbol{\theta}^c)\geq\max_{1\leq i\leq K}\mu_i(\boldsymbol{\theta}^c)-\epsilon_t$, where $\epsilon_t=\max_{j\in\mathcal{S},j\neq i^*}w_{i^*j,t}$.*

Corollary 1 implies that, before the procedure eliminates all the inferior designs, one can also make the selection at any intermediate stage t by choosing the remaining design that has the largest sample mean. Such a selection gives an ϵ_t-optimal design with confidence level $1-\alpha$. The value of ϵ_t depends on the confidence bands $w_{ij,t}$, which is the sum of $\{u_{ij,t}\}$ and $\{v_{ij,t}\}$ that account for SU and IU, respectively.

To gain more insight, consider a special case in which $m(\ell)\equiv m_0$ and $n_s(\ell)\equiv n_0$, that is, a constant batch size of data and simulation over different stages. Then, (11) and (12) can be simplified as

$$u_{ij,t}=2\frac{\bar{\sigma}_{ij}}{\sqrt{m_0}}\sqrt{\frac{\ln\left(\sqrt{\frac{K(K-1)\pi^2}{3\alpha}}t\right)}{t-t_\eta}}, \tag{16}$$

$$v_{ij,t}\approx\max_{1\leq s\leq S}\left\{\frac{2d\nu_s\bar{L}_i}{\sqrt{n_0}(1+\sqrt{\eta})}\sqrt{\frac{2\ln\left(\frac{2Sd_s\pi^2(t-t_\eta)^3}{3(1-\eta)^2\alpha}\right)}{t}}\right\}. \tag{17}$$

Thus, $u_{i,t}$ and $v_{i,t}$ are both of order $O(\sqrt{\ln t/t})$, which is common in SE procedures. Moreover, it can be seen from (16) that the width of $u_{i,t}$ primarily depends on the magnitude of $\bar{\sigma}_{ij}/\sqrt{m_0}$, that is, the discounted variance proxy after averaging m_0 output samples at each stage; clearly, a smaller $\bar{\sigma}_{ij}$ and a higher m_0 yield narrower bands. The drop rate η also matters: notice that $u_{ij,t}\to\infty$ as $\eta\to 1$, meaning that the confidence band needs to expand in order to cover inflated SU with more samples thrown away.

Also clear from (17) is that greater values of $\nu_s/\sqrt{n_0}$ and $\bar{L}_{ij}$, which capture the extent of IU and δ_{ij}'s sensitivity to IU, respectively, result in a wider $v_{ij,t}$. It can further be checked that $v_{ij,t}\to 0$ as $\eta\to 1$, which agrees with the intuition that the more stages we discard, the less we suffer from the biases $\{\delta_{ij}(\hat{\boldsymbol{\theta}}_t)-\delta_{ij}(\boldsymbol{\theta}^c)\}$ from previous stages.

In the SEIU procedure, we require full knowledge of the parameters $\bar{\sigma}_{i,j}, \nu_s, \bar{L}_{i,j}$, which is often unknown and needs to be estimated in practice. We propose some heuristic methods and discussions for estimating these unknown parameters, which can be found in the electronic companion, Section EC.1.

5. SEIU-MCB Procedures

In this section, we propose a more efficient SEIU-MCB procedure by deriving asymptotically valid confidence bands, which utilizes the result of MCB proposed by

Chang and Hsu (1992). MCB is extended by Song and Nelson (2019) to construct confidence bands accounting for IU, in which they exploit the jointly asymptotic normality of pairwise differences of performance estimators (referring to $\{\hat{\delta}_{ij,t}\}_{j\neq i}$) that consist of samples under the current estimated input distribution. We extend this jointly asymptotic normality to our moving average estimator. The major challenge of establishing the asymptotic normality result for our moving average estimator is that the estimator aggregates simulation outputs from different stages, which disallows the application of the central limit theorem by a direct partition of the total error into two parts caused, respectively, by IU and SU. Instead, we adopt the Lindeberg–Feller theorem (see proposition 2.2.7 in Van der Vaart 2000) to prove the asymptotic normality of our moving average estimator.

Compared with SEIU, the SEIU-MCB procedure requires much less restrictive assumptions and achieves higher efficiency. Furthermore, the MCB framework helps to avoid the usage of Boole's inequality across designs, and as a result, the asymptotic normality result yields much tighter confidence bands to control the cumulative error across stages. In addition, our result explicitly characterizes how η affects the trade-off between IU and SU and shows that the elimination between any two designs can be potentially boosted by optimizing η.

5.1. MCB

The MCB result is shown as the following theorem.

Theorem 2. *Let $\widehat{\mu}_{i,t}$ be the estimator of μ_i for $i = 1, 2, \ldots, K$, $x^+ = max(x, 0)$ and $x^- = -min(x, 0)$. If, for each fixed i,*

$$P\left\{\widehat{\mu}_{i,t} - \mu_i - (\widehat{\mu}_{j,t} - \mu_j) \geq -w_{ij,t} \textbf{ for } i \neq j\right\} = 1 - \alpha, \quad (18)$$

then we can make the joint probability statement

$$\mathbb{P}\left\{\mu_i - \max_{j\neq i} \mu_j \in [W_i^-, W_i^+] \quad \text{for } i = 1, 2, \ldots K\right\} = 1 - \alpha,$$

where W_i^- and W_i^+ can be computed as

$$W_i^+ = \left(\min_{j\neq i}[\widehat{\mu}_{i,t} - \widehat{\mu}_{j,t} + w_{ij,t}]\right)^+, \mathbb{G} = \{i : W_i^+ > 0\}$$

$$W_i^- = \begin{cases} 0 & \text{if } \mathbb{G} = \{i\} \\ -\left(\min_{j\neq i}[\widehat{\mu}_{i,t} - \widehat{\mu}_{j,t} - w_{ji,t}]\right)^- & \text{otherwise.} \end{cases} \quad (19)$$

Theorem 2 says that, if we are provided with the simultaneous pairwise confidence bands $w_{ij,t}$ for each i, then we can get a simultaneous confidence band for i that measures how the design i is compared with the remaining best. If the upper bound W_i^+ is zero, then we eliminate design i because it is inferior to the remaining best design. If W_i^- is zero, then we directly choose design i as the optimal design.

Now, the key to the problem remains as how to compute the simultaneous pairwise confidence band $w_{ij,t}$, which depends on the joint distribution of

$$\Delta_{i,t} \triangleq (\hat{\delta}_{i1,t} - \delta_{i1}(\boldsymbol{\theta}^c), \ldots, \hat{\delta}_{ii-1,t} - \delta_{ii-1}(\boldsymbol{\theta}^c), \hat{\delta}_{ii+1,t} - \delta_{ii+1}(\boldsymbol{\theta}^c), \ldots, \hat{\delta}_{iK,t} - \delta_{iK}(\boldsymbol{\theta}^c)).$$

However, usually the joint distribution is hardly computable, and even so, the computation can be expensive. Hence, instead of finding the exact joint distribution, we show the asymptotic joint distribution in the following Section 5.2.

5.2. Asymptotic Normality

In Section 4, a critical assumption underpinning the SEIU framework is the compactness of input distribution's parameter space Θ, which plays an important role in rigorously bounding the variation of simulation output in the presence of IU. We now relax this condition and shift to the following assumption.

Assumption 4. *For all $i \in \mathcal{I}$ and $s \in \{1, 2, \ldots, S\}$,*

i. $\Sigma_{D,s} := Cov(D_{s,1})$ *exists.*

ii. $\Sigma(\boldsymbol{\theta})$ *exists and is continuous for all $\boldsymbol{\theta} \in \Theta$, where $\Sigma_{ij}(\boldsymbol{\theta}) = Cov[X_i(\boldsymbol{\theta}), X_j(\boldsymbol{\theta})]$.*

iii. $\mu_i(\cdot)$ *is twice continuously differentiable in Θ.*

iv. $\{n_s(t)\}$ *and* $\{m(t)\}$ *are uniformly bounded. Furthermore, there exist positive constants $\bar{n}_s$ and $\bar{m}$ such that $N_s(t)/t \to \bar{n}_s$ and $M(t)/t \to \bar{m}$ as $t \to \infty$.*

In Assumption 4, (i) and (ii), the existence of input data's covariance and simulation output's variance is far less stringent than the sub-Gaussianity conditions in Assumption 3 and holds in most real-world applications. The smoothness of $\mu_i(\cdot)$ in (iii) is also a reasonable assumption for many parametric families of $P_{\boldsymbol{\theta}}$. Moreover, (iv) only requires the limit of input data and simulation batch sizes to exist in the Cesàro sense, and the uniform boundedness of $\{n_s(t)\}$ and $\{m(t)\}$ is guaranteed in practice. With Assumption 4, we establish the following asymptotic result for $\Delta_{i,t}$.

Theorem 3. *Let Assumption 4 hold. Then, for any $\eta \in [0, 1)$ and any design i,*

$$\sqrt{t}\Delta_{i,t} \Rightarrow \mathcal{N}(0, \Sigma_{i,\infty}), \quad \text{as } t \to \infty, \quad (20)$$

where $\Rightarrow$ means convergence in distribution, $\mathcal{N}$ denotes the normal distribution, and

$$\Sigma_{i,\infty}(j, j') := \lambda_{I,\eta} \nabla\delta_{ij}(\boldsymbol{\theta}^c)^\top \bar{\Sigma}_D \nabla\delta_{ij'}(\boldsymbol{\theta}^c) + \lambda_{S,\eta} \bar{m}^{-1} \boldsymbol{Cov}\left(X_i(\boldsymbol{\theta}^c) - X_j(\boldsymbol{\theta}^c), X_i(\boldsymbol{\theta}^c) - X_{j'}(\boldsymbol{\theta}^c)\right), \quad (21)$$

in which $\bar{\Sigma}_D := diag(\bar{n}_1^{-1}\Sigma_{D,1}, \ldots, \bar{n}_S^{-1}\Sigma_{D,S})$ *and*

$$\lambda_{I,\eta} := \frac{2}{1-\eta} + \frac{2\eta \ln \eta}{(1-\eta)^2}, \quad \lambda_{S,\eta} := \frac{1}{1-\eta}. \tag{22}$$

Theorem 3 shows that the asymptotic covariance matrix of $\Delta_{i,t}$ is the weighted sum of two components,

$$\Sigma_{i,IU}(j,j') := \nabla\delta_{ij}(\boldsymbol{\theta}^c)^\top \bar{\Sigma}_D \nabla\delta_{ij'}(\boldsymbol{\theta}^c),$$
$$\Sigma_{i,SU}(j,j') := \bar{m}^{-1}\mathbf{Cov}\Big(X_i(\boldsymbol{\theta}^c) - X_j(\boldsymbol{\theta}^c), X_i(\boldsymbol{\theta}^c) - X_{j'}(\boldsymbol{\theta}^c)\Big), \tag{23}$$

which quantify the variance induced by IU and SU, respectively. Furthermore, if we let $\partial_{\boldsymbol{\theta}_s}\delta_{ij}(\boldsymbol{\theta}^c)$ denote the partial derivative $\partial\delta_{ij}(\boldsymbol{\theta}^c)/\partial\boldsymbol{\theta}_s$, then we can decompose $\Sigma_{i,IU}$ as

$$\Sigma_{i,IU}(j,j') = \sum_{s=1}^{S} \partial_{\boldsymbol{\theta}_s}\delta_{ij}(\boldsymbol{\theta}^c)^\top \Sigma_{D,s}\partial_{\boldsymbol{\theta}_s}\delta_{ij'}(\boldsymbol{\theta}^c)/\bar{n}_s,$$

which attributes the variance to each individual input distribution. The $O(1/\bar{n}_s)$ and $O(1/\bar{m})$ convergence rate in (21) are standard. However, the weights $\lambda_{I,\eta}$ and $\lambda_{S,\eta}$ are less common and, therefore, deserve a closer look.

The expressions of $\lambda_{I,\eta}$ and $\lambda_{S,\eta}$ are shaped by two major factors: (i) the structure imposed on $\hat{\boldsymbol{\theta}}_t$ (see Assumption 2) and (ii) the choice of estimator $\widehat{\mu}_{i,t}$. In particular, the sample average structure of $\hat{\boldsymbol{\theta}}_t$ makes asymptotic analysis more tractable because it allows the correlated biases $\{\delta_{ij}(\hat{\boldsymbol{\theta}}_t) - \delta_{ij}(\boldsymbol{\theta}^c)\}$ to be decoupled into a weighted sum of i.i.d. samples $D_{s,t}$. To get a sense of the limiting behavior of $\lambda_{I,\eta}$ and $\lambda_{S,\eta}$, we investigate two extreme cases:

Case 1. If $\eta \to 1$, then $\lambda_{I,\eta} \to 1$ and $\lambda_{S,\eta} \to \infty$, which coincides with the intuition that IU can be reduced by dropping samples, but there will be fewer simulation outputs to average out SU.

Case 2. If $\eta = 0$, then $\lambda_{I,\eta} = 2$ and $\lambda_{S,\eta} = 1$, that is,

$$\sqrt{t}\Big[\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)\Big] \Rightarrow \mathcal{N}(0, 2\Sigma_{i,IU} + \Sigma_{i,SU}), \quad \text{as } t \to \infty. \tag{24}$$

To put (24) in perspective, we cite an asymptotic normality result from Wu and Zhou (2017) for the case of static input data. Suppose that there is only a single batch of input data, which we use to compute the parameter estimate $\hat{\boldsymbol{\theta}}_t$ and then run simulations under $P_{\hat{\boldsymbol{\theta}}_t}$. Let $\tilde{\mu}_{i,t} = \frac{1}{M_i(t)}\sum_{r=1}^{M_i(t)} X_{i,r}(\hat{\boldsymbol{\theta}}_t)$ be the estimator for $\mu_i(\boldsymbol{\theta}^c)$. Under some mild conditions, we have

$$\sqrt{t}\Big[\tilde{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c)\Big] \Rightarrow \mathcal{N}(0, \Sigma_{i,IU} + \Sigma_{i,SU}), \quad \text{as} \quad t \to \infty. \tag{25}$$

By comparing (24) with (25), it can be seen that the weight of $\Sigma_{i,IU}$ is doubled in the case of streaming input data, whereas the weight of $\Sigma_{i,SU}$ remains the same, and the inflation of $\Sigma_{i,IU}$ is due to estimation error accumulated over multiple stages.

With Theorem 3, we approximate the joint distribution of $\Delta_{i,t}$ as

$$\Delta_{i,t} \overset{D}{\approx} \mathcal{N}(0, \Sigma_{i,\infty}/t).$$

Then, $\{w_{ij,t}, j \neq i\}$ is a $(K-1)$-multidimensional quantile. When given the confidence level $1-\alpha$, the choice of $K-1$-dimensional quantile is not unique because the degree of freedom is $K-1$. We can reduce the degree of freedom to one by letting

$$\mathcal{P}\Big(\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c) \geq -w_{ij,t}\Big)$$
$$= \mathcal{P}\Big(\hat{\delta}_{ij',t} - \delta_{ij'}(\boldsymbol{\theta}^c) \geq -w_{ij',t}\Big) \quad \forall j \neq j'.$$

That is, all the confidence bands result in the same probability coverage. This can be done easily under the asymptotic normality of $\hat{\delta}_{ij,t}$. To be specific, for each i, we find the smallest $w_{i,t}$, such that

$$\mathcal{P}\Big(\hat{\delta}_{ij,t} - \delta_{ij}(\boldsymbol{\theta}^c) \geq -w_{i,t}\sigma_{ij,t}, \quad \forall j \neq i\Big) \geq 1-\alpha,$$

where $\sigma_{ij,t}$ refers to the variance of $\hat{\delta}_{ij,t}$ and we assume $\hat{\delta}_{ij,t}$ approximately follows a normal distribution centered at $\delta_{ij}(\boldsymbol{\theta}^c)$. We present the SEIU-MCB procedure in the next section.

5.3. SEIU-MCB Procedure

With the elimination and selection rule stated in Section 5.1, we have the SEIU-MCB procedure as follows.

SEIU-MCB

- Initialization. $\alpha \in (0,1), \eta \in (0,1), t = 1, \mathcal{S} = \mathcal{I}$.
- Step 1. At stage t, update the estimate of θ^c using new input data and run additional simulations using CRN. For each system $i \in \mathcal{S}$, compute $\widehat{\mu}_{i,t}$ using (2).
- Step 2. Compute the simultaneous pairwise confidence bands $w_{ij,t}$ using (18) and (20) with confidence level set as $\frac{6\alpha}{\pi^2 t^2}$. Then, compute W_i^-, W_i^+ using (19).
- Step 3. For each i, if $W_i^- = 0$, then $\mathcal{S} = \{i\}$; else if $W_i^+ = 0$, $\mathcal{S} \leftarrow \mathcal{S}\backslash\{i\}$. If $|\mathcal{S}| = 1$, then go to Output. Increment t by one and go to step 1.
- Output. Output the only design in $\mathcal{S}$ as the optimal system.

As we can see from the procedure, compared with the SEIU in which we have the sub-Gaussian and Lipschitz continuous assumption, SEIU-MCB requires much less knowledge of those parameter estimates. This is attributed to the general asymptotic normality result established in Section 5.2, by which we can compute the confidence bands easily.

Moreover, from Section 5.2, we can see the drop rate balances the impact of input uncertainty and simulation

uncertainty characterized by the asymptotic covariance matrix. Detailed discussions on minimizing such impact through η are presented in Section EC.2.

6. Extension to the IZ Setting

In Sections 4 and 5, the SEIU and SEIU-MCB procedures are designed to proceed until there is only one design left in the remaining candidate set because of our goal to identify the unique optimal design. However, in practice, selecting one of the "near-optimal" designs is often sufficient, which is the motivation behind the indifference zone R&S procedures. To be more specific, an IZ R&S procedure aims to find an ϵ-optimal design with a given $\epsilon > 0$, that is, a design i^* such that $\mu_{i^*} \geq \max_{1\leq i\leq K}\mu_i - \epsilon$, where the ϵ is often referred to as the IZ preference. Such an ϵ compromise can sometimes dramatically save simulation effort, especially when the optimal performance $\mu_b(\boldsymbol{\theta}^c)$ is close to the second optimal performance $\mu_{b'}(\boldsymbol{\theta}^c)$. Inspired by Corollary 1, we can also extend our procedures to the IZ setting and terminate the procedure when either of these cases happens: (i) there is only one design remaining or (ii) $\epsilon_t < \epsilon$. Denote by SEIU(IZ) and SEIU-MCB(IZ), the procedures with the IZ setting. We add an additional step 4 in both procedures to verify whether an ϵ-optimal is found.

- Step 4 in SEIU(IZ). Let $i^* = \arg\max_{i\in\mathcal{S}_t}\widehat{\mu}_{i,t}$. If

$$\max_{j\in\mathcal{S},j\neq i^*} w_{i^*j,t} < \epsilon,$$

then set $\mathcal{S} \leftarrow \{i^*\}$ and go to Output; otherwise, set $t \leftarrow t+1$ and go to step 1.

- Step 4 in SEIU-MCB(IZ). Let $\mathcal{I} = \{i \in \mathcal{S} | -W_i^- < \epsilon\}$. If $|\mathcal{I}| \geq 1$, let $\hat{i}^* = \arg\max_{i\in\mathcal{I}}\widehat{\mu}_{i,t}$, set $\mathcal{S} \leftarrow \{\hat{i}^*\}$, and go to Output. Otherwise, increment t by one and go to step 1.

With similar intuition as Theorem 1, the following corollary provides the statistical validity of SEIU(IZ) and characterization of the number of stages needed to terminate the algorithm.

Corollary 2. *Let Assumption 3 hold and suppose that $v_s, \bar{\sigma}_{ij}$, and $\bar{L}_{ij}$ are known. Then, SEIU(IZ) selects an ϵ-optimal design with probability at least $1-\alpha$. Furthermore, let τ denote the number of stages it takes to terminate. Then,*

$$\tau \leq \inf\left\{t \middle| \max_{1\leq i<j\leq K} w_{ij,t} < \epsilon\right\} \quad a.s. \tag{26}$$

Corollary 2 shows that an advantage of using the IZ criterion is that we can characterize the number of stages needed to terminate in a stronger, almost sure sense. The intuition is that, with the IZ criterion, when the confidence band is small enough, we always either identify the best design or reach the specified accuracy (ϵ-optimal). Again, although we do not have a closed form of the upper bound on τ, it clearly depends on the width of the confidence bands, the true difference $\delta_{ij}(\boldsymbol{\theta}^c)$, and the IZ preference ϵ.

Another advantage of our IZ procedures compared with the well-known KN procedure (see Kim and Nelson 2001) is that our procedures compute the confidence bands independently of IZ preference and terminate when either the best selection is made or the IZ preference is satisfied. In contrast, the KN procedure computes the confidence bands by considering a so-called "slippage configuration" in which the best design has expected performance exactly δ better than the second best. If the actual difference is bigger than δ, the KN procedure computes a larger confidence band and makes redundant simulations. In our procedures, if we set δ smaller than the true difference of the expected performance between the best and second best designs, it does not affect the performance of the procedures because they act at least as good as the SEIU and SEIU-MCB procedures.

7. Numerical Results

The numerical study mainly consists of three parts: (i) comparing our procedures with the KN++ procedure, which does not consider input uncertainty; (ii) investigating the performance of all our proposed procedures; (iii) optimizing η to boost the efficiency of SEIU-MCB.

7.1. Test Problems

Two problems are used for numerical testing. One is to minimize a quadratic objective function with a single source of IU, and the other is a more complex production inventory optimization problem from Hong (2009) with multiple independent sources of IU. A detailed description of the two problems can be found in Sections EC.3.1 and EC.3.2. In all subsequent experiments, we consider the following settings unless otherwise stated:

i. Equal batch size: $n_s(t) = n, m(t) = m$ for some fixed $n, m \in \mathbb{Z}^+$.

ii. Random batch size: $n_s(t)$ and $m(t)$ are i.i.d. samples drawn from the uniform distribution on the lattice $\{\bar{k}, 2\bar{k}, \ldots, 5\bar{k}\}$ for some fixed $\bar{k}$.

More specifically, in the case of unequal batch size, the random samples of $n_s(t)$ and $m(t)$ are independent of input data and simulation outputs.

7.2. Experiment Results

In the following sections, we first use the simple quadratic example to show the necessity of accounting for input uncertainty by comparing the KN++ procedure with our two IZ-type procedures: SEIU(IZ) and SEIU-MCB(IZ). We then demonstrate the applicability and effectiveness of all our proposed procedures on the more general production inventory example, in which

we have multiple input distributions and allow different batch sizes for both input data and simulation budget. Finally, we show that the performance of SEIU-MCB can be further boosted by optimizing η.

7.2.1. Necessity of Considering IU. We first use the quadratic problem with equal batch size and $\mathcal{I} = \{\boldsymbol{\theta}^c + 0.3 \cdot i : i \in [-5, 5] \cap \mathcal{Z}\}$, to test the three procedures with the IZ setting: SEIU(IZ), SEIU-MCB(IZ), and KN++. The KN++ procedure (Kim and Nelson 2006) is an extension of the KN procedure that allows nonnormal observation data and updating the variance estimator. The KN++ procedure uses average batch mean to estimate the variance and updates the variance estimator as more simulation outputs are collected, and hence, it can be directly applied to our example with equal batch size. Specifically, it is applied by ignoring the existence of input uncertainty and carrying out simulations in each stage conditioned on the current estimated input parameter. Table 1 shows the empirical PCS and average termination stage along with their 95% confidence intervals based on 100 independent runs of the three procedures with different values of the IZ parameter δ and different input data batch sizes. The empirical PCS is computed as the percentage of algorithm runs that correctly select the true optimal design. The target PCS is set as 90%. Some observations can be made as follows:

1. The empirical PCS of the KN++ procedure is less than the target PCS 0.9 across all settings. The true difference between the best design and the second best is 0.16. Hence, with $\delta = 0.1$, it is sufficient to identify the best design. Furthermore, by examining the estimated input parameters when incorrect selection or elimination happens, we find that the KN++ procedure makes the false selection mainly when procedures terminate in earlier stages, in which input parameters are estimated more inaccurately. This implies the necessity of controlling the input error.

2. The SEIU-MCB(IZ) achieves the target PCS when $\delta = 0.1$ and shoots a higher PCS than KN++ in all settings with mostly earlier termination. This implies the high efficiency of the SEIU-MCB procedure even when it takes IU into consideration.

3. Comparing SEIU(IZ) and SEIU-MCB(IZ), the latter procedure requires much less simulation effort and achieves more accurate empirical PCS. It shows that the usage of asymptotic normality and MCB in SEIU-MCB significantly relieves the conservativeness in the SEIU procedure. The empirical PCS of SEIU(IZ) always overshoots the target PCS, revealing the conservativeness of the procedure. The conservativeness is mainly a result of the Bonferroni inequality used in deriving the confidence bands in the SEIU procedure.

7.2.2. Inventory Production Problem: A More General Setting. In this section, we show the applicability and effectiveness of all our proposed procedures using an inventory production example, in which we have multiple unknown input distributions with random batch sizes for both input and simulation data. We set the maximal production amount $R^* = 4$, holding cost $c_H = 0.5$, and backlog cost $c_B = 1$. We consider two scenarios with different input dimension S and input parameter θ^c: $S = 2$, $\boldsymbol{\theta}^c = [4,5]^\top$, and $S = 4$, $\boldsymbol{\theta}^c = [4,5,3,3]^\top$. The optimal order-up-to quantity is selected among $\mathcal{I} = \{1, 2, \ldots, 10\}$. For IZ procedures, we set the IZ preference $\delta = 0.3$. Table 2 shows the empirical PCS and average termination stage along with their 95% confidence intervals

Table 1. PCS and Termination Stage for Quadratic Example (Equal Batch Size)

$\eta = 0.5, m_s = n_s = 10$			
δ	SEIU(IZ)	SEIU-MCB(IZ)	KN++(IZ)
0.3	98% ± 1.7%	82% ± 5%	75% ± 8%
	77 ± 6.2	17 ± 1.9	15 ± 1.5
0.2	99% ± 0.6%	88% ± 4%	83% ± 7%
	111 ± 6.7	18 ± 1.4	21 ± 1.3
0.1	100% ± 0%	90% ± 2%	85% ± 6%
	521 ± 31	25 ± 0.9	34 ± 5.4
$\eta = 0.5, m_s = 10, \delta = 0.2$			
n_s	SEIU(IZ)	SEIU-MCB(IZ)	KN++(IZ)
5	99% ± 0.6%	87% ± 5%	74% ± 5%
	170 ± 9.7	25 ± 1.6	24 ± 1.5
10	99% ± 0.6%	88% ± 4%	83% ± 7%
	111 ± 6.7	18 ± 1.4	21 ± 1.3
15	97% ± 1.8%	89% ± 4%	86% ± 3%
	89 ± 5.0	16 ± 0.9	18 ± 1.0

Table 2. PCS and Termination Stage for Production Inventory Problem with Random Batch Size ($\eta = 0.5$)

$\theta^c = [4,5]^\top (S = 2)$				
$\bar{k}$	SEIU	SEIU(IZ)	SEIU-MCB	SEIU-MCB(IZ)
10	100% ± 0%	100% ± 0%	90% ± 4%	81% ± 6%
	364 ± 16	205 ± 13	44 ± 4.7	17 ± 1.2
20	100% ± 0%	100% ± 4%	94% ± 3%	87% ± 5%
	156 ± 7.9	90 ± 5.5	25 ± 2.4	11 ± 0.6
30	100% ± 0%	100% ± 4%	96% ± 2%	90% ± 4%
	102 ± 5.7	54 ± 3.8	19 ± 1.4	7.5 ± 0.4
$\theta^c = [4,5,3,3]^\top (S = 4)$				
$\bar{k}$	SEIU	SEIU(IZ)	SEIU-MCB	SEIU-MCB(IZ)
10	100% ± 0%	100% ± 0%	93% ± 3%	90% ± 4%
	1137 ± 152	528 ± 84	60 ± 9.6	30 ± 3.1
20	100% ± 0%	100% ± 4%	95% ± 2%	92% ± 3%
	522 ± 102	235 ± 58	42 ± 6	18 ± 1.6
30	100% ± 0%	100% ± 4%	96% ± 2%	90% ± 4%
	301 ± 33	139 ± 30	36 ± 4.1	14 ± 1.2

based on 100 independent runs of each procedure. We summarize our observations as follows:

1. The SEIU and SEIU(IZ) procedures obtain the 100% empirical PCS across all settings, overshooting the target PCS. The improved procedures SEIU-MCB and SEIU-MCB(IZ) significantly relieve the conservativeness and achieve an empirical PCS much closer to the target and terminating in a much smaller number of stages.

2. Comparing the two scenarios in which the number of unknown input distributions $S = 2$ and $S = 4$, the actual difference between the best design and the second best is similar, but all four procedures tend to terminate in later stages when $S = 4$ than $S = 2$. Specifically, the termination stages of SEIU and SEIU(IZ) increase roughly by a factor of three, whereas those of SEIU-MCB and SEIU-MCB(IZ) increase by a factor of less than two. Their empirical PCS also increase and overshoot the target PCS slightly when $S = 4$. This implies that a higher dimension of input uncertainty can cause a larger impact on both the PCS and the termination stage, and SEIU-MCB and SEIU-MCB(IZ) are more robust to this impact.

7.2.3. Optimizing the Drop Rate η. Finally, we explore boosting the performance of SEIU-MCB by optimizing η. The experiment setting and result can be found in Section EC3.3. The result shows the optimized η improves both the PCS and the termination stage.

8. Conclusion and Future Directions

We study ranking and selection under input uncertainty in which input data arrive in time-varying batches over time. A moving average estimator is proposed to aggregate the simulation outputs under different input models across time stages because of the limited simulation budget at each stage. We further propose the SEIU and SEIU-MCB procedures by, respectively, deriving the exact and asymptotic confidence bands in a substantially extended sequential elimination framework. We also derive the corresponding SEIU(IZ) and SEIU-MCB(IZ) procedures to extend our algorithms to the indifference zone setting. We analyze the impact of the drop rate of the moving average estimator on the procedures. Numerical results show the necessity of our proposed procedures under input uncertainty and the statistical validity to achieve the target PCS. The SEIU-MCB procedure is highly efficient for practical use and can be accelerated by optimizing the drop rate.

There are several future research directions. First, other methods can be used to update the input model and aggregate simulation outputs over time stages. Second, other R&S procedures can be extended to the setting of streaming input data, especially when (asymptotic) normality of the aggregated performance estimate could be established. Third, other types of performance measure, such as quantiles, can be considered in data-driven ranking and selection problems.

Acknowledgments

The authors thank the associate editor and anonymous reviewers for their careful reading of the paper and very constructive comments that led to a substantially improved paper.

References

Barton RR, Schruben LW (1993) Uniform and bootstrap resampling of empirical distributions. Evans GW, Mollaghasemi M, Russell EC, Biles WE, eds. *Proc. 1993 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 503–508.

Barton RR, Nelson BL, Xie W (2014) Quantifying input uncertainty via simulation confidence intervals. *INFORMS J. Comput.* 26(1):74–87.

Batur D, Kim SH (2006) Fully sequential selection procedures with a parabolic boundary. *IIE Trans.* 38(9):749–764.

Bechhofer RE (1954) A single-sample multiple decision procedure for ranking means of normal populations with known variances. *Ann. Math. Statist.* 25(1):16–39.

Chang JY, Hsu JC (1992) Optimal designs for multiple comparisons with the best. *J. Statist. Planning Inference* 30(1):45–62.

Cheng RC, Holland W (1997) Sensitivity of computer simulation experiments to errors in input data. *J. Statist. Comput. Simulation* 57(1–4):219–241.

Chick SE (2001) Input distribution selection for simulation experiments: Accounting for input uncertainty. *Oper. Res.* 49(5):744–758.

Corlu CG, Biller B (2013) A subset selection procedure under input parameter uncertainty. Pasupathy R, Kim S-H, Tolk A, Hill R, Kuhl ME, eds. *Proc. 2013 Winter Simulation* (IEEE, Piscataway, NJ), 463–473.

Corlu CG, Biller B (2015) Subset selection for simulations accounting for input uncertainty. Yilmaz L, Chan WKV, Moon I, Roeder TMK, Macal C, Rossetti MD, eds. *Proc. 2015 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 437–446.

Corlu CG, Akcay A, Xie W (2020) Stochastic simulation under input uncertainty: A review. *Oper. Res. Perspect.* 7:100162.

Eckman DJ, Feng MB (2018) Green simulation optimization using likelihood ratio estimators. Rabe M, Juan AA, Mustafee N, Skoogh A, Jain S, Johansson B, eds. *Proc. 2018 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 2049–2060.

Even-Dar E, Mannor S, Mansour Y (2002) PAC bounds for multi-armed bandit and Markov decision processes. Kivinen J, Sloan RH, eds. *Internat. Conf. Comput. Learn. Theory* (Springer, Berlin), 255–270.

Even-Dar E, Mannor S, Mansour Y (2006) Action elimination and stopping conditions for the multi-armed bandit and reinforcement learning problems. *J. Machine Learn. Res.* 7:1079–1105.

Fan W, Hong LJ, Nelson BL (2016) Indifference-zone-free selection of the best. *Oper. Res.* 64(6):1499–1514.

Fan W, Hong LJ, Zhang X (2020) Distributionally robust selection of the best. *Management Sci.* 66(1):190–208.

Feng BM, Song E (2019) Efficient input uncertainty quantification via green simulation using sample path likelihood ratios. Mustafee N, Bae K-HG, Lazarova-Molnar S, Rabe M, Szabo C, Haas P, Son Y-J, eds. *Proc. 2019 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 3693–3704.

Feng M, Staum J (2015) Green simulation designs for repeated experiments. Yilmaz L, Chan WKV, Moon I, Roeder TMK, Macal C, Rossetti MD, eds. *Proc. 2015 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 403–413.

Feng M, Staum J (2017) Green simulation: Reusing the output of repeated experiments. *ACM Trans. Model. Comput. Simulation* 27(4):1–28.

Frazier PI (2014) A fully sequential elimination procedure for indifference-zone ranking and selection with tight bounds on probability of correct selection. *Oper. Res.* 62(4):926–942.

Fu MC, ed. (2015) *Handbook of Simulation Optimization*, vol. 216 (Springer, Berlin).

Gabillon V, Ghavamzadeh M, Lazaric A (2012) Best arm identification: A unified approach to fixed budget and fixed confidence. Pereira F, Burges CJ, Bottou L, Weinberger KQ, eds. *Adv. Neural Inform. Processing Systems*, vol. 25 (MIT Press, Cambridge, MA), 3212–3220.

Gao S, Xiao H, Zhou E, Chen W (2017) Robust ranking and selection with optimal computing budget allocation. *Automatica* 81:30–36.

Garivier A, Kaufmann E (2016) Optimal best arm identification with fixed confidence. Feldman V, Rakhlin A, Shamir O, eds. *Conf. Learn. Theory* (PMLR, New York), 998–1027.

Goldsman D, Kim SH, Marshall WS, Nelson BL (2002) Ranking and selection for steady-state simulation: Procedures and perspectives. *INFORMS J. Comput.* 14(1):2–19.

Ho YC, Sreenivas R, Vakili P (1992) Ordinal optimization of DEDS. *Discrete Event Dynamic Systems* 2(1):61–88.

Hong LJ (2006) Fully sequential indifference-zone selection procedures with variance-dependent sampling. *Naval Res. Logist.* 53(5):464–476.

Hong LJ (2009) Estimating quantile sensitivities. *Oper. Res.* 57(1): 118–130.

Hong LJ, Nelson BL (2005) The tradeoff between sampling and switching: New sequential procedures for indifference-zone selection. *IIE Trans.* 37(7):623–634.

Hong LJ, Nelson BL (2007) Selecting the best system when systems are revealed sequentially. *IIE Trans.* 39(7):723–734.

Jamieson K, Nowak R (2014) Best-arm identification algorithms for multi-armed bandits in the fixed confidence setting. Czachórski T, Gelenbe E, Lent R, eds. *48th Annual Conf. Inform. Sci. Systems* (IEEE, Piscataway, NJ), 1–6.

Karnin Z, Koren T, Somekh O (2013) Almost optimal exploration in multi-armed bandits. Dasgupta S, McAllester D, eds. *Internat. Conf. Machine Learn.* (PMLR, New York), 1238–1246.

Kaufmann E, Cappé O, Garivier A (2016) On the complexity of best-arm identification in multi-armed bandit models. *J. Machine Learn. Res.* 17(1):1–42.

Kim SH, Nelson BL (2001) A fully sequential procedure for indifference-zone selection in simulation. *ACM Trans. Model. Comput. Simulation* 11(3):251–273.

Kim SH, Nelson BL (2006) On the asymptotic validity of fully sequential selection procedures for steady-state simulation. *Oper. Res.* 54(3):475–488.

Lam H, Qian H (2022) Subsampling to enhance efficiency in input uncertainty quantification. *Oper. Res.* 70(3):1891–1913.

Lam H, Zhou E (2017) The empirical likelihood approach to quantifying uncertainty in sample average approximation. *Oper. Res. Lett.* 45(4):301–307.

Lin Y, Song E, Nelson BL (2015) Single-experiment input uncertainty. *J. Simulation* 9(3):249–259.

Liu T, Zhou E (2019) Online quantification of input model uncertainty by two-layer importance sampling. Preprint, submitted December 24, https://arxiv.org/abs/1912.11172.

Liu T, Zhou E (2020) Simulation optimization by reusing past replications: Don't be afraid of dependence. Bae K-H, Feng B, Kim S, Lazarova-Molnar S, Zheng Z, Roeder T, Thiesing R, eds. *Proc. 2020 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 2923–2934.

Liu T, Lin Y, Zhou E (2021) A Bayesian approach to online simulation optimization with streaming input data. Kim S, Feng B, Smith K, Masoud S, Zheng Z, Szabo C, Loper M, eds. *Proc. 2021 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 1–12.

Liu T, Lin Y, Zhou E (2022) Bayesian stochastic gradient descent for stochastic optimization with streaming input data. Preprint, submitted February 15, https://arxiv.org/abs/2202.07581.

Luo J, Hong LJ, Nelson BL, Wu Y (2015) Fully sequential procedures for large-scale ranking-and-selection problems in parallel computing environments. *Oper. Res.* 63(5):1177–1194.

Mannor S, Tsitsiklis JN (2004) The sample complexity of exploration in the multi-armed bandit problem. *J. Machine Learn. Res.* 5:623–648.

Pichitlamken J, Nelson BL, Hong LJ (2006) A sequential procedure for neighborhood selection-of-the-best in optimization via simulation. *Eur. J. Oper. Res.* 173(1):283–298.

Russo D (2016) Simple Bayesian algorithms for best arm identification. Feldman V, Rakhlin A, Shamir O, eds. *Conf. Learn. Theory* (PMLR, New York), 1417–1418.

Song E, Nelson BL (2015) Quickly assessing contributions to input uncertainty. *IIE Trans.* 47(9):893–909.

Song E, Nelson BL (2019) Input–output uncertainty comparisons for discrete optimization via simulation. *Oper. Res.* 67(2): 562–576.

Song E, Shanbhag UV (2019) Stochastic approximation for simulation optimization under input uncertainty with streaming data. Mustafee N, Bae K-HG, Lazarova-Molnar S, Rabe M, Szabo C, Haas P, Son Y-J, eds. *Proc. 2019 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 3597–3608.

Van der Vaart AW (2000) *Asymptotic Statistics*, vol. 3 (Cambridge University Press, Cambridge, UK).

Wu D, Zhou E (2017) Ranking and selection under input uncertainty: A budget allocation formulation. Chan WKV, D'Ambrogio A, Zacharewicz G, Mustafee N, Wainer G, Page E, eds. *Proc. 2017 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 2245–2256.

Wu D, Zhou E (2019) Fixed confidence ranking and selection under input uncertainty. Mustafee N, Bae K-HG, Lazarova-Molnar S, Rabe M, Szabo C, Haas P, Son Y-J, eds. *Proc. 2019 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 3717–3727.

Xiao H, Gao S (2018) Simulation budget allocation for selecting the top-m designs with input uncertainty. *IEEE Trans. Automatic Control* 63(9):3127–3134.

Xiao H, Gao F, Lee LH (2020) Optimal computing budget allocation for complete ranking with input uncertainty. *IISE Trans.* 52(5):489–499.

Xie W, Nelson BL, Barton RR (2014) A Bayesian framework for quantifying uncertainty in stochastic simulation. *Oper. Res.* 62(6):1439–1452.

Xie W, Nelson BL, Barton RR (2016) Multivariate input uncertainty in output analysis for stochastic simulation. *ACM Trans. Model. Comput. Simulation* 27(1):1–22.

Zhou E, Liu T (2018) Online quantification of input uncertainty for parametric models. *Proc. 2018 Winter Simulation Conf.*, 1587–1598.

Zhou E, Xie W (2015) Simulation optimization when facing input uncertainty. Yilmaz L, Chan WKV, Moon I, Roeder TMK, Macal C, Rossetti MD, eds. *Proc. 2015 Winter Simulation Conf.* (IEEE, Piscataway, NJ), 3714–3724.

Zhu H, Liu T, Zhou E (2020) Risk quantification in stochastic simulation under input uncertainty. *ACM Trans. Model. Comput. Simulation* 30(1):1–24.

Zouaoui F, Wilson JR (2003) Accounting for parameter uncertainty in simulation input modeling. *IIE Trans.* 35(9):781–792.

Zouaoui F, Wilson JR (2004) Accounting for input-model and input-parameter uncertainties in simulation. *IIE Trans.* 36(11): 1135–1151.

Di Wu is an applied scientist in Amazon Web Services. He received his PhD degree in operations research from the H. Milton Stewart School of Industrial and Systems Engineering at Georgia Institute of Technology in 2019.

Yuhao Wang is a PhD student in the H. Milton Stewart School of Industrial and Systems Engineering at Georgia Institute of Technology. His research interests include simulation and stochastic optimization.

Enlu Zhou is an associate professor in the H. Milton Stewart School of Industrial and Systems Engineering at Georgia Institute of Technology. Her research interests include simulation optimization, stochastic optimization, and stochastic control.

Copyright 2024, by INFORMS, all rights reserved. Copyright of Operations Research is the property of INFORMS: Institute for Operations Research and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.

EC.1. Parameter Estimation

We propose some heuristic methods for estimating the unknown parameters. To begin with, if the transformed input data $\{D_t\}$ and the simulation outputs $\{X_i\}$ are uniformly bounded, then ν_s and $\bar{\sigma}_{ij}$ can be computed using explicit bounds of their support. For example, if $X_i(\theta), X_j(\theta) \in [a,b]$ for all $\theta \in \Theta$, then $(b-a)$ would be a valid value of $\bar{\sigma}_{ij}$, but it tends to be conservative. An alternative is to approximate ν and $\bar{\sigma}_{ij}$ using the sample standard deviation of D_t and $X_i - X_j$. For $\bar{L}_{ij}$, assuming that $\delta_{ij}(\cdot)$ is sufficiently smooth, by the mean value theorem and Hölder's inequality,

$$|\delta_{ij}(\theta_1) - \delta_{ij}(\theta_2)| = \nabla\delta_{ij}(\tilde{\theta})^\intercal(\theta_1 - \theta_2) \le \sup_{\theta\in\Theta} \|\nabla\delta_{ij}(\theta)\|_\infty \|\theta_1 - \theta_2\|_1,$$

where ∇ denotes the gradient operator and $\tilde{\theta}$ is a convex combination of θ_1 and θ_2. The issue is that $\sup_{\theta\in\Theta} \|\nabla\delta_{ij}(\theta)\|_\infty$ is both conservative and difficult to estimate. Empirically, we find that approximating $\bar{L}_{ij}$ with $\|\nabla\delta_{ij}(\hat{\theta})\|_\infty$ usually preserves statistical validity for the procedure.In addition, $\nabla\mu_i(\theta)$ can be estimated via the likelihood ratio estimator

$$\widehat{\nabla\mu}_i(\theta) = \frac{1}{n_0}\sum_{r=1}^{n_0} X_{i,r}(\theta)\frac{\nabla_\theta f(\xi_{i,r}(\theta)\,|\,\theta)}{f(\xi_{i,r}(\theta)\,|\,\theta)}, \quad j = 1,\dots,d_s, \tag{EC.1}$$

where n_0 is an initial sample size, $f(\cdot|\theta)$ is the probability density (or the probability mass function) of X under parameter θ, and $\xi(\theta)$ denotes simulation sample generated from P_θ. Then $\widehat{\nabla\delta}_{ij}(\theta) = \widehat{\nabla\mu}_i(\theta) - \widehat{\nabla\mu}_j(\theta)$Other methods for estimating the gradient can be found in Fu et al. (2015).

The impact of estimation errors on the procedures is discussed as follows. To begin with, underestimating ν_s, σ_{ij} or $\bar{L}_{ij}$ can cause premature stopping and undershooting the target PCS since the resulting confidence bands may fail to achieve desired coverage. However, as we shall see in Section 7, this risk is mitigated by the conservatism of Bonferroni inequality. On the other hand, overestimating the parameters can lead to longer runtime and overshooting the target PCS, and thus we do not recommend estimators that are typically positively biased (e.g., $\sup_{\theta\in\Theta} \|\nabla\hat{\delta}_{ij}(\theta)\|_\infty$).

EC.2. Boosting Through η: More Analysis about Impact of Drop Rate

To minimize the impact caused by IU and SU, the analysis becomes difficult since we want to minimize a matrix-valued function in some sense. However, since our goal is to reduce the running time, which is often dominated by the difference of expected performance between the best and the second best design, we can then minimize the variance $\sigma^2_{bb'}$ to boost the procedure. Here b and b' denote the true best and the second best design, respectively. For this purpose, let $\sigma^2_{ij,\infty}$ denote the asymptotic variance of $\hat{\delta}_{ij,t}$. Then by applying Theorem 3 with only two designs i and j, we have

$$\sigma^2_{ij,\infty} := \lambda_{I,\eta} \nabla\delta_{ij}(\theta^c)^\intercal \bar{\Sigma}_D \nabla\delta_{ij}(\theta^c) + \lambda_{S,\eta} \bar{m}^{-1} \sigma^2_{ij}(\theta^c).$$

Then, for any η between $(0,1)$, $\sigma^2_{ij,\infty}$, the limiting variance for design i and j, satisfies

$$\sigma^2_{ij,\infty} \propto \left\{ \frac{\kappa}{1-\eta} + \frac{2\eta \ln \eta}{(1-\eta)^2} \right\},$$

where $\propto$ denotes "is proportional to", and

$$\kappa := \frac{2\sigma^2_{ij,IU} + \sigma^2_{ij,SU}}{2\sigma^2_{ij,IU}},$$

in which $\sigma^2_{ij,IU} := \nabla\delta_{ij}(\theta^c)^\intercal \bar{\Sigma}_D \nabla\delta_{ij}(\theta^c)$ and $\sigma^2_{ij,SU} := \bar{m}^{-1}\sigma^2_{ij}(\theta^c)$. Notice that for two specific designs i and j, the larger $\sigma^2_{ij,\infty}$, the harder for us to distinguish between them since we will get a larger confidence band $w_{ij,t}$. Hence, if we want elimination between i and j happens earlier, we should minimize $\sigma^2_{ij,\infty}$. Minimizing $\sigma^2_{ij,\infty}$ is equivalent to solving the following optimization problem,

$$\inf_{\eta\in(0,1)} \psi_\kappa(\eta) := \frac{\kappa}{1-\eta} + \frac{\eta \ln \eta}{(1-\eta)^2}, \quad \kappa \geq 1, \tag{EC.2}$$

which does not have a closed-form solution. However, the following result characterizes some important properties of the optimal solution.

Proposition EC.1. *For any fixed $\kappa > 1$, ψ_κ is strictly convex in $(0,1)$ and always has a unique minimizer in $(0,1)$, denoted by $\eta^*(\kappa)$, which satisfies*

$$\eta^*(\kappa) \leq \kappa^{-1}, \quad \lim_{\kappa\to 1} \eta^*(\kappa) = 1. \tag{EC.3}$$

Furthermore,

$$\inf_{\kappa>1} \frac{\psi_\kappa(\eta^*(\kappa))}{\psi_\kappa(0)} = \frac{1}{2}. \tag{EC.4}$$

Proposition EC.1 guarantees the existence and uniqueness of a minimizer, and the convexity of ψ_κ makes it numerically easy to solve (EC.2). Moreover, connecting back to the definition of κ yields the following interpretation.

(i) If $\sigma^2_{ij,SU} \gg \sigma^2_{ij,IU}$, i.e., SU dominates IU, then κ is large and the optimal η is near 0. In other words, we should discard only a tiny portion of "stale" simulation outputs because averaging more outputs is the most effective way to reduce SU.

(ii) If $\sigma^2_{ij,SU} \ll \sigma^2_{ij,IU}$, i.e., IU overshadows SU, then $\kappa \approx 1$ and the optimal η would be near 1, meaning that it is more advantageous to discard most of the previous simulation output. Indeed, in the extreme case where there is no SU (i.e., the simulation output is $\mu_i(\hat{\theta}_t)$), there would be no need to keep anything but the latest output.

The lower bound (EC.4) indicates that compared with blindly setting $\eta = 0$, choosing the optimal η can reduce the asymptotic variance of $\hat{\bar{\delta}}_{ij,t}$ by at most a factor of 2. Again, this happens when there is no SU, i.e., $\kappa = 1$, since we have seen in Section 5.2 that retaining only the latest output halves the asymptotic variance resulted from averaging all outputs.

EC.3. Supplement Material for Numerical Experiment

EC.3.1. Quadratic problem

The first problem is to minimize the expected value of a quadratic function $X_i = (i - \xi)^2$, where the true distribution of ξ is Poisson with mean θ^c. Let $\mathcal{I}$ be all the designs to be evaluated. Also let $\theta^c = 1 \in \mathcal{I}$ so that the best design is $b = \theta^c = 1$.

EC.3.2. Production-inventory problem

We also test the procedures on a production-inventory problem, where the objective function does not have a closed form but needs to be evaluated via simulation.

Suppose that we are running a capacitated production system and we want to minimize the expected total cost over a finite number of stages. The decision variable is the order-up-to level, i.e.,

the quantity we should fill up to once the inventory falls below that level. Please note this is an offline planning problem, where we select the best inventory policy after finishing the simulation. Meanwhile, the production amount in each stage is capped. At the beginning of a stage, the amount produced in the previous stage arrives. Then, the demand is revealed and we fulfill the total demand (both backlog and current demand) to the best allowed by on-hand inventory, after which unfulfilled demand becomes the new backlog. Decision of the production amount is carried out at the end of the stage.

The variables are listed as follows: i is the order-up-to level, I_s is the inventory level at the end of the sth stage, ξ_s is the demand at the sth stage, and R_s is the production amount at the sth stage. Let $I_0 = i$ and $R_0 = 0$. Starting from $s = 1$, the system dynamics evolve according to the following equations,

$$I_s = I_{s-1} + R_{s-1} - \xi_s,$$
$$R_s = \min\{R^*, (i - I_s)^+\},$$

where $a^+ := \max\{0, a\}$ and R^* is the maximum production amount. Assume that the demand quantities are independent random variables, where each ξ_s follows an exponential distribution with mean θ_s^c. Let c_H be the holding cost per unit and c_B be the backlog cost per unit. Then, the cost at the sth stage is

$$c_s := c_H(R_{s-1} + I_s^+) + c_B I_s^-,$$

where $a^- := -\min\{a, 0\}$. The expected total cost over S stages is

$$\mu_i(\theta^c) = \mathbb{E}\left(\sum_{s=1}^{S} c_s\right).$$

The goal is to select the optimal order-up-to quantity among candidate set $\mathcal{I}$.

EC.3.3. Experiment result for optimizing η

For generality, we use random batch sizes in both the quadratic example in Section 7.2.1 and inventory-production example in Secton 7.2.2. We optimize η to minimize the termination stage.

Table EC.1 Performance of SEIU-MCB: $\eta = 0.1$ vs. optimized η.

	Quadratic example		Production-inventory example	
$\bar{k}$	$\eta = 0.1$	Optimized η	$\eta = 0.1$	Optimized η
10	$94\% \pm 3\%$ 47 ± 6.4	$91\% \pm 4\%$ 35 ± 4.8	$93\% \pm 3\%$ 55 ± 4.7	$89\% \pm 4\%$ 37 ± 3.8
20	$95\% \pm 4\%$ 25 ± 2.7	$92\% \pm 6\%$ 20 ± 3.7	$96\% \pm 3\%$ 28 ± 2.4	$94\% \pm 3\%$ 22 ± 2.1
30	$98\% \pm 3\%$ 20 ± 2.6	$96\% \pm 4\%$ 16 ± 2.5	$98\% \pm 1\%$ 21 ± 1.5	$98\% \pm 1\%$ 17 ± 1.3

Table EC.1 summarizes the average termination stage and the corresponding empirical PCS along with their 95% confidence intervals, based on 100 independent runs of each procedure.

From Table EC.1, we can conclude that the drop rate η balances the trade off between PCS and run time. In almost all settings, the optimal drop rate results in less run time of the procedure while achieving a relatively lower PCS. This is consistent with our intuition that less total amount of data (corresponding to earlier termination stage) increases input uncertainty and hence results in a lower PCS. It is worth noting that although the optimal drop rate results in a lower empirical PCS, it still achieves the target PCS. From a practical viewpoint, a good selection of the drop rate will boost the efficiency by reducing the run time while still satisfying the probabilistic guarantee.

Technical Proofs

EC.4. Proofs for SEIU

Proof of Proposition 1 Suppress i, j from the index for convenience. Letting

$$\bar{\delta}_t := \mathbb{E}[\hat{\delta}_t \mid \hat{\theta}_1, \ldots, \hat{\theta}_t] = [M(t) - M(t_\eta)]^{-1} \sum_{\ell=t_\eta+1}^{t} m(\ell)\delta(\hat{\theta}_\ell),$$

$$Y_r = Y_{ij,r} := X_{i,r} - X_{j,r},$$

we have

$$\mathbb{P}\left\{|\hat{\delta}_t - \delta(\theta^c)| > x+y\right\} \le \mathbb{P}\left\{|\hat{\delta}_t - \bar{\delta}_t| > x\right\} + \mathbb{P}\left\{|\bar{\delta}_t - \delta(\theta^c)| > y\right\} \tag{EC.5}$$

For the first term on the RHS of (EC.5),

$$\begin{aligned}\mathbb{P}\left\{|\hat{\delta}_t - \bar{\delta}_t| > x\right\} &= \mathbb{E}\left[\mathbb{P}\left\{\left|[M(t)-M(t_\eta)]^{-1}\sum_{\ell=t_\eta+1}^{t}\sum_{r=1}^{m(\ell)}[Y_r(\hat{\theta}_\ell)-\delta(\hat{\theta}_\ell)]\right| > x \,\middle|\, \hat{\theta}_1,\ldots,\hat{\theta}_t\right\}\right] \\ &\le 2\exp\left\{-\frac{[M(t)-M(t_\eta)]x^2}{2\bar{\sigma}^2}\right\},\end{aligned}$$

where the inequality holds since conditioned on $\hat{\theta}_1,\ldots,\hat{\theta}_t$, $\{Y_r(\hat{\theta}_\ell)-\delta(\hat{\theta}_\ell)\}_r$ are independent zero-mean $subG(\bar{\sigma}^2)$ random variables. For the second term on the RHS of (EC.5),

$$\begin{aligned}\mathbb{P}\left\{|\bar{\delta}_t - \delta(\theta^c)| > y\right\} &\le \mathbb{P}\left\{\left|[M(t)-M(t_\eta)]^{-1}\sum_{\ell=t_\eta+1}^{t} m(\ell)[\delta(\hat{\theta}_\ell)-\delta(\theta^c)]\right| > y\right\} \\ &\le \mathbb{P}\left\{\sum_{\ell=t_\eta+1}^{t} m(\ell)|\delta(\hat{\theta}_\ell)-\delta(\theta^c)| > [M(t)-M(t_\eta)]y\right\} \\ &\le \mathbb{P}\left\{\sum_{\ell=t_\eta+1}^{t} m(\ell)\|\hat{\theta}_\ell-\theta^c\|_1 > [M(t)-M(t_\eta)]\frac{y}{\bar{\bar{L}}}\right\} \\ &= \mathbb{P}\left\{\sum_{\ell=t_\eta+1}^{t} m(\ell)\sum_{s=1}^{S}\sum_{j=1}^{d_s}|\hat{\theta}_{s,j,\ell}-\theta^c_{s,j}| > [M(t)-M(t_\eta)]\frac{y}{\bar{\bar{L}}}\right\} \\ &\le \sum_{s=1}^{S}\sum_{j=1}^{d_s}\mathbb{P}\left\{\sum_{\ell=t_\eta+1}^{t} m(\ell)|\hat{\theta}_{s,j,\ell}-\theta^c_{s,j}| > [M(t)-M(t_\eta)]\frac{y}{d\bar{\bar{L}}}\right\} \\ &:= \sum_{s=1}^{S}\sum_{j=1}^{d_s} p_{s,j},\end{aligned}$$

where $\hat{\theta}_{s,j,\ell}$ denotes the jth coordinate of $\hat{\theta}_{s,\ell}$. Note that for any nonnegative sequence $\{\omega_{s,\ell}\}_{\ell=t_\eta+1}^{t}$ that satisfies $\sum_{\ell=t_\eta+1}^{t}\omega_{s,\ell}=1$, we have

$$\begin{aligned}p_{s,j} &\le \mathbb{P}\left\{\bigcup_{\ell=t_\eta+1}^{t}\left\{m(\ell)|\hat{\theta}_{s,j,\ell}-\theta^c_{s,j}| > \omega_{s,\ell}[M(t)-M(t_\eta)]\frac{y}{d\bar{\bar{L}}}\right\}\right\} \\ &\le \sum_{\ell=t_\eta+1}^{t}\mathbb{P}\left\{m(\ell)|\hat{\theta}_{s,j,\ell}-\theta^c_{s,j}| > \omega_{s,\ell}[M(t)-M(t_\eta)]\frac{y}{d\bar{\bar{L}}}\right\} \\ &\le \sum_{\ell=t_\eta+1}^{t} 2\exp\left\{-\frac{N_s(\ell)\omega_\ell^2[M(t)-M(t_\eta)]^2y^2}{2d^2\bar{\bar{L}}^2\nu_s^2[m(\ell)]^2}\right\},\end{aligned} \tag{EC.6}$$

where the second inequality follows from the fact that $\hat{\theta}_{s,j,\ell} \sim subG\left(\nu_s^2/N_s(\ell)\right)$. Take

$$\omega_{s,\ell} = \frac{m(\ell)/\sqrt{N_s(\ell)}}{\gamma_{s,\eta}(t)}$$

and we further have

$$p_{s,j} \le 2(t-t_\eta)\exp\left\{-\frac{[M(t)-M(t_\eta)]^2 y^2}{2d^2\bar{L}^2\nu_s^2\gamma_{s,\eta}^2(t)}\right\}. \tag{EC.7}$$

Plugging (EC.7) back into (EC.5) completes the proof. ■

Proof of Lemma 1 Similar to the proof of Proposition 1, let

$$\bar{\delta}_{i,t} := \mathbb{E}[\hat{\delta}_{i,t} \mid \hat{\theta}_1, \ldots, \hat{\theta}_t] = [M_i(t) - M_i(t_\eta)]^{-1} \sum_{\ell=t_\eta+1}^{t} m_i(\ell)\delta_i(\hat{\theta}_\ell)$$

and we have

$$\mathbb{P}\left\{\bigcup_{i<j}\left\{|\hat{\delta}_{ij,t} - \delta_{ij}(\theta^c)| > x_{ij} + y_{ij}\right\}\right\} \le \sum_{i<j}\mathbb{P}\left\{|\hat{\delta}_{i,t} - \bar{\delta}_{i,t}| > x_{ij}\right\} + \mathbb{P}\left\{\cup_{i<j}\mathcal{E}_{ij,t}\right\},$$

where $\mathcal{E}_{ij,t} := \left\{|\bar{\delta}_{ij,t} - \delta_{ij}(\theta^c)| > y_{ij}\right\}$. By the same conditioning argument in the proof of Proposition 1 and a sub-Gaussian tail bound, we have

$$\sum_{i<j}\mathbb{P}\left\{|\hat{\delta}_{ij,t} - \bar{\delta}_{ij,t}| > x_{ij}\right\} \le \sum_{i<j} 2\exp\left\{-\frac{[M(t)-M(t_\eta)]x_{ij}^2}{2\bar{\sigma_{ij}}^2}\right\}, \tag{EC.8}$$

which gives the first part of the desired bound. For $\mathcal{E}_{ij,t}$, in light of the Lipschitz continuity of δ_{ij},

$$\begin{aligned}
\mathcal{E}_{ij,t} &\subseteq \left\{\sum_{\ell=t_\eta+1}^{t} m(\ell)\|\hat{\theta}_\ell - \theta^c\|_1 > [M(t)-M(t_\eta)]\frac{y_{ij}}{\bar{L}_{ij}}\right\} \\
&= \left\{\sum_{j=1}^{d}\sum_{\ell=t_\eta+1}^{t} m(\ell)|\hat{\theta}_{s,j,\ell} - \theta^c_{s,j}| > [M(t)-M(t_\eta)]\frac{y_{ij}}{\bar{L}_{ij}}\right\} \\
&\subseteq \bigcup_{s=1}^{S}\bigcup_{j=1}^{d_s}\left\{\sum_{\ell=t_\eta+1}^{t} m(\ell)|\hat{\theta}_{s,j,\ell} - \theta^c_{s,j}| > [M(t)-M(t_\eta)]\frac{y_{ij}}{d\bar{L}_{ij}}\right\} \\
&\subseteq \bigcup_{s=1}^{S}\bigcup_{j=1}^{d_s}\bigcup_{\ell=t_\eta+1}^{t}\left\{|\hat{\theta}_{s,j,\ell} - \theta^c_{s,j}| > \frac{\omega_{s,\ell}[M(t)-M(t_\eta)]y_{ij}}{m(\ell)d\bar{L}_{ij}}\right\},
\end{aligned}$$

where $\omega_{s,\ell} = m(\ell)/\left[\sqrt{N_s(\ell)}\gamma_{s,\eta}(t)\right]$ satisfies $\sum_{\ell=t_\eta+1}^{t}\omega_{s,\ell} = 1$. Plugging in the value of $\omega_{s,\ell}$ and we further have

$$\cup_{i<j}\mathcal{E}_{ij,t} \subseteq \bigcup_{s=1}^{S}\bigcup_{j=1}^{d_s}\bigcup_{\ell=t_\eta+1}^{t}\left\{|\hat{\theta}_{s,j,\ell} - \theta^c_{s,j}| > \min_{i<j}\left\{\frac{[M(t)-M(t_\eta)]y_{ij}}{d\bar{L}_{ij}\sqrt{N_s(\ell)}\gamma_\eta(t)}\right\}\right\},$$

and it follows from the fact $\hat{\theta}_{s,j,\ell} \sim subG(\bar{\nu}^2/N_s(\ell))$ that

$$\mathbb{P}(\cup_{i<j}\mathcal{E}_{ij,t}) \le 2(t-t_\eta)\sum_{s=1}^{S} d_s \max_{i<j}\left\{\exp\left\{-\frac{[M(t)-M(t_\eta)]^2 y_{ij}^2}{2\nu_s^2 d^2 \bar{L}_{ij}^2 \gamma_{s,\eta}^2(t)}\right\}\right\}. \tag{EC.9}$$

Finally, combining (EC.8) and (EC.9) yields the result. ■

Proof of Theorem 1 According to (5), it suffices to show that

$$\mathbb{P}\left\{\bigcup_{i<j}\left\{|\hat{\delta}_{ij,t}-\delta_{ij}(\theta^c)|>w_{ij,t}\right\}\right\} \le \frac{6\alpha}{\pi^2 t^2}, \tag{EC.10}$$

which would imply that the confidence bands $\{w_{ij,t}\}$ jointly achieve at least $1-\alpha$ coverage probability. Apply Lemma 1 to obtain

$$\begin{aligned}
&\mathbb{P}\left\{\bigcup_{i<j}\left\{|\hat{\delta}_{ij,t}-\delta_{ij}(\theta^c)|>w_{ij,t}\right\}\right\} = \mathbb{P}\left\{\bigcup_{i<j}\left\{|\hat{\delta}_{ij,t}-\mu_{ij}(\theta^c)|>u_{ij,t}+v_{ij,t}\right\}\right\} \\
&\le \sum_{i<j} 2\exp\left\{-\frac{[M(t)-M(t_\eta)]u_{ij,t}^2}{2\bar{\sigma_{ij}}^2}\right\} \qquad (\dagger) \\
&\quad + 2(t-t_\eta)d\max_{i<j}\left\{\exp\left\{-\frac{[M(t)-M(t_\eta)]^2 v_{ij,t}^2}{2\nu_s^2 d^2 \bar{L}_{ij}^2 \gamma_{s,\eta}^2(t)}\right\}\right\} \qquad (\dagger\dagger)
\end{aligned}$$

We check (†) and (††) as follows.

$$\begin{aligned}
(\dagger) &= \sum_{i<j} 2\exp\left\{-\frac{M(t)-M(t_\eta)}{2\bar{\sigma}_{ij}^2}\cdot\frac{4\bar{\sigma}_{ij}^2\ln\left(\sqrt{\frac{K(K-1)\pi^2}{3\alpha}}t\right)}{M(t)-M(t_\eta)}\right\} \\
&= \sum_{i<j} 2\exp\left\{-\ln\left(\frac{K(K-1)\pi^2t^2}{3\alpha}\right)\right\} \\
&= K(K-1)\cdot\frac{3\alpha}{K(K-1)\pi^2t^2} = \frac{3\alpha}{\pi^2t^2}.
\end{aligned}$$

For (††), first note that

$$\exp\left\{-\frac{[M(t)-M(t_\eta)]^2 v_{ij,t}^2}{2\nu_s^2 d^2 \bar{L}_{ij}^2 \gamma_{s,\eta}^2(t)}\right\} \le \exp\left\{-\ln\left(\frac{2Sd_s\pi^2(1-\eta)t^3}{3\alpha}\right)\right\} = \frac{3\alpha}{2Sd_s\pi^2(1-\eta)t^3},$$

and it follows that

$$\begin{aligned}
(\dagger\dagger) &= 2(t-t_\eta)\sum_{s=1}^{S} d_s\frac{3\alpha}{2Sd_s\pi^2(1-\eta)t^3} \\
&= \frac{3\alpha}{\pi^2t^2}.
\end{aligned}$$

Combining (†) and (††) gives (EC.10). It remains to show the upper bound on $\mathbb{E}[\tilde{\tau}]$. We claim that if all $\delta_{ij}(\theta^c)$ fall within their corresponding confidence bands $[\hat{\delta}_{ij,t} - w_{ij,t}, \hat{\delta}_{ij,t} + w_{ij,t}]$, then any suboptimal design $i \neq b$ must be eliminated prior to stage $\min_{j\neq i} \tilde{\tau}_{ij}$. To see this, note that $\tilde{\tau}_{ij}$ is how long design i can last before getting eliminated by design j. Thus, $\min_{j\neq i} \tilde{\tau}_{ij}$ is the longest survival time of design i. Furthermore, the procedure must terminate before stage $\tilde{\tau}_* := \max_{i\neq b} \min_{j\neq i} \tilde{\tau}_{ij}$. In other words, if the procedure has not terminated by some stage $t > \tilde{\tau}_*$, then at least one design's confidence band has failed to cover its true mean at stage t, i.e.,

$$\{\tilde{\tau} > t\} \subseteq \bigcup_{i<j} \left\{ |\hat{\delta}_{i,t} - \delta_{ij}(\theta^c)| > w_{ij,t} \right\}, \quad \forall t > \tilde{\tau}_*.$$

Let $\mathcal{E}_t := \bigcup_{i<j} \left\{ |\hat{\delta}_{ij,t} - \delta_{ij}(\theta^c)| > w_{ij,t} \right\}$,

$$\begin{aligned}\mathbb{E}[\tilde{\tau}] &= \sum_{t=0}^{\infty} \mathbb{P}(\tilde{\tau} > t) = \sum_{t=0}^{\tilde{\tau}_*-1} \mathbb{P}(\tilde{\tau} > t) + \sum_{t=\tilde{\tau}_*}^{\infty} \mathbb{P}(\tilde{\tau} > t) \\ &\leq \tilde{\tau}_* + \sum_{t=\tilde{\tau}_*}^{\infty} \mathbb{P}(\mathcal{E}_t) \leq \tilde{\tau}_* + \sum_{t=1}^{\infty} \mathbb{P}(\mathcal{E}_t) \leq \tilde{\tau}_* + \alpha,\end{aligned}$$

where the last inequality follows from (EC.10). ■

Proof of Corollary 1 According to our SE framework in Section 3, we know that with probability as least $1-\alpha$, the confidence bands achieve perfect coverage. Hence at any stage t, the best design b remains in the remaining set $\mathcal{S}$.

Now, with probability at least $1-\alpha$, we have

$$\mu_{i^*}(\theta^c) - \mu_b(\theta^c) \geq \hat{\delta}_{i^*b,t} - w_{i^*b,t} \geq -w_{i^*b,t} \geq -\max_{j\in\mathcal{S}, j\neq i*} w_{i*j,t}.$$

Hence the current design with the largest sample mean is ϵ_t optimal, where $\epsilon_t = \max_{j\in\mathcal{S}, j\neq i*} w_{i*j,t}$.

Proof of Corollary 2 The validity of the SEIU(IZ) is the result of Theorem 1 and Corollary 1. For the termination time, notice that for $t > \tau$ and any $j \in \mathcal{S}, j \neq i^*$

$$w_{i*j,t} \leq \max_{i\neq j} w_{ij,t} < \epsilon$$

Hence, $\max_{j\in\mathcal{S}, j\neq i^*} w_{i^*j,t} < \epsilon$. The procedure must terminate before τ.

EC.5. Proofs for SEIU-MCB

We establish a series of Lemmas for proving Theorem 3.

LEMMA EC.1 **(The Lindeberg-Feller Theorem (Proposition 2.2.7 in Van der Vaart (2000)).**

For each n let $Y_{n,1},\ldots,Y_{n,k_n}$ be independent random vectors with finite variances such that

(i) $\sum_{i=1}^{k_n}\operatorname{Cov} Y_{n,i}\to\Sigma$,

(ii) $\sum_{i=1}^{k_n}\mathrm{E}\,\|Y_{n,i}\|^2\,1\{\|Y_{n,i}\|>\varepsilon\}\to 0,\quad$ *every* $\varepsilon>0$.

Then the sequence $\sum_{i=1}^{k_n}(Y_{n,i}-\mathrm{E}Y_{n,i})$ *converges in distribution to a normal* $N_s(0,\Sigma)$ *distribution.*

LEMMA EC.2. *Let* $\{X_n\}$ *be independent random vectors with* $\mathbb{E}X_n=0$. *If* $X_n\Rightarrow X$ *and* $\operatorname{Cov}X_n\to\operatorname{Cov}X$ *as* $n\to\infty$, *then for any* $\eta\in[0,1)$,

$$(n-n_\eta)^{-1/2}\sum_{i=n_\eta+1}^{n}X_i\Rightarrow\mathcal{N}(0,\operatorname{Cov}X)\qquad\text{as } n\to\infty.$$

Proof of Lemma EC.2. Let $Y_{n,i}:=0$ if $i\le n_\eta$ and $X_i/\sqrt{n-n_\eta}$ otherwise. We apply Lemma EC.1 to $Y_{n,i}$. Condition (i) is satisfied since

$$\sum_{i=1}^{n}\operatorname{Cov}Y_{n,i}=\sum_{i=n_\eta+1}^{n}\operatorname{Cov}X_i/(n-n_\eta)\to\operatorname{Cov}X.$$

In addition, by a generalized Dominated Convergence Theorem (note that the integrand is dominated by $||X_n||^2$ and $\operatorname{Cov}X_n\to\operatorname{Cov}X$),

$$\mathbb{E}\left[||X_n||^2\mathbb{1}_{\{||X_n||>\epsilon\sqrt{n-n_\eta}\}}\right]\to 0,\quad\text{as } n\to\infty$$

Thus, as $n\to\infty$,

$$\sum_{i=1}^{n}\mathbb{E}[||Y_{n,i}||^2\mathbb{1}_{\{||Y_{n,i}||>\epsilon\}}]=\frac{1}{n-n_\eta}\sum_{i=n_\eta+1}^{n}\mathbb{E}[||X_i||^2\mathbb{1}_{\{||X_i||>\epsilon\sqrt{n-n_\eta}\}}]\to 0,$$

which verifies condition (ii), and the result follows. ■

The next few lemmas revolve around the convergence properties of

$$\phi_{s,t}(k):=\sum_{\ell=k}^{t}m(\ell)/N_s(\ell),\quad k\le t,\tag{EC.11}$$

which plays a crucial role in the proof of Theorem 3.

LEMMA EC.3. *Under Assumption 4, $\phi_t(t_\eta) \to -\bar{m}\ln\eta/\bar{n}$ as $t\to\infty$.*

Lemma EC.3 We drop the index s for simplicity. Note that

$$\left|\phi_t(t_\eta) - \sum_{\ell=t_\eta}^{t} \frac{\bar{m}}{\bar{n}\ell}\right| \le \left|\phi_t(t_\eta) - \sum_{\ell=t_\eta}^{t} \frac{\bar{m}}{N(\ell)}\right| + \left|\sum_{\ell=t_\eta}^{t} \frac{\bar{m}}{N(\ell)} - \sum_{\ell=t_\eta}^{t} \frac{\bar{m}}{\bar{n}\ell}\right|.$$

For the first term in the RHS,

$$\begin{aligned}\left|\phi_t(t_\eta) - \sum_{\ell=t_\eta}^{t} \frac{\bar{m}}{N_s(\ell)}\right| &= \left|\sum_{\ell=t_\eta}^{t} \frac{m(\ell)-\bar{m}}{N(\ell)}\right| \le \frac{1}{N(t_\eta)}\left|\sum_{\ell=t_\eta}^{t} m(\ell) - (t-t_\eta+1)\bar{m}\right| \\ &= \frac{t-t_\eta+1}{N(t_\eta)}\frac{1}{t-t_\eta+1}\left|\sum_{\ell=t_\eta}^{t} m(\ell) - (t-t_\eta+1)\bar{m}\right| \to 0, \quad \text{as } t\to\infty.\end{aligned}$$

For the second term,

$$\left|\sum_{\ell=t_\eta}^{t} \frac{\bar{m}}{N(\ell)} - \sum_{\ell=t_\eta}^{t} \frac{\bar{m}}{\bar{n}\ell}\right| = \frac{\bar{m}}{\bar{n}}\left|\sum_{\ell=t_\eta}^{t} \frac{\bar{n}\ell - N(\ell)}{N(\ell)\ell}\right| \le \frac{\bar{m}}{\bar{n}}\frac{1}{N(t_\eta)}\left|\sum_{\ell=t_\eta}^{t}\left(\bar{n} - \frac{N(\ell)}{\ell}\right)\right| \to 0, \quad \text{as } t\to\infty.$$

Therefore, we have

$$\lim_{t\to\infty}\phi_t(t_\eta) = \frac{\bar{m}}{\bar{n}}\lim_{t\to\infty}\sum_{\ell=t_\eta}^{t}\frac{1}{\ell} = \frac{\bar{m}}{\bar{n}}\lim_{t\to\infty}\int_{t_\eta}^{t}\frac{1}{x}dx = -\frac{\bar{m}\ln\eta}{\bar{n}},$$

which shows the desired limit. ■

LEMMA EC.4. *For $k\le t$,*

$$\sum_{\ell=1}^{k}\phi_t^2(\ell)n(\ell) = \sum_{\ell=1}^{k-1}\frac{m^2(\ell)}{N(\ell)} + 2\sum_{\ell=1}^{k-2}\frac{M(\ell)m(\ell+1)}{N(\ell+1)} + 2M(k-1)\phi_t(k) + N(k)\phi_t^2(k).$$

Proof of Lemma EC.4 By expanding $\phi_t(\ell)$ into $m(\ell)/N(\ell)+\phi_t(\ell+1)$ recursively, we have

$$\begin{aligned}\sum_{\ell=1}^{k}\phi_t^2(\ell)n(\ell) &= \phi_t^2(1)n(1) + \sum_{\ell=2}^{k}\phi_t^2(\ell)n(\ell) \\ &= n(1)\left[\frac{m(1)}{N(1)} + \phi_t(2)\right]^2 + \sum_{\ell=2}^{k}\phi_t^2(\ell)n(\ell) \\ &= \frac{m^2(1)}{N(1)} + 2m(1)\phi_t(2) + N(2)\phi_t^2(2) + \sum_{\ell=3}^{k}\phi_t^2(\ell)n(\ell) \\ &= \frac{m^2(1)}{N(1)} + 2m(1)\phi_t(2) + N(2)\left[\frac{m(2)}{N(2)} + \phi_t(3)\right]^2 + \sum_{\ell=3}^{k}\phi_t^2(\ell)n(\ell) \\ &= \cdots \\ &= \sum_{\ell=1}^{k-1}\frac{m^2(\ell)}{N(\ell)} + 2\sum_{\ell=1}^{k-1}m(\ell)\phi_t(\ell+1) + N(k)\phi_t^2(k). \qquad (*)\end{aligned}$$

Similarly, expanding the second term in (*) gives

$$\begin{aligned}\sum_{\ell=1}^{k-1} m(\ell)\phi_t(\ell+1) &= m(1)\left[\frac{m(2)}{N(2)}+\phi_t(3)\right]+\sum_{\ell=2}^{k-1} m(\ell)\phi_t(\ell+1)\\ &= \frac{m(1)m(2)}{N(2)}+M(2)\phi_t(3)+\sum_{\ell=3}^{k-1} m(\ell)\phi_t(\ell+1)\\ &= \frac{M(1)m(2)}{N(2)}+\frac{M(2)m(3)}{N(3)}+M(3)\phi_t(4)+\sum_{\ell=4}^{k-1} m(\ell)\phi_t(\ell+1)\\ &= \cdots\\ &= \sum_{\ell=1}^{k-2}\frac{M(\ell)m(\ell+1)}{N(\ell+1)}+M(\ell-1)\phi_t(k). \qquad (**)\end{aligned}$$

Plugging (**) into (*) yields the final result. ■

LEMMA EC.5. *Let Assumption 4 hold. Then,*

$$[M(t)-M(t_\eta)]^{-1}\left[\phi_t^2(t_\eta+1)N_s(t_\eta)+\sum_{\ell=t_\eta+1}^{t}\phi_t^2(\ell)n_s(\ell)\right]\to\beta_{s,\eta},\quad \text{as } t\to\infty, \tag{EC.12}$$

where

$$\beta_{s,\eta}=\frac{\bar{m}}{\bar{n}_s}\left(2+\frac{2\eta\ln\eta}{1-\eta}\right). \tag{EC.13}$$

Proof of Lemma EC.5 Fix s so that it can be suppressed from the indices. Define

$$\tilde{\beta}_\eta(t):=[M(t)-M(t_\eta)]^{-1}\left[\phi_t^2(t_\eta+1)N(t_\eta)+\sum_{\ell=t_\eta+1}^{t}\phi_t^2(\ell)n(\ell)\right]. \tag{EC.14}$$

The goal is to show that $\tilde{\beta}_\eta(t)\to\beta_\eta$ as $t\to\infty$. In light of Lemma EC.4,

$$\begin{aligned}\sum_{\ell=t_\eta+1}^{t}\phi_t^2(\ell)n(\ell) &= \sum_{\ell=t_\eta}^{t}\frac{m^2(\ell)}{N(\ell)}+2\sum_{\ell=t_\eta+1}^{t-2}\frac{M(\ell)m(\ell+1)}{N(\ell)}\\ &\quad+2M(t-1)\phi_t(t)-2M(t_\eta-1)\phi_t(t_\eta)\\ &\quad+N(t)\phi_t^2(t)-N(t_\eta)\phi_t^2(t_\eta)\end{aligned}$$

Plugging it into $\tilde{\beta}_\eta(t)$ and we have

$$\begin{aligned}\tilde{\beta}_\eta(t)=\frac{t-t_\eta}{M(t)-M(t_\eta)}\cdot\frac{1}{t-t_\eta}\cdot\Bigg\{&\phi_t^2(t_\eta+1)N(t_\eta)+\sum_{\ell=t_\eta}^{t}\frac{m^2(\ell)}{N(\ell)}\\ &+2\sum_{\ell=t_\eta+1}^{t-2}\frac{M(\ell)m(\ell+1)}{N(\ell)}+2M(t-1)\phi_t(t)-2M(t_\eta-1)\phi_t(t_\eta)\\ &+N(t)\phi_t^2(t)-N(t_\eta)\phi_t^2(t_\eta)\Bigg\}\end{aligned}$$

We will compute the limit of each term. To begin with,

$$\lim_{t\to\infty}\frac{t-t_\eta}{M(t)-M(t_\eta)}=\frac{1}{\bar{m}}.$$

Next, since $\lim_{t\to\infty}\phi_t(t_\eta)$ exists by Lemma EC.3, the first and last terms in the curly bracket cancel out in the limit:

$$\lim_{t\to\infty}\frac{1}{t-t_\eta}\left[\phi_t^2(t_\eta+1)N(t_\eta)-N(t_\eta)\phi_t^2(t_\eta)\right]=0.$$

Meanwhile, since $m(t)$ is bounded and $N(t)\to\infty$ as $t\to\infty$,

$$\begin{aligned}
&\lim_{t\to\infty}\frac{1}{t-t_\eta}\sum_{\ell=t_\eta}^{t}\frac{m^2(\ell)}{N(\ell)}=0,\\
&\lim_{t\to\infty}\frac{1}{t-t_\eta}2M(t-1)\phi_t(t)=\frac{2\bar{m}}{1-\eta}\lim_{t\to\infty}\frac{m(t)}{N(t)}=0\\
&\lim_{t\to\infty}\frac{1}{t-t_\eta}N(t)\phi_t^2(t)=\frac{\bar{n}}{1-\eta}\lim_{t\to\infty}\frac{m(t)}{N(t)}=0.
\end{aligned}$$

For the remaining terms, we further have

$$\lim_{t\to\infty}\frac{1}{t-t_\eta}2M(t_\eta-1)\phi_t(t_\eta)=\frac{\bar{m}^2}{\bar{n}}\frac{2\eta\ln\eta}{1-\eta}.$$

Also,

$$\lim_{t\to\infty}\frac{1}{t-t_\eta}\cdot 2\sum_{\ell=t_\eta+1}^{t-2}\frac{M(\ell)m(\ell+1)}{N(\ell)}=\frac{2\bar{m}^2}{\bar{n}},$$

which can be seen as follows. Since $M(t)/N(t)\to\bar{m}/\bar{n}$ as $t\to\infty$, for any $\epsilon>0$ there exists $T>0$ such that $|M(t)/N(t)-\bar{m}/\bar{n}|<\epsilon$ for all $t>T$. Hence, for all $t>T$,

$$\begin{aligned}
\left(\frac{\bar{m}}{\bar{n}}-\epsilon\right)\frac{1}{t-t_\eta}\sum_{\ell=t_\eta+1}^{t-2}m(\ell+1)&\le\frac{1}{t-t_\eta}\sum_{\ell=t_\eta+1}^{t-2}\frac{M(\ell)m(\ell+1)}{N(\ell)}\\
&\le\left(\frac{\bar{m}}{\bar{n}}+\epsilon\right)\frac{1}{t-t_\eta}\sum_{\ell=t_\eta+1}^{t-2}m(\ell+1),
\end{aligned}$$

where taking $t\to\infty$ and $\epsilon\to 0$ confirms the limit. Putting everything together,

$$\lim_{t\to\infty}\tilde{\beta}_\eta(t)=\frac{1}{\bar{m}}\left(\frac{2\bar{m}^2}{\bar{n}}+\frac{\bar{m}^2}{\bar{n}}\frac{2\eta\ln\eta}{1-\eta}\right)=\frac{\bar{m}}{\bar{n}}\left(2+\frac{2\eta\ln\eta}{1-\eta}\right)=\beta_\eta.$$

■

Proof of Theorem 3 It suffices to show

$$\sqrt{M(t)-M(t_\eta)}\Delta_{i,t} \Rightarrow \mathcal{N}(0,\tilde{\Sigma}_{i,\infty}), \quad \text{as } t\to\infty,$$

where

$$\tilde{\Sigma}_{i,\infty}(j,j') := \nabla\delta_{ij}(\theta^c)^\intercal\tilde{\Sigma}_D\nabla\delta_{ij'}(\theta^c) + \mathbf{Cov}\left(X_i(\theta^c)-X_j(\theta^c), X_i(\theta^c)-X_{j'}(\theta^c)\right),$$

in which $\tilde{\Sigma}_D := diag(\beta_{i,\eta}\Sigma_{D,1},\ldots,\beta_{S,\eta}\Sigma_{D,S})$ and β is defined in (EC.13). Let

$$\Delta_i(\theta) = (\delta_{i1}(\theta),\cdots,\delta_{ii-1}(\theta),\delta_{ii+1}(\theta),\cdots,\delta_{iK}(\theta)),$$

$$\hat{\Delta}_{i,t} = \left(\hat{\delta}_{i1,t},\cdots,\hat{\delta}_{ii-1,t},\hat{\delta}_{ii+1,t},\cdots,\hat{\delta}_{iK,t}\right),$$

$$\bar{\Delta}_{i,t} = \left(\bar{\delta}_{i1,t},\cdots,\bar{\delta}_{ii-1,t},\bar{\delta}_{ii+1,t},\cdots,\bar{\delta}_{iK,t}\right),$$

where

$$\bar{\delta}_{ij,t} := [M(t)-M(t_\eta)]^{-1}\sum_{\ell=t_\eta+1}^{t} m(\ell)[\mu_i(\hat{\theta}_\ell)-\mu_j(\hat{\theta}_\ell)].$$

We have the decomposition,

$$\begin{aligned}
&[M(t)-M(t_\eta)]^{1/2}[\hat{\Delta}_{i,t}-\Delta_i(\theta^c)]\\
=&\underbrace{[M(t)-M(t_\eta)]^{1/2}(\hat{\Delta}_{i,t}-\bar{\Delta}_{i,t})}_{Y_t} + \underbrace{[M(t)-M(t_\eta)]^{1/2}[\bar{\Delta}_{i,t}-\Delta_i(\theta^c)]}_{Z_t}
\end{aligned}$$

It suffices to show the respective convergence of Z_t and $Y_t \mid \mathcal{F}_t$, where $F_t := \sigma(\hat{\theta}_1,\cdots,\hat{\theta}_t)$, the filtration generated by input parameter estimates. We will show convergence of Y_t+Z_t through its characteristic function

$$\Phi_t(x) = \mathbb{E}[e^{\mathbf{i}x^T(Y_t+Z_t)}] = \mathbb{E}\left[e^{\mathbf{i}x^TZ_t}\mathbb{E}[e^{\mathbf{i}x^TY_t}\mid\mathcal{F}_t]\right].$$

If we can show the respective convergence of Z_t and $Y_t \mid \mathcal{F}_t$, then the result follows from the Dominated Convergence Theorem for complex-valued random variables.

(1) Convergence of $Y_t \mid \mathcal{F}_t$: Expand Y_t to get

$$\begin{aligned} Y_t =& [M(t) - M(t_\eta)]^{1/2}(\hat{\Delta}_{i,t} - \bar{\Delta}_{i,t}) \\ =& [M(t) - M(t_\eta)]^{-1/2} \sum_{\ell=t_\eta+1}^{t} \sum_{r=1}^{m(t)} \begin{pmatrix} X_{i,r}(\hat{\theta}_\ell) - \mu_i(\hat{\theta}_\ell) - (X_{1,r}(\hat{\theta}_\ell) - \mu_1(\hat{\theta}_\ell)) \\ \vdots \\ X_{i,r}(\hat{\theta}_\ell) - \mu_i(\hat{\theta}_\ell) - (X_{i-1,r}(\hat{\theta}_\ell) - \mu_{i-1}(\hat{\theta}_\ell)) \\ X_{i,r}(\hat{\theta}_\ell) - \mu_i(\hat{\theta}_\ell) - (X_{i+1,r}(\hat{\theta}_\ell) - \mu_{i+1}(\hat{\theta}_\ell)) \\ \vdots \\ X_{i,r}(\hat{\theta}_\ell) - \mu_i(\hat{\theta}_\ell) - (X_{K,r}(\hat{\theta}_\ell) - \mu_K(\hat{\theta}_\ell)) \end{pmatrix} \end{aligned}$$

where, conditioned on $\mathcal{F}_t$, $(X_{i,r}(\hat{\theta}_\ell) - \mu_i(\hat{\theta}_\ell - (X_{j,r}(\hat{\theta}_\ell) - \mu_j(\hat{\theta}_\ell))_{j\neq i}$ are independent random vectors with mean 0 and covariance matrix $\Sigma_i(\hat{\theta}_\ell)$. Since $\Sigma_{X,i}(\hat{\theta}_\ell) \to \Sigma_{X,i}(\theta^c)$ almost surely by the continuity of $\Sigma(\cdot)$. we can apply Lemma EC.2 to get (conditioned on $\mathcal{F}_t$)

$$[M(t)-M(t_\eta)]^{-1/2} \sum_{\ell=t_\eta+1}^{t} \sum_{r=1}^{m(t)} \begin{pmatrix} X_{i,r}(\hat{\theta}_\ell) - \mu_i(\hat{\theta}_\ell) - (X_{1,r}(\hat{\theta}_\ell) - \mu_1(\hat{\theta}_\ell)) \\ \vdots \\ X_{i,r}(\hat{\theta}_\ell) - \mu_i(\hat{\theta}_\ell) - (X_{i-1,r}(\hat{\theta}_\ell) - \mu_{i-1}(\hat{\theta}_\ell)) \\ X_{i,r}(\hat{\theta}_\ell) - \mu_i(\hat{\theta}_\ell) - (X_{i+1,r}(\hat{\theta}_\ell) - \mu_{i+1}(\hat{\theta}_\ell)) \\ \vdots \\ X_{i,r}(\hat{\theta}_\ell) - \mu_i(\hat{\theta}_\ell) - (X_{K,r}(\hat{\theta}_\ell) - \mu_K(\hat{\theta}_\ell)) \end{pmatrix} \Rightarrow \mathcal{N}(0, \Sigma_{X,i}(\theta^c)) \text{ as } t \to \infty$$

(2) Convergence of Z_t:

Since $\mu_j(\theta)$ is twice continuously differentiable for each j, so is $\Delta_i(\theta)$. By Taylor expansion,

$$\begin{aligned} Z_t =& [M(t) - M(t_\eta)]^{-1/2} \sum_{\ell=t_\eta+1}^{t} m(\ell)[\Delta_i(\hat{\theta}_\ell) - \Delta_i(\theta^c)] \\ =& [M(t) - M(t_\eta)]^{-1/2} \nabla(\Delta_i(\theta^c)) \sum_{\ell=t_\eta+1}^{t} m(\ell)(\hat{\theta}_\ell - \theta^c) \qquad (Z_{t,1}) \\ &+ [M(t) - M(t_\eta)]^{-1/2} \sum_{\ell=t_\eta+1}^{t} m(\ell)^\intercal \mathcal{O}(||\hat{\theta}_\ell - \theta^c||_2)(\hat{\theta}_\ell - \theta^c). \qquad (Z_{t,2}) \end{aligned}$$

We show the convergence of $Z_{t,1}$ and $Z_{t,2}$, respectively.

(2.1) Convergence of $Z_{t,1}$

Our goal is to show that

$$[M(t)-M(t_\eta)]^{-1/2}\sum_{\ell=t_\eta+1}^{t} m(\ell)(\hat{\theta}_\ell-\theta^c)\Rightarrow\mathcal{N}(0,\tilde{\Sigma}),\quad \text{as } t\to\infty.$$

Since the input distributions are mutually independent, it suffices to show that for any s,

$$[M(t)-M(t_\eta)]^{-1/2}\sum_{\ell=t_\eta+1}^{t} m(\ell)(\hat{\theta}_{s,\ell}-\theta_s^c)\Rightarrow\mathcal{N}(0,\beta_{s,\eta}\Sigma_{D,s}),\quad \text{as } t\to\infty. \tag{EC.15}$$

Note that $\{\hat{\theta}_{s,\ell}\}_{\ell=1}^{t}$ are linear combinations of $\{D_{s,j}\}_{j=1}^{N_s(t)}$, so we can rearrange the terms to obtain

$$\sum_{\ell=t_\eta+1}^{t} m(\ell)(\hat{\theta}_{s,\ell}-\theta_s^c)=\phi_{i,s,t}(t_\eta+1)\sum_{j=1}^{N_s(t_\eta)} D_{s,j}+\sum_{\ell=t_\eta+1}^{t}\phi_{i,s,t}(\ell)\sum_{j=N_s(\ell-1)+1}^{N_s(\ell)} D_{s,j},$$

where ϕ is defined in (EC.11). Since $\{D_{s,j}\}$ are i.i.d. and EC.12 holds by assumption, (EC.15) can be verified by invoking Lemma EC.1 (details omitted).

Hence,

$$Z_{t,1}\Rightarrow\mathcal{N}\left(0,\nabla(\Delta_{i,t}(\theta^c))^\intercal\beta_{s,\eta}\Sigma_{D,s}\nabla(\Delta_{i,t}(\theta^c))\right)\ \text{as } t\to\infty.$$

(2.2) Convergence of $Z_{t,2}$

It is left to show that the residual term $Z_{t,2}$ vanishes in probability. Since a.s.,

$$\|\hat{\theta}_\ell-\theta^c\|_2^2=\sum_{s=1}^{S}\|\hat{\theta}_{s,\ell}-\theta_s^c\|_2^2\le M_2\sum_{s=1}^{S}\frac{\log\log N_s(\ell)}{N_s(\ell)}\le M_2 S\log(U\ell)/\ell,$$

for some $M_2>0, U\ge\sup_\ell\{\sup_s n_s(\ell), m(\ell)\}$ by the Law of the Iterated Logarithm. Then we have a.s. $\exists M_1>0$,

$$\begin{aligned}
||Z_{t,2}||_2 &\le [M(t)-M(t_\eta)]^{-1/2}\sum_{\ell=t_\eta+1}^{t} m(\ell)M_1||\hat{\theta}_\ell-\theta^c||_2^2\\
&\le [M(t)-M(t_\eta)]^{-1/2}\sum_{\ell=t_\eta+1}^{t} m(\ell)M_1M_2S\log(U\ell)/\ell\\
&\le (t-t_\eta)^{-1/2}UM_1M_2S\log(Ut)\sum_{\ell=t_\eta+1}^{t}1/\ell\\
&\le (t-t_\eta)^{-1/2}UM_1M_2S\log(Ut)\log(t),
\end{aligned}$$

which converges to 0 as $t \to \infty$. Hence we get

$$Z_t \Rightarrow \mathcal{N}\left(0, \nabla(\Delta_{i,t}(\theta^c))^\intercal \beta_{s,\eta} \Sigma_{D,s} \nabla(\Delta_{i,t}(\theta^c))\right) \text{ as } t \to \infty.$$

Together with the convergence of $Y_t|\mathcal{F}_t$, we complete the proof.

Proof of Proposition EC.1 The proof is developed in three steps.

1. Strict convexity of ψ_κ. The first-order and second-order derivatives of ψ_κ are given by

$$\begin{aligned} \psi'_\kappa(\eta) &= \frac{(1+\kappa)(1-\eta)+(1+\eta)\ln\eta}{(1-\eta)^3}, \\ \psi''_\kappa(\eta) &= \frac{2(1+\kappa)(1-\eta)+(4+2\eta)\ln\eta-\eta+\eta^{-1}}{(1-\eta)^4}. \end{aligned} \tag{EC.16}$$

We claim that $\psi''_\kappa(\eta) > 0$ on $(0,1)$. To see this, denote the numerator of $\psi''_\kappa(\eta)$ as

$$g_\kappa(\eta) := 2(1+\kappa)(1-\eta)+(4+2\eta)\ln\eta-\eta+\eta^{-1}.$$

It can be verified that

$$g'_\kappa(\eta) = -(1+2\kappa)+4\eta^{-1}-\eta^{-2}+2\ln\eta, \quad g''_\kappa(\eta) = 2\eta^{-3}(\eta-1)^2.$$

We have $g'_\kappa(1) = 2(1-\kappa) < 0$ and $g''_\kappa(\eta) > 0$ for $\eta \in (0,1)$, which implies that $g'_\kappa(\eta) < 0$ for $\eta \in (0,1)$. Furthermore, since $g_\kappa(1) = 0$, we know that $g_\kappa(\eta) > 0$ for $\eta \in (0,1)$ and therefore ψ_κ is strictly convex in $(0,1)$. Furthermore, since $\lim_{\eta\to 0}\psi'_\kappa(\eta) = -\infty$ and $\lim_{\eta\to 1}\psi'_\kappa(\eta) = 1+\kappa$, ψ_κ must always have a unique minimizer in $(0,1)$, which we shall denote by $\eta^*(\kappa)$.

2. Limiting behavior of $\eta^*(\kappa)$ as κ approaches 0 or 1. To begin with, according to (EC.16), $\eta^*(\kappa)$ is the solution to the following equation.

$$\varphi_\kappa(\eta) := (1+\kappa)(1-\eta)+(\eta+1)\ln\eta = 0.$$

(i) $\eta^*(\kappa) \le \kappa^{-1}$.

Note that φ_κ is concave since $\varphi''_\kappa(\eta) = \eta^{-1}-\eta^{-2} < 0$ for $\eta \in (0,1)$. Together with the fact that $\lim_{\eta\to 0}\varphi_\kappa(\eta) = -\infty$ and $\lim_{\eta\to 1}\varphi_\kappa(\eta) = 1+\kappa$, φ_κ can cross 0 only once. Thus, for any $\eta' \in (0,1)$ such that $\varphi_\kappa(\eta') \ge 0$, we must have $\eta^*(\kappa) \le \eta'$. Finally,

$$\begin{aligned} \varphi_\kappa(\kappa^{-1}) &= \kappa-\kappa^{-1}-(\kappa^{-1}+1)\ln\kappa \\ &\ge \kappa-\kappa^{-1}-(\kappa^{-1}+1)(\kappa-1) = 0, \end{aligned}$$

where the last inequality follows from the fact that $\ln\eta \le \eta - 1$ for all $\eta > 0$. This implies that $\eta^*(\kappa) \le \kappa^{-1}$.

(ii) $\lim_{\kappa\to 1}\eta^*(\kappa) = 1$.

By a similar argument, for any $\epsilon \in (0,1)$ such that $\varphi_\kappa(1-\epsilon) \le 0$, we would have $\eta^*(\kappa) \ge 1-\epsilon$. Therefore, it suffices to show that

$$\liminf_{\kappa\to 1}\varphi_\kappa(1-\epsilon) \le 0, \quad \forall \epsilon \in (0,1). \tag{EC.17}$$

as this would imply that $\liminf_{\kappa\to 1}\eta^*(\kappa) \ge 1-\epsilon$ for any fixed $\epsilon \in (0,1)$. Note that for any fixed ϵ,

$$\liminf_{\kappa\to 1}\varphi_\kappa(1-\epsilon) = 2\epsilon + (2-\epsilon)\ln(1-\epsilon) := h(\epsilon),$$

and note that $h(0) = 0$ and $h(\epsilon) \to -\infty$ as $\epsilon \to 1$. Furthermore,

$$\begin{aligned} h'(\epsilon) &= 2 - \ln(1-\epsilon) - \frac{2-\epsilon}{1-\epsilon} \\ &\le 2 - \left(1 - \frac{1}{1-\epsilon}\right) - \frac{2-\epsilon}{1-\epsilon} = 0, \end{aligned}$$

where the last inequality follows from the fact that $\ln x \ge 1 - x^{-1}$ for all $x > 0$. We conclude that $h(\epsilon) \le 0$ for any $\epsilon \in (0,1)$.

Step 3. Maximum variance reduction rate. We first show that

$$\inf_{\kappa>1}\frac{\psi_\kappa(\eta^*(\kappa))}{\psi_\kappa(0)} \le \frac{1}{2}. \tag{EC.18}$$

Note that $\psi_\kappa(0) = \kappa$ for any $\kappa > 1$. Consider $\psi_\kappa(\kappa^{-k}), k \in \mathbb{Z}^+$, for which we have

$$\psi_\kappa(\kappa^{-k}) = \frac{\kappa^k[\kappa(\kappa^k - 1) - k\ln\kappa]}{(\kappa^k - 1)^2},$$

where sending κ to 1 and applying L'Hospital's Rule gives

$$\inf_{\kappa>1}\frac{\psi_\kappa(\eta^*(\kappa))}{\psi_\kappa(0)} \le \lim_{\kappa\to 1}\frac{\psi_\kappa(\kappa^{-k})}{\psi_\kappa(0)} = \frac{1}{2} + \frac{1}{k}.$$

This implies (EC.18) as k can be arbitrarily large. To prove the opposite direction of (EC.18), note that for any fixed $\eta \in (0,1)$, $\psi_\kappa(\eta)/\psi_\kappa(0)$ is a decreasing function in κ. Therefore,

$$\inf_{\kappa>1}\frac{\psi_\kappa(\eta^*(\kappa))}{\psi_\kappa(0)} \ge \inf_{\kappa>1}\frac{\psi_1(\eta^*(\kappa))}{\psi_1(0)} \ge \inf_{\eta\in(0,1)}\psi_1(\eta) = \inf_{\eta\in(0,1)}\left\{\frac{1}{1-\eta} + \frac{\eta\ln\eta}{(1-\eta)^2}\right\} = \frac{1}{2},$$

which completes the proof of (EC.4). ■

6. 混流装配线：**Mixed-model assembly lines with variable takt and open stations (POM, 2023)**

来源：Production and Operations Management. Vol. 32, No. 3, 2023, pp. 704–722.
https://doi.org/10.1111/poms.13893

作者：Arnd Huchzermeier, Tobias Mönch

关键词：混流排序；开放工位；可变投放速率；生产线长度约束

推荐人：盛舟山

推荐理由：产品定制与工时差异使混流装配线的固定投放节拍产生等待和超负荷劳动。本文首先给出理想的可变投放情形：在工时平衡、工位长度一致等条件下，可以消除闲置和辅助劳动。随后，作者建立统一的混合整数模型，比较固定与可变节拍以及开放与封闭工位，同时计入劳动低效和延长生产线的成本，并为异质工时与工位长度设计启发式。工业基准实验表明，开放工位与可变节拍的组合能够降低劳动成本、缩短生产线及通过时间。

Check for updates

Received: 5 January 2021 | Accepted: 24 August 2022

DOI: 10.1111/poms.13893

POMS

ORIGINAL ARTICLE

Mixed-model assembly lines with variable takt and open stations

Arnd Huchzermeier | **Tobias Mönch**

WHU – Otto Beisheim School of Management, Burgplatz 2, Vallendar, Germany

Correspondence
Arnd Huchzermeier, WHU – Otto Beisheim School of Management, Burgplatz 2, 56179 Vallendar, Germany.
Email: Arnd.Huchzermeier@whu.edu

Handling Editor: Panos Kouvelis

Abstract

Industry trends such as product customization, radical innovation, and local production accelerate the adoption of mixed-model assembly lines (MMALs) that can cope with a widening gap between model processing times and true build to order capabilitiy. The existing high work content deviations on such assembly lines stress production planning, especially the assembly line sequencing. Most manufacturers set the launching rate for all assembly line products to a fixed launching rate resulting in rising utility work and idle time when system load increases. We present an "ideal" variable rate launching (VRL) case resulting in minimal computation and achieving 100% productivity (full elimination of idle time and utility work) for balanced assembly times and homogeneous station lengths. Managers should foster the ideal circumstances where operators need not wait for a preceding task to be completed and product sequence restrictions are eliminated, thus enabling unmatched production flexibility. Furthermore, we present a mixed-integer model to analyze both closed and open workstations on an MMAL for fixed rate launching and VRL. This model incorporates costs not only for labor inefficiencies but also for extending the line length. We present a heuristic solution method when process times and station lengths are heterogeneous and demonstrate that the variable takt dominates the fixed takt. In a numerical, industrial benchmark study, we illustrate that a VRL strategy with open stations has significantly lower labor costs as well as a substantially reduced total line length and thus lower throughput time.

KEYWORDS
line length restriction, mixed-model sequencing, open stations, variable rate launching

1 | INTRODUCTION

During our visits with finalists in the INSEAD–WHU Industrial Excellence Award (IEA) competition (see www.industrial-excellence-award.eu), numerous executives from the automotive, machine tool, and electrical engineering industries emphasized that their respective businesses are being disrupted by three major trends: (i) increased product customization with a strong focus on high-end products (e.g., sport utility vehicles); (ii) radical innovation, as when electrical vehicles must be assembled alongside conventional vehicles; and (iii) natural hedging—that is, increased local production of their broader product portfolios because of tariff and tax issues, carbon emissions, pandemics, and/or the need to mitigate risk by matching foreign revenue with local costs (Cohen et al., 2018, 2022; Knowledge@Wharton, 2019).

This dynamic business environment exerts pressure on existing linear and inflexible assembly lines to adopt a more efficient layout and to handle a wider range of products at a single site (oftentimes "brownfield" plants with an existing assembly setup). Many executives voiced the concern that assembly lines must be expanded considerably if they are to handle such product diversity. Most of these manufacturers strive to fulfill unique customer requirements in a true build-to-order environment—one of the predominant variants of mass customization—where the next unit scheduled for assembly is linked to the last incoming customer order

Accepted by Panos Kouvelis, after two revision.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.
© 2022 The Authors. *Production and Operations Management* published by Wiley Periodicals LLC on behalf of Production and Operations Management Society.

(Agrawal et al., 2001; Wang et al., 2017). The central question that arises in this context is whether existing assembly line systems can effectively cope with such rapid changes and with the ever rising demand for greater flexibility and volume (Bloomberg, 2019).

Leading production systems are today based on four crucial objectives: takt, flow, zero defects, and pull; all of these derive from the Toyota Production System. Takt time (or cycle time) is the clock speed of every assembly process, which is determined by the available operating hours and the output per period that is needed to satisfy demand; it is the key lever of productivity (Monden, 2011). So when a new model with significantly greater work content is scheduled for production, plant managers can either plan a separate assembly line or adjust the existing line's parameters—that is, increase the fixed takt time. Yet, in either case, productivity suffers. Furthermore, the greater the difference in the workload of units sequenced on the line, the more difficult it is to assemble in a continuous flow of workload where the operator's individual workload is balanced—one of the cornerstones of lean. When assembling highly variable, customer-specific products in a fixed takt time, such a steady flow cannot be achieved. Mönch et al. (2022) show how variable takt time groups enable a convergence toward a perfect assembly flow, independent of workload differences within the product portfolio. It follows that determining the optimal launching strategy is critical when introducing multiple models with unequal work content onto a given assembly line. The approach most commonly used to determine the launching interval (λ) is fixed rate launching (FRL), where the interval between two adjacent units on the line approximates the fixed takt time (including all inefficiencies) (Bard et al., 1992). Our paper presents the concept of variable rate launching (VRL), under which models are placed onto the conveyor depending on their processing time. When this procedure is followed, labor inefficiencies are substantially reduced.

In 2017, Fendt—the German subsidiary of AGCO and an innovation leader within the agricultural machinery market—implemented such a VRL system, the "VarioTakt." Adopting this approach allowed Fendt to assemble all 10 of its product series, including an unlimited number of model configurations, on a single assembly line and with relatively few concerns about sequence restrictions or product complexity (Huchzermeier et al., 2020a). Note that in 2020, the World Economic Forum and McKinsey & Company selected Fendt's Marktoberdorf plant as one of their global digital lighthouse factories (Betti et al., 2020). The four cornerstones of our "ideal" case are based on Fendt's current operations (see Theorem 1 in Section 5). Without the VarioTakt, Fendt would not have been able to cope with the increased customization resulting from greater competition and its expansion into new markets (e.g., North America) (Bebersdorf, 2017). A recent process innovation enabled by this novel approach is that the "Fendt 1000 Vario"—the largest premium tractor in the market, with 738% more horsepower than Fendt's smallest tractor—is now assembled with all other models on the same assembly line (Bebersdorf, 2017). The complex tasks of sequencing products can be omitted as models are perfectly balanced over the entire set of stations and all stations have the same length. These cornerstones of the ideal case are discussed in Section 5. This is particulary of great interest for manufacturing executives during times when the product mix changes quickly, for example, due to recessions or pandemics. Huchzermeier et al. (2020b) provide an in-depth description of the Fendt VarioTakt and how Fendt achieves the underlying requirement of perfectly balancing station processing times. Also, Mönch et al. (2021) describe how variable launching rates simplify the assembly line balancing problem and achieve high operator utilization levels by almost perfectly balancing assembly times.

Most research that explores how best to cope with mixed models in assembly has focused on minimizing idle time and utility work (or work overload) independently (Bolat & Yano, 1992b; Dar-El & Cother, 1975; Okamura & Yamashina, 1979; Thomopoulos, 1967; Tsai, 1995; Yano & Rachamadugu, 1991) or on minimizing overall line length (Bard et al., 1992; Dar-El & Cother, 1975; Dar-El & Cucuy, 1977). Utility work arises when processing times on a unit exceed the operators capacity. Idle time occurs when the operator cannot immediately start work on the next unit. We provide a more detailed description of both concepts in Section 4. The paper by Dar-El (1978) is the first to address overall line length and throughput time while considering both FRL and VRL. The numerical studies of Bard et al. (1992), Sarker and Pan (2001), and Fattahi and Salehi (2009) show that VRL outperforms FRL in terms of the costs associated with idle time and utility work—but only for the case of closed workstations. There has been hardly any research that models the VRL strategy with open stations, and no empirical studies have discussed the requirements and practical benefits of implementing a VRL strategy on assembly lines with open stations.

Our paper's contribution is to formulate a mixed-integer model of the optimization problem for both closed and open workstations on a mixed-model assembly line (MMAL) with FRL and VRL. We refer to this formulation as the *general* case, and it accounts for circumstances that have not previously been considered together. In addition to incorporating the costs associated with operator idleness and utility work, our analysis also considers—given the space restrictions commonly encountered in manufacturing sites—the cost of extending the assembly line length. Unlike most previous research, this study also accounts for sequence-dependent setup times when switching between the production of two distinct product models. Following the real-world example of Fendt, we identify several prerequisites for the *ideal* case: implementation of a VRL strategy under which employees are always busy, yet there are no workload peaks. This setting features 100% operator utilization and station lengths that are reduced to a bare minimum; these goals can be achieved by balancing assembly tasks and allowing parallel work on products as long as workers do not interfere with each other. Most importantly, this approach does not require extensive

computations and is flexible enough to handle almost any change in product mix at nearly any time. We also present crucial insights on the practical application of VRL. In particular, we show how a firm can switch easily from a fixed takt system requiring high computational effort to a variable takt system with full flexibility, 100% productivity, and hardly any computational burden.

The rest of our paper proceeds as follows. In Section 2, we describe the characteristics of an MMAL, after which Section 3 reviews the literature on FRL and VRL with closed and open stations. Our unifying model formulation—which we refer to as the general case—is proposed in Section 4. Section 5 then discusses the modeling and implementation of the ideal case for a VRL system. In Section 6, we develop heuristic solution procedures and conduct a benchmarking study. We conclude in Section 7 with summary remarks and managerial insights.

2 | MIXED-MODEL ASSEMBLY LINE SYSTEMS

Assembly lines are classified into single-model assembly lines (SMALs) and mixed-model assembly lines (MMALs), depending on the number of models being manufactured on the same line (Al-Hawari et al., 2015; Becker & Scholl, 2006). On SMALs, a single model is assembled. Whereas, on MMALs, multiple different product models are assembled with setup times between different models (Becker & Scholl, 2006; Boysen, Fliedner, & Scholl, 2008). This section briefly outlines the basic structure of the MMAL systems considered in our paper (for a more detailed description of MMALs, see Buxey et al., 1973). Such assembly lines are used to assemble different workpieces (or products, or units), which vary in terms of assembly tasks and work content. A product model (model, hereafter) is defined by the same underlying design and physical characteristics as well as bill of materials (e.g., the BMW 3 series or the BMW 7 series). Thus, two workpieces can be of the same model. In this definition, model configurations, for example, interior upgrades, are not included. However, our approach also works for model configurations, where each workpiece belongs to a model configuration and not a model. Generally speaking, MMALs are partitioned into J work stations that are connected by a conveyor belt moving—from upstream to downstream—at a constant velocity v. There exists no buffer inventory between stations, and workpieces directly move into the next station. When looking to manufacture multiple models on a single assembly line, a manufacturer faces two closely related problems: the assembly line balancing problem and the assembly line sequencing problem. Line balancing defines the assignment of total workload among stations as evenly as possible (Boysen et al., 2007; Kilbridge & Wester, 1962; Thomopoulos, 1967). When a manufacturer decides to assemble models with differing work content on a single assembly line, the medium-term line balancing problem is linked to the short-term assembly line sequencing problem (Thomopoulos 1967). Since assembly times differ between models and stations, it follows that operators are either underutilized or overloaded in terms of the assembly line's average work content. Once the best possible line balance is achieved, line sequencing determines the assembly order of products on the line based on leveling operator workload across total demand. The optimal assembly sequence has the effect of leveling workload deviations among the models based on a given assembly line balance, that is, by alternating single products (or batches) that have high and low levels of utilization. In the following, the focus is solely on the assembly line sequencing problem and we assume that stations are "balanced" in the sense that they reflect the best possible assignment of assembly tasks to stations. Then, the number of work stations is given and we do not consider the possibility of rebalancing the assembly line. Mönch et al. (2021) show how variable takt times reduce the complexity of the assembly line balancing problem. Moreover, the workpiece sequence is predetermined before assembly starts. Workpieces are processed on a first-come, first-serve basis and in the same sequence as they are launched onto the line. Different product model types m are assembled in a fixed station order, and the workpieces in the sequence are denoted by i. One operator is assigned to each station j; that operator moves downstream with the conveyor while performing an assembly task on one workpiece at a time. Material supply is located at the upstream station boundary. We consider not only the deterministic assembly time t_{mj} but also sequence-dependent setup times s_{rmj}. The former is also seen as deterministic in practice, where a factor for unexpected downtimes and errors is included in the overall time. The latter occur when, at station j, an operator switches from processing a model r to a model m. Sequence-independent setup times (i.e., times that depend exclusively on the job to be processed) are included in the assembly time. To simplify the analysis, we assume that setup is performed by an operator while traveling with the workpiece on the conveyor. This assumption does not limit the generalizability of our model. In most real-world assembly lines, for example, in automotive, operators travel with the unit on the constantly paced assembly line. The timing with which models are launched onto the conveyor reflects one of three methods: (i) taking the maximum takt time over all models and stations; (ii) using the (weighted) average takt time; or (iii) dynamically adjusting the takt time—that is, adopting variable rate launching. These strategies are illustrated in Figure 1. The first two approaches necessarily lead to inefficiencies caused by idle time (Cases i and ii) and utility work (Case ii only). We shall demonstrate that, in the ideal case, idle time and utility work can be completely eliminated by varying the launching rate between successive units of production (Case iii). Note, that the utilization of operators is linked to operator idleness, where operator utilization is zero during idle time.

Depending on the nature of the task and on the assembly system's physical layout, station boundaries can be either open or closed. When stations are *closed*, the operator is not

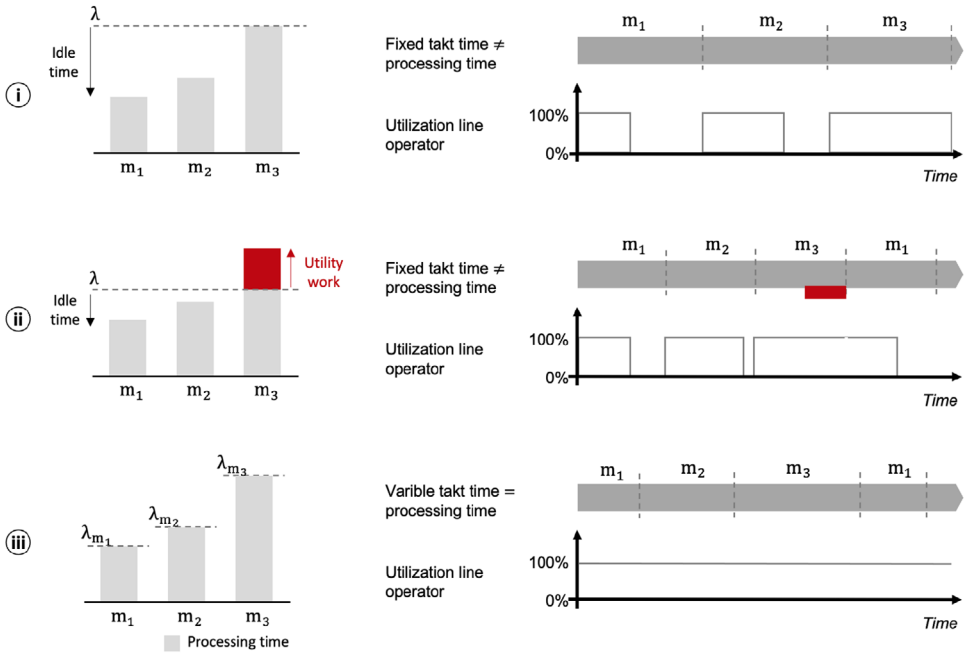


FIGURE 1 Launching rate strategies: Using the maximum process time (Case i), the average process time (Case ii), or variable process times (Case iii) [Color figure can be viewed at wileyonlinelibrary.com]

allowed to cross either its upstream or downstream boundary. This restriction might be needed, for instance, to preclude unproductive traveling or when technological or safety issues require that assigned tasks be performed within the station boundary—for example, in heating chambers or paint booths (Bard et al., 1992). Yet, when a workstation is *open*, the operator can cross its upstream boundary (to start work on a product outside that boundary) and/or its downstream boundary (to finish work on a product). Hence, the overlap or "drift" area between two adjacent stations extends an operator's allowable work area: by L_j^{UO} and L_j^{DO} for, respectively, the upstream overlap and downstream overlap (recall, however that two adjacent operators cannot interfere with each other's work). Even in open stations, the maximum overlap between subsequent stations can be restricted—as might be necessary when power tools, material handling equipment, or robotic devices have a limited range. Work overload can occur even in the presence of drift areas, but our model makes it easier to minimize overload. A hybrid MMAL is one that includes both closed and open work stations (Dar-El, 1978). In contrast to most researchers, we do not assume that assembly lines have homogeneous station characteristics. In case of work overload, specialized utility workers assist line operators facing utility work within the station. Another design criterion for MMALs is the strategy (FRL or VRL) for launching workpieces onto the paced conveyor. The chosen launching strategy depends on the takt time and the resulting distance between two successive units on the conveyor. If a constant-speed conveyor links all stations and if there are no buffers between those stations, then the launching interval (a.k.a. the feed-in rate) is equal to the takt time. In an assembly sequence, the distance between two consecutive workpieces i and $i+1$ is determined by two factors: the launching interval λ_i and the conveyor velocity v (Bard et al., 1992). In the FRL scenario, workpieces are spaced equally on the assembly line irrespective of model type. The fixed launching interval is usually calculated as the weighted average of the total assembly time of all products over all stations (Dar-El, 1978; Kim et al., 1996; Sarker & Pan, 2001). If several models with assembly times that greatly exceed the average follow one another, then the operator is moved to the station's downstream boundary—increasing the risk of work overload or even line stoppage. Since the *fixed* launching interval is the same for all pairs of successive workpieces, it follows that $\lambda = \lambda_i$ for all $i \in I$. In contrast, the *variable* launching interval λ_i allows for launching time variations in the placement of two adjacent products onto the conveyor regardless of the model type: $\lambda_i \gtreqless \lambda_{i+1}$ for all $i \in I$ (Bard et al., 1992). This paper examines the VRL strategy. Our model's assumptions are summarized in Table 1.

Note also that the product mix is predetermined and that sequences are built using the minimal part set (MPS). The notion of an MPS is widely used in the literature on MMAL sequencing, and it is defined as the smallest part set needed for the proportion of models that matches overall demand D_M (Bard et al., 1992).

Figure 2 presents an operator movement diagram for the case of open stations and VRL. This particular three-station scenario can be viewed as one component of a larger assembly line system. The horizontal dimension represents the operator's movement, and the vertical position corresponds to unit i's location in the sequence. Here, we assume that the MPS is $(1, 1, 1)$; the station length is L_j. Bold vertical lines in

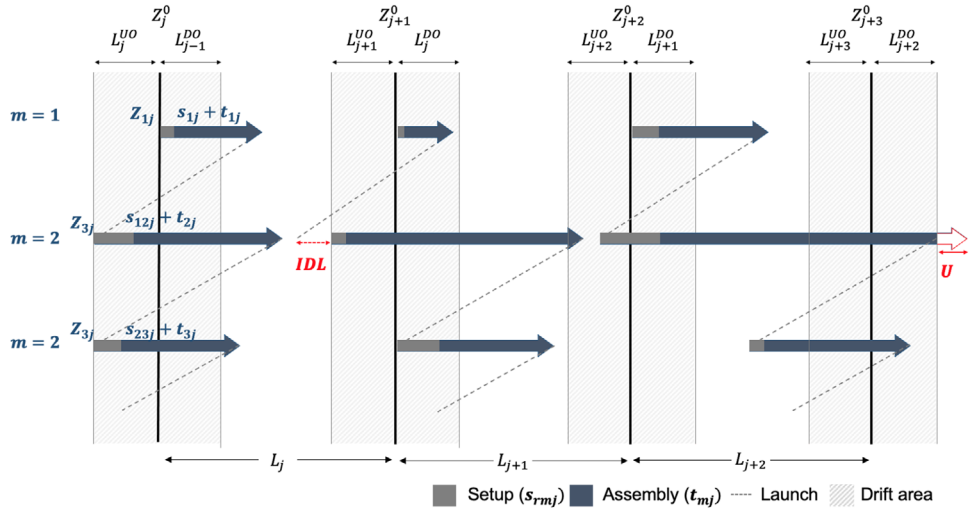


FIGURE 2 Operator movement diagram for the open-station case with variable launching rates [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Assumptions of the model

- The workstations of the assembly line can be open, closed, or hybrid
- No feeding lines or preassembly stations/lines are considered
- A moving conveyor with uniform speed carries the workpieces forward
- Each workstation is occupied by a single operator
- An operator can perform tasks on only one workpiece at a time
- No buffers exist within a workstation or between two adjacent workstations
- Extending the current assembly line is costly
- The workpiece order on the conveyor is fixed and does not vary between workstations
- Sequence-dependent setup times are not part of operator's assembly time
- Material supply is located near the upstream station boundary
- Utility workers assist operators facing utility work within the station

Figure 2 represent the station boundaries, where Z_j^0 represents the station upstream boundary (also called "zero reference point"). The thin vertical lines mark the maximum permitted upstream overlap (L_j^{UO}) and downstream overlap (L_j^{DO}) of station j; thus stations $j-1$ and $j+1$ enclose the operator drift area. Finally, Z_{ij} defines the starting position of product i at station j.

In the figure, a horizontal arrow starting at location Z_{ij} shows the "length equivalent" of the processing time ($t_{mj} + s_{rmj}$) needed to finish the work content of workpiece i at station j. The launching rate's length equivalent $\lambda_i v$ between two consecutive workpieces is represented by a dashed line. Once the operator has either completed the workpiece task(s) or reached the downstream limit ($Z_j^0 + L_j + L_j^{\mathrm{DO}}$), she returns upstream to continue work on the next item. We assume for the sake of simplicity that the operator moves instantaneously (i.e., in a negligible amount of time) from the completion of work on one product ZF to the starting point of work on the next product Z (while picking up needed material near Z_j^0). Thus, the operator immediately starts working on the next product at its current location. If the operator is unable to finish work on a product within the station limits, then utility work U comes into play. Such utility work is the first source of inefficiency; the second source is operator idleness (Thomopoulos, 1967). Idle time (IDL) occurs when the operator either finishes work early on a unit or cannot immediately work on the next unit. When there is no idle time, the station operator remains continuously occupied during the entire sequence.

3 | LITERATURE REVIEW

Over the past decades, a large amount of academic research has addressed the design criteria and configuration of MMAL systems. Thomopoulos (1967) distinguishes between two related barriers to an effective utilization of MMAL systems: the assembly line *balancing* problem and the assembly line *sequencing* problem. Line balancing involves the assignment of tasks (i.e., with and without precedence constraints) to a line's workstations so that the lowest takt time—which is usually the same for all products—is achieved. An overview of the line balancing problem can be found, for example, in Baybars (1986), Becker and Scholl (2006), Boysen et al. (2007), and Battaïa and Dolgui (2013). Mönch et al. (2021) present a generalizable mixed-integer programming model that regards variable takt times and random customization, that is, random processing times, when solving the assembly line balancing problem. Moreover, Mönch et al. (2022) present a preliminary step to the line balancing and sequencing problem by defining variable takt times groups, where units with similar assembly times are assigned the same takt time. The results show that defining takt time groups can significantly reduce the effort connected to solving the assembly line

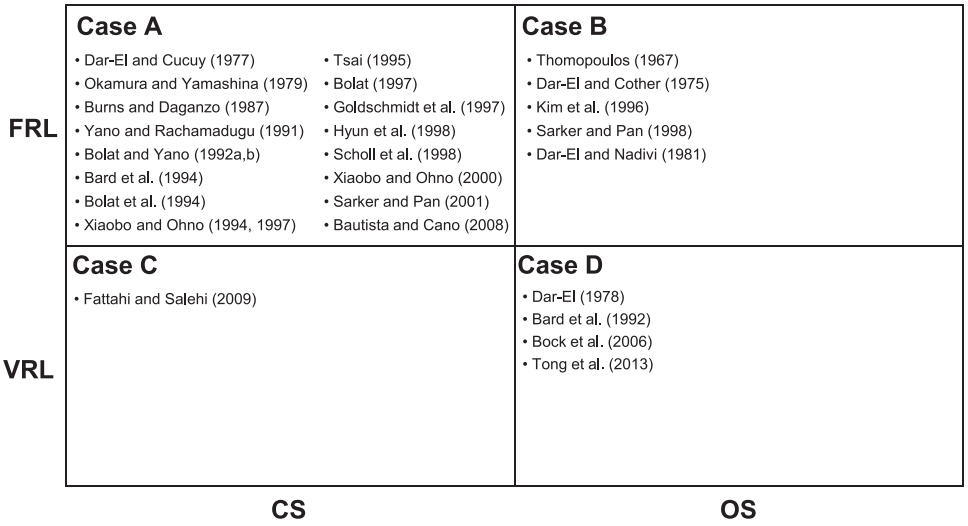


FIGURE 3 Classification of MMAL research based on type of work station and type of launching rate

balancing and sequencing problem. Our paper focuses on the line sequencing problem (for an extensive overview of literature on MMAL sequencing, see Boysen et al., 2009).

In the area of MMAL product sequencing, scholars have highlighted the development of algorithms and (meta-) heuristics to ensure that parts are used at a constant rate and to minimize idle time, utility work, line length, throughput time, total setup costs, and the risk of stopping the conveyor. Because this paper focuses on the multiple objectives of minimizing idle time and utility work while accommodating line length restrictions, here we review only that literature devoted to optimizing one or more of these objectives. In this endeavor, we classify the research on MMAL sequencing in terms of the launching rate (i.e., fixed or variable) and the chosen station type: open station (OS) or closed station (CS) boundaries. For each of the resulting four cases, we discuss the related literature (as illustrated schematically in Figure 3) in turn.

Case A. Dar-El and Cucuy (1977) present, for the single-station case, an optimization algorithm for minimizing the overall assembly line length with zero idle time and utility work. Okamura and Yamashina (1979) introduce a heuristic that aims to minimize—via interchanging product pairs—the maximum downstream displacement that an operator faces during task completion. Burns and Daganzo (1987) use grouping and spacing rules to find an efficient sequence for minimizing setup and capacity costs. Yano and Rachamadugu (1991) develop a mathematical programming model that aims to minimize the work overload at a single station when operators are never idle and perform one of two distinct tasks. Bolat and Yano (1992a, 1992b) describe a "greedy" procedure based on spacing rules; this procedure identifies an optimal solution for minimizing total utility work at a single station that is responsible for two distinct tasks. The dual-criteria minimization model of Bard et al. (1994) uses weighted factors to analyze the trade-off between line length and parts usage. Bolat et al. (1994) formulate a mixed-integer mathematical programming model that minimizes utility work and setup costs. Xiaobo and Ohno (1994, 1997) introduce a simulated annealing approach for large-scale problems, thereby minimizing the risk of conveyor stoppage. Tsai (1995) proves that the problems of minimizing conveyor stoppage risk and utility work are both NP-hard in the strong sense; for the (closed) single-station case with FRL, where all assembly times take one of two distinct values, he proposes an algorithm that reaches an optimal solution in $O(\log N)$ time. Bolat (1997) presents a computational study of a simulated annealing procedure under which units in the sequence are exchanged not randomly but rather based on specific knowledge about the problem, thus minimizing utility work. Goldschmidt et al. (1997) are the first to report complexity results for the sequencing problem of minimizing overall line length so that the operator can complete the required work on all products; the sequencing problem is strongly NP-complete not only for the single-station case but also for the case of multiple stations. Hyun et al. (1998) develop a genetic algorithm that aims to achieve the objectives of reducing utility work, achieving a constant rate of parts usage, and minimizing total setup costs; solutions generated by the algorithm are Pareto optimal (or nearly so). Scholl et al. (1998) develop a Tabu search algorithm that consists of a vocabulary-building strategy to minimize total work overload on the assembly line. Xiaobo and Ohno (2000) consider the operator's upstream walking time in their heuristic for minimizing the risk of stopping the conveyor. The model of Sarker and Pan (2001) is based on mixed-integer programming and seeks to minimize total costs of idle time and utility work by determining the optimal line parameters: launching rate, station length, operator starting point of work, upstream walking distance, and locus of the operator's movement. These authors use a sensitivity analysis to show that total costs for idle time and utility work decrease with increasing line length. Bautista and Cano (2008) extend the mathematical model of Okamura and Yamashina (1979) in proposing a heuristic algorithm

that yields a cyclical pattern of operator displacement while minimizing work overload and idleness.

Case B. Thomopoulos (1967) develops a simple approximate algorithm for determining a sequence that minimizes total costs of the inefficiencies—namely, those due to idleness, work deficiency, work congestion, and/or utility work—that result from sequencing an additional model onto the line. Dar-El and Cother (1975) present a selection heuristic that minimizes the overall station length with no operator interference (i.e., zero operator idle time and utility work). Kim et al. (1996) present a genetic algorithm for minimizing the total line length in hybrid workstations while considering sequence-dependent setup times. Sarker and Pan (1998) formulate a mathematical model, for both the open and closed station approach, that minimizes idle time and utility work; they show that the total costs of idle time and utility work are lower for open stations than for closed stations. Dar-El and Nadivi (2007) detail a step-by-step application of the Dar-El–Cother mixed-model sequencing algorithm to a telephone exchange–frame assembly line, thereby reducing the number of operators required from 160 to 117.

Case C. Fattahi and Salehi (2009) propose a hybrid metaheuristic, based on simulated annealing, to solve the sequencing problem. This approach incorporates their *launching interval between products* (LIBP) algorithm, which defines the launching intervals between two successive units in the sequence.

Case D. Dar-El (1978) classifies MMAL sequencing problems based on two objectives: minimizing (i) the overall line length or (ii) the throughput time. The optimal line length for zero idle time and utility work is found by defining the extreme operator movement locations as the station's upstream and downstream boundaries. Bard et al. (1992) introduce a mathematical framework to determine the optimal line length or throughput time for the mixed-model sequencing problem under the following design parameters: operator schedule, product mix, station boundaries, and launching strategy. Bard et al. establish that VRL is superior to FRL in terms of reducing line length and throughput time. Bock et al. (2006) extend the pure assembly line sequencing problem by incorporating specific real-time line balancing aspects and disturbances into an simulated annealing approach. Their model also allows assembly tasks to be moved from one workstation to a neighboring station during the execution process, which results in the movement of operators through several stations. Tong et al. (2013) present an advanced "scatter" search heuristic that yields a solution—for large-scale sequencing problems—that minimizes idle time and utility work.

The model that we develop minimizes restrictive modeling assumptions; in particular, it allows for (i) variable launching rates, (ii) open stations, (iii) unbalanced processing times, (iv) variable station lengths, (v) an unrestricted number of models, (vi) an unrestricted number of stations, and (vii) separation of setup and processing time. Most research considers setup time to be part of the processing time and hence independent of the sequence; however, Sun et al. (1999), Vallada and Ruiz (2011), and Yoshida and Hitomi (1997) are among the scholars who disagree with that approach. Moreover, we exploit structural properties—leading to the ideal case—that enable optimal solutions for VRL with open stations. If these properties do not exist, that is, inhomogeneous assembly times, we introduce the line length algorithm for open stations (LLAOS). Space restrictions play a dominant role in the layout planning of today's assembly lines; it is for this reason that our objective function accounts for extending the line length. By fixing the launching rate and/or setting both the upstream and downstream workstation overlaps to zero, our model can be reduced to Case A, B, or C. This paper is the first to benchmark all four of the cases described above.

4 | VARIABLE RATE LAUNCHING MODEL FOR OPEN STATIONS (VRLMOS)

In many manufacturing settings, competitiveness depends not only on inefficiency costs associated with idle time and utility work but also on space limitations that affect production planning and the facility layout. Hence, our model simultaneously considers all three drivers: idle time, utility work, and line length. This model also accounts for setup times by addressing assembly environments in which products are fundamentally different. We shall employ the notation summarized in Table 2 when describing the mathematical model and its solution procedure.

For a station j, one unit of idle time IDL_{ij} costs C_j^{IDL} and one unit of utility work U_{ij} costs C_j^U. Therefore, idle and utility costs at station j are equal to $C_j^{IDL} \sum_{i=1}^{I} IDL_{ij} + C_j^U \sum_{i=1}^{I} U_{ij}$. Over all stations, these costs amount to $\sum_{j=1}^{J}(C_j^{IDL} \sum_{i=1}^{I} IDL_{ij} + C_j^U \sum_{i=1}^{I} U_{ij})$. The objective function also contains a linear penalty term C^L for the additional length ΔL that is added (without loss of generality) to the line length's lower bound. To simplify the computations, we assume a linear relation between the length of a production line and its costs. When we account for the MPS' frequency F, the total cost of satisfying overall demand D is formally expressed as Equation (1)—whose additional constraints are given in what follows. In many manufacturing settings, competitiveness depends not only on inefficiency costs associated with idle time and utility work but also on space limitations that affect production planning and the facility layout. Hence, our model simultaneously considers all three drivers: idle time, utility work, and line length. This model also accounts for setup times by addressing assembly environments in which products are fundamentally different. We shall employ the notation summarized in Table 2 when describing the mathematical model and its solution procedure.

For a station j, one unit of idle time IDL_{ij} costs C_j^{IDL} and one unit of utility work U_{ij} costs C_j^U. Therefore, idle and utility costs at station j are equal to $C_j^{IDL} \sum_{i=1}^{I} IDL_{ij} +$

TABLE 2 Notation

Indices	
i	Position of a unit in the sequence, $i = 1, \dots, I$
j	Workstation number, $j = 1, \dots, J$
k	Workstation index, $k = 1, \dots, J$
m	Model type, $m = 1, \dots, M$
n	Model type, $n = 1, \dots, M$
r	Model type, $r = 1, \dots, M$
Input parameters	
C_j^U	Costs of utility work at station j
C_j^{IDL}	Costs of idle time at station j
C^L	Costs of an incremental increase in the line length
D	Overall demand
D_m	Demand for model m's minimal part set (MPS)
F	Number of times that MPS is repeated to meet overall demand
I	Number of workpieces in the MPS
J	Number of stations
$L_{\max}$	Upper bound of line length
M	Number of models within the MPS
s_{rmj}	Setup time required when changing from model r to model m at station j
t_{mj}	Assembly time of model m at station j
v	Conveyor velocity
β	Downstream shift area
ΔL	Line length increment
Computed parameters	
L_j^{UO}	Length of upstream overlap of station j
L_j^{DO}	Length of downstream overlap of station j
$L_j^{\min}$	Lower bound of station length
$L_{\min}$	Lower bound of line length
p_{mj}	Processing time of model m at station j
$p_{\max}$	Maximum processing time over all models and stations (incl. setup and assembly)
Z_j^0	Zero reference point for station j
Decision variables	
d_{ij}	Upstream walking distance of operator between workpiece i and workpiece $i+1$
IDL_{ij}^1	Idle time of operator due to waiting for workpiece i to enter station j
IDL_{ij}^2	Idle time of operator because workpiece i is not finished
IDL_{ij}	Total idle time at station j for workpiece i
L_j	Length of station j
U_{ij}	Utility work at station j for workpiece i
X_{irm}	1 if workpieces $i-1$ and i are models r and m, respectively; 0 otherwise
Z_{ij}	Starting position of work for operator on workpiece i at station j
ZF_{ij}	Finishing position of work for operator on workpiece i at station j
λ_i	Launching interval between workpiece $i-1$ and workpiece i

$C_j^U \sum_{i=1}^{I} U_{ij}$. Over all stations, these costs amount to $\sum_{j=1}^{J}(C_j^{IDL} \sum_{i=1}^{I} IDL_{ij} + C_j^U \sum_{i=1}^{I} U_{ij})$. The objective function also contains a linear penalty term C^L for the additional length ΔL that is added (without loss of generality) to the line length's lower bound. To simplify the computations, we assume a linear relation between the length of a production line and its costs. When we account for the MPS' frequency F, the total cost of satisfying overall demand D is formally expressed as Equation (1)—whose additional constraints are given in what follows:

$$\min F\left(\sum_{j=1}^{J}\left(C_j^{IDL}\sum_{i=1}^{I} IDL_{ij} + C_j^U \sum_{i=1}^{I} U_{ij}\right) + C^L \Delta L\right) \quad (1)$$

subject to

$$L_j^{\min} = \arg\min_{t_{mj}, \forall m} t_{mj} \quad \forall j; \quad (2)$$

$$L_{\min} = \sum_{j=1}^{J} L_j^{\min}; \quad (3)$$

$$L_{\min} + \Delta L \le L_{\max}; \quad (4)$$

$$\sum_{k=1}^{j-1} L_k = Z_j^0 \quad \text{for } j = 2, \dots, J,\ Z_1^0 = 0; \quad (5)$$

$$L_{\min} + \Delta L = \sum_{j=1}^{j} L_j; \quad (6)$$

$$\sum_{i=1}^{I}\sum_{r=1}^{M} X_{irm} = D_m \quad \forall m; \quad (7)$$

$$\sum_{m=1}^{M}\sum_{r=1}^{M} X_{irm} = 1 \quad \forall i; \quad (8)$$

$$Z_{ij} = \max\left\{Z_{(i-1)j} + v\sum_{m=1}^{M}\sum_{r=1}^{M} X_{(i-1)rm}(s_{rmj} + t_{mj} - U_{(i-1)j}) - d_{(i-1)j}, Z_j^0 - L_j^{UO}\right\} \text{ for } i = 2, \dots, I, \quad \forall j; \quad (9)$$

$$Z_{1j} = \max\left\{Z_{1(j-1)} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{1rm}(s_{rm(j-1)} + t_{m(j-1)} - U_{1(j-1)}), Z_j^0\right\} \text{ for } \quad j = 2,\dots,J; \quad (10)$$

$$d_{ij} = \min\left\{Z_{ij} - Z_j^0 + L_j^{UO} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{irm}(s_{rmj} + t_{mj} - U_{ij}), \lambda_i v, Z_{ij} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{irm}(s_{rmj} + t_{mj} - U_{ij}) - (Z_{(i+1)(j-1)} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{(i+1)rm}(s_{rm(j-1)} + t_{m(j-1)} - U_{(i+1)(j-1)})\right\} \quad \text{for}$$

$$i = 1,\dots,I-1, \quad j = 2,\dots,J; \quad (11)$$

$$d_{i1} = \min\left\{Z_{i1} - Z_1^0 + L_1^{UO} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{irm}(s_{rm1} + t_{m1} - U_{i1}), \lambda_i v\right\} \quad \text{for} \quad i = 1,\dots,I-1; \quad (12)$$

$$d_{Ij} = \min\left\{Z_{Ij} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{Irm}(s_{rmj} + t_{mj} - U_{Ij}), -Z_j^0, \lambda_I v\right\}\forall j; \quad (13)$$

$$IDL_{ij}^1 = \max\left\{0, \left(Z_j^0 - L_j^{UO} - \left(Z_{(i-1)j} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{(i-1)rm}(s_{rmj} + t_{mj} - U_{(i-1)j}) - \lambda_i v\right)\right)/v\right\} \quad \text{for} \quad i = 2,\dots,I-1, \forall j; \quad (14)$$

$$IDL_{Ij}^1 = \max\left\{0, Z_j^0 + \lambda_I v - \left(Z_{ij} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{Irm}(s_{rmj} + t_{mj} - U_{ij})\right)\right\} + IDL_{(I-1)j} \quad \forall j; \quad (15)$$

$$IDL_{ij}^2 = \max\left\{0, Z_{i(j-1)} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{irm}(s_{rm(j-1)} + t_{m(j-1)} - U_{i(j-1)}) - \left(Z_{(i-1)j} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{(i-1)rm}(s_{rmj} + t_{mj} - U_{(i-1)j}) - \lambda_i v\right) - IDL_j^1\right\}\forall i, \forall j; \quad (16)$$

$$IDL_{ij} = IDL_{ij}^1 + IDL_{ij}^2 \quad \forall i, \quad \forall j; \quad (17)$$

$$U_{ij} = \max\left\{0, \left(Z_{ij} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{irm}(s_{rmj} + t_{mj}) - \left(Z_j^0 + L_j + L_j^{DO}\right)\right)/v\right\} \quad \text{for} \quad i = 1,\dots,I-1, \quad \forall j; \quad (18)$$

$$U_{Ij} = \max\left\{0, \min\left\{(s_{rmj} + t_{mj}), \left(Z_{Ij} + v\sum_{m=1}^{M}\sum_{r=1}^{M}X_{Irm}(s_{rmj} + t_{mj}) - Z_j^0 - \min\{\lambda_I v, L_j - L_j^{DO}\}\right)/v\right\}\right\} \quad \forall j; \quad (19)$$

$$d_{ij} \geq 0,\ IDL_{ij}^1 \geq 0,\ IDL_{ij}^2 \geq 0, IDL_{ij} \geq 0, U_{ij} \geq 0, Z_{ij} \geq 0 \quad \forall i,j;$$

$$\lambda_i \geq 0 \quad \forall i; X_{irm} \in \{0,1\} \quad \forall i,r,m; \Delta L \geq 0.$$

4.1 | Line length and zero reference point

Most assembly line systems have a lower bound on the station length, $L_j^{\min}$, that reflects space requirements due to machines, robots, and so forth (Falkenauer, 2005). We define this lower bound on the station length as the minimum

assembly time over all models for that station (Equation 2). Hence, the lower bound on the overall line length, $L_{\min}$, is the sum of the lower bounds of all the station lengths (Equation 3). The upper bound on the available assembly line length, $L_{\max}$, is restricted by the available facility space (or possible extensions thereof). Since utility work decreases with increasing line length, it follows that the optimal line length for efficient operations lies between the lower and upper bounds on total line length. The existing total line length L is calculated as the line length increase ΔL plus the lower line length bound $L_{\min}$ (Equation 4). We define the zero reference point Z_j^0 as the upstream boundary of station j (Equation 5). For the first station, the zero reference point is set to zero: $Z_1^0 = 0$. Also, the line length is equal to the sum of the station lengths (Equation 6).

4.2 | MPS and model assignment

The MPS required to meet demand D_m for model m is satisfied by the sequence (Equation 7), and only one model m is assigned to a position i in the sequence (Equation 8). The term X_{irm} is set equal to 1 if model r is sequenced at position $i-1$ *and* model m is sequenced at position i; otherwise, $X_{irm} = 0$.

4.3 | Operator starting position

An operator starts her work within the allowable upstream station overlap area $Z_j^0 - L_j^{\mathrm{UO}}$, and the processing time that an operator works on the unit is given by $\sum_{m=1}^{M}\sum_{r=1}^{M} X_{irm}(s_{rmj} + t_{mj} - U_{ij})$. After finishing that task, the operator walks back the upstream distance d_{ij} to start work on the next item. Thus, the starting position Z_{ij} is either (a) the position of the operator after finishing her task on the previous product and returning upstream or (b) the allowable upstream station boundary (Equation 9). For $i = 1$, the operator's tasks start at the zero reference point (Z^0) if the previous task on the workpiece has already been completed; otherwise, the operator's starting position is the finishing position of the previous station's task (Equation 10). The zero reference point for the first station is set to 0 (i.e., $Z_{11} = 0$), a condition that ensures—provided all initial conditions remain the same—that every sequence is regenerative (Bolat, 1997).

4.4 | Operator's upstream walking distance

The operator's upstream walking distance d_{ij} is not always the length between two adjacent workpieces on the conveyor. When determining d_{ij}, three cases must be considered (Equation 11). In the first case, the next unit has entered the station and the previous operator has finished her task on it. In the second case, the workpiece has not yet arrived at the station boundary and so the operator must wait for it to enter the station. In the third case, the workpiece has entered the station even though the previous operator has not yet finished her task on the unit; hence the operator must wait until the prior task is finished. For $j = 1$, there can be no unfinished work from a prior station and so the upstream walking distance simplifies to Equation (12). For $i = I$, the upstream walking distance is either the launching interval or the length between the finishing position and the zero reference point (i.e., since the operator finishes every sequence at the station's zero reference point). In the latter case, the operator leaves unfinished work content for a utility worker (Equation 13).

4.5 | Idle time

Idle time (IDL_{ij}) arises in one of two scenarios. First, the operator may have to wait at the upstream station boundary for the workpiece to enter that station (IDL_{ij}^1); this case is captured by Equation (14). Second, the operator may be idle because the task in the prior station has not been finished (IDL_{ij}^2) or the operator of the prior station is currently working on the unit within the focal operator's station limits (Equation 16). In either case, the worker is idle while waiting to start assembly work on the next unit. For the first workpiece in the sequence, idle time is necessarily zero: $IDL_{1j}^1 = 0$. For $i = I$, any subsequent idle time IDL_{Ij}^1 is determined by the unit's own launching rate and that of the prior unit (Equation 15). Total idle time IDL_{ij} is the sum of IDL_{ij}^1 and IDL_{ij}^2 (Equation 17). If total idle time is zero, then (a) the operator's upstream walking distance is equal to the space between two successive units on the assembly line and (b) the operator utilization rate is 100%.

4.6 | Utility work

Utility work U_{ij} arises when the station operator must move outside the allowable downstream drift area $Z_j^0 + L_j + L_j^{\mathrm{DO}}$ to finish work content on the current unit (Equation 18). Such work can occur if the operator's finishing position, $Z_{ij} + v\sum_{m=1}^{M}\sum_{r=1}^{M} X_{irm}(s_{rmj} + t_{mj})$, exceeds the allowable downstream drift area. When an operator is about to reach her workstation's allowable downstream overlap, the work overload is assigned to a "utility operator" who assists the operator at that station (Hyun et al., 1998; Yano & Rachamadugu, 1991). Thus, the task is completed before the station downstream drift area ends, and the resulting work overload has no effect on the next station. For $i = I$, the operator ends the sequence at her station's zero reference point (Hyun et al., 1998). Because operators start at the zero reference point, this constraint guarantees that sequences are both comparable and replicable. If the distance between the workpiece's finishing position and the zero reference point exceeds the distance dictated by the launching rate, then additional utility work is incurred (Equation 19).

5 | IDEAL VRL APPROACH

In this section, we identify conditions that simplify the complex VRL model. These underlying assembly line requirements offer guidance for manufacturing companies—which face a wide variation in model process times—on where best to direct their investment and planning capacities. We show that the ideal VRL approach eliminates all labor inefficiencies (Theorem 1), almost entirely resolves the assembly line sequencing problem (Theorem 2), and minimizes the line length (Theorem 3). The conditions of Theorem 1 directly imply Theorem 2; yet given the latter's importance, we formulate it separately. For all three theorems, we set the upstream and downstream station overlaps to zero: $L_j^{\mathrm{UO}} = L_j^{\mathrm{DO}} = 0$ for all j. We shall use p_{ij} to abbreviate the station processing time, $\sum_{m=1}^{M} \sum_{r=1}^{M} X_{irm}(s_{rmj} + t_{mj})$; and $\widehat{p} = \arg\max_{p_{ij}, \forall i,j} p_{ij}$ denotes the maximum processing time (including setup times) over all units.

5.1 | Productivity

Theorem 1. *In the ideal VRL approach, idle time and utility work are completely eliminated and operator utilization is* 100% *under the following conditions:* (*i*) *Station lengths are set to the maximum processing time* (*over all models*) *multiplied by the conveyor speed,* $vp_{ij} \leq L_j$ *for all* i,j; (*ii*) *operators always start at the zero reference point,* $Z_{ij} = Z_j^0$ *for all* i,j; (*iii*) *the launching rate is equal to the prior processing time of unit* $i-1$, $p_{(i-1)j} = \lambda_i$ *for all* i,j; *and* (*iv*) *a model has an uniform processing time over all stations,* $p_{ij} = p_{ik}$ *for all* i,j,k.

Proof. Utility work. Using condition (ii), we can write $Z_{ij} = Z_j^0$ for all i,j and thereby reduce Equation (17) to $U_{ij} = \max\{0, p_{ij} - L_j/v\}$ for all i,j. Condition (i) states that no processing time can exceed the station length, $vp_{ij} \leq L_j$ for all i,j; hence $U_{ij} = 0$ for all i,j. It follows that no additional staffing is needed, since operators fulfill their tasks within the station boundary and are never overloaded. □

Idle time. Conditions (iii) and (iv) reduce Equation (14) to $IDL_{ij}^1 = \max\{0, U_{(i-1)j}\}$ for all i,j. Since $U_{ij} = 0$ for all i,j, we have $IDL_{ij}^1 = 0$ for all i,j. Now, by conditions (iii) and (iv) and given that $U_{ij} = 0$ for all i,j, Equation (15) yields $IDL_{ij}^2 = 0$ for all i,j. Therefore, operator idleness is eliminated because workers can immediately start their task on the next unit at the upstream boundary without having to wait for that unit to enter the station or for the prior operator to finish her task.

Single-station case. From conditions (ii) and (iv) it follows that all stations have the same length: $L_j = v\widehat{p}$ for all j. Moreover, condition (ii) ensures that all operators start at the upstream station boundary; hence all operators perform all of their work at their respective stations (condition (i)). Thus, the complex general model is reduced to a single-station problem—that is, since all stations and operators have the same characteristics and are independent of each other. Figure 4 illustrates the operator movement diagram for the single-station case. Additional stations can be added before or after this station because the station's operators then do not interfere with one another. Line productivity is 100% because there are no labor inefficiencies, all stations have the same length, and stations are connected without buffers.

Key to the applicability of the ideal case is the assumption that work can be perfectly balanced across workstations. This assumption sounds rather restrictive, but in practice, companies like Fendt achieve an almost perfect balance of work for their models of over 94%; where 100% is a perfect line balance (Huchzermeier et al., 2020b). Note, however, there is a practical need for a balance in the mid-nineties range as this grants operators some slack in case of contingencies.

5.2 | Model-mix flexibility

Theorem 2. *The ideal VRL approach guarantees the highest efficiency for any product sequence, so models can be launched onto the assembly line—at any time and in any order—without causing idle time or utility work.*

Proof. We use the four conditions of Theorem 1 to prove Theorem 2. As shown in our model, the operator starting position Z_{ij} depends on the launching rate, the processing time, idle time, and the previous unit's utility work. However, condition (ii) of Theorem 1 sets $Z_j^0 = Z_{ij}$ for all i,j. So if we separate the sequencing problem into I independent parts—where I is the number of products in a sequence—then Z_{ij} can be decoupled from the previous workpiece. In this case, each unit i can be considered as an independent part of the sequence. Note also that the conditions of Theorem 1 eliminate both idle time ($IDL_{ij} = 0$ for all i,j) and utility work ($U_{ij} = 0$ for all i,j). Since all parts of the sequence are independent and since there is no factor that can induce labor inefficiencies, it follows that resequencing in any (even random) order itself precludes the absence of operator idleness and work overload.

For assembly systems with and without setup times, satisfying the conditions of Theorem 1 leads to a 100% workforce utilization—that is, because $\widehat{p}$ already includes the maximum setup time for a model. In other words, no maximum (resp., unbalanced) loading of the assembly line will overburden the assembly line (resp., its operators). If setup times are relevant, then a simple optimization will batch similar products and reduce setup effort. Because such batching proceeds in a straightforward manner, this myopic optimization is omitted from our model formulation. □

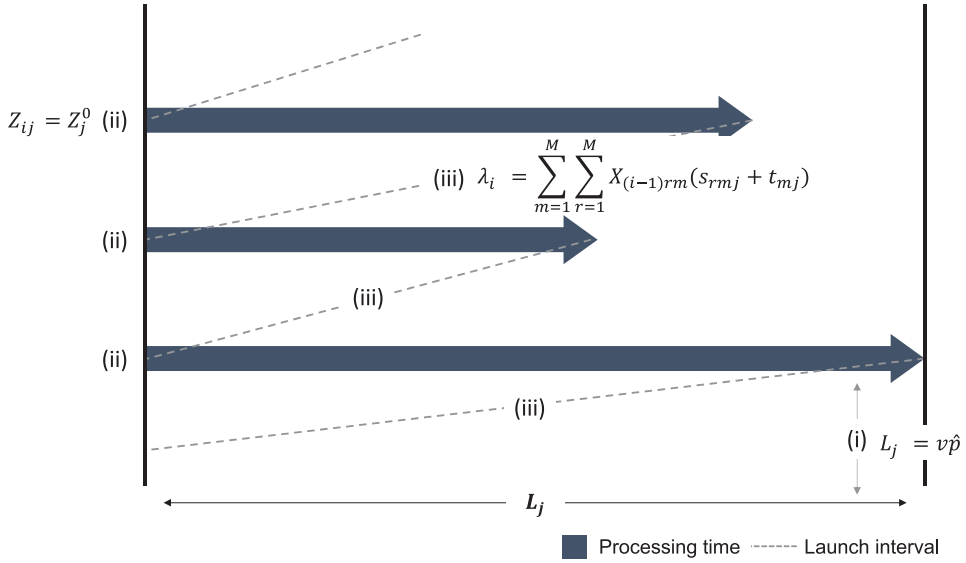


FIGURE 4 Operator movement under the ideal VRL approach for the single-station case [Color figure can be viewed at wileyonlinelibrary.com]

5.3 | Line length

Theorem 3. *When multiple operators are allowed to work simultaneously on a product, the optimal line length L is determined by one of the following cases:* (*a*) $L = v\hat{p}J$ *for* $\beta = 0$, (*b*) $L = v\hat{p}J - \beta(J-1)$ *for* $\beta \in (0, v\hat{p})$, *or* (*c*) $L = v\hat{p}$ for $\beta = v\hat{p}$, *when the four conditions of Theorem 1 hold also for Theorem 3. In the case of Fendt, this area β is negotiated with the works council and lies between* 10% *and* 15%.

Proof. The extent to which a workstation's task may be shifted to the previous station is determined by the factor $\beta \in [0, v\hat{p}]$, where β's value is the same for all stations. β determines the area in which operators are allowed to work simultaneously on the product. □

Case (a). If $\beta = 0$, then stations cannot be shifted into one another and operators are not allowed to work simultaneously on products. In this case, the line length is $L = v\hat{p}J$; in other words, it is the sum of all station lengths $L_j = v\hat{p}$ for all j (condition (i) of Theorem 1).

Case (c). For $\beta = v\hat{p}$, stations are completely shifted into one another and operators are allowed to work simultaneously on a product at any time. Since each operator starts her work at the upstream station boundary $Z_j^0 = Z_{ij}$ for all i, j (condition (ii) of Theorem 1), it follows that the line collapses to the shared operating area $L = v\hat{p}$. The locus of each operator is identical to that of the others because their respective processing times are uniform for a given product model (condition (iv) of Theorem 1). It should be clear that any further reduction of the line length would result in utility work (condition (i) of Theorem 1).

Case (b). For $\beta \in (0, v\hat{p})$, all stations are shifted into one another by the amount β. Thus, the overall line length L is somewhere between the extremes represented by Case (a) and Case (c).

6 | SOLUTION METHOD FOR THE GENERAL CASE

In this section, we discuss our heuristic solution method for those manufacturing environments in which it is not possible for the assembly process to be aligned with the ideal case's four conditions (Theorem 1). For the general case, the planning unit has to solve the NP-hard assembly line sequencing problem in order to reduce labor inefficiencies and line length. In our simulation, we show that—in terms of costs and line length—VRL with closed stations (Case C) dominates FRL with closed stations (Case A) in all of the scenarios considered. The same holds true for open stations.

6.1 | Line length algorithm for open stations (LLAOS)

We have established that idle time and utility work are substantially reduced when production switches from fixed to variable takt times. In general, VRL (with launching rate λ_i for unit i in a product sequence) at minimum matches the costs of FRL (with a launching rate λ) as VRL offers the freedom to change the launching rate λ_i between products. If $\lambda_i = \lambda_{i+1}$ for all i, then VRL is reduced to FRL; thus the latter is a subset of the former. Nevertheless, the assembly line is still not entirely efficient because of deviations in station lengths and drift areas and in model processing times across stations. Our LLAOS (Algorithm 1a) helps to identify, for the case of variable launching rates, a product sequence that minimizes the costs associated with idle time and utility work while accounting for space requirements.

The LLAOS. The general idea behind the LLAOS is to increase (gradually) the length of only one station, j^*, in every iteration. In this procedure, we start with the line

ALGORITHM 1a Line length algorithm for open stations (LLAOS)

0 $C^{\mathrm{IDL},U,L}_{\min} = \alpha$; $C^{\mathrm{IDL},U,L}_{l} = 0$; $C^{\mathrm{IDL},U}_{\min} = 0$; $C^{\mathrm{IDL},U} = 0$; $C^{L} = 1$; $C^{\mathrm{IDL}}_{j} = 0.2$, $C^{U}_{j} = 0.5\ \forall j$

1 For $l = L_{\min}$ **to** $L_{\max}$

2 For $i = 1$ **to** I

3 $\mathrm{ZF}_{ij} = \min\{\sum_{k=1}^{j} L_k + L_j^{\mathrm{DO}}, Z_{ij} + v\sum_{m=1}^{M}\sum_{r=1}^{M} X_{irm}(s_{rmj} + t_{mj})\}\ \forall j$

4 $p_i = \min\{(\mathrm{ZF}_{ij} - \sum_{k=1}^{j-1} L_k - \max\{0, L_{j-1}^{\mathrm{DO}}\})/v, (s_{rmj} + t_{mj})\}\ \forall j$

5 $\lambda_i = p_i$

6 $C^{\mathrm{IDL},U}_{\min} = \sum_{j=1}^{J} \mathrm{IDL}_{(i+1)j} C^{\mathrm{IDL}}_{j} + \sum_{j=1}^{J} U_{(i+1)j} C^{U}_{j}$

7 While ($C^{\mathrm{IDL},U}_{\min} == C^{\mathrm{IDL},U}$ **OR**($i = I$ **AND** $\lambda_i \le \min_{\forall j}\{Lj\}$)) Do

8 $\lambda_i = \lambda_i + 1$

9 $C^{\mathrm{IDL},U} = \sum_{j=1}^{J} \mathrm{IDL}_{(i+1)j} C^{\mathrm{IDL}}_{j} + \sum_{j=1}^{J} U_{(i+1)j} C^{U}_{j}$

10 If ($i \le I$ **AND** $C^{\mathrm{IDL},U} \le C^{\mathrm{IDL},U}_{\min}$) Then

11 $p_i = p_i + 1$

12 $C^{\mathrm{IDL},U}_{\min} = C^{\mathrm{IDL},U}$

13 ElseIf ($i = I$ **AND** $C^{\mathrm{IDL},U} \ge C^{\mathrm{IDL},U}_{\min}$ **AND**$\lambda_i \le \min_{\forall j}\{L_j\}$) Then

14 $p_i = p_i + 1$

15 End If

16 End While

17 $\lambda_i = p_i$

18 End For

19 $C^{\mathrm{IDL},U,L} = \sum_{j=1}^{J}\sum_{i=1}^{I} U_{ij} C^{U}_{j} + \sum_{j=1}^{J}\sum_{i=1}^{I} \mathrm{IDL}_{ij} C^{\mathrm{IDL}}_{j} + C^{L}\Delta L$

20 If ($C^{\mathrm{IDL},U,L}_{l} \le C^{\mathrm{IDL},U,L}_{\min}$) Then

21 $C^{\mathrm{IDL},U,L}_{\min} = C^{\mathrm{IDL},U,L}_{l}$

22 End If

23 If ($l + 1 \le L_{\max}$) Then

24 $L_{j^*} = L_{j^*} + 1$

25 End If

26 End For

length's lower bound, $L_{\min}$, and end with its upper bound, $L_{\max}$ (Step 1). The process requires that we determine the variable launching rates by using the *variable rate launching algorithm for open stations* (VRLAOS; Steps 2–18). For that purpose, we extend the LIBP algorithm (see Case C in Section 3) developed by Fattahi and Salehi (2009) to meet the requirements of our model in an open station setting. Fattahi and Salehi (2009) were the first to introduce a heuristic approach, the LIBP algorithm, for determining the variable launching intervals between successive units on MMALs with closed stations minimizing only labor costs; hence these authors do not consider variations in assembly line length and do not allow for operators drifting into other stations or costs for increasing line length. We extend this approach by allowing for open station boundaries while simultaneously minimizing labor costs and overall assembly line length. Also, we regard setup times enabling the adoption of our algorithm in a larger group of production environments. The VRLAOS algorithm's goal is to shift the operator starting position as far upstream as possible to avoid the risk of utility work. After the launching intervals are determined, we compute the total costs $C_l^{\mathrm{IDL},U,L}$ of the current iteration by summing up the costs for idle time, utility work, and longer line length (Step 19). These costs are then compared with the minimum costs over all iterations, $C^{\mathrm{IDL},U,L}_{\min}$, and result in an update if a reduction is achieved (Steps 20–22). Finally, the length of station j^* is increased by an increment of one line length (Steps 23–25).

The procedure for selecting j^* is presented in the next paragraph. The LLAOS allows overall costs to increase over iterations in the short term because, in some iterations, the cost reduction due to less utility work does not fully compensate for idle time and line length costs. However, this increase in overall costs may well be absorbed in subsequent iterations. We therefore choose the line length upper bound as a natural termination criterion, thereby limiting the number of iterations performed; thus the total cost calculated during each iteration is compared with the lowest cost calculated previously.

Procedure for selecting j^*. Here, we focus on practical, iterative improvement procedures for determining which station j^* should be increased in length within an iteration of the LLAOS. The first selection procedure (Option 1) is based on a local search approach that, before increasing the line length, evaluates all neighboring solutions. Thus, the neighborhood solution space is constructed by increasing the length of each station, one by one, and comparing the total cost for idle time and utility work that results; the station j^* that yields the greatest reduction in cost when its length is increased is then selected (Equation 21). Option 2 instead considers the current costs of idle time and utility work without regard to information on neighboring solutions, thereby reducing the computational effort required (Equation 22). Options 3 and 4 concentrate exclusively on utility work while ignoring idle time; the goal is a horizontal smoothing of the workload over every station. These latter two options aim either to reduce and align utility work at all stations (Equation 23) or to reduce the time that a utility worker is bound to one station (Equation 24).

Option 1. Selecting the station for which the *future* total cost of idle time and utility work would be highest:

$$j^* = \arg\max_{\forall j}\left(\sum_{i=1}^{I} U_{ij} C^{U} + \sum_{i=1}^{I} \mathrm{IDL}_{ij} C^{\mathrm{IDL}}\right) \quad \text{for} \quad L_j = L_j + 1. \tag{20}$$

Option 2. Selecting the station for which the *current* total cost of idle time and utility work is highest:

$$j^* = \arg\max_{\forall j}\left(\sum_{i=1}^{I} U_{ij} C^{U} + \sum_{i=1}^{I} \mathrm{IDL}_{ij} C^{\mathrm{IDL}}\right) \text{ for } L_j. \tag{21}$$

ALGORITHM 1b Extension of the LLAOS

27 For l in J
28 For x in $(-1, 1)$
29 $L_l = L_l + x$
30 VRLAOS
31 $C_l^{\mathrm{IDL},U,L} = \sum_{i=j}^{J} \sum_{i=1}^{I} \mathrm{IDL}_{ij} C_j^{\mathrm{IDL}} + \sum_{i=j}^{J} \sum_{i=1}^{I} U_{ij} C_j^{U} + C^{L} \Delta L$
32 If ($C_{\min}^{\mathrm{IDL},U,L} \geq C_l^{\mathrm{IDL},U,L}$) Then
33 $C_{\min}^{\mathrm{IDL},U,L} = C_l^{\mathrm{IDL},U,L}$
34 End If
35 End For
36 End For

Option 3. Selecting the station with the highest sum of utility work over *all units* in the sequence:

$$j^* = \arg\max_{\forall j} \sum_{i=1}^{I} U_{ij}. \tag{22}$$

Option 4. Selecting the station with the most utility work for *one unit* in the sequence:

$$j^* = \arg\max_{\forall j} (U_{1j}, \ldots, U_{Ij}). \tag{23}$$

Extending the LLAOS. We now discuss a simple procedure for improving the LLAOS even further. Starting with its solution $C_{\min}^{\mathrm{IDL},U,L}$, Algorithm 1b investigates the neighboring solution space by gradually increasing and decreasing—in one-length increments—the length of each station in turn (Step 28). For every entity in the solution space, we apply the VRLAOS (Step 30) and then calculate total costs (Step 32). Finally, the improvement yielding the greatest cost reduction is stored (Step 33). Our results when employing this simple extension are presented in the next section's benchmark analysis.

6.2 | Benchmark study and analysis

This section presents the findings derived from our large-scale analysis of 2070 product sequences. We benchmarked the LLAOS selection procedures for Cases A, B, C, and D with respect to total cost of inefficiencies and assembly line length.

6.2.1 | Data generation

The data set consists of 30 scenarios, each featuring a unique combination of the underlying MPS and model processing times. Each MPS is randomly generated under the condition that all models occur at least once in the sequence. Processing times are derived from a uniform distribution whose intervals are given in Table 3, and they include both assembly time and setup time. All scenarios incorporate the following line-related parameters: frequency, $F = 1$; number of products in the MPS, $I = 6$; number of stations, $J = 3$; number of models, $M = 3$; models, $m = (1, 2, 3)$; cost of idle time, $C_j^{\mathrm{IDL}} = 0.2$; cost of utility work, $C_j^{U} = 0.5$; cost of increasing the line length, $C^{L} = 1$; and conveyor velocity, $v = 1$. In addition, the upstream and downstream overlaps are set to 10% of the lower station bound and then rounded up to the next higher integer value. These parameters anchor in our review of real-world assembly systems and thus can be seen as representative for a real-world production setup, such as Fendt's production in Marktoberdorf, Germany. Furthermore, the three-station layout is treated as a subsection of a larger assembly line. Depending on the MPS, each scenario has 30, 60, or 90 sequences (or permutations); hence our benchmark analysis involves 2070 compared sequences. Rather than focusing on stochastic variations of processing time, this computational setup allows to solve the more relevant problem of implementing a total mix of sequences to be assembled on the same line and thus on the same space requirement. In today's assembly lines, stochastic processing times are rather rare since during the last decades, a great amount of effort has been deployed to reduce such deviations. Peter Bebersdorf, Director Manufacturing at AGCO-Fendt, mentioned that deviations may exist when a product is newly introduced on the line, but never reaches a value of 10% or higher. To the contrary, at AGCO-Fendt the workers council demands, at the minimum, three measurements of process times before the start of production. These three measurements set the baseline for an approval of the processing times by the workers council. In general, large deviations of process times raise red flags with the workers council and are addressed by management directly.

TABLE 3 Intervals for uniform distribution of processing times

	t_{m1}	t_{m2}	t_{m3}
Model 1	[8, 12]	[8, 14]	[8, 10]
Model 2	[4, 8]	[6, 9]	[4, 6]
Model 3	[5, 9]	[6, 11]	[5, 8]

6.2.2 | Benchmarking the LLAOS options

In order to identify the preferred LLAOS option, we compare the minimal costs of all 2070 sequences for each option and case. Table 4 reports the percentage of sequences for which the specified option yields the best result, where the value in parentheses is the percentage of sequences for which only this option achieves the single best outcome of all options. Option 1 is clearly the dominant choice for at least 84% of the sequences. In fact, for the most restrictive strategy (Case A), the first selection procedure always leads to the best sequence; and it leads to the single best solution in 57% of the cases. Another noteworthy observation is that Option 1 becomes less dominant as flexibility is increased—for

TABLE 4 Comparison of the LLAOS options by percentage of sequences yielding best (single best) result

	Case A	Case B	Case C	Case D
Option 1	100	98	92	84
	(57)	(57)	(37)	(32)
Option 2	33	39	52	63
	(0)	(1)	(7)	(12)
Option 3	27	23	26	30
	(0)	(1)	(0)	(1)
Option 4	15	15	12	34
	(0)	(0)	(0)	(0)

example, by opening up station boundaries and/or implementing VRL. Even so, Option 1 remains the best choice for flexible systems: For Case D, it delivers 84% of best solutions and 32% of the single best solutions.

These results can be improved if one undertakes the simple local search procedure described in Section 6.1. By examining neighboring solutions, our extended LLAOS for Option 1 (hereafter, LLAOS-1) reduces the costs associated with Cases B, C, and D in (respectively) 37%, 26%, and 52% of the sequences. We remark that LLAOS-1b does not yield any significant improvement for Case A; the reason is that it obtains the same station lengths (for most instances) as the LLAOS and thus results in the same inefficiency costs for each sequence.

With these parameters, a full enumeration over the line length was performed; starting with the minimal station lengths and iterating through all combinations until arriving at the maximum line length. The performed enumeration showed a runtime of roughly 2700 s. Moreover, the average optimality gap is 2.3% with one outlier of 8.5% (but still achieving better results than the fixed launching rate setting).

6.2.3 | Analysis of deviations in line length and total costs

Our purpose here is to show that shifting the assembly strategy from Case A to Case D reduces assembly line length and overall assembly costs. Figure 5 and Figure 6 plot the percentage of improvement in (respectively) line length and overall cost when the LLAOS-1 is implemented with variable takt times and open stations. As expected, Case A results in the highest average costs for all 30 scenarios. We can also see that Case A (resp., Case D) corresponds to the highest (resp., lowest) average cost and minimal cost for every scenario. These graphs show that changing from Case A to Case D reduces the line length of the best sequence by anywhere from approximately 4% to 22% (Figure 5) and also reduces total costs by approximately 11% to 37% (Figure 6). For any randomly chosen sequence, changing from Case A to Case D results in an average cost reduction of approximately 22%.

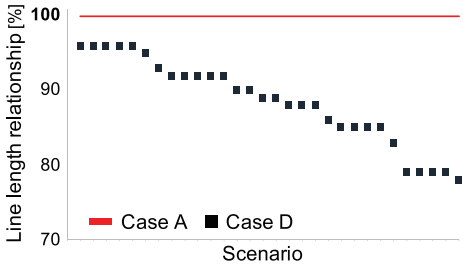


FIGURE 5 Line length improvement, from Case A to Case D, of best sequence [Color figure can be viewed at wileyonlinelibrary.com]

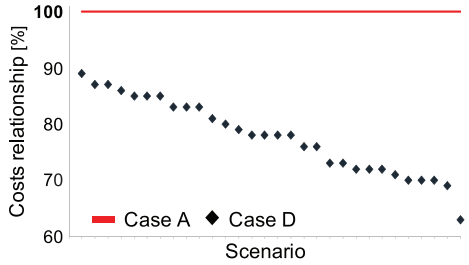


FIGURE 6 Cost reduction, from Case A to Case D, of best sequence [Color figure can be viewed at wileyonlinelibrary.com]

We next discuss whether manufacturing executives who seek to push, in a stepwise fashion, their Case A assembly process toward a more flexible approach should start by opening up station boundaries or rather by adopting VRL. From the *line length* perspective, opening up the station boundaries (Case B) has a significantly greater effect on the final length than does introducing variable takt times (Case C): In the former case, line length is reduced by approximately 12% on average. Of course, operators at open stations can start work earlier (than can those at closed stations), thanks to the adjacent station's drift area—thereby reducing the space required to fulfill their task. The combination of open stations and VRL has the greatest effect on line length: an average reduction of approximately 12%.

From the *cost* perspective, Case A and Case D can be seen as amounting to (respectively) an upper and lower bound on total inefficiency costs due to idle time, utility work, and line length—bounds that subsume the outcomes in Cases B and C. Unlike the results of our line length analysis, recommendations to open station boundaries or to introduce VRL are strongly dependent on the sequence.

Figure 7 plots the cost distribution of the sample scenario and its 90 sequences (the connecting lines are solely for the purpose of illustration). If the goal is to extend Case A, then the choice of opening stations versus establishing VRL clearly depends on the sequence—that is, since Case B yields a lower cost than does Case C in just over half (52%) of the sequences. This sequence dependency of production planning can be reduced significantly by the conditions presented in Section 5.

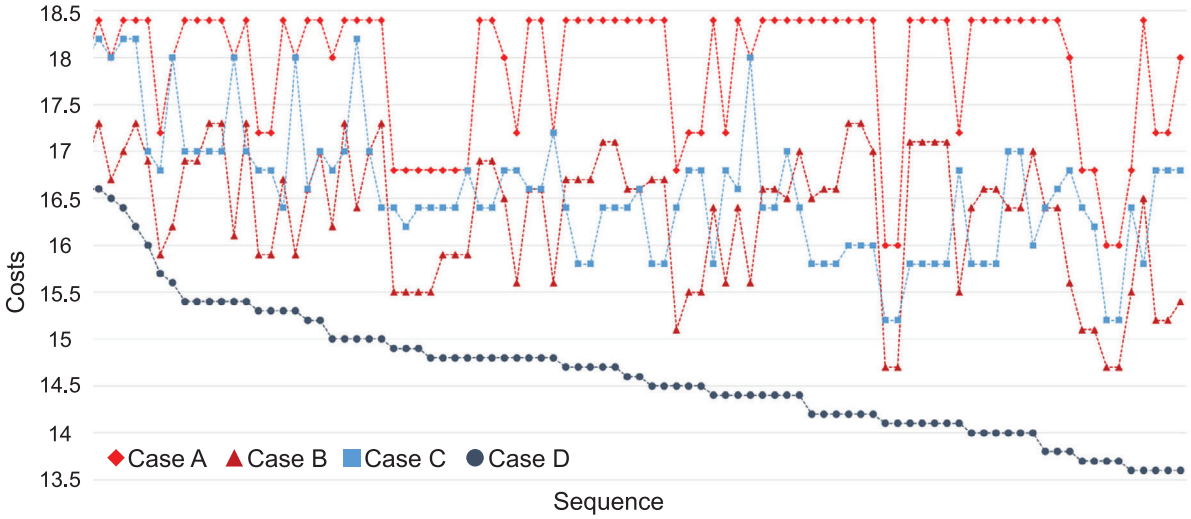


FIGURE 7 Total costs of every sequence in the sample scenario after implementing LLAOS-1 [Color figure can be viewed at wileyonlinelibrary.com]

7 | DISCUSSION AND PRACTICAL IMPLICATIONS

Our paper consists of three main parts. First, we introduce a mixed-integer programming model for minimizing labor inefficiencies and line length simultaneously on MMALs with variable takt times and open stations. Second, we discuss how companies can establish an ideal case, where labor inefficiencies are eliminated and operators are constantly utilized. Third, for manufacturing environments that are unable to align their operations with the ideal case, we present the LLAOS. In the rest of this section, we discuss the practical implications of all three elements and present managerial implications.

VRLMOS. Our general model extends the literature in that it is unrestricted with regard to launching rates and setup times as well as to variations in models, processing time, station length, and line length. This model applies to a large number of business cases, including those in which a leveled workload has proved to be elusive.

Ideal case. We present evidence that the general NP-hard problem of MMAL sequencing can be broken down into an "ideal" VRL approach that maximizes the benefits of a VRL strategy and overcomes most implementation barriers to employing variable takt times. Successful implementation requires that manufacturing executives focus on five key planning objectives. First, processing times for every model should be perfectly balanced over all stations so that a product model has the same processing time at each station. Second, station lengths should be set to the maximum processing time over all models. Third, operators should start work on each unit at the upstream station boundary. Fourth, the launching rate should be determined by the previous product's processing time at the first station. Fifth, when parallel work is feasible, then workstations should overlap as much as possible.

If enough assembly space is available to set the station lengths equal to the maximum processing time, then fulfilling the first two conditions should not be difficult. Yet, achieving the third condition requires a highly flexible conveyor system, one that allows the spacing between units to vary; for that purpose, products can be placed on automated guided vehicles. The last condition requires considerable planning effort. For example, Fendt established a highly productive assembly design with an almost perfect line balance by assigning single tasks or entire tractor components to preassembly stations and utility workers (Huchzermeier et al., 2020b; Mönch et al., 2021).

Implementing variable takt time leads to five benefits. First, worker productivity rises significantly because operators can immediately start work on the next unit; hence the manufacturer's workforce is constantly occupied. Second, a system with zero idle time and no utility work ensures the best possible throughput time. Third, eliminating sequence restrictions substantially reduces planning efforts. Fourth, the decrease in utility work naturally increases quality by lowering the stress level of operators; beneficial outcomes include lower error rates and less rework. Finally, dispensing with the restrictions due to sequence dependency greatly increases manufacturing flexibility to fulfill a broad range of different customer demands in real time. Our description and solution of the general case covers manufacturing environments that deviate from the ideal case in terms of one or more aspects.

Solution method for the general case. The general case applies for manufacturing companies at which station lengths vary and/or processing times cannot be perfectly balanced, circumstances that violate requirements of the ideal case. We are the first to formulate a mixed-model assembly optimization problem that accounts for both the cost of inefficiencies and line length. As discussed in the literature review, such combinatorial problems are solved efficiently by a variety of meta-heuristics. In our paper, we emphasize the ideal case

over the general case and thus do not focus on such standard solution methods. However, we suggest four novel and efficient search strategies—as defined above—for choosing the best station improvement for our MMAL problem. We tested these rules on a sample assembly line scenario, with three stations and three products. We develop a solution method for the general case—the *LLAOS*—that is an extension of Fattahi and Salehi's (2009) LIBP algorithm, which ignores open station boundaries and their effects on line length and labor costs. The LLAOS enables us to solve the unrestricted model by identifying the best sequence and its associated line length for open stations. We conduct a benchmark analysis to evaluate four station selection options for the LLAOS. This analysis involves comparing the minimum costs and line length of 2070 scenarios in terms of policies defined by the four cases that arise from the possible combinations of workstation type (closed or open) and launching rate (fixed or variable). To the best of our knowledge, this study is the first to compare the effects of all four cases on labor inefficiencies and line length. We find that the best outcome results from choosing, in every iteration of the LLAOS, the station predicted to yield the greatest reduction in both idle time and utility work. Another important finding is that, when the assembly strategy is changed from closed and fixed to open and variable, the line length and total costs are reduced by (respectively) 12% and 22% on average for the three-station scenario. Also, we performed a cost analysis for an assembly setup with 10 stations and compared the costs of Case A and Case D. (The processing times of the extension are based on the distribution presented in Table 3.) On average, switching from Case A to Case D leads to a cost reduction of approximately 9% over all scenarios. The range of cost savings over all scenarios lies between 5% and 16%. These results are in line with our observations made at Fendt: After switching to VRL, overall line length productivity increased by 9% and annual productivity savings rose to 6% (Bebersdorf, 2017). Moreover, we assess the optimality gap by performing a full enumeration over the possible line length, starting from the minimal station length. The results we obtain show that the optimality gap over 30 scenarios is a mere 2.3% with negligible run times. This states the effectiveness of the proposed algorithm. Our computational results indicate that companies should strive to implement variable launching rates. Clearly, the ideal case should be the "true north" for such assembly strategies. However, the more stations and models considered, the more difficult it will be to obtain optimal solutions if processing times are not balanced and space restrictions exit, certainly, there is no guarantee that idle time and utility work can be eliminated at all and most likely never will be. Even for small problem instances, for example, three stations, such an optimization proves difficult and does not come even close to the performance of the variable takt approach. The stark difference between an optimal result, that is, 100% utilization of frontline employees, with no need for any optimization—in the case of the variable takt—becomes really evident. This should be one of the key arguments for abandoning the fixed takt as it is unsuited for the optimization of an MMAL. Finally, the LLAOS results in the ideal case when processing times are balanced over all stations for fixed launching rates (for variable launching rates, see the approach presented by Mönch et al. (2021)—the results show that VRL is superior to FRL for uniform processing times as well). This is the case in many standardized assembly lines, for example, automotive final assembly, where all stations have the same work content. If model processing times are uniform over all stations and space is not limited (no costs for increasing stations lengths), then the stations length would increase up to the maximum processing time (multiplied by the conveyor speed) and variable launching rates would equal the prior processing times, thus resembling the ideal case conditions from Theorem 1.

Managerial implications. Mass customization, highly innovative products, and localized production facilities have increased the pressure on static manufacturing designs, even those that previously operated at high levels of productivity. The prevailing management view wrongly supposes that all models should be handled using the same fixed takt time and that sequencing will, in itself, somehow accommodate variations in workload—beliefs that stem from a decades-long quest to eliminate the variability from manufacturing systems (Monden, 1983). Yet, the inflexibility of employing a single takt time renders sequence optimization, a computationally expensive procedure. We want to point out that in a real-world assembly context, additional factors play a role: Planners in multistage operations strive to achieve an average workload that matches the time it takes for a line carrier to pass through a workstation. If the workload is (much) larger, workers tend to drift off, that is, in the downstream direction. Moreover, if this is the case, workers drift in the upstream direction unless they pause at the start of their workstation (Mönch et al., 2022). Therefore, process times tend to "not vary that much," at least they are bounded by a certain factor, which needs to be agreed by the workers council, worker representative, respectively (at Fendt, this factor is 15%). If process times are exceeding the time for passing of workstations, preassembly, suppliers, hybrid assembly (including matrix assembly), or flexible workers are scheduled. As one cannot ensure that workers use the entire space for assembly tasks, long process times have a high risk of significantly interfering with activities in the subsequent station and thus are often curtailed before the data are optimized. Conversely, operation tasks with little workload lead to idle time, which is also to be avoided (Bebersdorf & Huchzermeier, 2022). For those reasons, it is unreasonable, in our view, to simulate extreme realizations or variations in process times—as is typical for many operations research (OR) approaches. In our experiments, we took great care to assemble a meaningful data set for this specific context. Note that the variable takt solves the issue of accepting more variance in process times by leveling the work content of each station's operator even though the spread of that content widens. As each product is assigned its own takt, much larger variations can be accepted, and there is limited or no drifting off whatsoever. This can never be achieved, as outlined

above, with the fixed takt. It is not a matter of combinatorial optimization finesse; it is a matter of a fundamental change in approach.

For those who seek to move quickly from a fixed to a variable takt system, we suggest that product process times be rebalanced by distinguishing between tasks that must be performed directly on the production line and tasks that can be shifted to preassembly stations or lines located near the final assembly line (Huchzermeier et al., 2020b). In the medium term, it can be helpful to set up an interface between manufacturing and R&D; doing so will make it easier to integrate manufacturing objectives with the product development process. At Fendt, this is called Design for Takt (Bebersdorf & Huchzermeier, 2022).

Such integration, by enabling a perfect balance of workload across the assembly line for each product, can ease efforts to improve shop-floor planning and can also reduce assembly costs substantially. The typically time-intensive planning of the next sequence can thus be eliminated in systems without setup times, since all sequences yield the same efficiency. As for setup costs, an optimal batching of products in an MPS is both straightforward and independent of the assembly line status and layout. When combined with open stations, VRL not only significantly reduces operator idleness and work overload but also shortens the line length considerably.

We conclude that, with regard to future developments in assembly-intensive industries, the introduction of variable launching rates into MMALs is an efficient way to meet the challenges arising from mass customization and the need for more flexibility/responsiveness during turbulent times. Our results show that VRL outperforms FRL even when the conditions of the ideal case are violated. A major change in the manufacturing landscape is already evident in the form of new dynamic line configurations and layouts. The VRL strategy proposed here is a practical way to cope with the ever increasing demand for assembly line flexibility while keeping investment costs under control. As labor inefficiencies inevitably rise, we are certain that the VRL adoption rate will likewise increase. For example, during several visits at premium car manufacturers, we observed first experiments with variable takt times in subassembly stations, for example, in the assembly of two- and four-door cars. Employing variable takt times eliminates the major limitations of a fixed takt time system and yet retains all of the conventional assembly line's advantages.

REFERENCES

Agrawal, M., Mercer, G. A., & Kumaresh, T. V. (2001). The false promise of mass customization. *The McKinsey Quarterly*, *01*(3), 62–71.

Al-Hawari, T., Ali, M., Al-Araidah, O., & Mumani, A. (2015). Development of a genetic algorithm for multi-objective assembly line balancing using multiple assignment approach. *The International Journal of Advanced Manufacturing Technology*, *77*(5), 1419–1432.

Bard, J. F., Dar-El, E., & Shtub, A. (1992). An analytic framework for sequencing mixed model assembly lines. *International Journal of Production Research*, *30*(1), 35–48.

Bard, J. F., Shtub, A., & Joshi, S. B. (1994). Sequencing mixed-model assembly lines to level parts usage and minimize line length. *International Journal of Production Research*, *32*(10), 2431–2454.

Battaïa, O., & Dolgui, A. (2013). A taxonomy of line balancing problems and their solution approaches. *International Journal of Production Economics*, *142*(2), 259–277.

Bautista, J., & Cano, J. (2008). Minimizing work overload in mixed-model assembly lines. *International Journal of Production Economics*, *112*(1), 177–191.

Baybars, İ. (1986). A survey of exact algorithms for the simple assembly line balancing problem. *Management Science*, *32*(8), 909–932.

Bebersdorf, P. (2017). Strategy development at AGCO Fendt. http://de.industrial-excellence-award.eu/ie-konferenz/industrial-excellence-conference-2017/

Bebersdorf, P., & Huchzermeier, A. (2022). *Variable takt principle: Mastering variance with limitless product individualization*. Springer, 256 p.

Becker, C., & Scholl, A. (2006). A survey on problems and methods in generalized assembly line balancing. *European Journal of Operational Research*, *168*(3), 694–715.

Betti, F., de Boer, E., & Giraud, Y. (2020). *Industry's fast-mover advantage: Enterprise value from digital factories*. World Economic Forum and McKinsey & Company.

Bloomberg. (2019). *BMW takes on VW and Tesla with electric mini cooper*. https://www.industryweek.com/leadership/bmw-takes-vw-and-tesla-electric-mini-cooper

Bock, S., Rosenberg, O., & van Brackel, T. (2006). Controlling mixed-model assembly lines in real-time by using distributed systems. *European Journal of Operational Research*, *168*(3), 880–904.

Bolat, A. (1997). Stochastic procedures for scheduling minimum job sets on mixed model assembly lines. *Journal of the Operational Research Society*, *48*(5), 490–501.

Bolat, A., Savsar, M., & Al-Fawzan, M. A. (1994). Algorithms for real-time scheduling of jobs on mixed model assembly lines. *Computers & Operations Research*, *21*(5), 487–498.

Bolat, A., & Yano, C. A. (1992a). Scheduling algorithms to minimize utility work at a single station on a paced assembly line. *Production Planning & Control*, *3*(4), 393–405.

Bolat, A., & Yano, C. A. (1992b). A surrogate objective for utility work in paced assembly lines. *Production Planning & Control*, *3*(4), 406–412.

Boysen, N., Fliedner, M., & Scholl, A. (2007). A classification of assembly line balancing problems. *European Journal of Operational Research*, *183*(2), 674–693. https://doi.org/10.1016/j.ejor.2006.10.010

Boysen, N., Fliedner, M., & Scholl, A. (2008). Assembly line balancing: Which model to use when? *International Journal of Production Economics*, *111*(2), 509–528.

Boysen, N., Fliedner, M., & Scholl, A. (2009). Production planning of mixed-model assembly lines: Overview and extensions. *Production Planning & Control*, *20*(5), 455–471. https://doi.org/10.1080/09537280903011626

Burns, L. D., & Daganzo, C. F. (1987). Assembly line job sequencing principles. *International Journal of Production Research*, *25*(1), 71–99.

Buxey, G. M., Slack, N. D., & Wild, R. (1973). Production flow line system design—A review. *AIIE Transactions*, *5*(1), 37–48.

Cohen, M. A., Cui, S., Doetsch, S., Ernst, R., Huchzermeier, A., Kouvelis, P., Lee, H. L., Matsuo, H., & Tsay, A. (2022). Putting supply chain resilience theory into practise. *Management and Business Review*. Forthcoming.

Cohen, M. A., Cui, S., Ernst, R., Huchzermeier, A., Kouvelis, P., Lee, H. L., Matsuo, H., Steuber, M., & Tsay, A. A. (2018). Om forum—Benchmarking global production sourcing decisions: Where and why firms offshore and reshore. *Manufacturing & Service Operations Management*, *20*(3), 389–402.

Dar-El, E. M. (1978). Mixed-model assembly line sequencing problems. *Omega*, *6*(4), 313–323.

Dar-El, E. M., & Cother, R. F. (1975). Assembly line sequencing for model mix. *International Journal of Production Research*, *13*(5), 463–477.

Dar-El, E. M., & Cucuy, S. (1977). Optimal mixed-model sequencing for balanced assembly lines. *Omega*, *5*(3), 333–342.

Dar-El, E. M., & Nadivi, A. (2007). A mixed-model sequencing application. *International Journal of Production Research*, *19*(1), 69–84.

Falkenauer, E. (2005). Line balancing in the real world. *Proceedings of the International Conference on Product Lifecycle Management PLM*, (5), 360–370.

Fattahi, P., & Salehi, M. (2009). Sequencing the mixed-model assembly line to minimize the total utility and idle costs with variable launching interval. *The International Journal of Advanced Manufacturing Technology*, *45*(9-10), 987–998.

Goldschmidt, O., Bard, J. F., & Takvorian, A. (1997). Complexity results for mixed-model assembly lines with approximation algorithms for the single station case. *International Journal of Flexible Manufacturing Systems*, *9*(3), 251–272.

Huchzermeier, A., Mönch, T., & Bebersdorf, P. (2020a). The Fendt variotakt: Revolutionizing mixed-model assembly line production: Case article. *INFORMS Transactions on Education*, *20*(3), 134–140.

Huchzermeier, A., Mönch, T., & Bebersdorf, P. (2020b). The Fendt variotakt: Revolutionizing mixed-model assembly line production: Case study. *INFORMS Transactions on Education*, *20*(3), 141–153.

Hyun, C. J., Kim, Y., & Kim, Y. K. (1998). A genetic algorithm for multiple objective sequencing problems in mixed model assembly lines. *Computers & Operations Research*, *25*(7-8), 675–690.

Kilbridge, M., & Wester, L. (1962). A review of analytical systems of line balancing. *Operations Research*, *10*(5), 626–638.

Kim, Y. K., Hyun, C. J., & Kim, Y. (1996). Sequencing in mixed model assembly lines: A genetic algorithm approach. *Computers & Operations Research*, *23*(12), 1131–1145.

Knowledge@Wharton. (2019). *How the U.S. can regain its manufacturing edge*. https://knowledge.wharton.upenn.edu/article/future-of-us-manufacturing/

Mönch, T., Huchzermeier, A., & Bebersdorf, P. (2021). Variable takt times in mixed-model assembly line balancing with random customization. *International Journal of Production Research*, *59*(15), 4670–4689.

Mönch, T., Huchzermeier, A., & Bebersdorf, P. (2022). Variable takt time groups and workload equilibrium. *International Journal of Production Research*, *60*(5), 1535–1552.

Monden, Y. (1983). *Toyota production system: A practical approach to production management*. Industrial Engineering and Management.

Monden, Y. (2011). *Toyota production system: An integrated approach to just-in-time* (4th ed.). Productivity Press. https://doi.org/10.1201/b11731

Okamura, K., & Yamashina, H. (1979). A heuristic algorithm for the assembly line model-mix sequencing problem to minimize the risk of stopping the conveyor. *International Journal of Production Research*, *17*(3), 233–247.

Sarker, B. R., & Pan, H. (1998). Designing a mixed-model assembly line to minimize the costs of idle and utility times. *Computers & Industrial Engineering*, *34*(3), 609–628.

Sarker, B. R., & Pan, H. (2001). Design configuration for a closed-station, mixed-model assembly line: A filing cabinet manufacturing system. *International Journal of Production Research*, *39*(10), 2251–2270.

Scholl, A., Klein, R., & Domschke, W. (1998). Pattern based vocabulary building for effectively sequencing mixed-model assembly lines. *Journal of Heuristics*, *4*(4), 359–381.

Sun, X., Noble, J. S., & Klein, C. M. (1999). Single-machine scheduling with sequence dependent setup to minimize total weighted squared tardiness. *IIE Transactions*, *31*(2), 113–124.

Thomopoulos, N. T. (1967). Line balancing-sequencing for mixed-model assembly. *Management Science*, *14*(2), B–59–B–75.

Tong, K.-n., Xu, K.-l., & Zheng, Y.-q. (2013). Sequencing mixed-model flexible assembly lines with variable launching intervals. *Journal of Shanghai Jiaotong University (Science)*, *18*(4), 460–467.

Tsai, L.-H. (1995). Mixed-model sequencing to minimize utility work and the risk of conveyor stoppage. *Management Science*, *41*(3), 485–495.

Vallada, E., & Ruiz, R. (2011). A genetic algorithm for the unrelated parallel machine scheduling problem with sequence dependent setup times. *European Journal of Operational Research*, *211*(3), 612–622.

Wang, Y., Ma, H.-S., Yang, J.-H., & Wang, K.-S. (2017). Industry 4.0: A way from mass customization to mass personalization production. *Advances in Manufacturing*, *5*(4), 311–320.

Xiaobo, Z., & Ohno, K. (1994). A sequencing problem for a mixed-model assembly line in a JIT production system. *Computers & Industrial Engineering*, *27*(1-4), 71–74.

Xiaobo, Z., & Ohno, K. (1997). Algorithms for sequencing mixed models on an assembly line in a JIT production system. *Computers & Industrial Engineering*, *32*(1), 47–56.

Xiaobo, Z., & Ohno, K. (2000). Properties of a sequencing problem for a mixed model assembly line with conveyor stoppages. *European Journal of Operational Research*, *124*(3), 560–570.

Yano, C. A., & Rachamadugu, R. (1991). Sequencing to minimize work overload in assembly lines with product options. *Management Science*, *37*(5), 572–586.

Yoshida, T., & Hitomi, K. (1997). Optimal two-stage production scheduling with setup times separated. *AIIE Transactions*, *11*(3), 261–263.

How to cite this article: Huchzermeier, A., & Mönch, T. (2023). Mixed-model assembly lines with variable takt and open stations. *Production and Operations Management*, *32,* 704–722. https://doi.org/10.1111/poms.13893

7. 异质环境决策：Beyond IID: Data-Driven Decision Making in Heterogeneous Environments (MS, 2025)

来源：Management Science. Vol. 71, No. 12, 2025, pp. 10538–10555.
https://doi.org/10.1287/mnsc.2022.03448

作者：Omar Besbes, Will Ma, Omar Mouchtaki

关键词：分布偏移；数据驱动算法；最小最大遗憾；分布鲁棒优化；样本平均近似

推荐人：苏睿涵

推荐理由：本文放宽历史样本独立同分布的假设，允许样本来自未知且彼此不同的分布，并假定这些分布位于未来目标分布附近、半径已知的异质性集合中。作者分析样本平均近似及阶数最优策略的渐近最坏遗憾如何随异质性半径变化。通过定价、滑雪租赁和报童等典型问题，论文说明不同问题结构与分布距离会带来不同的性能保证，并建立数据驱动决策与分布鲁棒优化之间的联系。

https://pubsonline.informs.org/journal/mnsc

MANAGEMENT SCIENCE
Vol. 71, No. 12, December 2025, pp. 10538–10555
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Beyond IID: Data-Driven Decision Making in Heterogeneous Environments

Omar Besbes,[a] Will Ma,[a,*] Omar Mouchtaki[b]

[a] Graduate School of Business, Columbia University, New York, New York 10027; [b] NYU Stern School of Business, New York, New York 10012
*Corresponding author
Contact: ob2105@gsb.columbia.edu, https://orcid.org/0000-0002-2982-3794 (OB); wm2428@gsb.columbia.edu, https://orcid.org/0000-0002-2420-4468 (WM); om2166@stern.nyu.edu, https://orcid.org/0000-0002-1734-9247 (OM)

Received: November 7, 2022
Revised: June 11, 2024; January 1, 2025
Accepted: March 2, 2025
Published Online in Articles in Advance: May 7, 2025

https://doi.org/10.1287/mnsc.2022.03448

Copyright: © 2025 INFORMS

Abstract. How should one leverage historical data when past observations are not perfectly indicative of the future, for example, because of the presence of unobserved confounders which one cannot "correct" for? Motivated by this question, we study a data-driven decision-making framework in which historical samples are generated from unknown and different distributions assumed to lie in a heterogeneity ball with known radius and centered around the (also) unknown future (out-of-sample) distribution on which the performance of a decision will be evaluated. This work aims to analyze the performance of central data-driven policies and also near-optimal ones in these *heterogeneous environments*, and it aims to understand key drivers of performance. We establish a first result that allows us to upper bound the asymptotic worst-case regret of a broad class of policies. Leveraging this result, for any integral probability metric, we provide a general analysis of the performance achieved by sample average approximation (SAA) as a function of the radius of the heterogeneity ball. This analysis is centered around the approximation parameter, a notion of complexity we introduce to capture how the interplay between the heterogeneity and the problem structure impacts the performance of SAA. In turn, we illustrate, through several widely studied problems—for example, newsvendor, pricing—how this methodology can be applied and find that the performance of SAA varies considerably depending on the combinations of problem classes and heterogeneity. The failure of SAA for certain instances motivates the design of alternative policies to achieve rate optimality. We derive problem-dependent policies achieving strong guarantees for the illustrative problems described above and provide initial results toward a principled approach for the design and analysis of general rate-optimal algorithms.

History: Accepted by Vivek Farias, data science.
Supplemental Material: The online appendix is available at https://doi.org/10.1287/mnsc.2022.03448.

Keywords: data-driven algorithms • distribution shift • distributionally robust optimization • minimax regret • sample average approximation • pricing • newsvendor • ski rental

1. Introduction

In optimization under uncertainty, the desirability of a decision (e.g., inventory) depends on an unknown future outcome (e.g., demand). Typically, past data are collected to be indicative of the future and, hence, inform our decision. However, ideal data-driven decision making requires postulating beliefs about the reliability of the past data and, importantly, whether the future may *deviate* from it. In practice, past data may depend on contexts, some of which can be controlled for and some of which cannot (unobserved confounders). This may introduce data heterogeneity that is not "correctable."

In this paper, we consider a framework for modeling how the future may deviate from past data. At a high level, it accomplishes three goals: (i) capture different forms of future deviation, including no deviation, under the same umbrella and develop an approach for performance evaluation for general policies; (ii) understand the drivers of performance of a central policy, which makes the decision that optimizes the *average* objective value over past data; and (iii) if the central policy performs poorly, suggest modifications that are robust to different forms of anticipated deviation and illustrate the power of doing so through classical problems.

This framework is general, capturing different problems, and our analysis leads to insightful problem-specific conclusions. The only assumption is that the policy defined in (ii) above can be computed, which requires perfect counterfactual evaluation of any decision on all data points, as opposed to settings where past data are affected by previous actions (Besbes et al. 2014; Karnin and Anava 2016; Luo et al. 2018; Lykouris et al. 2018, 2021; Cheung et al. 2022). The policy described in

(ii) is widely studied across different fields, known as sample average approximation (SAA)[1] (Kleywegt et al. 2002, Kim et al. 2015, Lam 2021), although we emphasize that we also derive *new policies beyond SAA* when it falters.

1.1. Framework Description

1.1.1. Framework. In Section 2.1, we study a framework that models the future outcome as being drawn from an unknown distribution and past data as being drawn independently from (also unknown) "nearby" distributions. In this way, past data are indicative of the future, and we emphasize that the samples of data can be drawn from different nearby distributions. In accord, we call this setting a *heterogeneous environment*. Using varying definitions of "nearby distribution" (e.g., based on Kolmogorov and Wasserstein distances; see Section 4.1 for the exact definition), we analyze different forms of deviations possible between the past and future and let a radius ϵ bound the allowed deviation. When $\epsilon = 0$, our framework captures the classical independent and identically distributed (i.i.d.) setting in which past data are drawn from the same distribution as the future outcome (Vapnik and Chervonenkis 1974, Kleywegt et al. 2002, Bertsimas et al. 2018, Mohajerin Esfahani and Kuhn 2018). The framework we study presents a parametrization to interpolate between the i.i.d. setting and the adversarial one and shares conceptually similar goals to previous approaches considering algorithm analysis beyond worst case (Rakhlin et al. 2011; Bilodeau et al. 2020; Roughgarden 2021; Block et al. 2022; Haghtalab et al. 2022a, b).

1.1.2. Performance Measure. In Section 2.2, we consider the performance of different data-driven policies that map past data into a decision for a given problem; SAA defines one such feasible data-driven policy for any problem. Regret is measured as the difference in objective between the policy's decision and the optimal decision knowing the future distribution, taking an expectation over the draws of past data (which affect the policy's decision), any intrinsic randomness in the policy, and the outcome realization (which affects the objective of both the policy's decision and the optimal decision). We then take a worst-case regret over all possible distributions that could have been chosen for the future outcome and the data points to evaluate the performance of a fixed data-driven policy. This performance depends on the problem, the number of data points n, the definition of "nearby distribution," and the radius ϵ. In this paper, we adopt an asymptotic viewpoint in which the number of data points n tends to ∞, and we aim at understanding the worst-case regret of data-driven policies in this regime.

1.2. Contributions

1.2.1. Performance Evaluation for General Policies. Our first contribution (Section 3) lies in developing a general result to evaluate the asymptotic worst-case regret of a large class of data-driven policies. In particular, we establish in Theorem 1 the following: asymptotically, as the number of samples goes to ∞, the worst-case regret of a policy, whose decision only depends on the empirical distribution and not the number of samples, is upper-bounded by that of an alternative problem with two major simplifications: (ii) the worst-case is now taken over only a *single* distribution for historical environments as opposed to a sequence of heterogeneous distributions, and (ii) the decision maker (DM) has access to the exact historical distribution.

The upper bound we derive can be seen as a uniform distributionally robust optimization (DRO) problem *under a regret performance metric*, and it offers a systematic way to derive upper bounds for policies of interest. We leverage it to analyze both the SAA policy and also to derive guarantees on alternative policies when SAA falters. Moreover, we show that the assumption that the policy does not depend on the number of samples can be made without loss of optimality because, for any problem, there exists such a policy achieving the optimal asymptotic worst-case regret (see Proposition B.4 in Online Appendix B.1).

1.2.2. Performance Drivers for SAA and Illustration for Central Problems

1.2.2.1. Analysis of SAA for Integral Probability Metrics (IPMs). Our second contribution is the development of a general framework to bound the asymptotic regret of SAA under the assumption that the distance is an integral probability metric, as defined in Section 4.1. Notably, this broad class of metrics—which includes the Kolmogorov and Wasserstein distances—can be linked to specific generating classes of functions. We leverage this connection to introduce the "approximation parameter," a quantity that captures the complexity required to uniformly approximate the objective function using functions from the generating class of the IPM. This parameter not only provides a bound on the regret of SAA but also has a natural geometric interpretation. By utilizing standard topological results, we show that this approach enables meaningful regret bounds. Specifically, for the Kolmogorov and Wasserstein distances, our framework relates the worst-case regret of SAA to practical quantities such as the Lipschitz constant and the total variation of the objective function. Our analysis extends the results derived in the context of "the method of probability metrics" (Rachev and Römisch 2002). We discuss the relation of our work to these results in Remark 2.

1.2.2.2. Illustration of Our Method for Central Problems. To illustrate the versatility of our methodology, we study the performance of SAA for two central problems (newsvendor and pricing) and two distances defining the heterogeneity (Kolmogorov and Wasserstein). These problems have received significant attention in the literature across operations research, economics, and computer science in the known-prior, data-driven, distributionally robust, and advice-augmented settings.

We show in Section 4.3 that the upper bounds based on the approximation parameter directly imply bounds on the worst-case regret of SAA for the newsvendor under both heterogeneity types and for pricing under the Kolmogorov heterogeneity. Furthermore, we complement these results with lower bounds on the best achievable performance and show that for these three settings, SAA achieves the best possible dependence in ϵ, and its asymptotic worst-case regret scales linearly with ϵ. For pricing under Wasserstein distance, the picture is starkly different: the approximation parameter becomes infinite and, hence, does not allow us to derive any meaningful upper bound on the worst-case regret of SAA. As a matter of fact, we show that SAA performs extremely poorly and the asymptotic regret does not even vanish when ϵ goes to zero. Hence, the performance of SAA deteriorates considerably by slightly deviating from the i.i.d. regime for this class of problems.

These two problems illustrate that the approximation parameter is a powerful tool to derive tight bounds on the worst-case regret of SAA. Furthermore, whereas we do not derive a formal lower bound on the worst-case regret of SAA, which depends on the approximation parameter, the pricing problem with Wasserstein heterogeneity suggests that a large approximation parameter usually corresponds to a poor performance of SAA. This finding is consistent with a third problem we investigate in Online Appendix E, the ski rental problem, and for which SAA falters similarly when the approximation parameter is large.

We also illustrate in Online Appendix F how our framework can be generalized to tackle a multidimensional problem. In particular, we investigate the Bayesian mechanism design problem and show that the result derived by Brustle et al. (2020) for SAA under the Kolmogorov distance can be retrieved using our approximation parameter framework.

1.2.3. Design and Analysis of Policies Beyond SAA. The failure of SAA for the Wasserstein pricing problem naturally raises the question of designing and quantifying the performance of alternative policies to improve upon SAA and achieve near optimality. Our third main contribution (Section 5) consists of developing and analyzing policies that operate efficiently in heterogeneous environments.

1.2.3.1. Analyzing Policies Beyond SAA to Achieve Rate Optimality. In Section 5.1, we complete the picture for the pricing problem under Wasserstein heterogeneity. We propose a policy that appropriately deflates the price selected by SAA, and we show that this policy achieves a worst-case regret that has a $\sqrt{\epsilon}$ dependence in the radius of heterogeneity. We also show that this performance is rate optimal. To our understanding, analyzing these non-SAA policies (which are required for good performance in pricing) deviates from standard analyses used in learning theory and critically requires the reduction we derive in Theorem 1, as we elaborate on in Section 5.1. To derive our result, we leverage the structure of the objective function in pricing: whereas it is not continuous in general, it is ensured to be one-sided Lipschitz continuous (when deflating the price). We combine this observation with a critical relation between the Wasserstein distance (in one dimension) of two probability measures and their associated cumulative distribution functions to obtain the desired result. We believe that this problem-specific analysis may be of independent interest.

We also study the ski rental problem, a third central problem for which one needs to go beyond SAA to improve performance even for the Kolmogorov distance (cf. Online Appendix E). In this setting, we consider the policy proposed by Diakonikolas et al. (2021) and derive a lower bound on the best achievable performance that matches their upper bound up to logarithmic factors. For the Wasserstein distance, we show that, similarly to pricing, SAA incurs a worst-case regret that does not shrink to zero as ϵ goes to zero, but a policy that inflates the SAA decision appropriately achieves rate optimality. The proof techniques and results derived for ski rental leverage the structure of the problem and could be of independent interest.

1.2.3.2. Toward a General Design Principle for Data-Driven Policies. The examples of pricing and ski rental illustrate how critical Theorem 1 is to derive guarantees for a wide range of data-driven policies, but it leaves open the choice of the policy that should be analyzed. In Section 5.2, we take initial steps for the design of general policies with strong asymptotic worst-case regret guarantees. We first introduce a policy naturally suggested by Theorem 1, which minimizes the DRO regret objective and show that this policy is essentially minimax optimal. We also derive a general result regarding the performance of the classical DRO policy in Rahimian and Mehrotra (2019) (formally, a variant which optimizes the worst-case objective over randomized actions) and show in Proposition 9 that when the optimal value of the objective is not too sensitive to the structure of the heterogeneity,

the DRO policy achieves a rate-optimal regret guarantee. We further show that this condition holds for the pricing problem under Wasserstein heterogeneity. We note that the two general policies that we investigate could be challenging to compute, and we leave open the question regarding efficient implementation of these two policies.

1.2.4. Impact of Heterogeneity on Performance Across Some Classical Problems. The results above allow us to illustrate the interplay of heterogeneity and problem class. In Table 1, we present a high-level summary of the problem-specific results we derive. We assume that the support of the unknown environment is in $[0, M]$ for some positive real number M, which can be interpreted as parametrizing the "extent" of uncertainty one faces (e.g., maximal values of demand in newsvendor). We track the dependence on both the heterogeneity radius ϵ, as well as M. These results are for the asymptotic setting in which the number of samples grows large.

Our illustrative examples unveil a critical insight regarding the performance of data-driven policies in heterogeneous environments. It highlights the important fact that the quality of decisions made with SAA in heterogeneous environments is highly dependent on the combination of problem class and heterogeneity. Furthermore, the dependence of the best achievable performance in the heterogeneity radius ϵ varies considerably across problem classes and notions of heterogeneity. Therefore, our work implies that it is critical for a decision maker to understand the *nature of the heterogeneity* faced and the impact it may have on the objective of interest when making decisions.

1.3. Further Related Work

1.3.1. Connection to Offline Data Corruption Models. Different notions of offline data corruption and robustness have been previously studied in statistics. Huber (1992) introduces the so-called ϵ-contamination model in which an ϵ fraction of the observed samples is corrupted. They study the problem of location estimation for ϵ-contaminated Gaussian distributions and for approximations of Gaussian distributions that are ϵ away in Kolmogorov distance. They derive, in both cases, minimax estimators among the class of M-estimators. Hampel (1971) considers distributions that are ϵ away for the Prokhorov distance and establish that the classical empirical mean is a nonrobust estimator. These models have been extensively developed in learning theory (Haussler 1992, Kearns and Li 1993, Kearns et al. 1994, Servedio 2003, Klivans et al. 2009, Diakonikolas et al. 2019, Duchi and Namkoong 2021, Bennouna and Van Parys 2022, Blanc et al. 2022) and, more recently, in revenue management (Cai and Daskalakis 2017, Brustle et al. 2020, Guo et al. 2021, Chen and Wang 2022). Most approaches consider ϵ-contamination settings (Huber 1992) in which the adversary can shift a small proportion of data arbitrarily far from the true distribution. They consider either oblivious adversaries who fix a *single* corrupted distribution from which samples are generated or adaptive ones who observe samples before corrupting them. Blanc et al. (2022) establish that adaptive adversaries yield equivalent performance to oblivious ones in many settings. We note that our framework is general enough to derive guarantees on the regret under the ϵ-contamination model and the strong contamination model. We discuss in more detail connections between our work and previous contamination models in Online Appendix A.2.3.

Closest to us are Brustle et al. (2020) and Guo et al. (2021), who study optimal auctions. They assume the distribution is common across all past observations; the former focuses on regret for various distances and the latter on a ratio metric when data are generated from monotone hazard rate (MHR) and regular distributions close in Kolmogorov distance. The framework we develop is anchored around a setting that allows for *heterogeneous* "nearby" distributions for each past observation and is general in that it allows us to unify a variety of problems/metrics and highlight how these affect the levels of achievable performance. Related to auction design, Dütting and Kesselheim (2019) study the robustness of well-known posted pricing and prophet-inequality-based mechanisms when computed on distributions that differ from the true distributions for various distances.

Table 1. SAA and Achievable Performance for Different Problem Classes and Heterogeneity Balls

Problem class	Policy class	Kolmogorov	Wasserstein
Newsvendor	SAA	$\Theta(M\epsilon)$	$\Theta(\epsilon)$
	Best policy	$\Theta(M\epsilon)$	$\Theta(\epsilon)$
Pricing	SAA	$\Theta(M\epsilon)$	$\Omega(M)$
	Best policy	$\Theta(M\epsilon)$	$\Theta(\sqrt{M\epsilon})$
Ski rental	SAA	$\Theta(M\epsilon)$	$\Theta(1)$
	Best policy	$\tilde{\Theta}(\epsilon)$[a]	$\Theta(\sqrt{\epsilon})$

[a]The upper bound was proved in Diakonikolas et al. (2021), and we complement it by a matching lower bound up to logarithmic factors.

1.3.2. Connection to Online and Batch Learning in Nonstationary Environments. Our work also relates broadly to papers studying learnability in changing environment. Hanneke (2021) and Blanchard (2022) adopt a universal learning approach and analyze algorithms that are able to achieve vanishing long-term average loss for general data-generation processes. Mansour et al. (2009), Ben-David et al. (2010), and Mohri and Muñoz Medina (2012) derive generalization

guarantees in a setting where the training set distribution may differ from the test set distribution.

1.3.3. Connection to Robust Optimization and Stability in Stochastic Programming. Our work is also conceptually related to robust optimization in which one designs decisions for optimization problems that are robust to some variations in the parameters of the problem (see the survey by Bertsimas et al. 2011). When the variation is on the underlying probability measure, people refer to this approach as distributionally robust optimization (see references in Rahimian and Mehrotra 2019). Glasserman and Xu (2014) quantify the variation in risk that may result from distribution misspecification and discuss, for several applications in portfolio optimization, the shape and impact of the worst-case distribution shift within a Kullback-Leibler ball. In our work, we quantify how the problem structure *and the notion of heterogeneity* impact the performance. Another related line of papers use the *method of probability metrics* to understand when small perturbations of the distribution, with respect to a given metric, result in a small perturbation of the value of the stochastic problem and the set of optimal solutions (Römisch and Schultz 1991, Schultz 1996, Rachev and Römisch 2002, Dupačová et al. 2003, Xu and Zhang 2021, Pichler and Xu 2022). These papers fix the problem of interest (e.g., newsvendor, two-stage problems with recourse) and construct a "canonical IPM" under which the stochastic optimization problem is stable. Most of these works focus on the Wasserstein distance or the related Fortet-Mourier metric. Our paper differs from this literature along two critical dimensions. First, when analyzing SAA with IPMs, we take the IPM as given and do not construct one that ensures stability of the problem (see, for instance, pricing with Wasserstein distance). In that case, we derive a broader methodology that enables us to characterize the interplay between the problem structure and the heterogeneity structure (see detailed discussion in Remark 2). Second, we develop and analyze the performance of alternative data-driven policies in settings where SAA fails.

1.3.4. Prior Works on Our Illustrative Examples. Our paper also relates to previous work specifically addressing each of the prototypical problems we consider. The newsvendor problem is a foundational problem in operations research, with early focus on the setting where the demand distribution is known (see Morse and Kimball 1946), and since then, it has been studied in various distributionally robust settings (Scarf 1958, Gallego and Moon 1993, Perakis and Roels 2008) and also in data-driven ones (Levi et al. 2015, Cheung and Simchi-Levi 2019, Lin et al. 2022, Besbes and Mouchtaki 2023). The pricing problem is a central problem in economics, with early focus on the setting with known prior (see Myerson 1981, Riley and Zeckhauser 1983) and, more recently, in the distributionally robust (Eren and Maglaras 2010, Bergemann and Schlag 2011) and data-driven settings (Fu et al. 2015, Huang et al. 2018, Babaioff et al. 2018, Daskalakis and Zampetakis 2020, Allouah et al. 2022). Ski rental, on its end, is a foundational problem in computer science, critical to the development of competitive analysis (see Borodin and El-Yaniv 2005) that considers the *adversarial* setting where nothing is known and, recently, is also important to the development of *advice-augmented algorithms*, which receive a prediction on the length of the ski season (Purohit et al. 2018, Diakonikolas et al. 2021).

1.4. Preliminaries

We denote by $\mathbb{N}^*$ the set of positive integers and by $\mathbb{R}_+$ the set of nonnegative real numbers. For any set A, $\Delta(A)$ denotes the set of probability measures on A. For any deterministic sequences $(a_k)_{k\in\mathbb{N}}$ and $(b_k)_{k\in\mathbb{N}}$, both indexed by a common index k that goes to ∞, we say that $a_k = o(b_k)$ if $a_k/b_k \to 0$, $a_k = \mathcal{O}(b_k)$ if there exists a finite $M > 0$ such that $|a_k| \le M|b_k|$ for k large enough, $a_k = \omega(b_k)$ if $|a_k/b_k| \to \infty$, $a_k = \Omega(b_k)$ if there exist a finite $M > 0$ such that $|a_k| > M|b_k|$ for k large enough, $a_k = \Theta(b_k)$ if $a_k = \mathcal{O}(b_k)$ and $a_k = \Omega(b_k)$, and $a_k \sim b_k$ if $a_k/b_k \to 1$. For any metric spaces $(\mathcal{M}_1, d_1)$ and $(\mathcal{M}_2, d_2)$ and any mapping f from $\mathcal{M}_1$ to $\mathcal{M}_2$, we denote by $\mathrm{Lip}(f)$, the Lipschitz constant of f, that is, the largest $L \in [0,\infty) \cup \{\infty\}$ such that, for every $x, y \in \mathcal{M}_1$, we have $d_2(f(x), f(y)) \le L \cdot d_1(x,y)$. For a function f defined on a closed interval $[a,b] \subset \mathbb{R}$, we define the total variation of f as $\mathrm{V}(f) := \sup\{\sum_{i=1}^n |f(z_i) - f(z_{i+1})| \text{ s.t. } a = z_1 < z_2 < \cdots < z_{n+1} = b\}$. For any metric space, $(\mathcal{M}, d)$, we denote by $\mathrm{diam}(\mathcal{M}) := \sup_{x,y\in\mathcal{M}} d(x,y)$ the diameter of $\mathcal{M}$. For a set $\mathcal{S}$, we denote by $\mathrm{span}(\mathcal{S})$ the set generated by finite linear combinations of elements from $\mathcal{S}$; that is, $\mathrm{span}(\mathcal{S}) = \{\sum_{i=1}^k \lambda_i \cdot v_i | k \in \mathbb{N}, v_i \in \mathcal{S}, \ \lambda_i \in \mathbb{R}\}$. For any real-valued function defined on a set $\mathcal{S}$, we denote by $\|f\|_\infty = \sup_{x\in\mathcal{S}} |f(x)|$ the uniform norm.

All proofs of statements are deferred to the Online Appendix.

2. Problem Formulation

We consider a general stochastic optimization problem in which a DM wants to optimize an objective in a stochastic environment. Formally, we denote by $\mathcal{X}$ the decision space and by Ξ the space of environment realizations. The objective of the DM is represented by a function $g : \mathcal{X} \times \Xi \to \mathbb{R}$, which maps each pair of decision $x \in \mathcal{X}$ and environment realization $\xi \in \Xi$ to an objective value.

The DM faces a subset of possible probability measures $\mathcal{P} \subset \Delta(\Xi)$, where $\Delta(\Xi)$ denotes the space of probability measures supported on Ξ.[2] For every probability

measure $\mu \in \mathcal{P}$, the goal of the DM is to optimize the objective defined as

$$\mathcal{G}(\cdot, \mu) : \begin{cases} \mathcal{X} \to \mathbb{R} \\ x \mapsto \mathbb{E}_{\xi \sim \mu}[g(x, \xi)]. \end{cases} \quad (1)$$

We extend the definition of $\mathcal{G}(\cdot, \mu)$ to allow for randomized decisions $\rho \in \Delta(\mathcal{X})$, in which case we define $\mathcal{G}(\rho, \mu) = \mathbb{E}_{x \sim \rho}[\mathcal{G}(x, \mu)]$ to also take an expectation over the randomized decision.

We remark that depending on the nature of the problem, the objective could represent, for example, a function to maximize (profit) or to minimize (cost). To unify exposition, we consider that the decision maker aims at solving the maximization problem

$$\text{opt}(\mu) := \sup_{x \in \mathcal{X}} \mathcal{G}(x, \mu),$$

where $\mu \in \mathcal{P}$ is an unknown distribution. The value $\text{opt}(\mu)$ is the value achieved by an oracle who knows the demand distribution.

2.1. Heterogeneous Environments

In the absence of distributional knowledge on the underlying environment, a common approach consists of assuming that historical samples are independent and identically drawn from a fixed probability measure μ. We relax the assumption that the samples are identically distributed and introduce a framework with heterogeneous distributions based on a parameter ϵ.

Let $d : \mathcal{P} \times \mathcal{P} \to \mathbb{R}$ be a metric, which we refer to interchangeably as a distance, on $\mathcal{P}$. Given a target probability measure $\mu \in \mathcal{P}$ and a radius of heterogeneity ϵ, we define the heterogeneity ball anchored around μ with radius ϵ as

$$\mathcal{U}_\epsilon(\mu) := \{\nu \in \mathcal{P}, \text{ s.t } d(\mu, \nu) \le \epsilon\}.$$

The heterogeneity ball $\mathcal{U}_\epsilon(\mu)$ includes all distributions that are within a distance of ϵ from μ. Next, we formally define a data-driven decision-making problem in a heterogeneous environment.

Definition 1 (Data-Driven Decision-Making Problem in Heterogeneous Environment). An instance $\mathcal{I}$ of a data-driven decision-making problem in a heterogeneous environment is defined by the tuple: $\mathcal{I} = (\mathcal{X}, \Xi, \mathcal{P}, d, g)$, where $\mathcal{X}$ is the set of possible actions, Ξ is the environment space, $\mathcal{P}$ is the space of measures nature can select from, d is a metric on the space of measures, and g is an objective function.

2.1.1. Focal Metrics. The definition above is general, and the framework can be applied to a variety of metrics. In this work, we also aim to develop an understanding of the effect of heterogeneity under various metrics and various problem classes. To illustrate the versatility of our framework, we will focus in this work on two common distances: the Kolmogorov distance denoted by d_K and the Wasserstein distance denoted by d_W (see Section 4.1 for formal definitions). These distances describe different ways in which two distributions may differ and, as we will see, imply different trade-offs in the problems we analyze.

2.2. Data-Driven Policies and Performance

2.2.1. Data-Driven Policies. In most settings, the out-of-sample (or future) probability measure μ is unknown to the DM, and the optimal objective $\text{opt}(\mu)$ is not achievable. In turn, the DM only observes a sequence of historical samples $\xi_1, \xi_2, \ldots, \xi_n$ representing the previous environment realizations they observed. We assume that these samples are drawn independently from unknown distributions $\nu_1, \ldots, \nu_n$, respectively, and the only knowledge is that these distributions are "not far" from the future unknown distribution μ and belong to $\mathcal{U}_\epsilon(\mu)$.

In general, a data-driven policy is a mapping from the past samples to a (randomized) decision. In this work, we use an equivalent definition of data-driven policies through the empirical probability measure. Define the empirical probability measure $\hat{\nu}_{\boldsymbol{\xi}^n}$ as

$$\hat{\nu}_{\boldsymbol{\xi}^n}(A) := \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{\xi_i \in A\}, \quad \text{for every measurable set } A.$$

In a setting in which the order of the samples does not matter (as the one we consider), to represent a policy taking as input $\boldsymbol{\xi}^n := (\xi_1, \ldots, \xi_n)$, one can instead consider mappings that take as an input the number of samples and the empirical measure. Indeed, $(\xi_1, \ldots, \xi_n)$ is only characterized by the values taken by the previous samples and the count of each value. Furthermore, the empirical probability measure gives the frequency of each value, which can be multiplied by the sample size to obtain the actual count of appearances of values.

We therefore define a data-driven policy π as a mapping from $\mathbb{N}^* \times \mathcal{P}$ to $\Delta(\mathcal{X})$, the space of measures on the action space $\mathcal{X}$, that associates $(n, \hat{\nu})$ to a distribution of actions $\pi(n, \hat{\nu})$. We let Π denote the set of all such mappings. We assume that such a mapping can be made with knowledge of the metric d and radius ϵ. In Online Appendix A.1, we provide a detailed discussion about the relation between Π and the usual definition of data-driven policies.

2.2.2. Performance Through Worst-Case Regret. Given an out-of-sample distribution $\mu \in \mathcal{P}$, the best achievable objective is $\text{opt}(\mu)$. For a sequence of historical distributions $\nu_1, \ldots, \nu_n$ in $\mathcal{U}_\epsilon(\mu)$, the expected objective of a policy π is

$$\mathbb{E}_{\xi_i \sim \nu_i}[\mathcal{G}(\pi(n, \hat{\nu}_{\boldsymbol{\xi}^n}), \mu)].$$

In turn, we define the expected (absolute) regret of a policy $\pi \in \Pi$ against an out-of-sample distribution

$\mu \in \mathcal{P}$ and a sequence of n historical distributions $\nu_1, \ldots, \nu_n \in \mathcal{U}_\epsilon(\mu)$ as

$$\mathcal{R}_n(\pi, \mu, \nu_1, \ldots, \nu_n) := \mathrm{opt}(\mu) - \mathbb{E}_{\xi_i \sim \nu_i}[\mathcal{G}(\pi(n, \hat{\nu}_{\xi^n}), \mu)].$$

In this work, we will assess the performance of a policy based on its *worst-case expected regret*. Given an instance $\mathcal{I} = (\mathcal{X}, \Xi, \mathcal{P}, d, g)$, we define the worst-case expected regret of a data-driven policy $\pi \in \Pi$ for a radius of heterogeneity $\epsilon \geq 0$ as

$$\mathfrak{R}^{\pi}_{\mathcal{I},n}(\epsilon) := \sup_{\mu \in \mathcal{P}} \sup_{\nu_1, \ldots, \nu_n \in \mathcal{U}_\epsilon(\mu)} \mathcal{R}_n(\pi, \mu, \nu_1, \ldots, \nu_n). \tag{2}$$

We finally define the *worst-case asymptotic regret* of a data-driven policy $\pi \in \Pi$ in an environment with radius of heterogeneity ϵ as

$$\mathfrak{R}^{\pi}_{\mathcal{I},\infty}(\epsilon) := \limsup_{n \to \infty} \mathfrak{R}^{\pi}_{\mathcal{I},n}(\epsilon). \tag{3}$$

We sometimes write $\mathfrak{R}^{\pi}_{K,\infty}(\epsilon)$ or $\mathfrak{R}^{\pi}_{W,\infty}(\epsilon)$ instead of $\mathfrak{R}^{\pi}_{\mathcal{I},\infty}(\epsilon)$ when the problem class is specified to highlight the dependence on the Kolmogorov or Wasserstein metrics.

Remark 1 (Relations Between Different Notions of Heterogeneity). Note that many notions of distances or divergences are related to one another (Gibbs and Su 2002). We show in Online Appendix A.2.1 that, in general, by leveraging relations between different distances, one may relate the worst-case regret of data-driven decision-making instances in heterogeneous environments that have the same ϵ but differ along the type of distance used. In particular, when Ξ is a bounded subset of $\mathbb{R}$, we obtain, by specifying our result to the Kolmogorov and the Wasserstein distances, that for any data-driven policy $\pi \in \Pi$ and any sample size n and any radius of heterogeneity ϵ,

$$\mathfrak{R}^{\pi}_{K,n}(\epsilon) \leq \mathfrak{R}^{\pi}_{W,n}(\mathrm{diam}(\Xi) \cdot \epsilon). \tag{4}$$

This relation follows from the fact that these distances are related as for every μ_1 and $\mu_2 \in \mathcal{P}$,

$$d_W(\mu_1, \mu_2) \leq \mathrm{diam}(\Xi) \cdot d_K(\mu_1, \mu_2).$$

3. Reduction for Policy Evaluation

When trying to understand the performance of a particular data-driven policy, a significant challenge in the analysis of Problem (3) stems from nature's problem. This problem requires us to optimize over a general set of product measures that represent the possible sequences of historical probability measures $\nu_1, \ldots, \nu_n$ and of an out-of-sample measure μ.

When considering Problem (3), a natural approach would be to try to characterize the worst-case product measure as a function of n and, in turn, the limiting behavior of this product measure as the product becomes infinite. Our next result shows that this approach is a priori challenging, as nature will attempt to use different distributions for each of the past environments.

Proposition 1. *There exists a data-driven decision problem in a heterogeneous environment $\mathcal{I} = (\mathcal{X}, \Xi, \mathcal{P}, d, g)$ and a radius of heterogeneity $\epsilon > 0$ such that the SAA policy defined in Section 4 satisfies*

$$\mathfrak{R}^{\pi^{\mathrm{SAA}}}_{\mathcal{I},2}(\epsilon) > \sup_{\mu \in \mathcal{P}} \sup_{\nu \in \mathcal{U}_\epsilon(\mu)} \mathcal{R}_2(\pi^{\mathrm{SAA}}, \mu, \nu, \nu).$$

Proposition 1 formalizes that the performance of SAA, a central data-driven policy formally introduced in Section 4, may be strictly worse when allowing for various historical distributions than when restricting attention to a fixed one. We prove this result by considering a variant of the ski rental problem in which we restrict Ξ to be three points. In this setting, we argue that the performance of SAA is the worse when observing two samples with different values and show that selecting two different point mass distributions is the unique worst-case sequence of historical distributions.

Whereas the previous result shows that nature will take full advantage of heterogeneity by selecting different distributions in sample that are also different from the out-of-sample distribution, we will next establish that under fairly general conditions, one can upper bound the asymptotic worst-case regret of a broad class of data-driven algorithms by considering a surrogate problem in which nature selects a common distribution for all of the past environments. Recall that a data-driven policy is a general mapping from $\mathbb{N}^* \times \mathcal{P}$ to a potentially randomized decision in $\mathcal{X}$. We say that π is sample size agnostic if its decision does not depend on n. Formally, a policy is sample size agnostic if, for any $\hat{\nu} \in \mathcal{P}$, we have that $\pi(n_1, \hat{\nu}) = \pi(n_2, \hat{\nu})$ for any $n_1, n_2 \in \mathbb{N}^*$, in which case we write $\pi(\hat{\nu})$ for brevity. This condition implies that the policy only needs information on the empirical probability measure (i.e., the frequency of appearance of past samples) as opposed to the actual count of appearance. The class of sample-size-agnostic policies encompasses central policies such as the central SAA policy analyzed in Section 4 and also the DRO policies commonly analyzed in the literature, which will be defined in Section 5.2. We denote by Π_{agn} the subclass of sample-size-agnostic policies.

We define, for a sample-size-agnostic policy $\pi \in \Pi_{\mathrm{agn}}$, the following optimization problem:

$$\mathfrak{R}^{\pi}_{\mathcal{I},\mathrm{DRO}}(\epsilon) := \sup_{\mu \in \mathcal{P}} \sup_{\nu \in \mathcal{U}_\epsilon(\mu)} \{\mathrm{opt}(\mu) - \mathcal{G}(\pi(\nu), \mu)\}. \tag{5}$$

Whereas Problem (5) resembles Problem (2), it differs on two crucial dimensions: (i) in Problem (5), nature selects the *same* distribution ν in the heterogeneity ball for all past environments; and (ii) in Problem (5), the

policy π is assumed to know the actual distribution of the past environments ν, whereas in Problem (2), the policy only observed the empirical distribution of past realizations.

Next, we will establish that Problems (2) and (5) are tightly connected under some mild assumptions. We define the following conditions on the distance d defining the heterogeneity.

Definition 2 (Empirical Triangular Convergence). We say that a distance d on the space of probability measures satisfies the empirical triangular convergence (ETC) property if, for every triangular array sequence of probability measures $(\mu_{i,n})_{1\le i\le n, n\in\mathbb{N}^*}$, all belonging to $\mathcal{P}$, we have

$$\lim_{n\to\infty} d(\hat{\mu}_n, \bar{\mu}_n) = 0 \qquad \text{a.s.},$$

where, for every measurable set A, $\hat{\mu}_n(A) := \frac{1}{n}\sum_{i=1}^n \mathbb{1}\{\xi_{i,n} \in A\}$ for samples $\xi_{i,n} \sim \mu_{i,n}$ and $\bar{\mu}_n = \frac{1}{n}\sum_{i=1}^n \mu_{i,n}$.

Definition 3 (Convexity). A distance d on a set S is convex if, for any x in S, $d(\cdot, x)$ is convex.

The ETC property requires that the sequence of empirical distributions converges to the average of ground truth distributions for arbitrary triangular arrays. This assumption is satisfied by many common distances, and we show in Online Appendix A.2.2 that this property is satisfied by the Kolmogorov and Wasserstein distances. Convexity is also trivially satisfied by these two distances.

We now state our first main result.

Theorem 1 (Performance Evaluation Through Upper Bound). *Let $\mathcal{I} = (\mathcal{X}, \Xi, \mathcal{P}, d, g)$ be a data-driven decision problem in a heterogeneous environment, and let $\pi \in \Pi$ be a sample-size-agnostic policy. Assume that g is bounded on $\mathcal{X} \times \Xi$ and that d is a convex distance that satisfies the empirical triangular convergence property. Then,*

$$\mathfrak{R}^{\pi}_{\mathcal{I},\infty}(\epsilon) \le \lim_{\eta\to\epsilon^+} \mathfrak{R}^{\pi}_{\mathcal{I},\mathrm{DRO}}(\eta). \tag{6}$$

Theorem 1 establishes that under some mild conditions, in the asymptotic regime, one may restrict attention to Problem (5) to evaluate a policy through an upper bound on its regret. This reduction significantly simplifies the analysis of sample-size-agnostic policies, as Problem (5) does not involve any form of randomness in the input received by the policy and only allows the adversary to select a single distribution. As discussed in Proposition 1, the asymptotic regime is a key driver of this result because the reduction to a single distribution does not hold for finite samples. We also remark that considering sample-size-agnostic policies is without loss of optimality (see Proposition B.4 in Online Appendix B.1).

We note that Mohri and Muñoz Medina (2012, equation (7)) derive a finite-sample bound on the regret of SAA in a similar setting to ours, which can be used to derive an asymptotic bound for SAA. We discuss, in more detail, the relation between our result, when applied to SAA, and theirs in Section 4. Quite importantly, their result is only applicable to SAA, whereas our reduction applies to *any* policy in the wide range of sample-size-agnostic policies. In particular, this will be critical to derive policies that achieve minimax optimal asymptotic regret rates when SAA fails and is not (rate) optimal (see Section 5).

We prove Theorem 1 through a sample path analysis. We show that almost surely (over all possible historical samples observed), the asymptotic worst-case regret of a sample-size-agnostic policy can be bounded by the right-hand side of (6). In particular, we use the ETC property of the distance to show that asymptotically, the empirical cdf observed by the policy belongs to the heterogeneity ball of radius ϵ. The proof also relies on the formalism we develop to define data-driven policies, as discussed in Online Appendix A.1.

We note that in settings where the ETC property fails, one may derive results on the performance of data-driven policies by using relations between different distances, as discussed in Remark 1. We discuss in Online Appendix A.2.1 how one can use the results established in the main body for the Kolmogorov and Wasserstein distances to other metrics and ϕ-divergences, such as the total variation or the KL-divergence.

4. Analysis of SAA for Integral Probability Metrics

In the i.i.d. setting, that is, when $\epsilon = 0$, a widely studied policy that achieves excellent performance across a vast range of data-driven decision-making problems is SAA. SAA is usually defined for empirical probability measures, but for the purpose of this work, we extend its definition for arbitrary probability measures.[3] For every $\nu \in \mathcal{P}$, SAA is defined as

$$\pi^{\mathrm{SAA}}(\nu) = \arg\max_{x\in\mathcal{X}} \mathcal{G}(x, \nu),$$

whenever the extremum of the optimization problem is achieved. This is the case whenever the function $\mathcal{G}(\cdot, \nu)$ is upper semicontinuous, and the action space is compact. In the case of multiple optimal solutions, our results hold for an arbitrary choice of optimal solution.

Given the centrality of SAA both in theory and practice, it is natural to ask whether this policy performs well for data-driven problems in heterogeneous environments. In this section, we provide a general methodology to analyze SAA and provide an understanding of some of the key drivers of its worst-case regret.

The analysis in this section is general and is motivated by the following upper bound on the worst-case regret of SAA:

$$\mathfrak{R}^{\pi^{\mathrm{SAA}}}_{\mathcal{I},\infty}(\epsilon) \le 2 \sup_{\mu\in\mathcal{P}} \sup_{\nu\in\mathcal{U}_\epsilon(\mu)} \mathrm{Unif}_{\mathcal{I}}(\mu, \nu), \tag{7}$$

where, for every $\mu, \nu \in \mathcal{P}$,

$$\text{Unif}_{\mathcal{I}}(\mu,\nu) = \sup_{x\in\mathcal{X}} |\mathcal{G}(x,\mu) - \mathcal{G}(x,\nu)|. \tag{8}$$

To explain, the Inequality (7) is a trivial corollary of Theorem 1 whenever the ETC property holds and the distance is convex (see the proof of Proposition 4). The *uniform deviation* quantity defined in Equation (8) is a quantity of interest in the statistical learning literature, which is commonly analyzed (in the special case where ν is an empirical distribution generated from i.i.d. samples of μ) to prove that the finite-sample generalization error of a learning algorithm vanishes at a certain rate. Motivated by (7), we will analyze the uniform deviation for arbitrary μ and ν to be able to derive bounds on the asymptotic worst-case regret of SAA in heterogeneous environments.

As mentioned in Section 3, Mohri and Muñoz Medina (2012) develop a finite-sample analysis of SAA in the heterogeneous setting and provide a finite-sample bound that involves a sequential notion of Rademacher complexity. In the settings where the Rademacher complexity vanishes, their bound implies that (7) holds asymptotically.

In the rest of this section, we derive a methodology to bound the performance of SAA for data-driven problems in heterogeneous environments when the distance defining the heterogeneity belongs to the class of integral probability metrics, formally introduced in Section 4.1. In Section 4.2, we develop a general analysis to bound $\text{Unif}_{\mathcal{I}}$ for IPMs. We introduce the approximation parameter, a key quantity that enables us to derive guarantees across problem classes and distances. We finally translate in Section 4.3 our bounds on $\text{Unif}_{\mathcal{I}}$ into bounds on the asymptotic worst-case regret of SAA and illustrate our methodology in the context of the newsvendor and the pricing problems.

As mentioned in Section 1.3, the results developed in this section are conceptually close to the method of probability metrics (see, e.g., Rachev and Römisch 2002). We discuss in detail the relation to this work in Remark 2.

4.1. Integral Probability Metrics: Definition and Examples

We focus on a broad class of metrics often referred to as *probability metrics with a ζ-structure* (Zolotarev 1984) or *integral probability metrics* (Müller 1997, Sriperumbudur et al. 2010). In this work, we will use the latter denomination. These metrics are defined as follows:

Definition 4 (Integral Probability Metrics). Given a class of functions $\mathcal{F}$ from Ξ to $\mathbb{R}$, which we will refer to as a generating class of functions, the associated integral probability metric $d_{\mathcal{F}}$ is defined for every $\mu, \nu \in \mathcal{P}$ as

$$d_{\mathcal{F}}(\mu,\nu) = \sup_{f\in\mathcal{F}} \left| \int_{\xi\in\Xi} f(\xi) d\mu(\xi) - \int_{\xi\in\Xi} f(\xi) d\nu(\xi) \right|. \tag{9}$$

We note that different classes of functions may induce the same metric. Furthermore, many common metrics (that we define below), such as the Wasserstein distance or the Kolmogorov distance, are, in fact, integral probability metrics.

4.1.1. Wasserstein Distance. For any metric space (Ξ, d_Ξ), the Wasserstein distance, denoted as d_W, is defined for all probability measures μ, ν as

$$d_W(\mu,\nu) = \inf_{\gamma\in\Gamma(\mu,\nu)} \int_{\xi,\xi'\in\Xi} d_\Xi(\xi,\xi') d\gamma(\xi,\xi'),$$

where $\Gamma(\mu,\nu)$ is the set of measures on $\Xi\times\Xi$ with marginal distributions μ and ν.

The Wasserstein distance also has a dual representation as an integral probability metric (Kantorovich and Rubinshtein 1958, Kantorovich and Akilov 2016). Recall that, for any function, we denote by $\text{Lip}(f)$ its Lipschitz constant and consider the generating class, $\mathcal{W} = \{f : \Xi \to \mathbb{R} \text{ s.t. } \text{Lip}(f) \le 1\}$. One can equivalently describe the Wasserstein distance as an integral probability metric by setting $\mathcal{F} = \mathcal{W}$ in (9).

4.1.2. Kolmogorov Distance. When Ξ is a subset of $\mathbb{R}$, one may consider the Kolmogorov distance, denoted as d_K and defined for all probability measures μ and ν as

$$d_K(\mu,\nu) = \sup_{t\in\Xi} \left| \int_{-\infty}^{t} d\mu(\xi) - \int_{-\infty}^{t} d\nu(\xi) \right|.$$

The Kolmogorov distance is therefore the uniform distance between the cumulative distribution functions associated with μ and ν. Furthermore, it is clear that by considering $\mathcal{K} := \{\xi \in \mathbb{R} \longmapsto \mathbb{1}\{\xi \le t\} | t \in \mathbb{R}\}$, one may equivalently define the Kolmogorov as an integral probability metric by setting $\mathcal{F} = \mathcal{K}$ in (9). In what follows, whenever we refer to an instance in which the distance is the Kolmogorov one, we implicitly assume that the space Ξ is a subset of $\mathbb{R}$.

4.2. Bound on $\text{Unif}_{\mathcal{I}}$ Under IPM Heterogeneity

In this section, we develop a general analysis of $\text{Unif}_{\mathcal{I}}$ defined in (8) for different instances of data-driven decision-making problems under heterogeneous environments by relating it to the generating class of the integral probability metric of interest.

4.2.1. Bound on $\text{Unif}_{\mathcal{I}}$ Through the Approximation Parameter. Let $\mathcal{F}$ be a class of functions and let $d_{\mathcal{F}}$ be the induced integral probability metric. We develop an upper bound for every $\mu, \nu \in \mathcal{P}$ on $\text{Unif}_{\mathcal{I}}(\mu,\nu)$, defined in (8), by relating the objective function g to the generating class of functions $\mathcal{F}$.

First, remark that if the objective function $g(x,\cdot) : \Xi \to \mathbb{R}$ lies in $\mathcal{F}$ for all actions $x \in \mathcal{X}$, then by definition,

$\sup_{x\in\mathcal{X}}|\mathcal{G}(x,\mu)-\mathcal{G}(x,\nu)|\le d_{\mathcal{F}}(\mu,\nu)$. More generally, for an action $x\in\mathcal{X}$, if there exist $f\in\mathcal{F}$ and $\delta\ge 0$ such that $\|f-g(x,\cdot)\|_\infty\le\delta$, we have that, for any μ and ν,

$$\begin{aligned}
&|\mathcal{G}(x,\mu)-\mathcal{G}(x,\nu)|\\
&=\left|\int_{\xi\in\Xi}g(x,\xi)d\mu(\xi)-\int_{\xi\in\Xi}g(x,\xi)d\nu(\xi)\right|\\
&\le\left|\int_{\xi\in\Xi}(g(x,\xi)-f(\xi))d\mu(\xi)-\int_{\xi\in\Xi}(g(x,\xi)-f(\xi))d\nu(\xi)\right|\\
&\quad+\left|\int_{\xi\in\Xi}f(\xi)d\mu(\xi)-\int_{\xi\in\Xi}f(\xi)d\nu(\xi)\right|\\
&\le 2\|f-g(x,\cdot)\|_\infty+\sup_{\tilde f\in\mathcal{F}}\left|\int_{\xi\in\Xi}\tilde f(\xi)d\mu(\xi)-\int_{\xi\in\Xi}\tilde f(\xi)d\nu(\xi)\right|\\
&\le 2\delta+d_{\mathcal{F}}(\mu,\nu),
\end{aligned}$$

where the last inequality holds because $\|f-g(x,\cdot)\|_\infty\le\delta$ and by definition of $d_{\mathcal{F}}$.

Therefore, $\mathrm{Unif}_{\mathcal{I}}$ can be bounded by studying how well the function class $\mathcal{F}$ approximates the family of objective functions $(g(x,\cdot))_{x\in\mathcal{X}}$. We formalize this notion by introducing the approximation parameter.

Definition 5 (Approximation Parameter). Given a class of functions $\mathcal{F}$, an objective function g, an action $x\in\mathcal{X}$, and a real number $\delta\ge 0$, we define the approximation parameter $\lambda_x(\mathcal{F},g,\delta)$ as

$$\lambda_x(\mathcal{F},g,\delta):=\inf_{(\lambda_i)_{i\in\mathbb{N}},(f_i)_{i\in\mathbb{N}}}\ \sum_{i=1}^{\infty}|\lambda_i| \tag{10a}$$

$$\text{subject to}\quad \limsup_{m\to\infty}\left\|\sum_{i=1}^{m}\lambda_i\cdot f_i-g(x,\cdot)\right\|_\infty\le\delta, \tag{10b}$$

$$f_i\in\mathcal{F},\ \lambda_i\in\mathbb{R}. \tag{10c}$$

Furthermore, we let $\lambda(\mathcal{F},g,\delta)=\sup_{x\in\mathcal{X}}\lambda_x(\mathcal{F},g,\delta)$.

The approximation parameter quantifies the complexity required to approximate, within a uniform distance of at most δ, all functions $(g(x,\cdot))_{x\in\mathcal{X}}$ with functions from the generating class $\mathcal{F}$.[4] The objective function $\sum_{i=1}^{\infty}|\lambda_i|$ represents the notion of complexity, and the Constraint (10b) enforces that $g(x,\cdot)$ is well approximated. In the special case where $\delta=0$ and $g(x,\cdot)\in\mathcal{F}$, we note that $\lambda_x(\mathcal{F},g,\delta)\le 1$ (which can be obtained by setting $\lambda_1=1$, $\lambda_i=0$ for $i\ge 2$ and letting $f_1=g(x,\cdot)$). Furthermore, the Constraint (10b) is looser as δ grows; therefore, $\lambda(\mathcal{F},g,\delta)$ is decreasing in δ. Finally, we note that when Problem (10) is infeasible for some $x\in\mathcal{X}$, then $\lambda_x(\mathcal{F},g,\delta)=+\infty$ (which is the standard value of an infeasible minimization problem); hence, in that case, $\lambda(\mathcal{F},g,\delta)=+\infty$.

Our next main result relates the uniform deviation $\mathrm{Unif}_{\mathcal{I}}$ to the approximation parameter of the problem.

Theorem 2 (Bounding Uniform Deviation Under IPM Heterogeneity). *Let $\mathcal{F}$ be a class of functions, and let $d_{\mathcal{F}}$ be the associated integral probability metric. Let $\mathcal{I}=(\mathcal{X},\Xi,\mathcal{P},d_{\mathcal{F}},g)$ be a data-driven decision-making problem under heterogeneity; then, for every $\mu,\nu\in\mathcal{P}$, we have that*

$$\mathrm{Unif}_{\mathcal{I}}(\mu,\nu)\le\inf_{\delta\ge 0}\{\lambda(\mathcal{F},g,\delta)\cdot d_{\mathcal{F}}(\mu,\nu)+2\delta\}.$$

Theorem 2 establishes an upper bound on $\mathrm{Unif}_{\mathcal{I}}$ by relating the objective function to the structure of the heterogeneity distance through the approximation parameter. Therefore, the analysis of $\mathrm{Unif}_{\mathcal{I}}$ (which will imply bounds on the performance of SAA) can be reduced to bounding the approximation parameter. We next show how to derive simple upper bounds on the approximation parameter for common IPMs.

4.2.2. Bounds on the Approximation Parameter Through the Maximal Generator.

For two generating classes of functions $\mathcal{F}\subset\mathcal{F}'$, an objective g and $\delta\ge 0$, we note that the approximation parameter satisfies $\lambda(\mathcal{F}',g,\delta)\le\lambda(\mathcal{F},g,\delta)$. Furthermore, we note that the generating class is not unique for a given distance. Therefore, this motivates the study of the largest class of functions generating a particular distance in order to derive the tightest possible bounds. The notion of "largest class" has been extensively analyzed in Müller (1997) and is formally defined below.

Definition 6 (Maximal Generator). Let $\mathcal{F}$ be a generating class, and let $d_{\mathcal{F}}$ be the associated integral probability metric. The set $\mathfrak{M}_{\mathcal{F}}$ of all functions f satisfying

$$\left|\int_{\xi\in\Xi}f(\xi)d\mu(\xi)-\int_{\xi\in\Xi}f(\xi)d\nu(\xi)\right|\le d_{\mathcal{F}}(\mu,\nu)\quad\text{for all }\mu\text{ and }\nu\in\mathcal{P}$$

is called a maximal generator.

Müller (1997) characterizes the maximal generator for many common integral probability metrics. In particular, Müller shows the following:

Lemma 1 (Common Maximal Generators (Müller 1997)).
1. *$\mathfrak{M}_{\mathcal{K}}$ is the set of one-dimensional measurable functions f with total variation* $\mathrm{V}(f)\le 1$.
2. *$\mathfrak{M}_{\mathcal{W}}$ is the set of measurable functions f with Lipschitz constant* $\mathrm{Lip}(f)\le 1$.

We note that the maximal generator for the Kolmogorov distance is different from the initial generating class $\mathcal{K}$, whereas the maximal generator for the Wasserstein distance corresponds to the initial class $\mathcal{W}$.

Furthermore, remark that for any function f with bounded variation and bounded Lipschitz constant, we have that $\frac{f}{\mathrm{V}(f)}\in\mathfrak{M}_{\mathcal{K}}$ and $\frac{f}{\mathrm{Lip}(f)}\in\mathfrak{M}_{\mathcal{W}}$. Therefore, Proposition 1 immediately implies the following bounds on the approximation parameter for common distances.

Corollary 1. *For any objective function g, we have that*

$$\lambda(\mathfrak{M}_{\mathcal{W}},g,0)\le\sup_{x\in\mathcal{X}}\mathrm{Lip}(g(x,\cdot)).$$

Furthermore, if $\Xi = [0, M]$ *for some* $M > 0$*, we have that*

$$\lambda(\mathfrak{M}_{\mathcal{K}}, g, 0) \le \sup_{x \in \mathcal{X}} \mathrm{V}(g(x, \cdot)).$$

Corollary 1 enables us to derive bounds on $\mathrm{Unif}_{\mathcal{I}}$ by solely understanding analytical properties of each function in the family $(g(x,\cdot))_{x\in\mathcal{X}}$. We note that whereas we stated this result for the two IPMs we are analyzing in this paper, this methodology readily generalizes to other distances such as the total variation distance, the Dudley metric, or for IPMs for which the function class $\mathcal{F}$ is the unit ball in a reproducing kernel Hilbert space (Sriperumbudur et al. 2010). These other IPMs can be related to alternative analytical properties of the functions $(g(x,\cdot))_{x\in\mathcal{X}}$ (e.g., differentiability, continuity).

On the flip side, Corollary 1 does not enable us to leverage the full generality of Theorem 2, as it only applies to settings where for all $x \in \mathcal{X}$, $\frac{1}{L} g(x,\cdot) \in \mathfrak{M}_{\mathcal{F}}$ for some constant L and $\lambda(\mathfrak{M}_{\mathcal{F}}, g, 0) < \infty$. Whereas this result is already sufficient to handle a broad class of problems, we highlight that there exist problem instances such that $\lambda(\mathfrak{M}_{\mathcal{F}}, g, 0) = \infty$ and for which one can still derive meaningful bounds on $\mathrm{Unif}_{\mathcal{I}}$ by bounding $\lambda(\mathfrak{M}_{\mathcal{F}}, g, \delta)$ for $\delta > 0$. For sake of completeness, we next state a more general result, which enables us to derive a stronger geometric connection between the approximation parameter and the maximal generator. Recall that for a set $\mathcal{S}$, we denote by $\mathrm{span}(\mathcal{S})$ the set generated by finite linear combinations of elements from $\mathcal{S}$. Given a maximal generator $\mathfrak{M}_{\mathcal{F}}$, we define $\overline{\mathrm{span}(\mathfrak{M}_{\mathcal{F}})}$ as the closure of the set $\mathrm{span}(\mathfrak{M}_{\mathcal{F}})$ (where the closure is taken with respect to the topology induced by the supremum norm $\|\cdot\|_\infty$). The following result provides a more detailed connection between the analytical properties of $g(x,\cdot)$, the geometry of the generating class of function $\mathcal{F}$, and the approximation parameter.

Proposition 2 (Geometric Interpretation of the Approximation Parameter). *Let* $\mathcal{F}$ *be a class of functions, let* $d_{\mathcal{F}}$ *be the associated integral probability metric, and let* g *be an objective function. Then,* $g(x,\cdot) \in \overline{\mathrm{span}(\mathfrak{M}_{\mathcal{F}})}$ *if and only if, for every* $\delta > 0$, $\lambda_x(\mathcal{F}, g, \delta) < +\infty$.

Proposition 2 shows that finiteness of the approximation parameter at x for arbitrarily small δ is equivalent to the fact that $g(x,\cdot)$ is in the closure of $\mathrm{span}(\mathfrak{M}_{\mathcal{F}})$, which topologically means that $g(x,\cdot)$ is well approximated by a linear combination of functions in $\mathfrak{M}_{\mathcal{F}}$. We note that Proposition 2 considerably generalizes the simpler argument requiring that $g(x,\cdot) \in \mathrm{span}(\mathfrak{M}_{\mathcal{F}})$ to obtain that $\lambda_x(\mathfrak{M}_{\mathcal{F}}, g, 0) < \infty$. The next example illustrates how Proposition 2 can be used to derive more general bounds by using analytical properties of $g(x,\cdot)$.

Example 1. Consider the maximal generator $\mathfrak{M}_{\mathcal{K}}$ associated with the Kolmogorov distance. Corollary 1 implies that if $g(x,\cdot)$ has bounded variation, that is, $V(g(x,\cdot)) < \infty$, then $\lambda(\mathfrak{M}_{\mathcal{K}}, g, 0) < \infty$. This corresponds to the case where $g(x,\cdot) \in \mathrm{span}(\mathfrak{M}_{\mathcal{K}})$. However, if $g(x,\cdot)$ is only assumed to be continuous without any assumption on its variation, one can show that $g(x,\cdot) \in \overline{\mathrm{span}(\mathfrak{M}_{\mathcal{K}})}$ and conclude from Proposition 2 that $\lambda_x(\mathfrak{M}_{\mathcal{K}}, g, \delta) < \infty$ for all $\delta > 0$. This argument, in combination with Theorem 2, leads to bounds on $\mathrm{Unif}_{\mathcal{I}}$ in this setting. We expand more on this in Online Appendix C.3.

We next complement our upper bound results from a lower bound, which allows us to further relate the uniform deviation and the approximation parameter.

Proposition 3. *Let* $\mathcal{F}$ *be a class of functions, and let* $d_{\mathcal{F}}$ *be the associated integral probability metric. Let* $\mathcal{I} = (\mathcal{X}, \Xi, \mathcal{P}, d_{\mathcal{F}}, g)$ *be a data-driven decision-making problem under heterogeneity. Assume that* $\sup_{x\in\mathcal{X}} \|g(x,\cdot)\|_\infty < \infty$*. If* $\lambda(\mathfrak{M}_{\mathcal{F}}, g, 0) = +\infty$*; then,*

$$\limsup_{\epsilon \to 0} \frac{1}{\epsilon} \cdot \sup_{\mu \in \mathcal{P}} \sup_{\nu \in \mathcal{U}_\epsilon(\mu)} \mathrm{Unif}_{\mathcal{I}}(\mu, \nu) = \infty.$$

Proposition 3 shows that, to some extent, the approximation parameter provides a characterization of the behavior of the worst-case uniform deviation for small ϵ. Indeed, when the approximation parameter is infinite for $\delta = 0$, the worst-case uniform deviation is at least $\omega(\epsilon)$ when ϵ goes to zero. We note that Proposition 3 does not preclude the uniform deviation to converge to zero at a slower rate than ϵ. As a matter of fact, this can happen by bounding the approximation parameter for $\delta > 0$, as discussed in Example 1. We also acknowledge that Proposition 3 only provides a lower bound on the uniform deviation, which does not provide a lower bound on the original regret of interest—it only establishes a limitation of using the uniform deviation to analyze regret.

Remark 2 (Comparison with the Method of Probability Metrics). Rachev and Römisch (2002) derive a result conceptually close to the result we have established in this section. In particular, given a generating class $\mathcal{F}$, they establish that whenever $(g(x,\cdot))_{x\in\mathcal{X}} \subset \kappa \cdot \mathcal{F}$ for some $\kappa > 0$, one can bound the uniform deviation $\mathrm{Unif}_{\mathcal{I}}(\mu, \nu)$ by $\kappa \cdot d_{\mathcal{F}}(\mu, \nu)$. Our analysis considerably expands this reasoning by (i) considering the maximal generator of a generating class, and (ii) allowing for $g(x,\cdot)$ to be in the topological closure (with respect to the uniform convergence norm). Point (i) allows us to assess whether the uniform deviation is sensitive to the heterogeneity structure through analytical properties of the objective function. Furthermore, the lower bound in Proposition 3 shows that the maximal generator is in fact a driver of the behavior of $\mathrm{Unif}_{\mathcal{I}}$. Finally, (ii) is required to establish guarantees in settings where the objective function needs to be approximated by the generating class, as discussed in Online Appendix C.3.

4.3. Bounding Scheme for SAA Under IPMs and Applications

We next leverage the relation between the performance of SAA and the quantity $\mathrm{Unif}_{\mathcal{I}}$ (see (7)), along with the analysis developed in Section 4.2, to derive a bound on the asymptotic worst-case regret of SAA. In particular, we obtain the following results:

Proposition 4. *Let $\Xi = [0, M]$ for some $M > 0$. Let $\mathcal{I}_K = (\mathcal{X}, \Xi, \mathcal{P}, d_K, g)$ and $\mathcal{I}_W$ be instances of the data-driven decision problem under the Kolmogorov and Wasserstein distances, respectively. Then, for every $\epsilon \geq 0$,*

$$\mathfrak{R}^{\pi^{\mathrm{SAA}}}_{W,\infty}(\epsilon) \leq 2 \sup_{x\in\mathcal{X}} \mathrm{Lip}(g(x,\cdot)) \cdot \epsilon$$
$$\mathfrak{R}^{\pi^{\mathrm{SAA}}}_{K,\infty}(\epsilon) \leq 2 \sup_{x\in\mathcal{X}} \mathrm{V}(g(x,\cdot)) \cdot \epsilon.$$

Proposition 4 translates the bounds derived on $\mathrm{Unif}_{\mathcal{I}}$ through the approximation parameter in Section 4.2 into bounds on the worst-case asymptotic regret of SAA. It implies that for a broad class of problems, the asymptotic worst-case regret of SAA vanishes as the radius of the heterogeneity ball goes to zero. More specifically, the asymptotic worst-case regret scales linear for the Kolmogorov distance whenever the objective function g has uniformly bounded variation and for the Wasserstein distance whenever the objective function g is uniformly Lipschitz continuous.

We prove this result by first showing that (7) holds using Theorem 1. We then leverage the results developed in Section 4.2.

We next illustrate the application of Proposition 4 for two central problems.

4.3.1. Newsvendor Problem: Performance of SAA.

We first consider the newsvendor problem, a widely studied model in operations. Using $M > 0$ to denote an upper bound on demand, we let $\Xi = \mathcal{X} = [0, M]$, and our regret bounds may (or may not) depend on M. The objective function is the newsvendor cost, which is parameterized by two quantities: c_u, the underage cost, and c_o, the overage cost. The cost of carrying an inventory level $x \in \mathcal{X}$ and then observing demand $\xi \in \Xi$ is

$$g(x, \xi) = c_u(\xi - x)^+ + c_o(x - \xi)^+, \tag{11}$$

where $(\cdot)^+ := \max\{\cdot, 0\}$ is the positive part operator. The measure space $\mathcal{P}$ is the set of probability measures on Ξ without restrictions, and for every $\mu \in \mathcal{P}$, we define the cost of the oracle as $\mathrm{opt}(\mu) = \inf_{x\in\mathcal{X}} \mathcal{G}(x, \mu)$.[5]

We note that for the newsvendor objective g defined in (11), we have that g lies in the linear span of the maximal generator class for Kolmogorov and Wasserstein distances. In particular,

$$\sup_{x\in\mathcal{X}} \mathrm{V}(g(x,\cdot)) = \max(c_u, c_o) \cdot M \quad \text{and}$$
$$\sup_{x\in\mathcal{X}} \mathrm{Lip}(g(x,\cdot)) = \max(c_u, c_o),$$

where the suprema are attained at some action $x \in \{0, M\}$. Therefore, one can derive a vanishing regret guarantee for SAA under these two forms of heterogeneity by exploiting Proposition 4. Our next result formalizes this and presents upper bounds on the asymptotic worst-case regret of SAA. We also derive matching lower bounds on the minimax achievable regret by any data-driven policy.

Proposition 5 (Heterogeneous Newsvendor). *For the data-driven newsvendor problem in a heterogenous environment under the Kolmogorov and the Wasserstein distances, we have for ϵ small enough that*

$$\frac{c_u + c_o}{2} \cdot M \cdot \epsilon \leq \inf_{\pi\in\Pi} \mathfrak{R}^{\pi}_{K,\infty}(\epsilon) \leq \mathfrak{R}^{\pi^{\mathrm{SAA}}}_{K,\infty}(\epsilon) \leq 2\max(c_u, c_o) \cdot M \cdot \epsilon,$$
$$\frac{c_u + c_o}{2} \cdot \epsilon \leq \inf_{\pi\in\Pi} \mathfrak{R}^{\pi}_{W,\infty}(\epsilon) \leq \mathfrak{R}^{\pi^{\mathrm{SAA}}}_{W,\infty}(\epsilon) \leq 2\max(c_u, c_o) \cdot \epsilon.$$

Proposition 5 first shows that SAA is robust to deviation under multiple forms of heterogeneity and that the regret scales linearly with the radius of the heterogeneity ball for both distances. We show, furthermore, that the linear dependence is, for the newsvendor problem, the best rate achievable for any data-driven policy across these forms of heterogeneity.

The upper bounds derived in Proposition 5 naturally follow from Proposition 4. We also note that the lower bound obtained for the newsvendor depends linearly on the total variation and the Lipschitz constant through the factor $(c_u + c_o)$. This therefore shows that these characterizing quantities for the function g are indeed respectively driving the regret of SAA in this case.

It is also worth noting that for the newsvendor problem, there is a significant difference between considering an initial "naïve" generating class for an integral probability metric and the maximal generator. For instance, when considering the Kolmogorov distance, it would have been more challenging to remark that the newsvendor cost can be well approximated by the generating class $\mathcal{K} = \{\xi \in \mathbb{R} \longmapsto \mathbb{1}\{\xi \leq t\} | t \in \mathbb{R}\}$, whereas it is straightforward when considering $\mathfrak{M}_{\mathcal{K}}$ the set of functions with bounded variation.

Whereas SAA achieves vanishing regret for both forms of heterogeneity in the newsvendor problem, we next analyze the performance of SAA for pricing and show that SAA may fail in some cases.

4.3.2. Pricing Problem: Performance of SAA and Potential Failure.

We next consider the classical data-driven pricing. The environment space $\Xi = [0, M]$ represents the set of possible values. The set of possible prices $\mathcal{X}$ is $[0, M]$. The set of considered probability measures $\mathcal{P}$ is the whole set of probability measures on $[0, M]$ without restriction. The revenue generated by the DM with a price decision $x \in \mathcal{X}$ when facing a

customer with value $\xi \in \Xi$ is

$$g(x,\xi) = x \cdot \mathbb{1}\{\xi \geq x\}. \tag{12}$$

Therefore, for any probability measure $\mu \in \mathcal{P}$, we have that $\mathcal{G}(x,\mu) = x \cdot \mathbb{P}_\mu(\xi \geq x)$. The goal of the DM is to set a price x in order to maximize its revenue. Equivalently, the goal is to minimize the regret against the oracle which posts the optimal price given a willingness-to-pay (wtp) distribution.

We note that for the pricing objective g defined in (12), we have that g lies in the linear span of the maximal generator class for Kolmogorov. In particular,

$$\sup_{x\in\mathcal{X}} \mathrm{V}(g(x,\cdot)) = M,$$

where the supremum for $\mathrm{V}(g(x,\cdot))$ is attained for $x = M$. Therefore, Proposition 4 implies directly an upper bound on the asymptotic worst-case performance of SAA, which we complement by a lower bound on the performance of any data-driven policy.

Proposition 6 (Kolmogorov Pricing). *For the data-driven pricing problem in a heterogenous environment under the Kolmogorov distance, for any ϵ small enough, we have*

$$\frac{M\cdot\epsilon}{2} \leq \inf_{\pi\in\Pi} \mathfrak{R}^{\pi}_{K,\infty}(\epsilon) \leq \mathfrak{R}^{\pi^{\mathrm{SAA}}}_{K,\infty}(\epsilon) \leq 2M\cdot\epsilon.$$

4.3.3. Failure of SAA. Note that the pricing problem $g(x,\cdot)$ is not Lipschitz continuous for any $x > 0$ (in fact, it is not even continuous) and, hence, does not lie in $\mathrm{span}(\mathfrak{M}_{\mathcal{W}})$. Therefore, for this problem, $\lambda(\mathfrak{M}_{\mathcal{W}}, g, 0) = \infty$, and Proposition 3 implies that $\mathrm{Unif}_{\mathcal{I}}$ is at least supralinear for small ϵ. Whereas Proposition 3 does not provide a lower bound on SAA itself (because $\mathrm{Unif}_{\mathcal{I}}$ is an upper bound on the performance of SAA), we formally derive in our next result a lower bound on the asymptotic worst-case regret of SAA.

Proposition 7 (Failure of SAA for Wasserstein Pricing). *For the data-driven pricing problem in a heterogenous environment under the Wasserstein distance, we have for $\epsilon > 0$ that*

$$\mathfrak{R}^{\pi^{\mathrm{SAA}}}_{W,\infty}(\epsilon) = M.$$

Proposition 7 shows that for the pricing problem, the asymptotic regret of SAA under Wasserstein heterogeneity does not shrink to zero as ϵ goes to zero. As a matter of fact, the regret of M is the worst possible, as the revenue of the oracle is bounded above by M, and the revenue of any data-driven policy is bounded below by zero.

The key challenge in pricing is that by shifting the willingness-to-pay distribution from a point mass $\nu = \delta_1$ to a point mass $\mu = \delta_{1-\epsilon}$, the revenue obtained by posting the price \$1 drops from \$1 to \$0. Furthermore, the Wasserstein distance between two point masses that are close enough is small (as opposed to the Kolmogorov distance, which is one in that case). Quite notably, this result holds for any radius of heterogeneity ϵ, and an arbitrarily small deviation from the i.i.d. case leads to extremely poor performance of SAA, even with infinite data.

4.3.4. Additional Applications. Whereas the purpose of the above concrete examples was to establish the variety of cases that many emerge, the framework readily applies to other problem classes such as the classical ski rental problem or multidimensional problems such as Bayesian mechanism design. We elaborate on these in Online Appendices E and F, respectively.

5. Alternative Policies: Design and Analysis

As seen in Section 4.3.2, the performance of SAA is poor for certain instances, even with infinitely many samples. Our goal in this section is twofold: first, we complement the discussion sparked by Proposition 7 by investigating whether there exists a policy that can ensure a vanishing regret for the pricing problem under Wasserstein heterogeneity. Then, we discuss open directions toward a principled approach to design and analyze policies beyond SAA.

5.1. Rate-Optimal Policy for the Wasserstein Pricing Problem

Recall the pricing problem with Wasserstein distance introduced in Section 4.3.2. For this problem, we have seen that SAA incurs an asymptotic worst-case regret that is not vanishing as ϵ goes to zero. We next present an alternative sample-size-agnostic policy for which the asymptotic worst-case vanishes as ϵ goes to zero. Furthermore, we characterize the worst-case performance of that policy, showing that it has the best possible dependence with respect to ϵ.

To be able to design and analyze these alternative policies, Theorem 1 is critical because (i) any analysis involving the uniform deviation $\mathrm{Unif}_{\mathcal{I}}$ will lead to a nonvanishing upper bound (because the worst-case regret of SAA is a lower bound on $\mathrm{Unif}_{\mathcal{I}}$), and (ii) we need to derive upper bounds on $\mathfrak{R}^{\pi}_{W,\infty}(\epsilon)$ for policies π for which standard upper bounding schemes are not present in the literature (the analysis of Mohri and Muñoz Medina 2012 only applies to SAA).

Fix $\delta \in \mathbb{R}$. For every $\nu \in \mathcal{P}$ and $n \in \mathbb{N}^*$, we define the δ-SAA policy as

$$\pi^{\mathrm{SAA}(\delta)}(n,\nu) = \pi^{\mathrm{SAA}}(\nu) - \delta,$$

where we project the decision on the closest point in $\mathcal{X}$ if it is outside the action space. Given that SAA is a sample-size-agnostic policy, it is clear that $\pi^{\mathrm{SAA}(\delta)}$ is also sample size agnostic. Therefore, in what follows, we drop the dependence in the sample size. Our next

result shows that for an appropriate choice of δ, the policy $\pi^{\mathrm{SAA}(\delta)}$ can achieve a rate-optimal performance.

Proposition 8 (Rate-Optimal Regret for Wasserstein Pricing). *Consider the data-driven pricing problem in a heterogenous environment under the Wasserstein distance. For any ϵ small enough, let $\delta = \sqrt{M \cdot \epsilon}$. Then,*

$$\mathfrak{R}^{\pi^{\mathrm{SAA}(\delta)}}_{W,\infty}(\epsilon) \le 4 \cdot \sqrt{M \cdot \epsilon}.$$

Furthermore,

$$\frac{\sqrt{M \cdot \epsilon}}{4} \le \inf_{\pi \in \Pi} \mathfrak{R}^{\pi}_{W,\infty}(\epsilon).$$

Proposition 8 shows that the asymptotic worst-case regret of $\pi^{\mathrm{SAA}(\delta)}$ scales as $\mathcal{O}(\sqrt{M \cdot \epsilon})$ when ϵ goes to zero. It highlights the need to adapt to the heterogeneity structure in pricing. Quite notably, the significant loss of performance of SAA can be mitigated by slightly robustifying the SAA decision and by deflating the SAA price.

Proposition 8 also implies that the dependence of the asymptotic worst-case regret of $\pi^{\mathrm{SAA}(\delta)}$ in terms of ϵ is optimal across all data-driven policies. Therefore, this very mild modification of the decision of SAA does not only improve performance but actually leads to a near-optimal one. We finally remark that the $\pi^{\mathrm{SAA}(\delta)}$ policy improves over SAA when one can make use of the knowledge of ϵ and d. It is worth noting that SAA does not need such knowledge. A natural question, which is beyond the scope of the current paper, would be to understand the best achievable performance without knowledge of ϵ and/or d.

5.1.1. Proof Sketch. To prove Proposition 8, we first leverage Theorem 1 to reduce the analysis of $\pi^{\mathrm{SAA}(\delta)}$ to the DRO regret problem. We then decompose the DRO regret objective for any policy π as follows:

$$\begin{aligned}\mathrm{opt}(\mu) - \mathcal{G}(\pi(\nu), \mu) = {} & [\mathrm{opt}(\mu) - \mathrm{opt}(\nu)] \\ & + [\mathrm{opt}(\nu) - \mathcal{G}(\pi(\nu), \mu)]. \end{aligned} \tag{13}$$

The decomposition in (13) starkly differs from the bound derived in (7) to analyze SAA. Indeed, the latter involves the uniform difference $\mathrm{Unif}_{\mathcal{I}}(\mu, \nu)$, quantifying the difference in performance when changing the distribution from μ to ν for a *fixed action* x, whereas the former involves two differences, where both the action and the distribution change.

To bound the difference in objectives when changing both actions and distributions, we develop a structural result that leverages properties of the pricing objective. In particular, the key observation is to remark that for the pricing problem, the revenue cannot abruptly *decrease* when decreasing prices (a property that is false when increasing prices). Using this notable fact, one can control the difference in revenues at two different prices. Formally, the following bound holds.

Lemma 2. *Consider the pricing objective. Let μ and $\nu \in \mathcal{P}$ be two different distributions, and let x_1 and x_2 be two different prices such that $x_1 < x_2$. We have that*

$$\mathcal{G}(x_2, \nu) - \mathcal{G}(x_1, \mu) \le (x_2 - x_1) + M \cdot \frac{d_W(\mu, \nu)}{x_2 - x_1}.$$

Lemma 2 unveils an interesting trade-off faced by a DM who prices in a heterogeneous environment. By selecting appropriately the difference between two prices x_1 and x_2, one can ensure that the gap in revenues for two different distributions can be controlled by the Wasserstein distance. Intuitively, one should think of ν as the historical empirical distribution, μ as the out-of-sample distribution, x_2 as the optimal decision when facing ν, and x_1 as the actual price posted by the DM. In that case, Lemma 2 implies that by deflating appropriately, the revenue generated by the posted price x_1 against the out-of-sample distribution μ cannot be much lower than the optimal revenue generated for distribution ν.

We leverage Lemma 2 to show that

$$\mathrm{opt}(\mu) - \mathrm{opt}(\nu) \le 2\sqrt{M \cdot d_W(\mu, \nu)}. \tag{14}$$

We then conclude the proof by using Lemma 2 again to show that for an appropriately chosen δ, $\mathrm{opt}(\nu) - \mathcal{G}(\pi^{\mathrm{SAA}(\delta)}(\nu), \mu)$ is also order $\mathcal{O}(\sqrt{M \cdot d_W(\mu, \nu)})$, which concludes the proof.

We note that related arguments allow us to design alternative policies for the ski rental with Wasserstein distance for which SAA performs poorly. Furthermore, quite notably, the performance of SAA can be significantly improved upon for the ski rental problem, even under the Kolmogorov distance. We discuss in detail the derivation of alternative policies for ski rental in Online Appendix E.

5.2. Toward a General Method To Design Policies

In Section 5.1, we have seen that there are problems, such as the pricing problem with Wasserstein distance, for which one can obtain significantly better performance than SAA by using a specific alternate policy to SAA. In this section, we propose initial steps toward a general paradigm to design and analyze policies other than SAA in heterogeneous environments.

5.2.1. Regret DRO Policy. Theorem 1 suggests a systematic way to design policies that perform well in heterogeneous environments when the number of samples becomes large—minimize the upper bound constructed in Theorem 1. This implies solving, for

every $\epsilon \geq 0$, the problem

$$\inf_{\pi\in\Pi_{\text{agn}}} \mathfrak{R}^{\pi}_{\mathcal{I},\text{DRO}}(\epsilon) = \inf_{\pi\in\Pi_{\text{agn}}} \sup_{\mu\in\mathcal{P}} \sup_{\nu\in\mathcal{U}_\epsilon(\mu)} |\text{opt}(\mu) - \mathcal{G}(\pi(\nu),\mu)| = \sup_{\nu\in\mathcal{P}} \inf_{\pi\in\Pi_{\text{agn}}} \sup_{\mu\in\mathcal{U}_\epsilon(\nu)} |\text{opt}(\mu) - \mathcal{G}(\pi(\nu),\mu)|,$$

where the last equality holds because the policy observes the historical distribution. In settings where the infimum is achieved, the previous equality suggests the following sample-size-agnostic policy. For every $\nu \in \mathcal{P}$,

$$\pi^{\text{RDRO}}(\nu) = \arg\min_{\rho\in\Delta(\mathcal{X})} \sup_{\mu\in\mathcal{U}_\epsilon(\nu)} \{\text{opt}(\mu) - \mathcal{G}(x,\mu)\}. \tag{15}$$

We note that π^{RDRO} resembles the common DRO policies studied in the literature (and formally defined in (16)) to the difference that the problem in (15) involves a *regret objective*. We show in Proposition D.6 (see Online Appendix D.3) that the Regret-DRO (RDRO) policy π^{RDRO}, which emerges from optimizing the surrogate DRO regret, is, in fact, optimal for the asymptotic worst-case regret $\mathfrak{R}_{\mathcal{I},\infty}$. We note that π^{RDRO} is a priori challenging to compute, as it requires us to solve a minimax nonconvex, infinite-dimensional optimization problem. We leave as open the exciting computational question regarding efficient implementations of (15).

5.2.2. Absolute DRO Policy. A natural policy in a heterogeneous environment that has been widely studied in the literature and resembles the policy defined in (15) is the DRO policy defined as

$$\pi^{\text{DRO}}(\nu) = \arg\max_{\rho\in\Delta(\mathcal{X})} \inf_{\mu\in\mathcal{U}_\epsilon(\nu)} \mathcal{G}(\rho,\mu). \tag{16}$$

Methods to compute the solution of the max-min problem when the maximization is over deterministic actions[6] have been developed under different assumptions on $\mathcal{G}$ and for different structures of the sets $\mathcal{U}_\epsilon$ (Rahimian and Mehrotra 2019). Our next result shows that the standard π^{DRO} policy has an asymptotic worst-case regret "close" to the one of the minimax optimal policy.

Proposition 9. *Let d be a convex distance satisfying the ETC property, and let $\mathcal{I} = (\mathcal{X},\Xi,\mathcal{P},d,g)$ be a data-driven decision problem in a heterogeneous environment. Then, for every $\epsilon \geq 0$, we have*

$$\lim_{\eta\to\epsilon^-} \mathfrak{R}^{\pi^{\text{DRO}}}_{\mathcal{I},\infty}(\eta) \leq \inf_{\pi\in\Pi} \mathfrak{R}^{\pi}_{\mathcal{I},\infty}(\epsilon) + \sup_{\mu\in\mathcal{P}} \sup_{\nu\in\mathcal{U}_{2\epsilon}(\mu)} \{\text{opt}(\mu) - \text{opt}(\nu)\}.$$

Proposition 9 shows that the difference between the performance of the commonly studied policy π^{DRO} and the best asymptotic performance among all data-driven policies is at most $\sup_{\mu\in\mathcal{P}} \sup_{\nu\in\mathcal{U}_{2\epsilon}(\mu)}\{\text{opt}(\mu) - \text{opt}(\nu)\}$ (when $\mathfrak{R}^{\pi^{\text{DRO}}}_{\mathcal{I},\infty}$ is continuous at ϵ). Furthermore, Proposition 9 suggests a general methodology to prove that the π^{DRO} policy achieves the best dependence in ϵ when ϵ is small. Indeed, assume that one can establish that there exists some constant $C > 0$ such that for every ϵ small enough,

$$\sup_{\mu\in\mathcal{P}} \sup_{\nu\in\mathcal{U}_\epsilon(\mu)} \{\text{opt}(\mu) - \text{opt}(\nu)\} \leq C \cdot \inf_{\pi\in\Pi} \mathfrak{R}^{\pi}_{\mathcal{I},\infty}(\epsilon). \tag{17}$$

Then, one can conclude that for ϵ small enough, the asymptotic worst-case regret of π^{DRO} is at most a constant factor away from the optimal achievable performance. Thus, in that case, π^{DRO} has the best possible dependence in terms of ϵ achievable by any data-driven policy.

We note that Proposition 9 can be used to prove that the DRO policy achieves rate optimality for the pricing problem with Wasserstein distance. Indeed, the lower bound derived in Proposition 8 along with the bound established in (14) implies that (17) holds.

This section highlights that π^{DRO} is a strong candidate to be a general rate-optimal policy. Similarly to π^{RDRO}, we leave as open the question of being able to *compute* the π^{DRO} policy.

6. Open Questions and Concluding Remarks

All in all, our framework enables us to develop a common language to compare achievable performance across central classes of problems and to unveil novel insights about the performance of the SAA policy when slightly deviating from the widely studied i.i.d. regime. In settings where SAA fails, we also design algorithms achieving rate-optimal asymptotic guarantees. A key takeaway of this analysis across a broad class of problems and for different heterogeneity structures is that it is necessary to understand the structure of the problem we are facing but also the nature of the heterogeneity when designing data-driven policies that are robust to these environments. Whereas our work mainly focuses on the Kolmogorov and Wasserstein distances to illustrate our framework, we note that one can naturally extend it for ϕ-divergences or study alternative metrics. We discuss in Online Appendix A.2.1 how our results for these two distances imply results for the total variation and the KL-divergence. This work also opens many additional questions. We summarize below what we believe are exciting future research directions.

First, Theorem 1 applied to SAA establishes an upper bound that is at least as tight as the previous one present in the literature (Mohri and Muñoz Medina 2012), which involves the uniform deviation (see (7)). In this work, we mainly used Theorem 1 to analyze non-SAA policies for which no bound were previously established, but our general analysis of SAA in Section 4 still goes through the uniform deviation. Whereas this was sufficient to derive tight guarantees on the rate of the

worst-case regret, a natural question is whether using the tighter upper bound in Theorem 1 can lead to improved guarantees. We show in Online Appendix G that for the newsvendor problem under Kolmogorov heterogeneity, the bound derived by Theorem 1 is *strictly* tighter than (7), which leaves open the question of using our theorem to derive problem-specific guarantees on SAA with better constant factors.

Whereas the theory developed in Section 4 was sufficient to establish tight regret guarantees for the problem classes and heterogeneity structures explored in this work, an important question remains unanswered: does having a low approximation parameter truly constitute a necessary condition for SAA to perform well? The general lower bound presented in Proposition 3 does not fully address this, leaving open the opportunity for future research to clarify the sensitivity of empirical optimization methods to heterogeneity.

Furthermore, our work focuses mainly on the evaluation of data-driven policies and the understanding of the achievable performance across heterogeneity structures and problem classes. It leaves open the question of designing *general* alternative data-driven policies that can be efficiently computed and that perform well. Section 5.2 presents results that suggest that both the RDRO and the DRO policies are very good candidate policies in terms of achievable performance, but it leaves open the question of computing these policies or deriving approximations that would still perform well.

Finally, this paper opens up broader questions. Our analysis characterizes the performance in the asymptotic regime where the sample size grows large and could be complemented by quantifying the performance of policies with finite samples in heterogeneous environments. Another key question would be to understand whether there exists a policy that does not use the knowledge of the type of heterogeneity and performs well across heterogeneity types. We believe that additional exciting and practical complements to this work include incorporating contexts to historical samples, deriving statistical tests to characterize the type of heterogeneity along with its radius, and potentially providing an empirical validation of such procedures.

Acknowledgments

The authors thank the editor, the associate editor, as well as three anonymous referees for valuable suggestions that led to significant improvements in the final version of the paper.

Endnotes

[1] It is also known under different names, for example, empirical optimization or empirical risk minimization.

[2] Rigorously, one should specify a σ-algebra $\mathcal{A}$ on Ξ. We do not specify explicitly these when not necessary to define the objects.

[3] We refer the reader to Online Appendix A.1 for a discussion regarding the definition of policies on probability measures that may not be empirical ones.

[4] We note that the sequence of functions $(f_i)_{i\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$ in Definition 5 is restricted to be countable. Formalizing an optimization problem over an uncountable set of functions as variables remains an open question.

[5] Our framework was introduced for maximization problems (see Section 2), but it readily extends to minimization problems.

[6] In general, taking randomized actions in the max-min problem can strictly improve the performance of the policy (Delage et al. 2019).

References

Allouah A, Bahamou A, Besbes O (2022) Pricing with samples. *Oper. Res.* 70(2):1088–1104.

Babaioff M, Gonczarowski YA, Mansour Y, Moran S (2018) Are two (samples) really better than one? *EC'18: Proc. 2018 ACM Conf. Econom. Comput.* (Association for Computing Machinery, New York), 175–175.

Ben-David S, Blitzer J, Crammer K, Kulesza A, Pereira F, Vaughan JW (2010) A theory of learning from different domains. *Machine Learn.* 79(1):151–175.

Bennouna A, Van Parys B (2022) Holistic robust data-driven decisions. Preprint, submitted July 19, https://arxiv.org/abs/2207.09560.

Bergemann D, Schlag K (2011) Robust monopoly pricing. *J. Econom. Theory* 146(6):2527–2543.

Bertsimas D, Brown DB, Caramanis C (2011) Theory and applications of robust optimization. *SIAM Rev.* 53(3):464–501.

Bertsimas D, Gupta V, Kallus N (2018) Robust sample average approximation. *Math. Programming* 171(1):217–282.

Besbes O, Mouchtaki O (2023) How big should your data really be? Data-driven newsvendor: Learning one sample at a time. *Management Sci.* 69(10):5848–5865.

Besbes O, Gur Y, Zeevi A (2014) Stochastic multi-armed-bandit problem with non-stationary rewards. *NIPS'14: Proc. 28th Internat. Conf. Neural Inform. Processing Systems*, vol. 1 (MIT Press, Cambridge, MA), 199–207.

Bilodeau B, Negrea J, Roy DM (2020) Relaxing the I.I.D. assumption: Adaptively minimax optimal regret via root-entropic regularization. Preprint, submitted July 13, https://arxiv.org/abs/2007.06552.

Blanc G, Lange J, Malik A, Tan L-Y (2022) On the power of adaptivity in statistical adversaries. *Proc. 35th Conf. Learn. Theory (COLT)*, vol. 178 (PMLR, New York), 5030–5061.

Blanchard M (2022) Universal online learning: An optimistically universal learning rule. *Proc. 35th Conf. Learn. Theory (COLT)*, vol. 178 (PMLR, New York), 1077–1125.

Block A, Dagan Y, Golowich N, Rakhlin A (2022) Smoothed online learning is as easy as statistical learning. *Proc. 35th Conf. Learn. Theory (COLT)*, vol. 178 (PMLR, New York), 1716–1786.

Borodin A, El-Yaniv R (2005) *Online Computation and Competitive Analysis* (Cambridge University Press, Cambridge, UK).

Brustle J, Cai Y, Daskalakis C (2020) Multi-item mechanisms without item-independence: Learnability via robustness. *EC'20: Proc. 21st ACM Conf. Econom. Comput.* (Association for Computing Machinery, New York), 715–761.

Cai Y, Daskalakis C (2017) Learning multi-item auctions with (or without) samples. *2017 IEEE 58th Annual Sympos. Foundations Comput. Sci. (FOCS)* (IEEE, Piscataway, NJ), 516–527.

Chen X, Wang Y (2022) Robust dynamic pricing with demand learning in the presence of outlier customers. *Oper. Res.* 71(4):1362–1386.

Cheung WC, Simchi-Levi D (2019) Sampling-based approximation schemes for capacitated stochastic inventory control models. *Math. Oper. Res.* 44(2):668–692.

Cheung WC, Simchi-Levi D, Zhu R (2022) Hedging the drift: Learning to optimize under nonstationarity. *Management Sci.* 68(3): 1696–1713.

Daskalakis C, Zampetakis M (2020) More revenue from two samples via factor revealing SDPs. *EC'20: Proc. 21st ACM Conf. Econom. Comput.* (Association for Computing Machinery, New York), 257–272.

Delage E, Kuhn D, Wiesemann W (2019) "Dice"-sion–making under uncertainty: When can a random decision reduce risk? *Management Sci.* 65(7):3282–3301.

Diakonikolas I, Kontonis V, Tzamos C, Vakilian A, Zarifis N (2021) Learning online algorithms with distributional advice. *Proc. 38th Internat. Conf. Machine Learn. (ICML)*, vol. 139 (PMLR, New York), 2687–2696.

Diakonikolas I, Kamath G, Kane D, Li J, Moitra A, Stewart A (2019) Robust estimators in high-dimensions without the computational intractability. *SIAM J. Comput.* 48(2):742–864.

Duchi JC, Namkoong H (2021) Learning models with uniform performance via distributionally robust optimization. *Ann. Statist.* 49(3):1378–1406.

Dupačová J, Gröwe-Kuska N, Römisch W (2003) Scenario reduction in stochastic programming. *Math. Programming* 95(3):493–511.

Dütting P, Kesselheim T (2019) Posted pricing and prophet inequalities with inaccurate priors. *Proc. 2019 ACM Conf. Econom. Comput.* (Association for Computing Machinery, New York), 111–129.

Eren SS, Maglaras C (2010) Monopoly pricing with limited demand information. *J. Revenue Pricing Management* 9(1–2):23–48.

Fu H, Immorlica N, Lucier B, Strack P (2015) Randomization beats second price as a prior-independent auction. *Proc. 16th ACM Conf. Econom. Comput.* (Association for Computing Machinery, New York), 323–323.

Gallego G, Moon I (1993) The distribution free newsboy problem: Review and extensions. *J. Oper. Res. Soc.* 44(8):825–834.

Gibbs AL, Su FE (2002) On choosing and bounding probability metrics. *Internat. Statist. Rev.* 70(3):419–435.

Glasserman P, Xu X (2014) Robust risk measurement and model risk. *Quant. Finance* 14(1):29–58.

Guo W, Jordan M, Zampetakis E (2021) Robust learning of optimal auctions. *NIPS'21: Proc. 34th Internat. Conf. Neural Inform. Processing Systems* (Curran Associates, Red Hook, NY), 21273–21284.

Haghtalab N, Roughgarden T, Shetty A (2022a) Smoothed analysis with adaptive adversaries. *2021 IEEE 62nd Annual Sympos. Foundations Comput. Sci. (FOCS)* (IEEE, Piscataway, NJ), 942–953.

Haghtalab N, Han Y, Shetty A, Yang K (2022b) Oracle-efficient online learning for smoothed adversaries. *NeurIPS'22: Proc. 35th Internat. Conf. Neural Inform. Processing Systems* (Curran Associates, Red Hook, NY), 4072–4084.

Hampel FR (1971) A general qualitative definition of robustness. *Ann. Math. Statist.* 42(6):1887–1896.

Hanneke S (2021) Learning whenever learning is possible: Universal learning under general stochastic processes. *J. Machine Learn. Res.* 22(130):1–116.

Haussler D (1992) Decision theoretic generalizations of the PAC model for neural net and other learning applications. *Inform. Comput.* 100(1):78–150.

Huang Z, Mansour Y, Roughgarden T (2018) Making the most of your samples. *SIAM J. Comput.* 47(3):651–674.

Huber PJ (1992) Robust estimation of a location parameter. Kotz S, Johnson NL, eds. *Breakthroughs in Statistics*, Springer Series in Statistics (Springer, New York), 492–518.

Kantorovich LV, Akilov GP (2016) *Functional Analysis* (Elsevier, Amsterdam).

Kantorovich LV, Rubinshtein S (1958) On a space of totally additive functions. *Vestnik St. Petersburg Uni. Math.* 13(7):52–59.

Karnin ZS, Anava O (2016) Multi-armed bandits: Competing with optimal sequences. *NIPS'16: Proc. 30th Internat. Conf. Neural Inform. Processing Systems* (Curran Associates, Red Hook, NY), 199–207.

Kearns M, Li M (1993) Learning in the presence of malicious errors. *SIAM J. Comput.* 22(4):807–837.

Kearns MJ, Schapire RE, Sellie LM (1994) Toward efficient agnostic learning. *Machine Learn.* 17(2):115–141.

Kim S, Pasupathy R, Henderson SG (2015) A guide to sample average approximation. Fu M, ed. *Handbook of Simulation Optimization*, International Series in Operations Research & Management Science, vol. 216 (Springer, New York), 207–243.

Kleywegt AJ, Shapiro A, Homem-de Mello T (2002) The sample average approximation method for stochastic discrete optimization. *SIAM J. Optim.* 12(2):479–502.

Klivans AR, Long PM, Servedio RA (2009) Learning halfspaces with malicious noise. *J. Machine Learn. Res.* 10(12):2715–2740.

Lam H (2021) On the impossibility of statistically improving empirical optimization: A second-order stochastic dominance perspective. Preprint, submitted May 27, https://arxiv.org/abs/2105.13419.

Levi R, Perakis G, Uichanco J (2015) The data-driven newsvendor problem: New bounds and insights. *Oper. Res.* 63(6):1294–1306.

Lin M, Huh WT, Krishnan H, Uichanco J (2022) Data-driven newsvendor problem: Performance of the sample average approximation. *Oper. Res.* 70(4):1996–2012.

Luo H, Wei C-Y, Agarwal A, Langford J (2018) Efficient contextual bandits in non-stationary worlds. *Proc. 31st Conf. Learn. Theory (COLT)*, vol. 75 (PMLR, New York), 1739–1776.

Lykouris T, Mirrokni V, Paes Leme R (2018) Stochastic bandits robust to adversarial corruptions. *STOC 2018: Proc. 50th Annual ACM SIGACT Sympos. Theory Comput.* (Association for Computing Machinery, New York), 114–122.

Lykouris T, Simchowitz M, Slivkins A, Sun W (2021) Corruption-robust exploration in episodic reinforcement learning. *Proc. 34th Conf. Learn. Theory (COLT)*, vol. 134 (PMLR, New York), 3242–3245.

Mansour Y, Mohri M, Rostamizadeh A (2009) Domain adaptation: Learning bounds and algorithms. Preprint, submitted February 9, https://arxiv.org/abs/0902.3430.

Mohajerin Esfahani P, Kuhn D (2018) Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Math. Programming* 171(1):115–166.

Mohri M, Muñoz Medina A (2012) New analysis and algorithm for learning with drifting distributions. Bshouty NH, Stoltz G, Vayatis N, Zeugmann T, eds. *Algorithmic Learn. Theory. ALT 2012*, Lecture Notes in Computer Science, vol. 7568 (Springer, Berlin, Heidelberg), 124–138.

Morse PM, Kimball GE (1946) Methods of operations research. Technical report, Operations Evaluation Group, Center for Naval Analyses, Alexandria, VA.

Müller A (1997) Integral probability metrics and their generating classes of functions. *Adv. Appl. Probab.* 29(2):429–443.

Myerson RB (1981) Optimal auction design. *Math. Oper. Res.* 6(1): 58–73.

Perakis G, Roels G (2008) Regret in the newsvendor model with partial information. *Oper. Res.* 56(1):188–203.

Pichler A, Xu H (2022) Quantitative stability analysis for minimax distributionally robust risk optimization. *Math. Programming* 191(1):47–77.

Purohit M, Svitkina Z, Kumar R (2018) Improving online algorithms via ML predictions. *NIPS'18: Proc. 32nd Internat. Conf. Neural Inform. Processing Systems* (Curran Associates, Red Hook, NY), 9684–9693.

Rachev ST, Römisch W (2002) Quantitative stability in stochastic programming: The method of probability metrics. *Math. Oper. Res.* 27(4):792–818.

Rahimian H, Mehrotra S (2019) Distributionally robust optimization: A review. Preprint, submitted August 13, https://arxiv.org/abs/1908.05659.

Rakhlin A, Sridharan K, Tewari A (2011) Online learning: Stochastic, constrained, and smoothed adversaries. *Proc. 25th Internat. Conf. Neural Inform. Processing Systems (NIPS'11)* (Curran Associates Inc., Red Hook, NY), 1764–1772.

Riley J, Zeckhauser R (1983) Optimal selling strategies: When to haggle, when to hold firm. *Quart. J. Econom.* 98(2):267–289.

Römisch W, Schultz R (1991) Stability analysis for stochastic programs. *Ann. Oper. Re.* 30(1):241–266.

Roughgarden T (2021) *Beyond the Worst-Case Analysis of Algorithms* (Cambridge University Press, Cambridge, UK).

Scarf H (1958) A min-max solution of an inventory problem. Arrow KJ, Karlin S, Scarf H, eds. *Studies in the Mathematical Theory of Inventory and Production* (Stanford University Press, Redwood City, CA), 201–209.

Schultz R (1996) Rates of convergence in stochastic programs with complete integer recourse. *SIAM J. Optim.* 6(4):1138–1152.

Servedio RA (2003) Smooth boosting and learning with malicious noise. *J. Machine Learn. Res.* 4:633–648.

Sriperumbudur BK, Gretton A, Fukumizu K, Schölkopf B, Lanckriet GR (2010) Hilbert space embeddings and metrics on probability measures. *J. Machine Learn. Res.* 11(50):1517–1561.

Vapnik V, Chervonenkis A (1974) *Theory of Pattern Recognition* (Nauka, Moscow). [In Russian.]

Xu H, Zhang S (2021) Quantitative statistical robustness in distributionally robust optimization models. *Pacific J. Optim.* 19(2):335–361.

Zolotarev VM (1984) Probability metrics. *Theory Probab. Appl.* 28(2): 278–302.

Copyright of Management Science is the property of INFORMS: Institute for Operations Research & the Management Sciences and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.